TM Forum Specification

Network as a Service (NaaS) Component Suite API REST Specification

TMF909
Release 21.5
October 2021

Latest Update: TM Forum Release 21.5.0	Member Evaluation
Version 4.0.0	IPR Mode: RAND

©TM Forum 2021. All Rights Reserved.

NOTICE

Copyright © TM Forum 2021. All Rights Reserved.

This document and translations of it may be copied and furnished to others, and derivative works that comment on or otherwise explain it or assist in its implementation may be prepared, copied, published, and distributed, in whole or in part, without restriction of any kind, provided that the above copyright notice and this section are included on all such copies and derivative works. However, this document itself may not be modified in any way, including by removing the copyright notice or references to TM FORUM, except as needed for the purpose of developing any document or deliverable produced by a TM FORUM Collaboration Project Team (in which case the rules applicable to copyrights, as set forth in the TM FORUM IPR Policy, must be followed) or as required to translate it into languages other than English.

The limited permissions granted above are perpetual and will not be revoked by TM FORUM or its successors or assigns.

This document and the information contained herein is provided on an "AS IS" basis and TM FORUM DISCLAIMS ALL WARRANTIES, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO ANY WARRANTY THAT THE USE OF THE INFORMATION HEREIN WILL NOT INFRINGE ANY OWNERSHIP RIGHTS OR ANY IMPLIED WARRANTIES OF MERCHANTABILITY OR FITNESS FOR A PARTICULAR PURPOSE.

Direct inquiries to the TM Forum office:

4 Century Drive, Suite 100
Parsippany, NJ 07054, USA
Tel No. +1 973 944 5100
Fax No. +1 973 998 7196
TM Forum Web Page: www.tmforum.org

Table of Contents

NOTICE 2
Table of Contents 3
List of Tables ..... **Error! Bookmark not defined.**
Introduction ..... **Error! Bookmark not defined.**
SAMPLE USE CASES 8
Support of polymorphism and extension patterns ..... **Error! Bookmark not defined.**
RESOURCE MODEL ..... **Error! Bookmark not defined.**
Managed Entity and Task Resource Models 11
FIRST resource ..... **Error! Bookmark not defined.**
Notification Resource Models 149
First Notification ..... **Error! Bookmark not defined.**
API OPERATIONS 171
VERB url ..... **Error! Bookmark not defined.**
API NOTIFICATIONS 346
Register listener 346
Unregister listener 348
Publish Event to listener 348
Acknowledgements 350
Release History 350
Contributors to Document 350

© TM Forum 2021. All Rights Reserved

LIST OF TABLES

N/A

© TM Forum 2021. All Rights Reserved

Executive Summary

The scope of this document is to define the API Component Suite in support of a set of Operational Domains exposing and managing the full lifecycle of “Network” Services. These services are referred to as Network as a Service (NaaS) and include all services offered from service providers including connectivity, end points, media services, etc.

An Operational Domain supports a set of functions and capabilities responsible for the complete lifecycle of services and resources within its domain, including exposure of services supported by SLAs, interfacing via standard TMF APIs with the BSS systems, other operational domains, and external developers. The API set supports the management of services from the design to order, configuration and activation, inventory, test, problem and usage management.

The first aim and objectives of the NaaS component suite is to set the API control plane contract between OSS/BSS (IT systems) and operational domains as well as between operational domains to operational domains and in the future to external developers and partners.

Whilst the API standards in this document define the framework and common header information for exposed services, it does not describe the service payload details which are to be defined by the individual operational domains.

© TM Forum 2021. All Rights Reserved

INTRODUCTION

The following document is the specification of the REST APIs for the Network as a Service (NaaS) Component Suite. This component suite covers the operations required to provide the functionality required by Operational Domains interworking with OSS/BSS applications and/or other domains from one service provider or from 3rd parties. One of the key requirements is the re-use of an API functionality against any technology to avoid creating many custom APIs reducing the initial and maintenance costs from suppliers.

The APIs:

- Will not be defining customer products or offers
- Will be implemented using the Hypermedia APIs with dynamic payloads that make the services self-descriptive wherever possible
- Will be operating on customer facing services or private services, the latter services only exposed towards other operational domains
- Will perform lifecycle actions to the expose network services (generally CFS)
- Communicate what the BSS needs back from the Network Services
- Will not be dealing with CRM, or Billing and will not have to care about a network function being virtualized or physical (That’s the domain responsibility) unless it is mandatory (such as a service end point).

© TM Forum 2021. All Rights Reserved

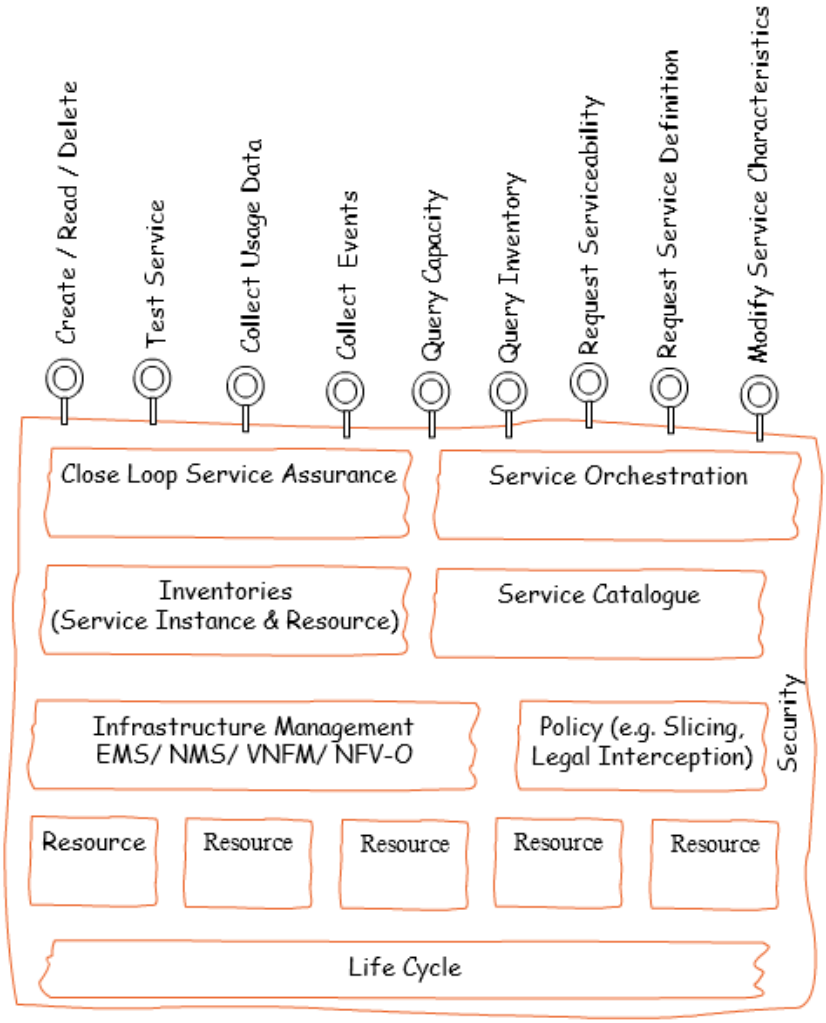


Figure 1. Operational Domain Management Functions and Capabilities

The NaaS API component suite supports the lifecycle functions required to manage the network capabilities exposed as Network as a Service and managed by operational domains.

© TM Forum 2021. All Rights Reserved

REQUIREMENTS AND USE CASES

This section describes the functions that are needed to deliver Network as a Service.

While “service” can be the service specification or service instance, within this document the use of unqualified service will always mean service instance. Service specification will also be fully qualified.

A network service typically has a long-life span and can undergo numerous changes throughout its life. A service may be created, fail, be restored, be made inactive (maybe for non-payment), be made active again, fail, and be restored, etc. throughout its life. As such this standard does not try to provide all the possible ordering of actions against a service.

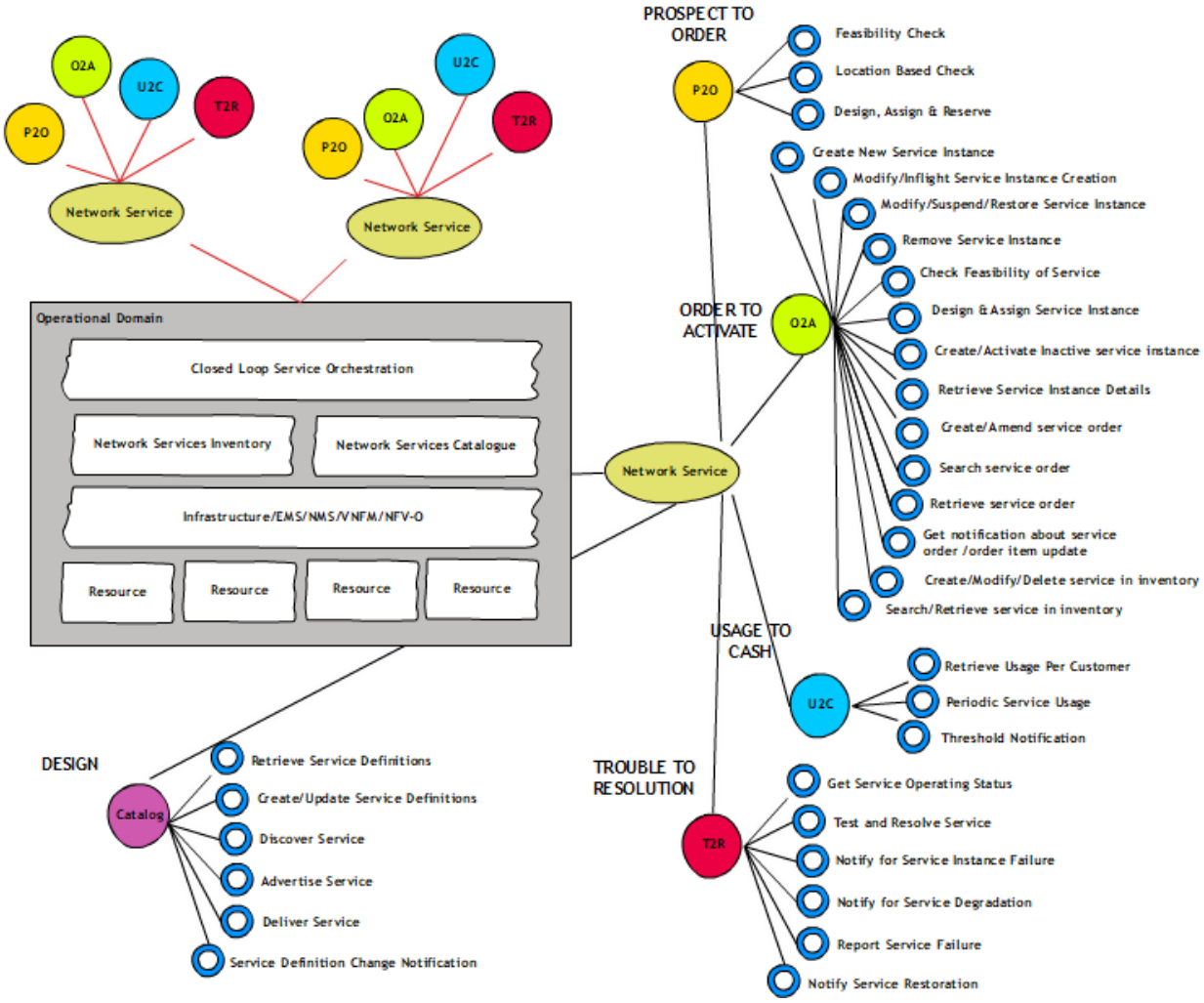
There is some inherent flexibility in the TMF Open API design to support different business models. One area where this flexibility is clear is whether the network exposes single services and service order processing is fully a BSS responsibility, or the network may accept the full or partial service order and service order processing is a network responsibility. The decision drives the usage of TMF 641 and/or 640 and is at the discretion of each CSP. You can find NaaS best practice recommendations in IG1224 NaaS Service Fulfillment guidelines https://www.tmforum.org/resources/reference/ig1224-naas-service-fulfillment-guidelines-v4-0-0/

Requirements

The NaaS API component suite includes APIs classified in 5 main lifecycle areas:

- Prospect to Order (P2O)
  - TMF645 Service Qualification
  - TMF640 Service Activation and Configuration
- Order to Activate (O2A)
  - TMF640 Service Activation and Configuration
  - TMF641 Service Ordering
- Trouble to Resolution (T2R)
  - TMF653 Service Test
  - TMF656 Service Problem
- Usage to Payment (also referred to as U2C - Usage to Cash)
  - TMF677 Usage Consumption
- Concept to Market
  - The output of service design process are the service specifications. The existence of the service specifications in the TMF Service Catalog (TMF633) is a pre-condition for this component suite standard

© TM Forum 2021. All Rights Reserved



© TM Forum 2021. All Rights Reserved

Support of polymorphism and extension patterns

Support of polymorphic collections and types and schema-based extension is provided by means of a list of generic meta-attributes that we describe below.

Generic support of polymorphism and pattern extensions is described in the TMF API Guidelines v3.0 Part 2 document.

The @type attribute provides a way to represent the actual class type of an entity. All resources and sub-resources of this API have a @type attributes that can be provided when this is useful. All resources and sub-resources of these APIs have a @type attributes that can be provided when this is useful.

The @referredType can be used within reference entities (like for instance ServiceRef) to explicitly denote the actual entity type of the referred class. Notice that in reference entities the @type, when used, denotes the class types of the reference itself, such as ServiceRef, and not the class type of the referred object. However, since reference classes are rarely sub-classed, @type is generally not useful in reference objects.

The @schemaLocation property can be used in resources to allow specifying user-defined properties of an Entity or to specify the expected characteristics of an entity.

The @baseType attribute gives a way to provide explicitly the base of class of a given resource that has been extended.

Notice that because these meta-attributes have a generic meaning, we will not repeat their definition in the resource description tables of each resource and each sub-resource.

© TM Forum 2021. All Rights Reserved

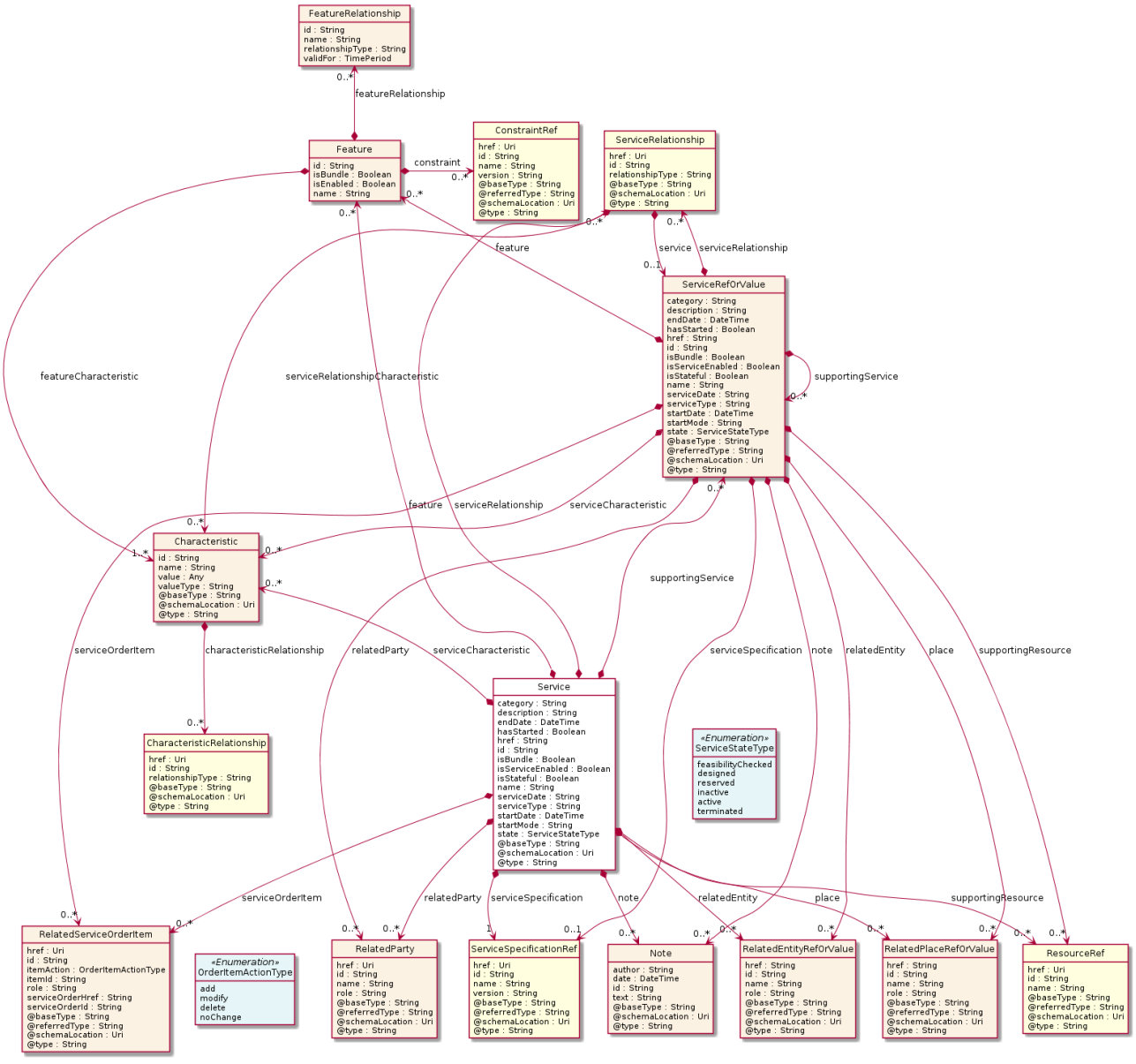
RESOURCE MODEL

Managed Entity and Task Resource Models

Service resource

Service is a base class for defining the Service hierarchy. All Services are characterized as either being possibly visible and usable by a Customer or not. This gives rise to the two subclasses of Service: CustomerFacingService and ResourceFacingService.

Resource model



Field descriptions

Service fields

© TM Forum 2021. All Rights Reserved

href	A string. Reference of the service.
id	A string. Unique identifier of the service.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
category	A string. Is it a customer facing or resource facing service.
description	A string. Free-text description of the service.
endDate	A date time (DateTime). Date when the service ends.
feature	A list of features (Feature [*]). A list of feature associated with this service.
hasStarted	A boolean. If TRUE, this Service has already been started.
href	A string. Reference of the service.
id	A string. Unique identifier of the service.
isBundle	A boolean. If true, the service is a ServiceBundle which regroup a service hierachy. If false, the service is a 'atomic' service (hierachy leaf).
isServiceEnabled	A boolean. If FALSE and hasStarted is FALSE, this particular Service has NOT been enabled for use - if FALSE and hasStarted is TRUE then the service has failed.
isStateful	A boolean. If TRUE, this Service can be changed without affecting any other services.
name	A string. Name of the service.
note	A list of notes (Note [*]). A list of notes made on this service.
place	A list of related place ref or values (RelatedPlaceRefOrValue [*]). A list of places (Place [*]). Used to define a place useful for the service (for example a geographical place whre the service is installed).
relatedEntity	A list of related entity ref or values (RelatedEntityRefOrValue [*]). A list of related entity in relationship with this service.
relatedParty	A list of related parties (RelatedParty [*]). A list of related party references (RelatedParty [*]). A related party defines party or party role linked to a specific entity.
serviceCharacteristic	A list of characteristics (Characteristic [*]). A list of characteristics that characterize this service (ServiceCharacteristic [*]).

© TM Forum 2021. All Rights Reserved

serviceDate	A string. Date when the service was created (whatever its status).
serviceOrderItem	A list of related service order items (RelatedServiceOrderItem [*]). A list of service order items related to this service.
serviceRelationship	A list of service relationships (ServiceRelationship [*]). A list of service relationships (ServiceRelationship [*]). Describes links with other service(s) in the inventory.
serviceSpecification	A service specification reference (ServiceSpecificationRef). The specification from which this service was instantiated.
serviceType	A string. Business type of the service.
startDate	A date time (DateTime). Date when the service starts.
startMode	A string. This attribute is an enumerated integer that indicates how the Service is started, such as: 0: Unknown; 1: Automatically by the managed environment; 2: Automatically by the owning device; 3: Manually by the Provider of the Service; 4: Manually by a Customer of the Provider; 5: Any of the above.
state	A service state type (ServiceStateType). The life cycle state of the service, such as designed, reserved, active, etc...
supportingResource	A list of resource references (ResourceRef [*]). A list of supporting resources (SupportingResource [*]).Note: only Service of type RFS can be associated with Resources.
supportingService	A list of service ref or values (ServiceRefOrValue [*]). A list of supporting services (SupportingService [*]). A collection of services that support this service (bundling, link CFS to RFS).

Characteristic sub-resource

Describes a given characteristic of an object or entity through a name/value pair.

@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
characteristicRelationship	A list of characteristic relationships (CharacteristicRelationship [*]). Another Characteristic that is related to the current Characteristic;.
id	A string. Unique identifier of the characteristic.
name	A string. Name of the characteristic.
value	An any (Any). The value of the characteristic.

© TM Forum 2021. All Rights Reserved

valueType	A string. Data type of the value of the characteristic.

CharacteristicRelationship sub-resource

Another Characteristic that is related to the current Characteristic;.

href	An uri (Uri). Hyperlink reference.
id	A string. Unique identifier of the characteristic.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
relationshipType	A string. The type of relationship.

Feature sub-resource

Configuration feature.

constraint	A list of constraint references (ConstraintRef [*]). This is a list of feature constraints.
featureCharacteristic	A list of characteristics (Characteristic [1..*]). This is a list of Characteristics for a particular feature.
featureRelationship	A list of feature relationships (FeatureRelationship [*]). Configuration feature.
id	A string. Unique identifier of the feature.
isBundle	A boolean. True if this is a feature group. Default is false.
isEnabled	A boolean. True if this feature is enabled. Default is true.
name	A string. This is the name for the feature.

FeatureRelationship sub-resource

Configuration feature.

id	A string. Unique identifier of the target feature.
name	A string. This is the name of the target feature.
relationshipType	A string. This is the type of the feature relationship.
validFor	A time period. The period for which this feature relationship is valid.

Note sub-resource

Extra information about a given entity.

© TM Forum 2021. All Rights Reserved

@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
author	A string. Author of the note.
date	A date time (DateTime). Date of the note.
id	A string. Identifier of the note within its containing entity.
text	A string. Text of the note.

RelatedEntityRefOrValue sub-resource

A reference to an entity, where the type of the entity is not known in advance. A related entity defines a entity described by reference or by value linked to a specific entity. The polymorphic attributes @type, @schemaLocation & @referredType are related to the Entity and not the RelatedEntityRefOrValue class itself.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	A string. Reference of the related entity.
id	A string. Unique identifier of a related entity.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
role	A string.

RelatedParty sub-resource

Related Party reference. A related party defines party or party role linked to a specific entity.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.

© TM Forum 2021. All Rights Reserved

@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
role	A string. Role played by the related party.

RelatedPlaceRefOrValue sub-resource

Related Entity reference. A related place defines a place described by reference or by value linked to a specific entity. The polymorphic attributes @type, @schemaLocation & @referredType are related to the place entity and not the RelatedPlaceRefOrValue class itself.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. A user-friendly name for the place, such as [Paris Store], [London Store], [Main Home].
href	A string. Unique reference of the place.
id	A string. Unique identifier of the place.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
role	A string.

RelatedServiceOrderItem sub-resource

RelatedServiceOrderItem (a ServiceOrder item) .The service order item which triggered service creation/change/termination.

href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
@referredType	A string. The actual type of the target instance when needed for disambiguation.
itemAction	An order item action type (OrderItemActionType). Action of the order item for this service.

© TM Forum 2021. All Rights Reserved

itemId	A string. Identifier of the order item where the service was managed.
role	A string. role of the service order item for this service.
serviceOrderHref	A string. Reference of the related entity.
serviceOrderId	A string. Unique identifier of a related entity.

ServiceRefOrValue sub-resource

A Service to be created defined by value or existing defined by reference. The polymorphic attributes @type, @schemaLocation & @referredType are related to the Service entity and not the RelatedServiceRefOrValue class itself.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the service.
href	A string. Reference of the service.
id	A string. Unique identifier of the service.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
category	A string. Is it a customer facing or resource facing service.
description	A string. Free-text description of the service.
endDate	A date time (DateTime). Date when the service ends.
hasStarted	A boolean. If TRUE, this Service has already been started.
isBundle	A boolean. If true, the service is a ServiceBundle which regroup a service hierachy. If false, the service is a 'atomic' service (hierachy leaf).
isServiceEnabled	A boolean. If FALSE and hasStarted is FALSE, this particular Service has NOT been enabled for use - if FALSE and hasStarted is TRUE then the service has failed.
isStateful	A boolean. If TRUE, this Service can be changed without affecting any other services.
serviceDate	A string. Date when the service was created (whatever its status).
serviceType	A string. Business type of the service.
startDate	A date time (DateTime). Date when the service starts.

© TM Forum 2021. All Rights Reserved

startMode	A string. This attribute is an enumerated integer that indicates how the Service is started, such as: 0: Unknown; 1: Automatically by the managed environment; 2: Automatically by the owning device; 3: Manually by the Provider of the Service; 4: Manually by a Customer of the Provider; 5: Any of the above.
feature	A list of features (Feature [*]). A list of feature associated with this service.
note	A list of notes (Note [*]). A list of notes made on this service.
place	A list of related place ref or values (RelatedPlaceRefOrValue [*]). A list of places (Place [*]). Used to define a place useful for the service (for example a geographical place whre the service is installed).
relatedEntity	A list of related entity ref or values (RelatedEntityRefOrValue [*]). A list of related entity in relationship with this service.
relatedParty	A list of related parties (RelatedParty [*]). A list of related party references (RelatedParty [*]). A related party defines party or party role linked to a specific entity.
serviceCharacteristic	A list of characteristics (Characteristic [*]). A list of characteristics that characterize this service (ServiceCharacteristic [*]).
serviceOrderItem	A list of related service order items (RelatedServiceOrderItem [*]). A list of service order items related to this service.
serviceRelationship	A list of service relationships (ServiceRelationship [*]). A list of service relationships (ServiceRelationship [*]). Describes links with other service(s) in the inventory.
serviceSpecification	A service specification reference (ServiceSpecificationRef). The specification from which this service was instantiated.
state	A service state type (ServiceStateType). The life cycle state of the service, such as designed, reserved, active, etc...
supportingResource	A list of resource references (ResourceRef [*]). A list of supporting resources (SupportingResource [*]).Note: only Service of type RFS can be associated with Resources.
supportingService	A list of service ref or values (ServiceRefOrValue [*]). A list of supporting services (SupportingService [*]). A collection of services that support this service (bundling, link CFS to RFS).

ServiceRelationship sub-resource

href	An uri (Uri). Hyperlink reference.

© TM Forum 2021. All Rights Reserved

id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
relationshipType	A string.
service	A service ref or value (ServiceRefOrValue). A Service to be created defined by value or existing defined by reference. The polymorphic attributes @type, @schemaLocation & @referredType are related to the Service entity and not the RelatedServiceRefOrValue class itself.
serviceRelationshipCharacteristic	A list of characteristics (Characteristic [*]). Describes a given characteristic of an object or entity through a name/value pair.

ConstraintRef relationship

Constraint reference. The Constraint resource represents a policy/rule applied to an entity or entity spec.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
version	A string. constraint version.

ResourceRef relationship

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.

© TM Forum 2021. All Rights Reserved

@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.

ServiceSpecificationRef relationship

Service specification reference: ServiceSpecification(s) required to realize a ProductSpecification.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
version	A string. Service specification version.

Json representation sample

We provide below the json representation of an example of a 'Service' resource object

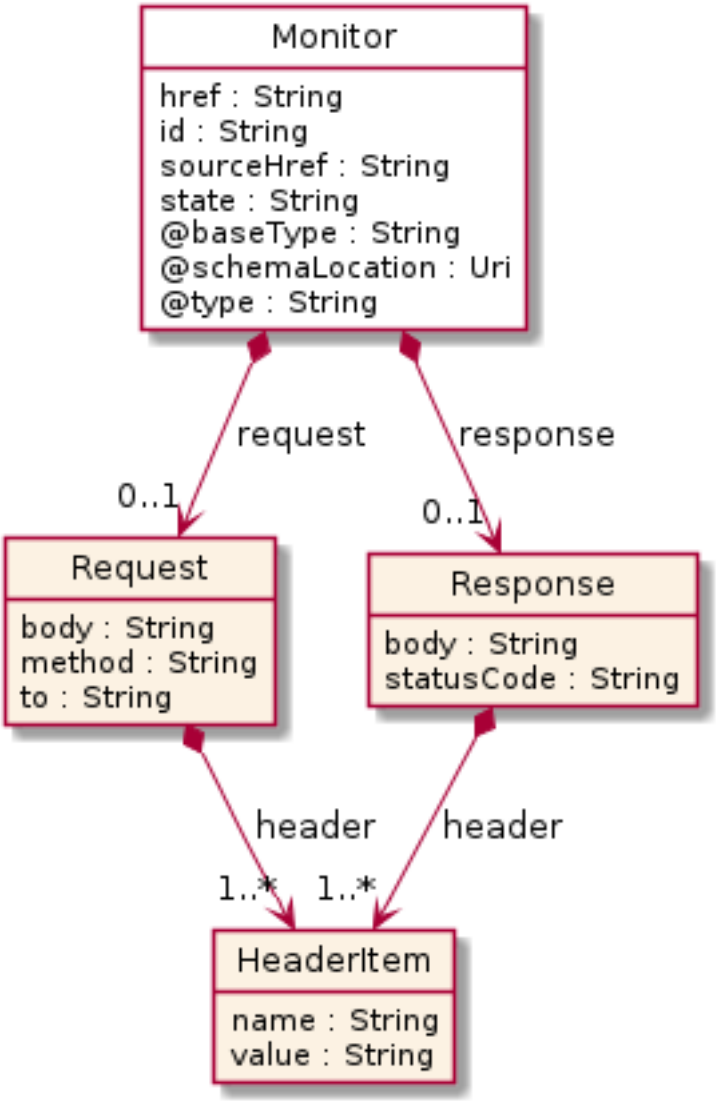

```
{
  "id": "5351",
  "href": "https://mycsp.com:8080/tmf-api/ServiceActivationAndConfiguration/v4/service/5351",
  "serviceType": "Cloud",
  "state": "Active",
  "serviceDate": "2018-01-15T12:26:11.747Z",
  "startDate": "2018-01-15T12:26:11.747Z",
  "serviceCharacteristic": [
    {
      "name": "vCPE_IP",
      "valueType": "object",
      "value": {
        "@type": "IPAddress",
        "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Service/IPAddress.schema.json",
        "address": "193.218.236.21"
      }
    }
  ],
  "@type": "vCPE",
  "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Service/vCPE.schema.json",
  "@baseType": "Service"
}
```


© TM Forum 2021. All Rights Reserved

Monitor resource

Monitoring of resources.

Resource model



Field descriptions

Monitor fields

href	A string. reference to this monitor.
id	A string. Identifier of an instance of the monitor. Required to be unique within the resource type. Used in URIs as the identifier for specific instances of a type.
@baseType	A string. When sub-classing, this defines the super-class.

© TM Forum 2021. All Rights Reserved

@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
href	A string. reference to this monitor.
id	A string. Identifier of an instance of the monitor. Required to be unique within the resource type. Used in URIs as the identifier for specific instances of a type.
request	A request (Request). Represents the request.
response	A response (Response). Represents the response.
sourceHref	A string. The monitored resource href.
state	A string. The Monitor state of the resource. InProgress, InError, Completed.

HeaderItem sub-resource

An item typically included in a request or response.

name	A string. The name of the header item, e.g. locale.
value	A string. The value of the header item, e.g. en-us.

Request sub-resource

A response to a request.

body	A string. The body of the request. For example for an HTTP request might contain content of a form .
header	A list of header items (HeaderItem [1..*]). Items included in the header of the request. For example for an HTTP request might contain requested locale, basic authentication.
method	A string. The protocol of the request, e.g. http.
to	A string. The target of the request, e.g. a URL for an HTTP request.

Response sub-resource

A response to a request.

body	A string. The body of the response. For example for an HTTP response might contain HTML for rendering.
header	A list of header items (HeaderItem [1..*]). Items included in the header of the response. For example for an HTTP response might contain negotiated locale.

© TM Forum 2021. All Rights Reserved

statusCode	A string. The status of the response. For example for an HTTP response would be codes such as 200, 400, etc.

Json representation sample

We provide below the json representation of an example of a 'Monitor' resource object

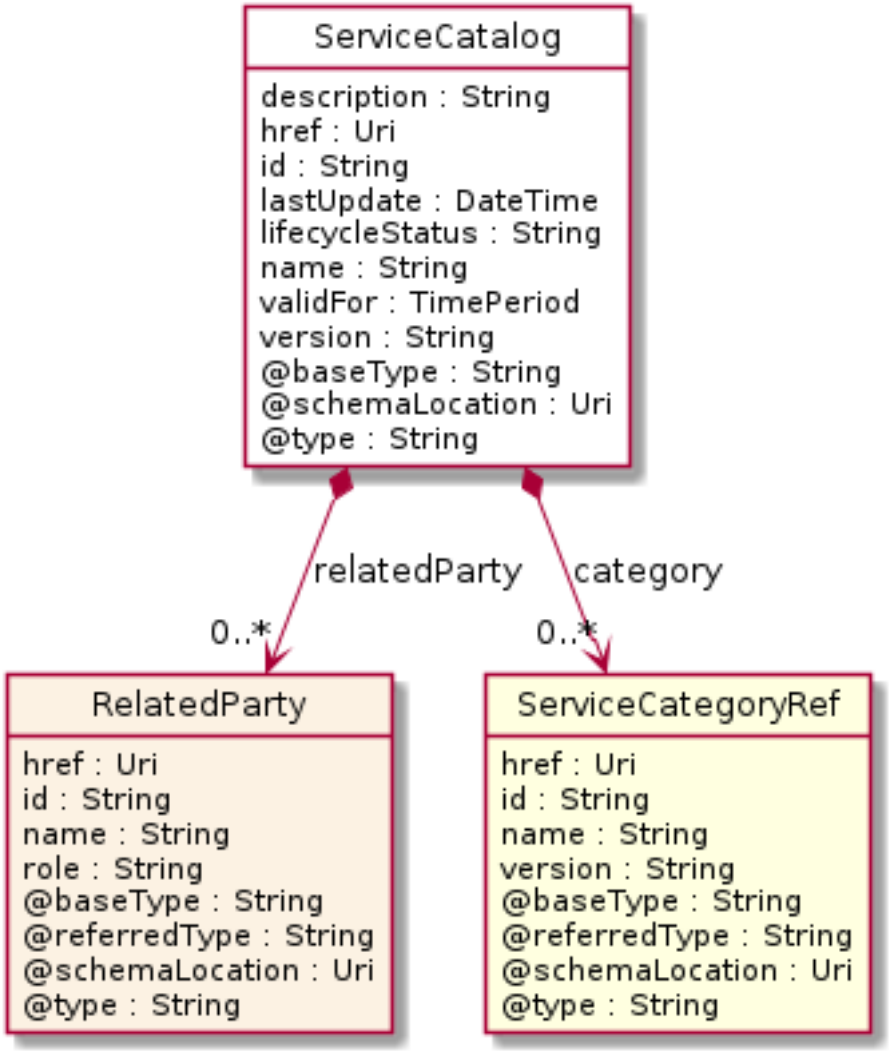

```
{
  "id": "44555",
  "href": "https://mycsp.com:8080/tmf-api/ServiceActivationAndConfiguration/v4/monitor/44555",
  "state": "In Progress",
  "request": {
    "method": "http",
    "to": "https://mycsp.com:5050/netAPIs/activate",
    "body": "ACTIVATE BBAND 144445",
    "header": [
      {
        "name": "locale",
        "value": "en-us"
      }
    ]
  },
  "response": {
    "statusCode": "344",
    "body": "ACTIVATE BBAND 144445 REQUEST RECEIVED PENDING",
    "header": [
      {
        "name": "locale",
        "value": "en-us"
      }
    ]
  },
  "sourceHref": "https://mycsp.com:8080/tmf-api/ServiceActivationAndConfiguration/v4/service/5351",
  "@type": "Monitor"
}
```


Service Catalog resource

The root entity for service catalog management.
A service catalog is a group of service specifications made available through service candidates that an organization provides to the consumers (internal consumers like its employees or B2B customers or B2C customers).
A service catalog typically includes name, description and time period that is valid for. It will have a list of ServiceCandidate catalog items. A ServiceCandidate is an entity that makes a ServiceSpecification available to a catalog.
A ServiceCandidate and its associated ServiceSpecification may be "published" - made visible -in any number of ServiceCatalogs, or in none.

Resource model

© TM Forum 2021. All Rights Reserved



Field descriptions

ServiceCatalog fields

href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
category	A list of service category references (ServiceCategoryRef [*]). List of service categories associated with this catalog.
description	A string. Description of this catalog.

© TM Forum 2021. All Rights Reserved

lastUpdate	A date time (DateTime). Date and time of the last update.
lifecycleStatus	A string. Used to indicate the current lifecycle status.
name	A string. Name of the service catalog.
relatedParty	A list of related parties (RelatedParty [*]). List of parties or party roles related to this category.
validFor	A time period. The period for which the service catalog is valid.
version	A string. ServiceCatalog version.

RelatedParty sub-resource

Related Party reference. A related party defines party or party role linked to a specific entity.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
role	A string. Role played by the related party.

ServiceCategoryRef relationship

The (service) category resource is used to group service candidates in logical containers. Categories can contain other categories.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.

© TM Forum 2021. All Rights Reserved

@type	A string. When sub-classing, this defines the sub-class Extensible name.
version	A string. Category version.

Json representation sample

We provide below the json representation of an example of a 'ServiceCatalog' resource object

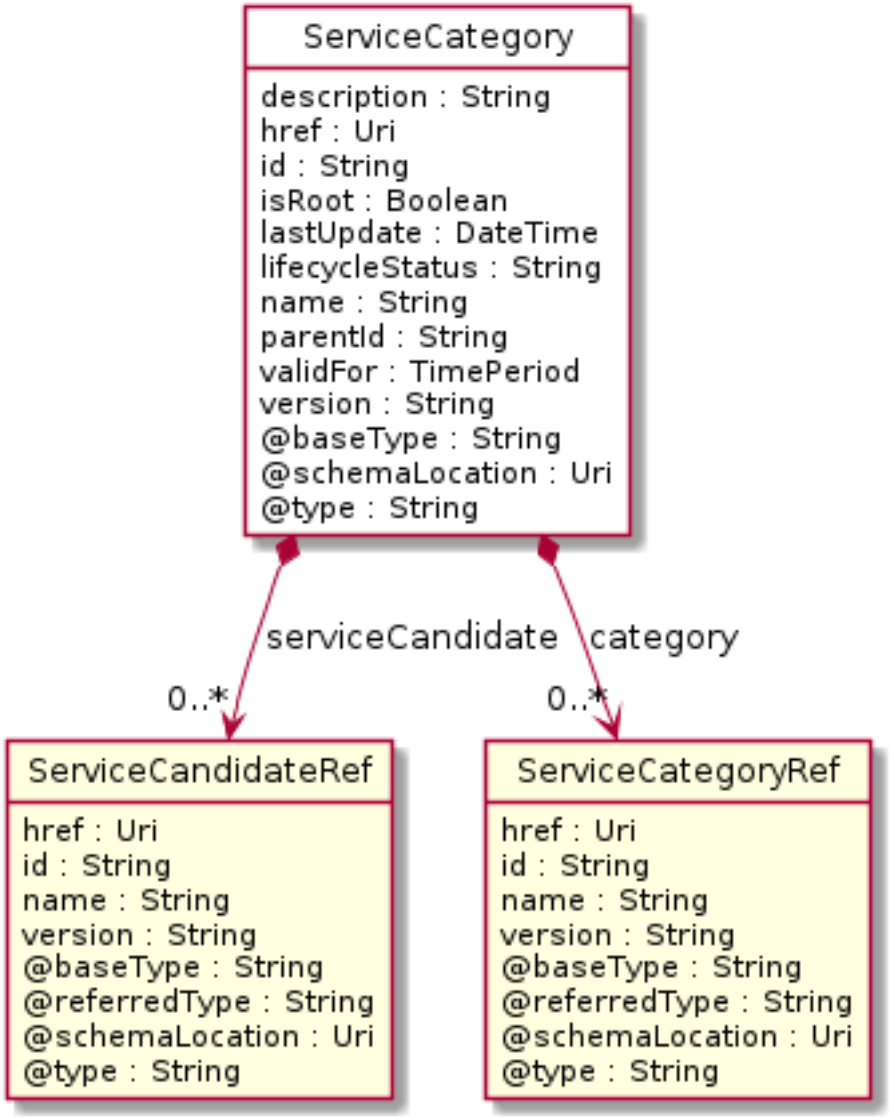

```
{
  "id": "3830",
  "href": "https://mycsp.com:8080/tmf-api/serviceCatalogManagement/v4/serviceCatalog/3830",
  "name": "Catalog Wholesale Business",
  "description": "This service catalog describes services that address the wholesale business segment.",
  "version": "1.0",
  "validFor": {
    "startDateTime": "2017-08-29T00:00:00Z",
    "endDateTime": "2018-03-25T00:00:00Z"
  },
  "lastUpdate": "2017-08-27T00:00:00Z",
  "lifecycleStatus": "Active",
  "relatedParty": [
    {
      "href": "https://mycsp.com:8080/tmf-api/partyManagement/v4/organization/3426",
      "id": "3426",
      "name": "Broadly Broad Ltd",
      "role": "vendor"
    }
  ],
  "category": [
    {
      "href": "https://mycsp.com:8080/tmf-api/serviceCatalogManagement/v4/category/7752",
      "id": "7752",
      "name": "IoT"
    }
  ],
  "@type": "ServiceCatalog",
  "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Service/ServiceCatalog.schema.json ",
  "@baseType": "Catalog"
}
```


Service Category resource

The (service) category resource is used to group service candidates in logical containers. Categories can contain other categories.

Resource model

© TM Forum 2021. All Rights Reserved



Field descriptions

ServiceCategory fields

href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
category	A list of service category references (ServiceCategoryRef [*]). List of child categories in the tree for in this category.

© TM Forum 2021. All Rights Reserved

description	A string. Description of the category.
isRoot	A boolean. If true, this Boolean indicates that the category is a root of categories.
lastUpdate	A date time (DateTime). Date and time of the last update.
lifecycleStatus	A string. Used to indicate the current lifecycle status.
name	A string. Name of the category.
parentId	A string. Unique identifier of the parent category.
serviceCandidate	A list of service candidate references (ServiceCandidateRef [*]). List of service candidates associated with this category.
validFor	A time period. The period for which the category is valid.
version	A string. ServiceCategory version.

ServiceCandidateRef relationship

ServiceCandidate reference. ServiceCandidate is an entity that makes a ServiceSpecification available to a catalog.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
version	A string. Version of the service candidate.

ServiceCategoryRef relationship

The (service) category resource is used to group service candidates in logical containers. Categories can contain other categories.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.

© TM Forum 2021. All Rights Reserved

@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
version	A string. Category version.

Json representation sample

We provide below the json representation of an example of a 'ServiceCategory' resource object

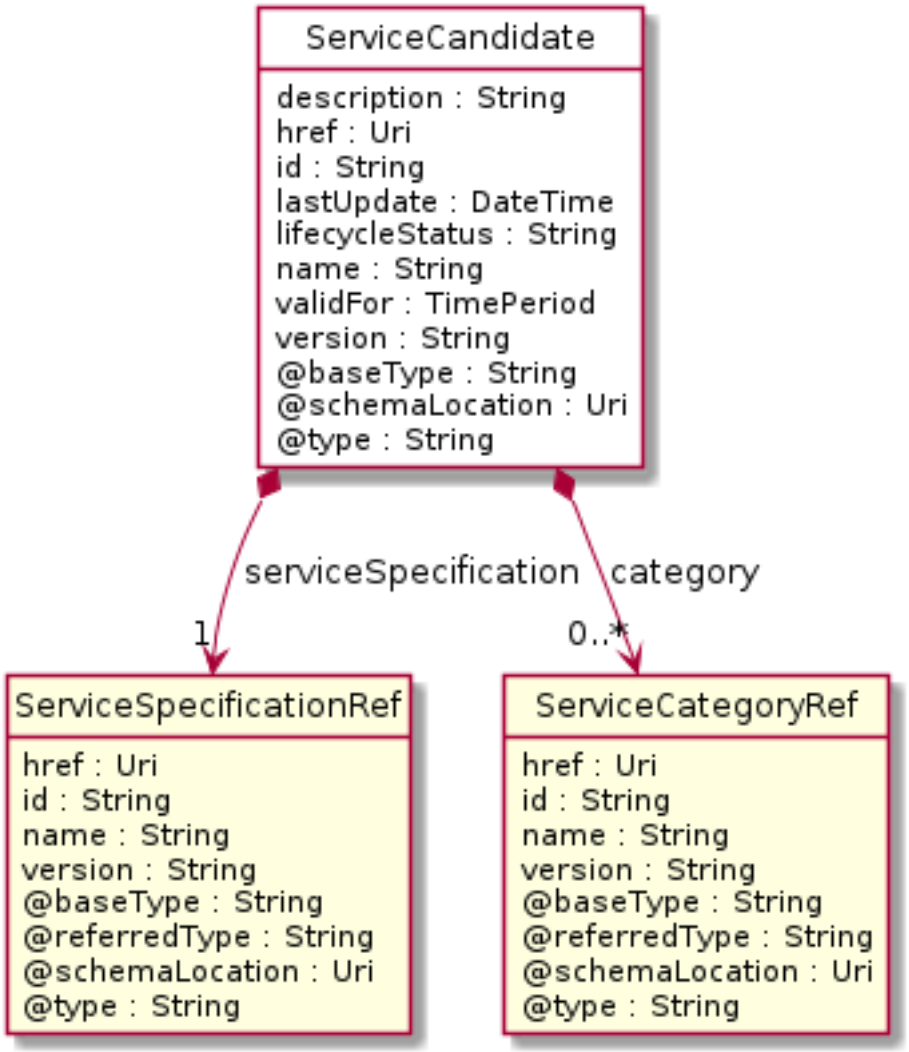

```
{
  "id": "1708",
  "href": "https://mycsp.com:8080/tmf-api/serviceCatalogManagement/v4/serviceCategory/1708",
  "name": "Cloud Services",
  "description": "A category to hold all available cloud service offers",
  "version": "1.0",
  "validFor": {
    "startDateTime": "2017-08-24T00:00:00Z",
    "endDateTime": "2018-03-25T00:00:00Z"
  },
  "lifecycleStatus": "Active",
  "lastUpdate": "2017-08-27T00:00:00Z",
  "isRoot": true,
  "serviceCandidate": [
    {
      "href": "https://mycsp.com:8080/tmf-api/serviceCatalogManagement/v4/serviceCandidate/5850",
      "id": "5850",
      "name": "Speed Max",
      "version": "1.1",
      "@referredType": "ServiceCandidate"
    }
  ],
  "category": [
    {
      "href": "https://mycsp.com:8080/tmf-api/serviceCatalogManagement/v4/category/6086",
      "id": "6086",
      "name": "Cloud",
      "@referredType": "ServiceCategory"
    }
  ],
  "@type": "ServiceCategory",
  "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Service/ServiceCategory.schema.json",
  "@baseType": "Category"
}
```


Service Candidate resource

ServiceCandidate is an entity that makes a service specification available to a catalog. A ServiceCandidate and its associated service specification may be published - made visible - in any number of service catalogs, or in none. One service specification can be composed of other service specifications.

© TM Forum 2021. All Rights Reserved

Resource model



Field descriptions

ServiceCandidate fields

href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
category	A list of service category references (ServiceCategoryRef [*]). List of categories for this candidate.

© TM Forum 2021. All Rights Reserved

description	A string. Description of this REST resource.
lastUpdate	A date time (DateTime). Date and time of the last update of this REST resource.
lifecycleStatus	A string. Used to indicate the current lifecycle status of the service candidate.
name	A string. Name given to this REST resource.
serviceSpecification	A service specification reference (ServiceSpecificationRef). The service specification implied by this candidate.
validFor	A time period. The period for which this REST resource is valid.
version	A string. the version of service candidate.

ServiceCategoryRef relationship

The (service) category resource is used to group service candidates in logical containers. Categories can contain other categories.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
version	A string. Category version.

ServiceSpecificationRef relationship

Service specification reference: ServiceSpecification(s) required to realize a ProductSpecification.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.

© TM Forum 2021. All Rights Reserved

@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
version	A string. Service specification version.

Json representation sample

We provide below the json representation of an example of a 'ServiceCandidate' resource object

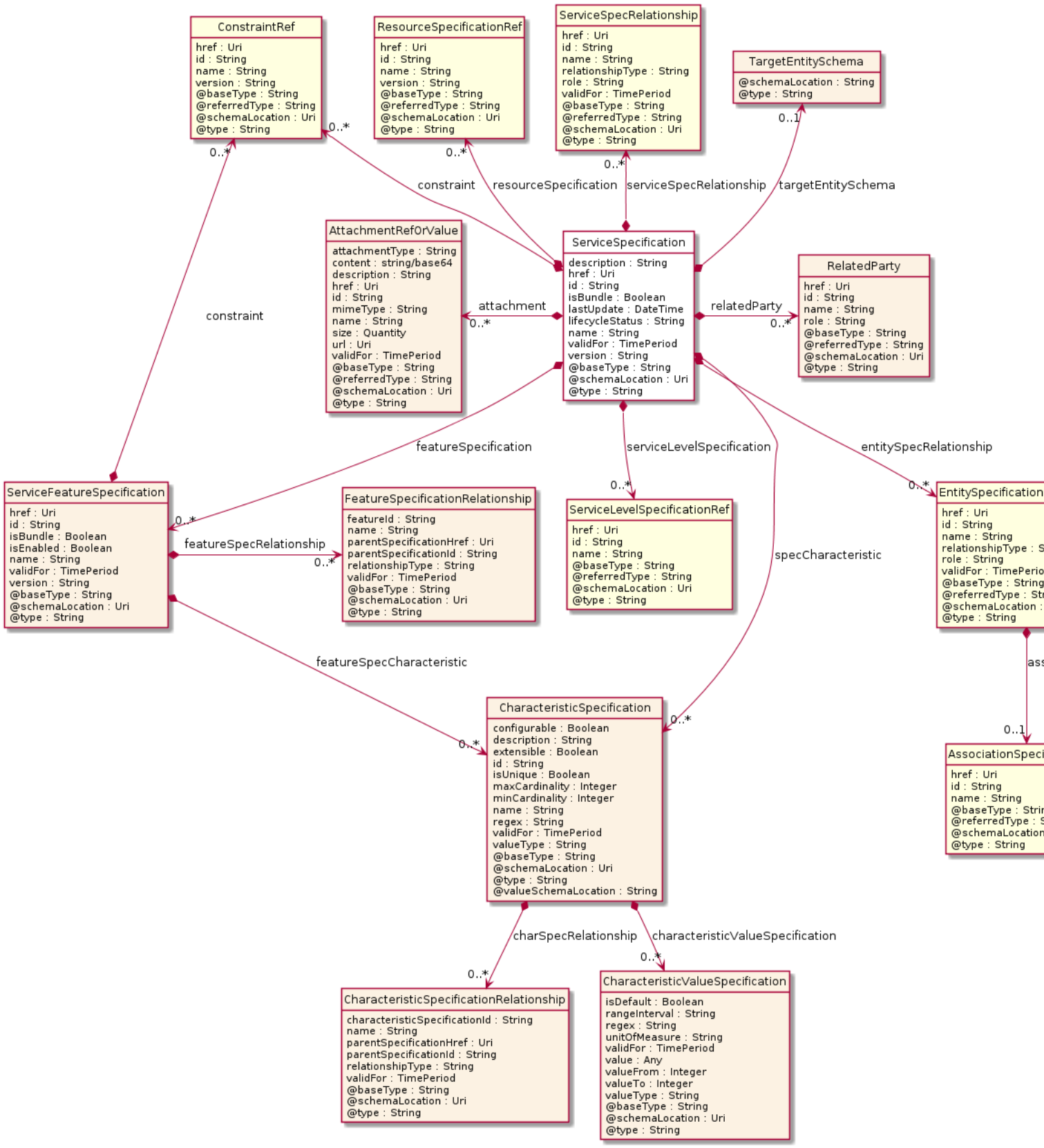

```
{
  "id": "4994",
  "href": "https://mycsp.com:8080/tmf-api/serviceCatalogManagement/v4/serviceCandidate/4994",
  "name": "TVServiceCandidate",
  "description": "This service candidate allows provision of TV service",
  "version": "2.1",
  "validFor": {
    "startDateTime": "2017-08-23T00:00:00Z",
    "endDateTime": "2018-03-25T00:00:00Z"
  },
  "lastUpdate": "2017-08-27T00:00:00Z",
  "lifecycleStatus": "Active",
  "category": [
    {
      "id": "5980",
      "href": "https://mycsp.com:8080/tmf-api/serviceCatalogManagement/v4/serviceCategory/5980",
      "version": "3.2",
      "name": "TV"
    }
  ],
  "serviceSpecification": {
    "id": "9600",
    "href": "https://mycsp.com:8080/tmf-api/serviceCatalogManagement/v4/serviceSpecification/9600",
    "version": "2.1",
    "name": "CFSS_TV",
    "@type": "CustomerFacingServiceSpecification"
  },
  "@type": "ServiceCandidate",
  "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Service/ServiceCandidate.schema.json ",
  "@baseType": ""
}
```


Service Specification resource

ServiceSpecification is a class that offers characteristics to describe a type of service.
Functionally, it acts as a template by which Services may be instantiated. By sharing the same specification, these services would therefore share the same set of characteristics.

Resource model

© TM Forum 2021. All Rights Reserved



Field descriptions

ServiceSpecification fields

© TM Forum 2021. All Rights Reserved

description	A string. Description of the specification.
isBundle	A boolean. isBundle determines whether specification represents a single specification (false), or a bundle of specifications (true).
lastUpdate	A date time (DateTime). Date and time of the last update of the specification.
lifecycleStatus	A string. Used to indicate the current lifecycle status of this catalog item.
name	A string. Name given to the specification.
version	A string. specification version.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
featureSpecification	A list of service feature specifications (ServiceFeatureSpecification [*]). A list of Features for this specification.
resourceSpecification	A list of resource specification references (ResourceSpecificationRef [*]). A list of resource specification references (ResourceSpecificationRef [*]). The ResourceSpecification is required for a service specification with type ResourceFacingServiceSpecification (RFSS).
serviceLevelSpecification	A list of service level specification references (ServiceLevelSpecificationRef [*]). A list of service level specifications related to this service specification, and which will need to be satisifiable for corresponding service instances; e.g. Gold, Platinum.
serviceSpecRelationship	A list of service spec relationships (ServiceSpecRelationship [*]). A list of service specifications related to this specification, e.g. migration, substitution, dependency or exclusivity relationship.
attachment	A list of attachment ref or values (AttachmentRefOrValue [*]). Attachments that may be of relevance to this specification, such as picture, document, media.
constraint	A list of constraint references (ConstraintRef [*]). This is a list of constraint references applied to this specification.
entitySpecRelationship	A list of entity specification relationships (EntitySpecificationRelationship [*]). Relationship to another specification.

© TM Forum 2021. All Rights Reserved

relatedParty	A list of related parties (RelatedParty [*]). Parties who manage or otherwise have an interest in this specification.
specCharacteristic	A list of characteristic specifications (CharacteristicSpecification [*]). List of characteristics that the entity can take.
targetEntitySchema	A target entity schema (TargetEntitySchema). Pointer to a schema that defines the target entity.
validFor	A time period. The period for which this REST resource is valid.

AttachmentRefOrValue sub-resource

An attachment by value or by reference. An attachment complements the description of an element, for example through a document, a video, a picture.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
description	A string. A narrative text describing the content of the attachment.
href	An uri (Uri). URI for this Attachment.
id	A string. Unique identifier for this particular attachment.
url	An uri (Uri). Uniform Resource Locator, is a web page address (a subset of URI).
name	A string. The name of the attachment.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
attachmentType	A string. Attachment type such as video, picture.
content	A base 6 4 (Base64). The actual contents of the attachment object, if embedded, encoded as base64.
mimeType	A string. Attachment mime type such as extension file for video, picture and document.
size	A quantity (Quantity). The size of the attachment.
validFor	A time period. The period of time for which the attachment is valid.

CharacteristicSpecification sub-resource

This class defines a characteristic specification.

© TM Forum 2021. All Rights Reserved

@valueSchemaLocation	A string. This (optional) field provides a link to the schema describing the value type.
configurable	A boolean. If true, the Boolean indicates that the target Characteristic is configurable.
description	A string. A narrative that explains the CharacteristicSpecification.
extensible	A boolean. An indicator that specifies that the values for the characteristic can be extended by adding new values when instantiating a characteristic for a resource.
id	A string. Unique ID for the characteristic.
isUnique	A boolean. An indicator that specifies if a value is unique for the specification. Possible values are; "unique while value is in effect" and "unique whether value is in effect or not".
maxCardinality	An integer. The maximum number of instances a CharacteristicValue can take on. For example, zero to five phone numbers in a group calling plan, where five is the value for the maxCardinality.
minCardinality	An integer. The minimum number of instances a CharacteristicValue can take on. For example, zero to five phone numbers in a group calling plan, where zero is the value for the minCardinality.
name	A string. A word, term, or phrase by which this characteristic specification is known and distinguished from other characteristic specifications.
regex	A string. A rule or principle represented in regular expression used to derive the value of a characteristic value.
valueType	A string. A kind of value that the characteristic can take on, such as numeric, text and so forth.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
charSpecRelationship	A list of characteristic specification relationships (CharacteristicSpecificationRelationship [*]). An aggregation, migration, substitution, dependency or exclusivity relationship between/among Specification Characteristics.
characteristicValueSpecification	A list of characteristic value specifications (CharacteristicValueSpecification [*]). A CharacteristicValueSpecification object is used to define a set of

© TM Forum 2021. All Rights Reserved

	attributes, each of which can be assigned to a corresponding set of attributes in a CharacteristicSpecification object. The values of the attributes in the CharacteristicValueSpecification object describe the values of the attributes that a corresponding Characteristic object can take on.
validFor	A time period. The period of time for which a characteristic is applicable.

CharacteristicSpecificationRelationship sub-resource

An aggregation, migration, substitution, dependency or exclusivity relationship between/among Characteristic specifications. The specification characteristic is embedded within the specification whose ID and href are in this entity, and identified by its ID.

@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
characteristicSpecificationId	A string. Unique identifier of the characteristic within the specification.
name	A string. Name of the target characteristic within the specification.
parentSpecificationHref	An uri (Uri). Hyperlink reference to the parent specification containing the target characteristic.
parentSpecificationId	A string. Unique identifier of the parent specification containing the target characteristic.
relationshipType	A string. Type of relationship such as aggregation, migration, substitution, dependency, exclusivity.
validFor	A time period. The period for which the object is valid.

CharacteristicValueSpecification sub-resource

specification of a value (number or text or an object) that can be assigned to a Characteristic.

@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
isDefault	A boolean. If true, the Boolean Indicates if the value is the default value for a characteristic.

© TM Forum 2021. All Rights Reserved

rangeInterval	A string. An indicator that specifies the inclusion or exclusion of the valueFrom and valueTo attributes. If applicable, possible values are "open", "closed", "closedBottom" and "closedTop".
regex	A string. A regular expression constraint for given value.
unitOfMeasure	A string. A length, surface, volume, dry measure, liquid measure, money, weight, time, and the like. In general, a determinate quantity or magnitude of the kind designated, taken as a standard of comparison for others of the same kind, in assigning to them numerical values, as 1 foot, 1 yard, 1 mile, 1 square foot.
validFor	A time period. The period of time for which a value is applicable.
value	An any (Any). A discrete value that the characteristic can take on, or the actual value of the characteristic.
valueFrom	An integer. The low range value that a characteristic can take on.
valueTo	An integer. The upper range value that a characteristic can take on.
valueType	A string. A kind of value that the characteristic value can take on, such as numeric, text and so forth.

EntitySpecificationRelationship sub-resource

A migration, substitution, dependency or exclusivity relationship between/among entity specifications.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
associationSpec	An association specification reference (AssociationSpecificationRef). A specification for an association used by this relationship.
relationshipType	A string. Type of relationship such as migration, substitution, dependency, exclusivity.
role	A string. The association role for this entity specification.
validFor	A time period. The period for which the entitySpecRelationship is valid.

© TM Forum 2021. All Rights Reserved

FeatureSpecificationRelationship sub-resource

Relationship between feature specifications.

@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
featureId	A string. Unique identifier of the target feature specification.
name	A string. This is the name of the target feature specification.
parentSpecificationHref	An uri (Uri). Hyperlink reference to the parent specification containing the target feature.
parentSpecificationId	A string. Unique identifier of the parent specification containing the target feature.
relationshipType	A string. This is the type of the feature specification relationship.
validFor	A time period. The period for which this feature spoecification relationship is valid.

Quantity sub-resource

An amount in a given unit.

amount	A float. Numeric value in a given unit.
units	A string. Unit.

RelatedParty sub-resource

Related Party reference. A related party defines party or party role linked to a specific entity.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
role	A string. Role played by the related party.

© TM Forum 2021. All Rights Reserved

ServiceFeatureSpecification sub-resource

Specification for service features.

href	An uri (Uri). Hyperlink reference.
id	A string. Identifier of the feature specification. Must be locally unique within the containing specification, thus allowing direct access to the feature spec.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
constraint	A list of constraint references (ConstraintRef [*]). A list of feature constraints/rules.
featureSpecCharacteristic	A list of characteristic specifications (CharacteristicSpecification [*]). This is a list of characteristics for a particular feature.
featureSpecRelationship	A list of feature specification relationships (FeatureSpecificationRelationship [*]). A dependency, exclusivity or aggratation relationship between/among feature specifications.
isBundle	A boolean. A flag indicating if this is a feature group (true) or not (false).
isEnabled	A boolean. A flag indicating if the feature is enabled (true) or not (false).
name	A string. Unique name given to the feature specification.
validFor	A time period. The period for which this feature specification is valid.
version	A string. Version of the feature specification.

ServiceSpecRelationship sub-resource

A dependency, substitution or exclusivity relationship between/among service specifications.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.

© TM Forum 2021. All Rights Reserved

@type	A string. When sub-classing, this defines the sub-class Extensible name.
relationshipType	A string. Type of relationship such as dependency, substitution or exclusivity.
role	A string. The association role for this service specification.
validFor	A time period. The period for which the serviceSpecRelationship is valid.

TargetEntitySchema sub-resource

The reference object to the schema and type of target entity which is described by a specification.

@schemaLocation	A string. This field provides a link to the schema describing the target entity.
@type	A string. Class type of the target entity.

AssociationSpecificationRef relationship

reference to an AssociationSpecification object.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.

ConstraintRef relationship

Constraint reference. The Constraint resource represents a policy/rule applied to an entity or entity spec.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.

© TM Forum 2021. All Rights Reserved

@type	A string. When sub-classing, this defines the sub-class Extensible name.
version	A string. constraint version.

ResourceSpecificationRef relationship

Resources are physical or non-physical components (or some combination of these) within an enterprise's infrastructure or inventory. They are typically consumed or used by services (for example a physical port assigned to a service) or contribute to the realization of a Product (for example, a SIM card). They can be drawn from the Application, Computing and Network domains, and include, for example, Network Elements, software, IT systems, content and information, and technology components.
A ResourceSpecification is an abstract base class for representing a generic means for implementing a particular type of Resource. In essence, a ResourceSpecification defines the common attributes and relationships of a set of related Resources, while Resource defines a specific instance that is based on a particular ResourceSpecification.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
version	A string. Resource Specification version.

ServiceLevelSpecificationRef relationship

A Service Level Specification represents a pre-defined or negotiated set of Service Level Objectives. In addition, certain consequences are associated with not meeting the Service Level Objectives. Service Level Agreements are expressed in terms of Service Level Specifications.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.

© TM Forum 2021. All Rights Reserved

@type	A string. When sub-classing, this defines the sub-class Extensible name.

Json representation sample

We provide below the json representation of an example of a 'ServiceSpecification' resource object


```
{
  "id": "7655",
  "href": "https://mycsp.com:8080/tmf-api/serviceCatalogManagement/v4/serviceSpecification/7655",
  "name": "Firewall Service",
  "description": "This service specification describes a firewall service that can be deployed in customer premise.",
  "version": "2.1",
  "validFor": {
    "startDateTime": "2017-08-23T00:00:00Z",
    "endDateTime": "2018-03-25T00:00:00Z"
  },
  "lastUpdate": "2017-08-27T00:00:00Z",
  "lifecycleStatus": "Active",
  "isBundle": false,
  "resourceSpecification": [
    {
      "href": "https://mycsp.com:8080/tmf-api/resourceCatalogManagement/v4/resourceSpecification/42",
      "id": "42",
      "name": "Firewall",
      "version": "1.0",
      "@referredType": "ResourceFunctionSpec"
    }
  ],
  "attachment": [
    {
      "description": "This attachment gives a block diagram of the firewall.",
      "href": "https://mycsp.com:8080/tmf-api/documentManagement/v4/attachment/22",
      "id": "22",
      "url": "https://mycsp.com:7070/docloader?docnum=3534536"
    }
  ],
  "serviceSpecCharacteristic": [
    {
      "name": "OperatingSystem",
      "description": "This characteristic describes the operating system run by the service",
      "valueType": "string",
      "configurable": true,
      "validFor": {
        "startDateTime": "2017-08-12T00:00:00Z",
        "endDateTime": "2018-03-07T00:00:00Z"
      },
      "@valueSchemaLocation": "",
      "minCardinality": 0,
      "maxCardinality": 1,
      "isUnique": true,
      "regex": "",
      "extensible": false,
      "serviceSpecCharacteristicValue": [
        {
          "valueType": "string",
```


© TM Forum 2021. All Rights Reserved

```
            "isDefault": true,
            "value": "Android KitKat",
            "validFor": {
               "startDateTime": "2017-08-06T00:00:00Z",
               "endDateTime": "2018-03-07T00:00:00Z"
            }
         }
      ],
      "serviceSpecCharRelationship": [
         {
            "relationshipType": "dependency",
            "role": "",
            "id": "4690",
            "href": "https://mycsp.com:8080/tmf-
api/serviceCatalogManagement/v4/serviceSpecificationCharacteristic/4690",
            "name": "OperatingSystem",
            "validFor": {
               "startDateTime": "2017-08-11T00:00:00Z",
               "endDateTime": "2018-03-07T00:00:00Z"
            }
         }
      ],
      "@type": "ServiceSpecCharacteristic",
      "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Service/ServiceSpecCharacteristic.schema.json"
   },
   {
      "name": "Scalability",
      "description": "Scalability parameters for this resource facing service spec ",
      "valueType": "CapabilityScalable",
      "configurable": true,
      "validFor": {
         "startDateTime": "2017-08-17T00:00:00Z",
         "endDateTime": "2018-03-12T00:00:00Z"
      },
      "@valueSchemaLocation": "https://mycsp.com:8080/tmf-api/schema/Service/CapabilityScalable.schema.json",
      "minCardinality": 0,
      "maxCardinality": 1,
      "isUnique": true,
      "regex": "",
      "extensible": true,
      "serviceSpecCharacteristicValue": [
         {
            "valueType": "object",
            "isDefault": true,
            "value": {
               "minInstances": 1,
               "maxInstances": 1000
            },
            "validFor": {
               "startDateTime": "2017-08-17T00:00:00Z",
               "endDateTime": "2018-03-12T00:00:00Z"
            },
            "@type": "CapabilityScalable",
            "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Service/CapabilityScalable.schema.json"
         }
      ],
```

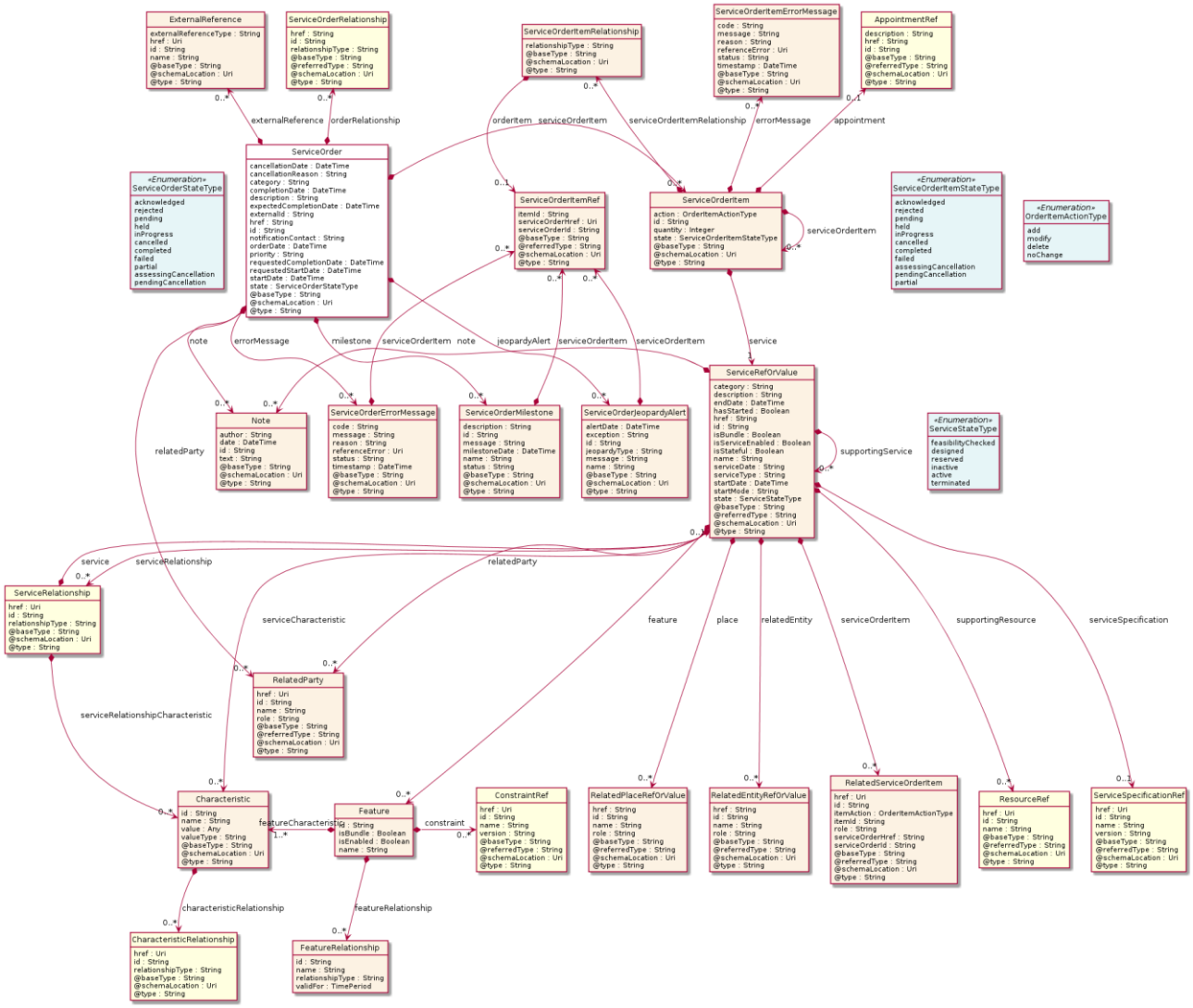

© TM Forum 2021. All Rights Reserved

```
        "serviceSpecCharRelationship": [],
        "@type": "ServiceSpecCharacteristic",
        "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Service/ServiceSpecCharacteristic.schema.json"
      }
    ],
    "relatedParty": [
      {
        "href": "https://mycsp.com:8080/tmf-api/partyManagement/v4/organization/3643",
        "id": "3643",
        "name": "Firewall Express",
        "role": "Supplier"
      }
    ],
    "serviceSpecRelationship": [
      {
        "relationshipType": "dependency",
        "role": "dependent",
        "id": "5563",
        "href": "https://mycsp.com:8080/tmf-api/serviceCatalogManagement/v4/serviceSpecification/5563",
        "name": "Points to the Deep Packet Inspection service on which this Firewall service depends",
        "validFor": {
          "startDateTime": "2017-08-25T00:00:00Z",
          "endDateTime": "2018-03-25T00:00:00Z"
        }
      }
    ],
    "serviceLevelSpecification": [
      {
        "href": "https://mycsp.com:8080/tmf-api/serviceQualityManagement/v4/serviceLevelSpecification/8899",
        "id": "8899",
        "name": "Platinum"
      }
    ],
    "targetServiceSchema": {
      "@type": "RFS",
      "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Service/RFS.schema.json"
    },
    "@type": "ResourceFacingServiceSpecification",
    "@schemaLocation": "https://mycsp.com:8080/tmf-
api/schema/Service/ResourceFacingServiceSpecification.schema.json",
    "@baseType": "ServiceSpecification"
}
```


Service Order resource

Resource model

© TM Forum 2021. All Rights Reserved



Field descriptions

ServiceOrder fields

href	A string. Hyperlink to access the order.
id	A string. ID created on repository side.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
cancellationDate	A date time (DateTime). Date when the order is cancelled. This is used when order is cancelled.
cancellationReason	A string. Reason why the order is cancelled. This is used when order is cancelled.

© TM Forum 2021. All Rights Reserved

category	A string. Used to categorize the order, useful for the OM system, such as: Broadband, TVOption.
completionDate	A date time (DateTime). Effective delivery date amended by the provider.
description	A string. A free-text description of the service order.
errorMessage	A list of service order error messages (ServiceOrderErrorMessage [*]). the error(s) cause an order status change.
expectedCompletionDate	A date time (DateTime). Expected delivery date amended by the provider.
externalId	A string. ID given by the consumer to facilitate searches.
externalReference	A list of external references (ExternalReference [*]). External reference of the individual or reference in other system.
href	A string. Hyperlink to access the order.
id	A string. ID created on repository side.
jeopardyAlert	A list of service order jeopardy alerts (ServiceOrderJeopardyAlert [*]). A list of jeopardy alerts related to this order.
milestone	A list of service order milestones (ServiceOrderMilestone [*]). A list of milestones related to this order.
note	A list of notes (Note [*]). Extra-information about the order; e.g. useful to add extra delivery information that could be useful for a human process.
notificationContact	A string. Contact attached to the order to send back information regarding this order.
orderDate	A date time (DateTime).
orderRelationship	A list of service order relationships (ServiceOrderRelationship [*]). A list of service orders related to this order (e.g. prerequisite, dependent on).
priority	A string. Can be used by consumers to prioritize orders in a Service Order Management system.
relatedParty	A list of related parties (RelatedParty [*]). A list of parties which are involved in this order and the role they are playing.
requestedCompletionDate	A date time (DateTime). Requested delivery date from the requestors perspective.
requestedStartDate	A date time (DateTime). Order start date wished by the requestor.
serviceOrderItem	A list of service order items (ServiceOrderItem [*]). A list of service order items to be processed by this order.

© TM Forum 2021. All Rights Reserved

startDate	A date time (DateTime). Date when the order was started for processing.
state	A service order state type (ServiceOrderStateType). State of the order: described in the state-machine diagram.

Characteristic sub-resource

Describes a given characteristic of an object or entity through a name/value pair.

@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
characteristicRelationship	A list of characteristic relationships (CharacteristicRelationship [*]). Another Characteristic that is related to the current Characteristic;.
id	A string. Unique identifier of the characteristic.
name	A string. Name of the characteristic.
value	An any (Any). The value of the characteristic.
valueType	A string. Data type of the value of the characteristic.

CharacteristicRelationship sub-resource

Another Characteristic that is related to the current Characteristic;.

href	An uri (Uri). Hyperlink reference.
id	A string. Unique identifier of the characteristic.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
relationshipType	A string. The type of relationship.

ExternalReference sub-resource

External reference of the individual or reference in other system.

href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.

© TM Forum 2021. All Rights Reserved

@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
externalReferenceType	A string. Type of the external reference.
name	A string. External reference name.

Feature sub-resource

Configuration feature.

constraint	A list of constraint references (ConstraintRef [*]). This is a list of feature constraints.
featureCharacteristic	A list of characteristics (Characteristic [1..*]). This is a list of Characteristics for a particular feature.
featureRelationship	A list of feature relationships (FeatureRelationship [*]). Configuration feature.
id	A string. Unique identifier of the feature.
isBundle	A boolean. True if this is a feature group. Default is false.
isEnabled	A boolean. True if this feature is enabled. Default is true.
name	A string. This is the name for the feature.

FeatureRelationship sub-resource

Configuration feature.

id	A string. Unique identifier of the target feature.
name	A string. This is the name of the target feature.
relationshipType	A string. This is the type of the feature relationship.
validFor	A time period. The period for which this feature relationship is valid.

Note sub-resource

Extra information about a given entity.

@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.

© TM Forum 2021. All Rights Reserved

author	A string. Author of the note.
date	A date time (DateTime). Date of the note.
id	A string. Identifier of the note within its containing entity.
text	A string. Text of the note.

RelatedEntityRefOrValue sub-resource

A reference to an entity, where the type of the entity is not known in advance. A related entity defines a entity described by reference or by value linked to a specific entity. The polymorphic attributes @type, @schemaLocation & @referredType are related to the Entity and not the RelatedEntityRefOrValue class itself.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	A string. Reference of the related entity.
id	A string. Unique identifier of a related entity.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
role	A string.

RelatedParty sub-resource

Related Party reference. A related party defines party or party role linked to a specific entity.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
role	A string. Role played by the related party.

© TM Forum 2021. All Rights Reserved

RelatedPlaceRefOrValue sub-resource

Related Entity reference. A related place defines a place described by reference or by value linked to a specific entity. The polymorphic attributes @type, @schemaLocation & @referredType are related to the place entity and not the RelatedPlaceRefOrValue class itself.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. A user-friendly name for the place, such as [Paris Store], [London Store], [Main Home].
href	A string. Unique reference of the place.
id	A string. Unique identifier of the place.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
role	A string.

RelatedServiceOrderItem sub-resource

RelatedServiceOrderItem (a ServiceOrder item) .The service order item which triggered service creation/change/termination.

href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
@referredType	A string. The actual type of the target instance when needed for disambiguation.
itemAction	An order item action type (OrderItemActionType). Action of the order item for this service.
itemId	A string. Identifier of the order item where the service was managed.
role	A string. role of the service order item for this service.
serviceOrderHref	A string. Reference of the related entity.

© TM Forum 2021. All Rights Reserved

serviceOrderId	A string. Unique identifier of a related entity.

ServiceOrderErrorMessage sub-resource

A ServiceOrderErrorMessage represents an error that causes a status change in a service order.

code	A string. error code.
message	A string. More details and corrective actions related to the error.
reason	A string. Explanation of the reason for the error.
referenceError	An uri (Uri). URI of documentation describing the error.
status	A string. error code extension like sys-ABC-2001.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
serviceOrderItem	A list of service order item references (ServiceOrderItemRef [*]). A list of order item references corresponded to this error.
timestamp	A date time (DateTime). Date when the error happened.

ServiceOrderItem sub-resource

@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
action	An order item action type (OrderItemActionType). The action to be carried out on the Service. Can be: add, modify, delete, noChange.
appointment	An appointment reference (AppointmentRef). An appointment that was set up with a related party for this order item.
errorMessage	A list of service order item error messages (ServiceOrderItemErrorMessage [*]). the error(s) cause an order item status change.
id	A string. Identifier of the individual line item.
quantity	An integer. Quantity ordered.

© TM Forum 2021. All Rights Reserved

service	A service ref or value (ServiceRefOrValue). The Service to be acted on by the order item.
serviceOrderItem	A list of service order items (ServiceOrderItem [*]). A list of order items embedded to this order item.
serviceOrderItemRelationship	A list of service order item relationships (ServiceOrderItemRelationship [*]). A list of order items related to this order item.
state	A service order item state type (ServiceOrderItemStateType). State of the order item: described in the state machine diagram. This is the requested state.

ServiceOrderItemErrorMessage sub-resource

AServiceOrderItemErrorMessage represents an error that causes a status change in a service order item.

code	A string. error code.
message	A string. More details and corrective actions related to the error.
reason	A string. Explanation of the reason for the error.
referenceError	An uri (Uri). URI of documentation describing the error.
status	A string. error code extension like sys-ABC-2001.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
timestamp	A date time (DateTime). Date when the error happened.

ServiceOrderItemRelationship sub-resource

Linked service order item to the one containing this attribute.

@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
orderItem	A service order item reference (ServiceOrderItemRef). A service order item in relationship with this order item.
relationshipType	A string. The type of related order item, can be: dependency if the order item needs to be not started until another order item is complete.

© TM Forum 2021. All Rights Reserved

ServiceOrderJeopardyAlert sub-resource

A ServiceOrderJeopardyAlert represents a predicted exception during a service order processing that would brings risk to complete successfully the ordetr.

alertDate	A date time (DateTime). A date time(DateTime). The date that the alert issued.
exception	A string. The exception associated with this jeopardy alert.
id	A string. identifier of the JeopardyAlert.
jeopardyType	A string. A string represents the type of jeopardy/risk like Normal, Hazard, Critical, ...
message	A string. A string represents the message of the alert.
name	A string. A string used to give a name to the jeopardy alert.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
serviceOrderItem	A list of service order item references (ServiceOrderItemRef [*]). A list of order item references corresponded to this alert.

ServiceOrderMilestone sub-resource

ServiceOrderMilestone represents an action or event marking a significant change or stage in processing of a service order.

description	A string. free-text description of the Milestone.
id	A string. identifier of the Milestone.
message	A string. A string represents the message of the milestone.
milestoneDate	A date time (DateTime). A date time(DateTime). The date that the milestone happens.
name	A string. A string used to give a name to the milestone.
status	A string. The milestone status.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.

© TM Forum 2021. All Rights Reserved

serviceOrderItem	A list of service order item references (ServiceOrderItemRef [*]). A list of order item references corresponded to this milestone.

ServiceOrderRelationship sub-resource

Linked service order to the one containing this attribute.

href	A string. A hyperlink to the related order.
id	A string. The id of the related order.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
@referredType	A string. The entity type of the related order.
relationshipType	A string. The type of related order, such as: [dependency] if the order needs to be [not started] until another order item is complete (a service order in this case) or [cross-ref] to keep track of the source order (a productOrder).

ServiceRefOrValue sub-resource

A Service to be created defined by value or existing defined by reference. The polymorphic attributes @type, @schemaLocation & @referredType are related to the Service entity and not the RelatedServiceRefOrValue class itself.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the service.
href	A string. Reference of the service.
id	A string. Unique identifier of the service.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
category	A string. Is it a customer facing or resource facing service.
description	A string. Free-text description of the service.
endDate	A date time (DateTime). Date when the service ends.

© TM Forum 2021. All Rights Reserved

hasStarted	A boolean. If TRUE, this Service has already been started.
isBundle	A boolean. If true, the service is a ServiceBundle which regroup a service hierachy. If false, the service is a 'atomic' service (hierachy leaf).
isServiceEnabled	A boolean. If FALSE and hasStarted is FALSE, this particular Service has NOT been enabled for use - if FALSE and hasStarted is TRUE then the service has failed.
isStateful	A boolean. If TRUE, this Service can be changed without affecting any other services.
serviceDate	A string. Date when the service was created (whatever its status).
serviceType	A string. Business type of the service.
startDate	A date time (DateTime). Date when the service starts.
startMode	A string. This attribute is an enumerated integer that indicates how the Service is started, such as: 0: Unknown; 1: Automatically by the managed environment; 2: Automatically by the owning device; 3: Manually by the Provider of the Service; 4: Manually by a Customer of the Provider; 5: Any of the above.
feature	A list of features (Feature [*]). A list of feature associated with this service.
note	A list of notes (Note [*]). A list of notes made on this service.
place	A list of related place ref or values (RelatedPlaceRefOrValue [*]). A list of places (Place [*]). Used to define a place useful for the service (for example a geographical place whre the service is installed).
relatedEntity	A list of related entity ref or values (RelatedEntityRefOrValue [*]). A list of related entity in relationship with this service.
relatedParty	A list of related parties (RelatedParty [*]). A list of related party references (RelatedParty [*]). A related party defines party or party role linked to a specific entity.
serviceCharacteristic	A list of characteristics (Characteristic [*]). A list of characteristics that characterize this service (ServiceCharacteristic [*]).
serviceOrderItem	A list of related service order items (RelatedServiceOrderItem [*]). A list of service order items related to this service.
serviceRelationship	A list of service relationships (ServiceRelationship [*]). A list of service relationships (ServiceRelationship [*]). Describes links with other service(s) in the inventory.
serviceSpecification	A service specification reference (ServiceSpecificationRef). The specification from which this service was instantiated.

© TM Forum 2021. All Rights Reserved

state	A service state type (ServiceStateType). The life cycle state of the service, such as designed, reserved, active, etc...
supportingResource	A list of resource references (ResourceRef [*]). A list of supporting resources (SupportingResource [*]).Note: only Service of type RFS can be associated with Resources.
supportingService	A list of service ref or values (ServiceRefOrValue [*]). A list of supporting services (SupportingService [*]). A collection of services that support this service (bundling, link CFS to RFS).

ServiceRelationship sub-resource

href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
relationshipType	A string.
service	A service ref or value (ServiceRefOrValue). A Service to be created defined by value or existing defined by reference. The polymorphic attributes @type, @schemaLocation & @referredType are related to the Service entity and not the RelatedServiceRefOrValue class itself.
serviceRelationshipCharacteristic	A list of characteristics (Characteristic [*]). Describes a given characteristic of an object or entity through a name/value pair.

AppointmentRef relationship

Refers an appointment, such as a Customer presentation or internal meeting or site visit.

href	A string. The reference of the appointment.
id	A string. The identifier of the referred appointment.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.

© TM Forum 2021. All Rights Reserved

@referredType	A string. The actual type of the target instance when needed for disambiguation.
description	A string. An explanatory text regarding the appointment made with a party.

ConstraintRef relationship

Constraint reference. The Constraint resource represents a policy/rule applied to an entity or entity spec.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
version	A string. constraint version.

ResourceRef relationship

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.

ServiceOrderItemRef relationship

@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.

© TM Forum 2021. All Rights Reserved

@type	A string. When sub-classing, this defines the sub-class Extensible name.
@referredType	A string. The actual type of the target instance when needed for disambiguation.
itemId	A string. Identifier of the line item.
serviceOrderHref	An uri (Uri). Link to the order to which this item belongs to.
serviceOrderId	A string. Identifier of the order that this item belongs to.

ServiceSpecificationRef relationship

Service specification reference: ServiceSpecification(s) required to realize a ProductSpecification.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
version	A string. Service specification version.

Json representation sample

We provide below the json representation of an example of a 'ServiceOrder' resource object


```
{
  "id": "42",
  "href": "http://serverlocation:port/serviceOrderingManagement/v4/serviceOrder/42",
  "externalId": "OrangeBSS747",
  "priority": "1",
  "description": "Service order description",
  "category": "TMF resource illustration",
  "state": "acknowledged",
  "orderDate": "2018-01-12T09:37:40.508Z",
  "completionDate": "",
  "requestedStartDate": "2018-01-15T09:37:40.508Z",
  "requestedCompletionDate": "2018-01-15T09:37:40.508Z",
  "expectedCompletionDate": "2018-01-15T09:37:40.508Z",
  "startDate": "2018-01-12T09:37:40.508Z",
  "@type": "ServiceOrder",
  "note": [
    {
      "date": "2018-01-15T09:37:40.508Z",
      "author": "Jean Pontus",
```


© TM Forum 2021. All Rights Reserved

```
      "text": "Some text"
    }
  ],
  "relatedParty": [
    {
      "id": "456",
      "href": "http://serverlocation:port/partyManagement/v4/party/456",
      "role": "requester",
      "name": "Jean Pontus",
      "@referredType": "Individual"
    }
  ],
  "orderItem": [
    {
      "id": "1",
      "action": "add",
      "state": "acknowledged",
      "service": {
        "@type": "vCPE",
        "@schemaLocation": "http//my.schemas/vCpe.schema.json",
        "state": "active",
        "serviceType": "CFS",
        "serviceCharacteristic": [
          {
            "name": "vCPE_IP",
            "valueType": "object",
            "value": {
              "@type": "IPAddress",
              "@schemaLocation": "http://my.schemas/IPAddress.schema.json",
              "vCPE_IP": "193.218.236.21"
            }
          }
        ],
        "serviceSpecification": {
          "@type": "ONAPServiceSpec",
          "@schemaLocation": "http//my.schemas/OnapServiceSpec.schema.json",
          "id": "12",
          "href": "http://.../serviceSpecification/12",
          "name": "vCPE",
          "version": "1",
          "invariantUUID": "456-852-357",
          "toscaModelURL": "http//...",
          "targetServiceSchema": {
            "@type": "vCPE",
            "@schemaLocation": "http//my.schemas/vCpe.schema.json"
          }
        }
      }
    },
    {
      "id": "2",
      "action": "modify",
      "state": "acknowledged",
      "service": {
        "id": "456",
        "href": "http://serverlocation:port/serviceInventoryManagement/v4/service/456",
```


© TM Forum 2021. All Rights Reserved

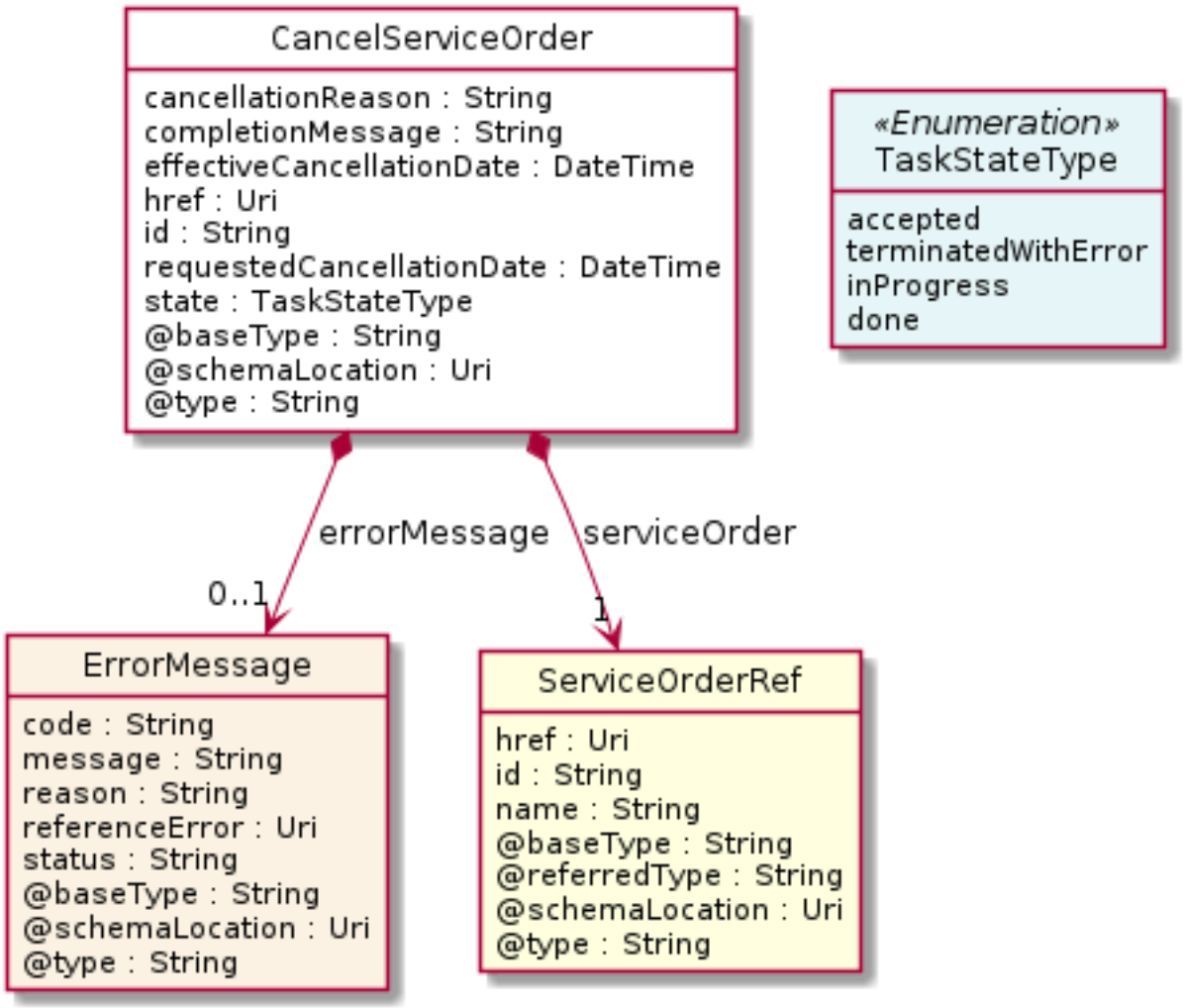
```
          "state": "active",
          "serviceType": "CFS",
          "serviceCharacteristic": [
            {
              "name": "Characteristic1",
              "value": "newValue"
            }
          ],
          "supportingResource": [
            {
              "id": "3456_DFG5-H690",
              "href": "http://...",
              "@referredType": "CloudResource"
            }
          ]
        }
      },
      {
        "id": "3",
        "action": "add",
        "state": "acknowledged",
        "service": {
          "state": "active",
          "serviceType": "CFS",
          "serviceRelationship": [
            {
              "relationshipType": "reliesOn",
              "service": {
                "href": "https://.../serviceManagement/v4/service/45",
                "id": "45"
              }
            }
          ],
          "serviceSpecification": {
            "id": "48",
            "href": "http//..../48",
            "name": "genericService48",
            "version": "2"
          }
        }
      },
      {
        "id": "4",
        "action": "modify",
        "state": "acknowledged",
        "service": {
          "id": "12",
          "href": "http://serverlocation:port/serviceInventoryManagement/v4/service/12",
          "state": "inactive"
        }
      }
    ]
}
```


Cancel Service Order resource

© TM Forum 2021. All Rights Reserved

Request for cancellation an existing Service order.

Resource model



Field descriptions

CancelServiceOrder fields

cancellationReason	A string. Reason why the order is cancelled.
effectiveCancellationDate	A date time (DateTime). Date when the order is cancelled.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
requestedCancellationDate	A date time (DateTime). Date when the submitter wants the order to be cancelled.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.

© TM Forum 2021. All Rights Reserved

@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
completionMessage	A string. an optional message describing the completion of the task if it is done as expected or it is denied for a reason (like order in an state of PoNR).
errorMessage	An error message (ErrorMessage). the error(s) cause the termination of cancelServiceOrder (in TerminatedWithError state).
serviceOrder	A service order reference (ServiceOrderRef). Useful to understand the which was the Service order through which the service was instantiated in the service inventory.
state	A task state type (TaskStateType). Tracks the lifecycle status of the cancellation request, such as Acknowledged, Rejected, InProgress, Pending and so on.

ErrorMessage sub-resource

represents an Error.

@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
code	A string. error code.
message	A string. More details and corrective actions related to the error.
reason	A string. Explanation of the reason for the error.
referenceError	An uri (Uri). URI of documentation describing the error.
status	A string. error code extension like sys-ABC-2001.

ServiceOrderRef relationship

Service Order reference. Useful to understand the which was the Service order through which the service was instantiated in the service inventory.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related service order.
href	An uri (Uri). Hyperlink reference.

© TM Forum 2021. All Rights Reserved

id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.

Json representation sample

We provide below the json representation of an example of a 'CancelServiceOrder' resource object

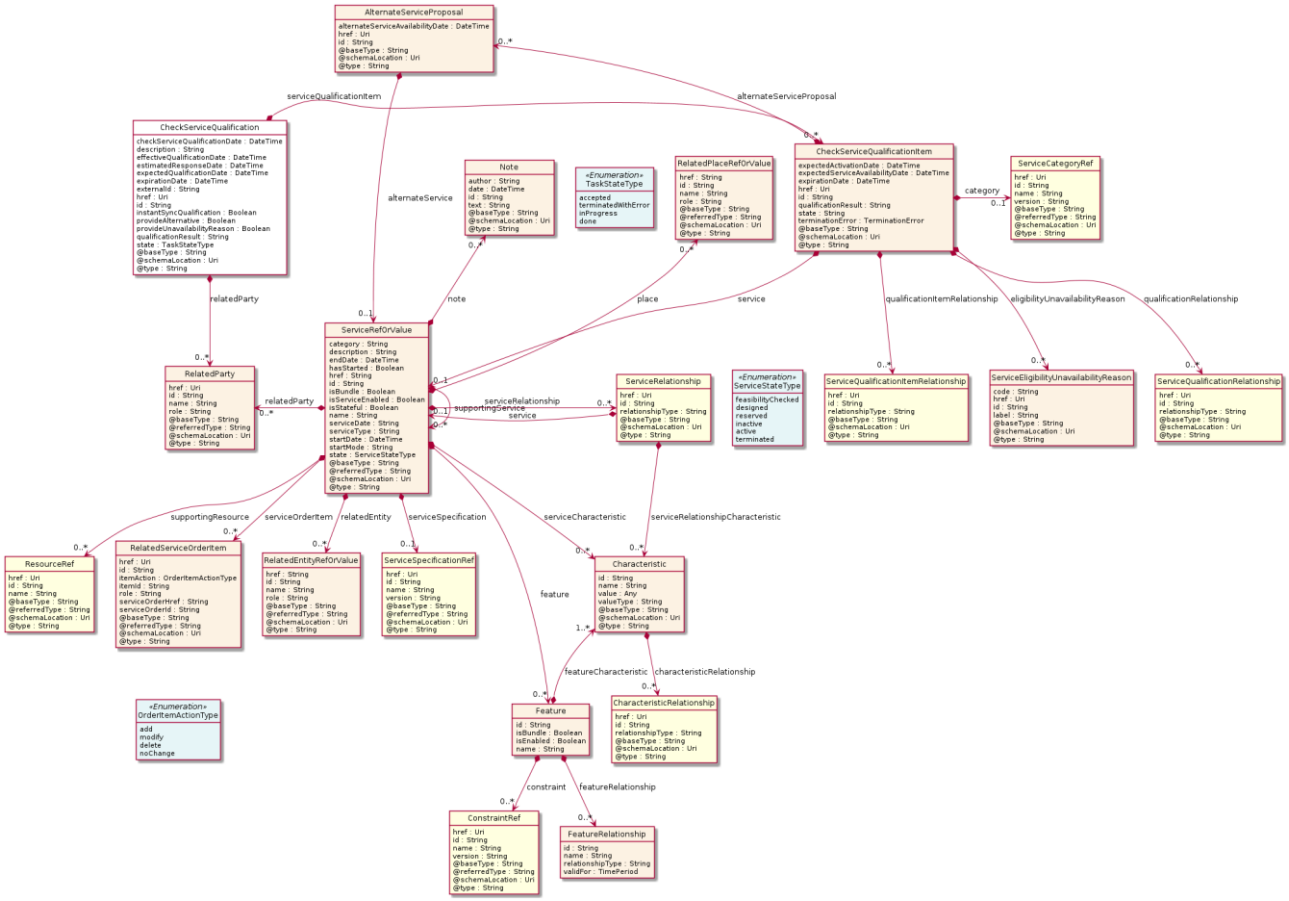

```
{
  "id": "789-fsds5-kjp",
  "href": "https://host:port/serviceOrdering/v4/cancelServiceOrder/789-fsds5-kjp",
  "cancellationReason": "Duplicate service order",
  "effectiveCancellationDate": "2020-08-27T12:58:35.489Z",
  "requestedCancellationDate": "2020-08-27T12:56:21.931Z",
  "serviceOrder": {
    "id": "45f-98f-ss45",
    "href": "https://host:port/serviceOrdering/v4/serviceOrder/45f-98f-ss45",
    "@referredType": "ServiceOrder"
  },
  "state": "done",
  "@type": "CancelServiceOrder"
}
```


Check Service Qualification resource

CheckServiceQualification is used to perform a technical eligibility on service configuration(s). It allows to retrieve services that are technically available in the context of the interaction (place, party, service characteristics, ...).

Resource model

© TM Forum 2021. All Rights Reserved



Field descriptions

CheckServiceQualification fields

description	A string. Description of the serviceQualification.
effectiveQualificationDate	A date time (DateTime). Effective date to serviceQualification completion.
estimatedResponseDate	A date time (DateTime). Date when the requester expect to provide an answer for the qualification request.
expectedQualificationDate	A date time (DateTime). A date (DateTime). Deadline date when the requester expected a qualification answer.
expirationDate	A date time (DateTime). Date when the qualification response expires.
externalId	A string. Identifier provided by the requester.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
instantSyncQualification	A boolean. An indicator which when the value is "true" means that requester expects to get qualifcation result immediately in the response. If the indicator is

© TM Forum 2021. All Rights Reserved

	true then the response code of 200 indicates the operation is successful otherwise a task is created with a response 201.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
checkServiceQualificationDate	A date time (DateTime). Date when the serviceQualification was submitted.
provideAlternative	A boolean. When the value is TRUE means that alternative solutions should be provided.
provideUnavailabilityReason	A boolean. When the value is TRUE means that unavailability reason are expected for non available service.
qualificationResult	A string. Qualification result for this serviceQualification. It could be: qualified (all qualification item are qualified), alternate (At least one item alternate and no item with unqualified), unqualified (At least one item unqualified).
serviceQualificationItem	A list of check service qualification items (CheckServiceQualificationItem [*]). A list of service qualification items.
relatedParty	A list of related parties (RelatedParty [*]). A list of related party references, defines party or party role linked to this request.
state	A task state type (TaskStateType). State of the serviceQualification (acknowledged, inProgress, terminatedWithError, done).

AlternateServiceProposal sub-resource

Alternate service proposal is used when the requested service is not available with characteristic and date asked for. An alternate proposal could be a distinct serviceSpecification close to requested one or same as requested but with a different activation date.

href	An uri (Uri). Hyperlink reference.
id	A string. Identifier of a alternate service proposal.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.

© TM Forum 2021. All Rights Reserved

@type	A string. When sub-classing, this defines the sub-class Extensible name.
alternateService	A service ref or value (ServiceRefOrValue). A Service to be created defined by value or existing defined by reference. The polymorphic attributes @type, @schemaLocation & @referredType are related to the Service entity and not the RelatedServiceRefOrValue class itself.
alternateServiceAvailabilityDate	A date time (DateTime). Alternate availability date in case seller is not able to meet requested expected availability date for the service.

Characteristic sub-resource

Describes a given characteristic of an object or entity through a name/value pair.

@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
characteristicRelationship	A list of characteristic relationships (CharacteristicRelationship [*]). Another Characteristic that is related to the current Characteristic;.
id	A string. Unique identifier of the characteristic.
name	A string. Name of the characteristic.
value	An any (Any). The value of the characteristic.
valueType	A string. Data type of the value of the characteristic.

CharacteristicRelationship sub-resource

Another Characteristic that is related to the current Characteristic;.

href	An uri (Uri). Hyperlink reference.
id	A string. Unique identifier of the characteristic.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
relationshipType	A string. The type of relationship.

CheckServiceQualificationItem sub-resource

A ServiceQualificationItem relates to a specific service being checked in a qualification operation.

© TM Forum 2021. All Rights Reserved

expectedActivationDate	A date time (DateTime). The date when the service is expected to be activated.
expectedServiceAvailabilityDate	A date time (DateTime). Date when the requester looks for service availability.
expirationDate	A date time (DateTime). Date when the qualification item response expires.
id	A string. unique identifier.
href	An uri (Uri). Hyperlink reference.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
alternateServiceProposal	A list of alternate service proposals (AlternateServiceProposal [*]). Alternate service proposal is used when the requested service is not available with characteristic and date asked for. An alternate proposal could be a distinct service Spec close to requested one or same as requested but with a different availability date.
eligibilityUnavailabilityReason	A list of service eligibility unavailability reasons (ServiceEligibilityUnavailabilityReason [*]). A list of eligibility unavailability reasons (EligibilityUnavailabilityReason [*]). Reason for eligibility result if the serviceQualification result is no (meaning the service is not available).
qualificationItemRelationship	A list of service qualification item relationships (ServiceQualificationItemRelationship [*]). A list of qualification item relationships used to describe relationship between serviceQualification item from the same serviceQualification.
qualificationRelationship	A list of service qualification relationships (ServiceQualificationRelationship [*]). Structure used to describe relationship between serviceQualification item from the same serviceQualification.
qualificationResult	A string. Qualification result for serviceQualification item. It could be: - qualified (request service are available), - unqualified (requested not available and not alternate available), - alternate (requested not available but proposal available).
state	A string. State of the serviceQualification item (acknowledged, inProgress, terminatedWithError, done).

© TM Forum 2021. All Rights Reserved

terminationError	A list of termination errors (TerminationError [*]). If qualificationItem has not been done properly this lists the error(s) that caused termination of the qualification.
category	A service category reference (ServiceCategoryRef). The category resource is used to group product offerings, service and resource candidates in logical containers. Categories can contain other categories and/or product offerings, resource or service candidates.
service	A service ref or value (ServiceRefOrValue). Configure the service characteristics (only configurable characteristics and necessary only if a non default value is selected) and/or identify the service that needs to be modified/deleted.

Feature sub-resource

Configuration feature.

constraint	A list of constraint references (ConstraintRef [*]). This is a list of feature constraints.
featureCharacteristic	A list of characteristics (Characteristic [1..*]). This is a list of Characteristics for a particular feature.
featureRelationship	A list of feature relationships (FeatureRelationship [*]). Configuration feature.
id	A string. Unique identifier of the feature.
isBundle	A boolean. True if this is a feature group. Default is false.
isEnabled	A boolean. True if this feature is enabled. Default is true.
name	A string. This is the name for the feature.

FeatureRelationship sub-resource

Configuration feature.

id	A string. Unique identifier of the target feature.
name	A string. This is the name of the target feature.
relationshipType	A string. This is the type of the feature relationship.
validFor	A time period. The period for which this feature relationship is valid.

Note sub-resource

Extra information about a given entity.

@baseType	A string. When sub-classing, this defines the super-class.

© TM Forum 2021. All Rights Reserved

@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
author	A string. Author of the note.
date	A date time (DateTime). Date of the note.
id	A string. Identifier of the note within its containing entity.
text	A string. Text of the note.

RelatedEntityRefOrValue sub-resource

A reference to an entity, where the type of the entity is not known in advance. A related entity defines a entity described by reference or by value linked to a specific entity. The polymorphic attributes @type, @schemaLocation & @referredType are related to the Entity and not the RelatedEntityRefOrValue class itself.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	A string. Reference of the related entity.
id	A string. Unique identifier of a related entity.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
role	A string.

RelatedParty sub-resource

Related Party reference. A related party defines party or party role linked to a specific entity.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.

© TM Forum 2021. All Rights Reserved

@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
role	A string. Role played by the related party.

RelatedPlaceRefOrValue sub-resource

Related Entity reference. A related place defines a place described by reference or by value linked to a specific entity. The polymorphic attributes @type, @schemaLocation & @referredType are related to the place entity and not the RelatedPlaceRefOrValue class itself.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. A user-friendly name for the place, such as [Paris Store], [London Store], [Main Home].
href	A string. Unique reference of the place.
id	A string. Unique identifier of the place.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
role	A string.

RelatedServiceOrderItem sub-resource

RelatedServiceOrderItem (a ServiceOrder item) .The service order item which triggered service creation/change/termination.

href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
@referredType	A string. The actual type of the target instance when needed for disambiguation.
itemAction	An order item action type (OrderItemActionType). Action of the order item for this service.

© TM Forum 2021. All Rights Reserved

itemId	A string. Identifier of the order item where the service was managed.
role	A string. role of the service order item for this service.
serviceOrderHref	A string. Reference of the related entity.
serviceOrderId	A string. Unique identifier of a related entity.

ServiceEligibilityUnavailabilityReason sub-resource

Reason for eligibility result if the ServiceQualification result is no (meaning the Service is not available).

href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
code	A string. Unavailability reason code.
label	A string. Unavailability reason label.

ServiceQualificationItemRelationship sub-resource

Structure used to describe relationship between serviceQualification items from the same serviceQualification.

href	An uri (Uri). Hyperlink reference.
id	A string. Id of the serviceQualificationItem (from the same serviceQualificationItem).
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
relationshipType	A string. Type of relationship (ex: reliesOn, connectedTo, etc...).

ServiceQualificationRelationship sub-resource

href	An uri (Uri). Hyperlink reference.
id	A string. The id of the target qualification pointed to by this relationship.

© TM Forum 2021. All Rights Reserved

@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
relationshipType	A string. The type of relationship.

ServiceRefOrValue sub-resource

A Service to be created defined by value or existing defined by reference. The polymorphic attributes @type, @schemaLocation & @referredType are related to the Service entity and not the RelatedServiceRefOrValue class itself.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the service.
href	A string. Reference of the service.
id	A string. Unique identifier of the service.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
category	A string. Is it a customer facing or resource facing service.
description	A string. Free-text description of the service.
endDate	A date time (DateTime). Date when the service ends.
hasStarted	A boolean. If TRUE, this Service has already been started.
isBundle	A boolean. If true, the service is a ServiceBundle which regroup a service hierachy. If false, the service is a 'atomic' service (hierachy leaf).
isServiceEnabled	A boolean. If FALSE and hasStarted is FALSE, this particular Service has NOT been enabled for use - if FALSE and hasStarted is TRUE then the service has failed.
isStateful	A boolean. If TRUE, this Service can be changed without affecting any other services.
serviceDate	A string. Date when the service was created (whatever its status).
serviceType	A string. Business type of the service.
startDate	A date time (DateTime). Date when the service starts.

© TM Forum 2021. All Rights Reserved

startMode	A string. This attribute is an enumerated integer that indicates how the Service is started, such as: 0: Unknown; 1: Automatically by the managed environment; 2: Automatically by the owning device; 3: Manually by the Provider of the Service; 4: Manually by a Customer of the Provider; 5: Any of the above.
feature	A list of features (Feature [*]). A list of feature associated with this service.
note	A list of notes (Note [*]). A list of notes made on this service.
place	A list of related place ref or values (RelatedPlaceRefOrValue [*]). A list of places (Place [*]). Used to define a place useful for the service (for example a geographical place whre the service is installed).
relatedEntity	A list of related entity ref or values (RelatedEntityRefOrValue [*]). A list of related entity in relationship with this service.
relatedParty	A list of related parties (RelatedParty [*]). A list of related party references (RelatedParty [*]). A related party defines party or party role linked to a specific entity.
serviceCharacteristic	A list of characteristics (Characteristic [*]). A list of characteristics that characterize this service (ServiceCharacteristic [*]).
serviceOrderItem	A list of related service order items (RelatedServiceOrderItem [*]). A list of service order items related to this service.
serviceRelationship	A list of service relationships (ServiceRelationship [*]). A list of service relationships (ServiceRelationship [*]). Describes links with other service(s) in the inventory.
serviceSpecification	A service specification reference (ServiceSpecificationRef). The specification from which this service was instantiated.
state	A service state type (ServiceStateType). The life cycle state of the service, such as designed, reserved, active, etc...
supportingResource	A list of resource references (ResourceRef [*]). A list of supporting resources (SupportingResource [*]).Note: only Service of type RFS can be associated with Resources.
supportingService	A list of service ref or values (ServiceRefOrValue [*]). A list of supporting services (SupportingService [*]). A collection of services that support this service (bundling, link CFS to RFS).

ServiceRelationship sub-resource

href	An uri (Uri). Hyperlink reference.

© TM Forum 2021. All Rights Reserved

id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
relationshipType	A string.
service	A service ref or value (ServiceRefOrValue). A Service to be created defined by value or existing defined by reference. The polymorphic attributes @type, @schemaLocation & @referredType are related to the Service entity and not the RelatedServiceRefOrValue class itself.
serviceRelationshipCharacteristic	A list of characteristics (Characteristic [*]). Describes a given characteristic of an object or entity through a name/value pair.

TerminationError sub-resource

This indicates an error that caused a qualificationItem to be terminated.

id	A string. Unique identifier of the termination error.
value	A string. Text to describe the termination error - for example: Unable to proceed to qualification because incomplete information provided.

ConstraintRef relationship

Constraint reference. The Constraint resource represents a policy/rule applied to an entity or entity spec.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
version	A string. constraint version.

ResourceRef relationship

© TM Forum 2021. All Rights Reserved

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.

ServiceCategoryRef relationship

The (service) category resource is used to group service candidates in logical containers. Categories can contain other categories.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
version	A string. Category version.

ServiceSpecificationRef relationship

Service specification reference: ServiceSpecification(s) required to realize a ProductSpecification.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.

© TM Forum 2021. All Rights Reserved

@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
version	A string. Service specification version.

Json representation sample

We provide below the json representation of an example of a 'CheckServiceQualification' resource object


```
{
  "id": "44",
  "href": "http://serverlocation:port/serviceQualificationManagement/v4/checkServiceQualification/456",
  "checkServiceQualificationDate": "2020-04-10T19:20:39.239Z",
  "description": "check service qualification illustration",
  "effectiveQualificationDate": "2020-04-11T15:20:39.239Z",
  "expectedQualificationDate": "2020-04-11T15:20:39.239Z",
  "expirationDate": "2020-04-24T19:20:39.239Z",
  "externalId": "CSQ1",
  "provideAlternative": true,
  "provideUnavailabilityReason": true,
  "instantSyncQualification": false,
  "qualificationResult": "alternate",
  "relatedParty": [
    {
      "id": "456",
      "href": "http://serverlocation:port/partyManagement/v4/party/456",
      "name": "Jean Pontus",
      "role": "Buyer contact",
      "@type": "RelatedParty",
      "@referredType": "Individual"
    }
  ],
  "serviceQualificationItem": [
    {
      "id": "1",
      "expectedActivationDate": "2020-04-16T19:20:39.239Z",
      "expectedServiceAvailabilityDate": "2020-04-11T19:20:39.239Z",
      "expirationDate": "2020-04-24T19:20:39.239Z",
      "qualificationResult": "alternate",
      "state": "done",
      "alternateServiceProposal": [
        {
          "id": "1",
          "alternateServiceAvailabilityDate": "2020-04-15T19:20:39.239Z",
          "alternateService": {
            "serviceCharacteristic": [
              {
                "id": "1",
                "name": "BroadcastQuality",
                "valueType": "string",
                "value": "HD",
                "@type": "ServiceCharacteristic"
              }
```


© TM Forum 2021. All Rights Reserved

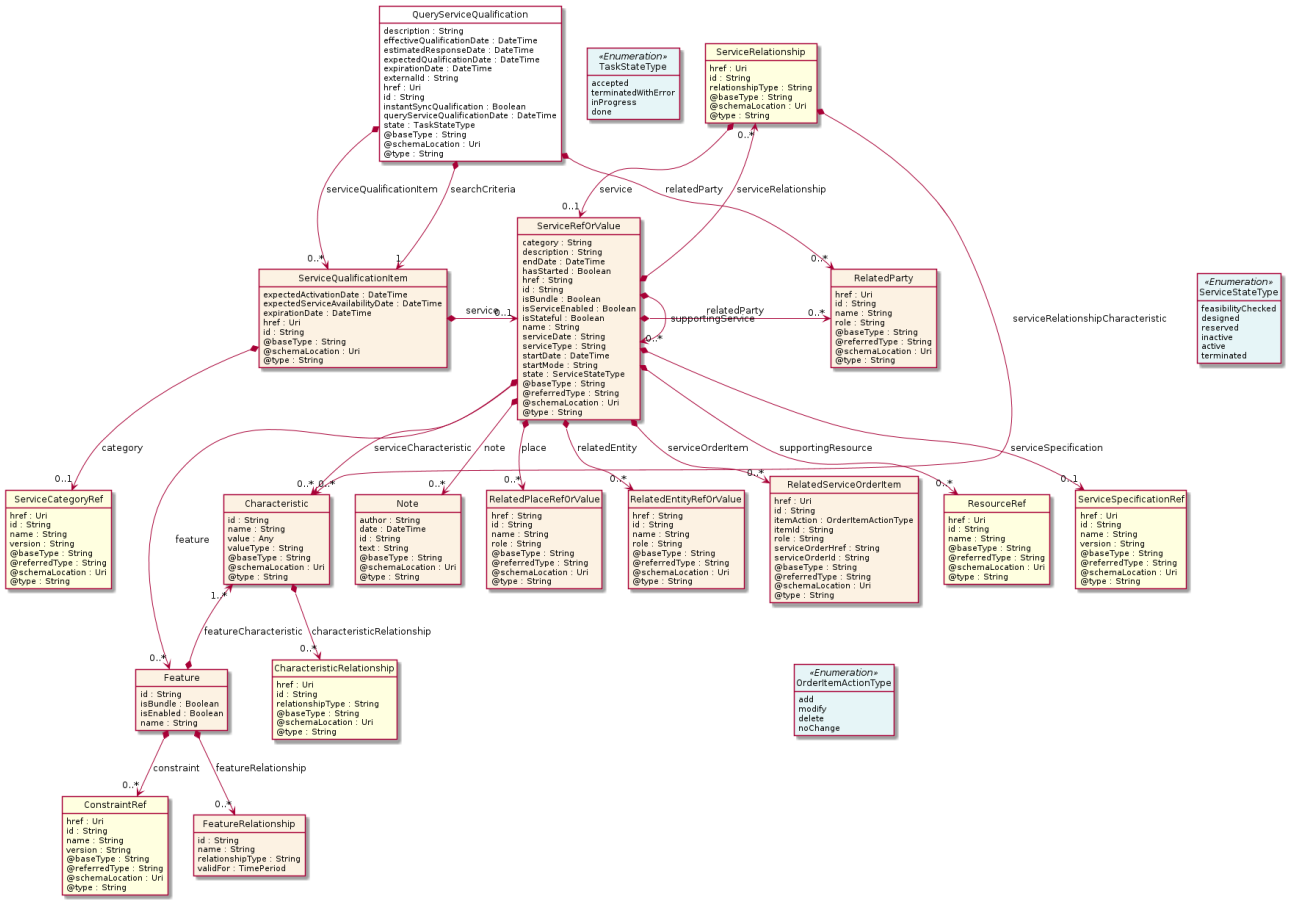
```
                ],
                "serviceSpecification": {
                  "id": "23",
                  "href": "string",
                  "name": "TVAccess",
                  "@type": "ServiceSpecification"
                },
                "@type": "Service"
              },
              "@type": "AlternateServiceProposal"
            }
          ],
          "eligibilityUnavailabilityReason": [
            {
              "code": "26-001",
              "label": "not enough bandwith at this place",
              "@baseType": "string",
              "@schemaLocation": "string",
              "@type": "string"
            }
          ],
          "service": {
            "place": [
              {
                "id": "11",
                "href": "http://serverlocation:port/geographicSiteManagement/v4/geographicSite/11",
                "name": "Jean Pontus home",
                "role": "Installation place",
                "@referredType": "GeographicSite"
              }
            ],
            "serviceCharacteristic": [
              {
                "id": "1",
                "name": "BroadcastQuality",
                "valueType": "string",
                "value": "4K"
              }
            ],
            "serviceSpecification": {
              "id": "23",
              "href": "string",
              "name": "TVAccess",
              "@type": "ServiceSpecification"
            },
            "@type": "Service"
          },
          "@type": "ServiceQualificationItem"
        }
      ],
      "state": "done",
      "@type": "CheckServiceQualification"
    }
```


Query Service Qualification resource

© TM Forum 2021. All Rights Reserved

QueryServiceQualification is used to retrieve a list of services that are technically available in the context of the interaction (place, party, service characteristics, ...).

Resource model



Field descriptions

QueryServiceQualification fields

description	A string. Description of the serviceQualification.
effectiveQualificationDate	A date time (DateTime). Effective date to serviceQualification completion.
estimatedResponseDate	A date time (DateTime). Date when the requester expect to provide an answer for the qualification request.
expectedQualificationDate	A date time (DateTime). A date (DateTime). Deadline date when the requester expected a qualification answer.
expirationDate	A date time (DateTime). Date when the qualification response expires.
externalId	A string. Identifier provided by the requester.
href	An uri (Uri). Hyperlink reference.

© TM Forum 2021. All Rights Reserved

id	A string. unique identifier.
instantSyncQualification	A boolean. An indicator which when the value is "true" means that requester expects to get qualifcation result immediately in the response. If the indicator is true then the response code of 200 indicates the operation is successful otherwise a task is created with a response 201.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
queryServiceQualificationDate	A date time (DateTime). Date when the serviceQualification was submitted.
searchCriteria	A service qualification item (ServiceQualificationItem). Search structure to retrieve eligible service specification.
serviceQualificationItem	A list of service qualification items (ServiceQualificationItem [*]).
relatedParty	A list of related parties (RelatedParty [*]). A list of related party references, defines party or party role linked to this request.
state	A task state type (TaskStateType). State of the serviceQualification (acknowledged, inProgress, terminatedWithError, done).

Characteristic sub-resource

Describes a given characteristic of an object or entity through a name/value pair.

@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
characteristicRelationship	A list of characteristic relationships (CharacteristicRelationship [*]). Another Characteristic that is related to the current Characteristic;.
id	A string. Unique identifier of the characteristic.
name	A string. Name of the characteristic.
value	An any (Any). The value of the characteristic.

© TM Forum 2021. All Rights Reserved

valueType	A string. Data type of the value of the characteristic.

CharacteristicRelationship sub-resource

Another Characteristic that is related to the current Characteristic;.

href	An uri (Uri). Hyperlink reference.
id	A string. Unique identifier of the characteristic.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
relationshipType	A string. The type of relationship.

Feature sub-resource

Configuration feature.

constraint	A list of constraint references (ConstraintRef [*]). This is a list of feature constraints.
featureCharacteristic	A list of characteristics (Characteristic [1..*]). This is a list of Characteristics for a particular feature.
featureRelationship	A list of feature relationships (FeatureRelationship [*]). Configuration feature.
id	A string. Unique identifier of the feature.
isBundle	A boolean. True if this is a feature group. Default is false.
isEnabled	A boolean. True if this feature is enabled. Default is true.
name	A string. This is the name for the feature.

FeatureRelationship sub-resource

Configuration feature.

id	A string. Unique identifier of the target feature.
name	A string. This is the name of the target feature.
relationshipType	A string. This is the type of the feature relationship.
validFor	A time period. The period for which this feature relationship is valid.

Note sub-resource

Extra information about a given entity.

© TM Forum 2021. All Rights Reserved

@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
author	A string. Author of the note.
date	A date time (DateTime). Date of the note.
id	A string. Identifier of the note within its containing entity.
text	A string. Text of the note.

RelatedEntityRefOrValue sub-resource

A reference to an entity, where the type of the entity is not known in advance. A related entity defines a entity described by reference or by value linked to a specific entity. The polymorphic attributes @type, @schemaLocation & @referredType are related to the Entity and not the RelatedEntityRefOrValue class itself.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	A string. Reference of the related entity.
id	A string. Unique identifier of a related entity.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
role	A string.

RelatedParty sub-resource

Related Party reference. A related party defines party or party role linked to a specific entity.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.

© TM Forum 2021. All Rights Reserved

@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
role	A string. Role played by the related party.

RelatedPlaceRefOrValue sub-resource

Related Entity reference. A related place defines a place described by reference or by value linked to a specific entity. The polymorphic attributes @type, @schemaLocation & @referredType are related to the place entity and not the RelatedPlaceRefOrValue class itself.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. A user-friendly name for the place, such as [Paris Store], [London Store], [Main Home].
href	A string. Unique reference of the place.
id	A string. Unique identifier of the place.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
role	A string.

RelatedServiceOrderItem sub-resource

RelatedServiceOrderItem (a ServiceOrder item) .The service order item which triggered service creation/change/termination.

href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
@referredType	A string. The actual type of the target instance when needed for disambiguation.
itemAction	An order item action type (OrderItemActionType). Action of the order item for this service.

© TM Forum 2021. All Rights Reserved

itemId	A string. Identifier of the order item where the service was managed.
role	A string. role of the service order item for this service.
serviceOrderHref	A string. Reference of the related entity.
serviceOrderId	A string. Unique identifier of a related entity.

ServiceQualificationItem sub-resource

A ServiceQualificationItem relates to a specific service being checked in a qualification operation.

href	An uri (Uri). Hyperlink reference.
id	A string. Id of the Service Qualification Item.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
category	A service category reference (ServiceCategoryRef). The category resource is used to group product offerings, service and resource candidates in logical containers. Categories can contain other categories and/or product offerings, resource or service candidates.
expectedActivationDate	A date time (DateTime). The date when the service is expected to be activated.
expectedServiceAvailabilityDate	A date time (DateTime). Date when the requester looks for service availability.
expirationDate	A date time (DateTime). Date when the qualification item response expires.
service	A service ref or value (ServiceRefOrValue). Configure the service characteristics (only configurable characteristics and necessary only if a non default value is selected) and/or identify the service that needs to be modified/deleted.

ServiceRefOrValue sub-resource

A Service to be created defined by value or existing defined by reference. The polymorphic attributes @type, @schemaLocation & @referredType are related to the Service entity and not the RelatedServiceRefOrValue class itself.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the service.

© TM Forum 2021. All Rights Reserved

href	A string. Reference of the service.
id	A string. Unique identifier of the service.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
category	A string. Is it a customer facing or resource facing service.
description	A string. Free-text description of the service.
endDate	A date time (DateTime). Date when the service ends.
hasStarted	A boolean. If TRUE, this Service has already been started.
isBundle	A boolean. If true, the service is a ServiceBundle which regroup a service hierachy. If false, the service is a 'atomic' service (hierachy leaf).
isServiceEnabled	A boolean. If FALSE and hasStarted is FALSE, this particular Service has NOT been enabled for use - if FALSE and hasStarted is TRUE then the service has failed.
isStateful	A boolean. If TRUE, this Service can be changed without affecting any other services.
serviceDate	A string. Date when the service was created (whatever its status).
serviceType	A string. Business type of the service.
startDate	A date time (DateTime). Date when the service starts.
startMode	A string. This attribute is an enumerated integer that indicates how the Service is started, such as: 0: Unknown; 1: Automatically by the managed environment; 2: Automatically by the owning device; 3: Manually by the Provider of the Service; 4: Manually by a Customer of the Provider; 5: Any of the above.
feature	A list of features (Feature [*]). A list of feature associated with this service.
note	A list of notes (Note [*]). A list of notes made on this service.
place	A list of related place ref or values (RelatedPlaceRefOrValue [*]). A list of places (Place [*]). Used to define a place useful for the service (for example a geographical place whre the service is installed).
relatedEntity	A list of related entity ref or values (RelatedEntityRefOrValue [*]). A list of related entity in relationship with this service.

© TM Forum 2021. All Rights Reserved

relatedParty	A list of related parties (RelatedParty [*]). A list of related party references (RelatedParty [*]). A related party defines party or party role linked to a specific entity.
serviceCharacteristic	A list of characteristics (Characteristic [*]). A list of characteristics that characterize this service (ServiceCharacteristic [*]).
serviceOrderItem	A list of related service order items (RelatedServiceOrderItem [*]). A list of service order items related to this service.
serviceRelationship	A list of service relationships (ServiceRelationship [*]). A list of service relationships (ServiceRelationship [*]). Describes links with other service(s) in the inventory.
serviceSpecification	A service specification reference (ServiceSpecificationRef). The specification from which this service was instantiated.
state	A service state type (ServiceStateType). The life cycle state of the service, such as designed, reserved, active, etc...
supportingResource	A list of resource references (ResourceRef [*]). A list of supporting resources (SupportingResource [*]).Note: only Service of type RFS can be associated with Resources.
supportingService	A list of service ref or values (ServiceRefOrValue [*]). A list of supporting services (SupportingService [*]). A collection of services that support this service (bundling, link CFS to RFS).

ServiceRelationship sub-resource

href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
relationshipType	A string.
service	A service ref or value (ServiceRefOrValue). A Service to be created defined by value or existing defined by reference. The polymorphic attributes @type, @schemaLocation & @referredType are related to the Service entity and not the RelatedServiceRefOrValue class itself.

© TM Forum 2021. All Rights Reserved

serviceRelationshipCharacteristic	A list of characteristics (Characteristic [*]). Describes a given characteristic of an object or entity through a name/value pair.

ConstraintRef relationship

Constraint reference. The Constraint resource represents a policy/rule applied to an entity or entity spec.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
version	A string. constraint version.

ResourceRef relationship

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.

ServiceCategoryRef relationship

The (service) category resource is used to group service candidates in logical containers. Categories can contain other categories.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.

© TM Forum 2021. All Rights Reserved

href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
version	A string. Category version.

ServiceSpecificationRef relationship

Service specification reference: ServiceSpecification(s) required to realize a ProductSpecification.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
version	A string. Service specification version.

Json representation sample

We provide below the json representation of an example of a 'QueryServiceQualification' resource object


```
{
  "id": "54",
  "href": "http://serverlocation:port/serviceQualificationManagement/v4/queryServiceQualification/54",
  "description": "Query Service Qualification Illustration - 2 serviceSpec available at this place for this category",
  "effectiveQualificationDate": "2020-04-13T13:51:41.218Z",
  "estimatedResponseDate": "2020-04-13T14:00:00.218Z",
  "expectedQualificationDate": "2020-04-13T13:51:41.218Z",
  "expirationDate": "2020-04-15T13:51:41.218Z",
  "externalId": "QSQ-456",
  "instantSyncQualification": true,
  "queryServiceQualificationDate": "2020-04-13T13:51:41.218Z",
  "relatedParty": [
    {
      "id": "456",
      "href": "http://serverlocation:port/partyManagement/v4/party/456",
      "name": "Jean Pontus",
```


© TM Forum 2021. All Rights Reserved

```
      "role": "Buyer contact",
      "@type": "RelatedParty",
      "@referredType": "Individual"
    }
  ],
  "searchCriteria": {
    "id": "1",
    "category": {
      "id": "1",
      "href": "http://serverlocation:port/serviceCatalogManagement/v4/category/1",
      "name": "Fiber Access",
      "@type": "Category"
    },
    "service": {
      "place": [
        {
          "role": "Installation Place",
          "@type": "GeographicAddress",
          "city": "San Francisco",
          "postcode": "94121",
          "streetName": "25th",
          "streetType": "Avenue",
          "streetNr": "437",
          "geographicAddress": [
            {
              "levelType": "floor",
              "levelNumber": "3"
            }
          ]
        }
      ]
    },
    "@type": "SearchCriteria"
  },
  "serviceQualificationItem": [
    {
      "id": "1",
      "service": {
        "serviceSpecification": {
          "id": "78",
          "href": "http://serverlocation:port/serviceCatalogManagement/v4/serviceSpecification/78",
          "name": "X-Fiber Service",
          "@type": "ServiceSpecification"
        },
        "@type": "Service"
      },
      "@type": "ServiceQualificationItem"
    },
    {
      "id": "2",
      "service": {
        "serviceSpecification": {
          "id": "87",
          "href": "http://serverlocation:port/serviceCatalogManagement/v4/serviceSpecification/87",
          "name": "Standard Fiber Service",
          "@type": "ServiceSpecification"
```

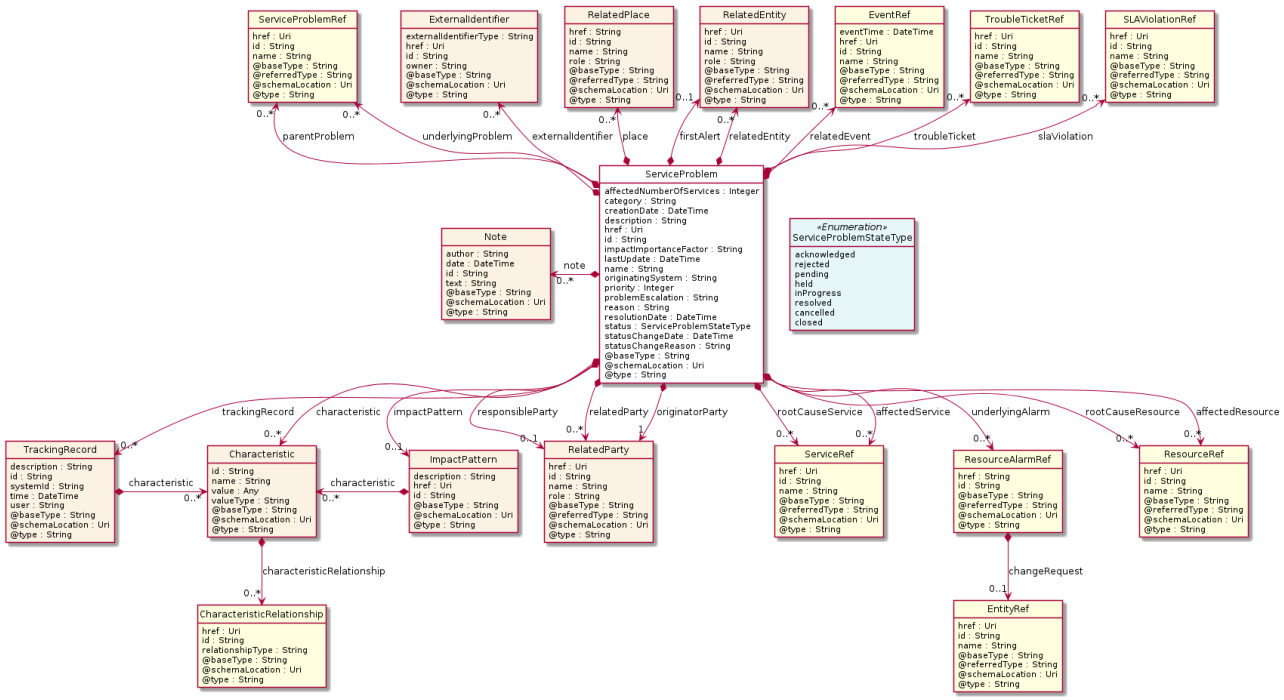

© TM Forum 2021. All Rights Reserved

```
            },
            "@type": "Service"
          },
          "@type": "ServiceQualificationItem"
        }
    ],
    "state": "done",
    "@type": "QueryServiceQualification"
}
```

Service Problem resource

The problem information for Middle B which is abstracted in the service layer from the issued event information by First B.

Resource model



Field descriptions

ServiceProblem fields

href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.

© TM Forum 2021. All Rights Reserved

affectedNumberOfServices	An integer. Number of affected services.
affectedResource	A list of resource references (ResourceRef [*]). A list of the resources affected by the problem. At least one of affectedResource, affectedService or affectedLocation should be present.
affectedService	A list of service references (ServiceRef [*]). List of affected services. At least one of affectedResource, affectedService or affectedLocation should be present.
category	A string. Classifier for the problem. Settable. For example, this is used for distinguish the category of problem originator in [role].[category] format. Example: serviceProvider.declarer, supplier.originated, system.originated.
characteristic	A list of characteristics (Characteristic [*]). Describes a given characteristic of an object or entity through a name/value pair.
creationDate	A date time (DateTime). Time the problem was created.
description	A string. Free form text describing the Service Problem.
externalIdentifier	A list of external identifiers (ExternalIdentifier [*]). An identification of an entity that is owned by or originates in a software system different from the current system, for example a ProductOrder handed off from a commerce platform into an order handling system. The structure identifies the system itself, the nature of the entity within the system (e.g. class name) and the unique ID of the entity within the system. It is anticipated that multiple external IDs can be held for a single entity, e.g. if the entity passed through multiple systems on the way to the current system. In this case the consumer is expected to sequence the IDs in the array in reverse order of provenance, i.e. most recent system first in the list.
firstAlert	A related entity (RelatedEntity). Indicates what first alerted the system to the problem. It is not the root cause of the Service Problem. Examples: Threshold crossing alert.
impactImportanceFactor	A string. Impact Importance is characterized by an Impact Importance Factor: overall importance of the impact of all the affected services, e.g. 0 (zero impact) to 100 (worst impact). The Impact Importance is a calculated field which is set by the OSS determining the impact.
impactPattern	An impact pattern (ImpactPattern). Define the patterns of impact (optional)- e.g. other service characteristics- Used when defining impact through another pattern than the predefined attributes.
lastUpdate	A date time (DateTime). Time the problem was last changed.
name	A string. Name of the Service Problem.
note	A list of notes (Note [*]). A list of comments or notes made on the problem.

© TM Forum 2021. All Rights Reserved

originatingSystem	A string. Indicates where the problem was generated.
originatorParty	A related party (RelatedParty). Individual or organization that created the problem.
parentProblem	A list of service problem references (ServiceProblemRef [*]). The parent problem to which this problem is attached.
place	A list of related places (RelatedPlace [*]). A list of the locations affected by the problem. At least one of affectedResource, affectedService or affectedLocation should be present.
priority	An integer. An indication varying from 1 (highest) to 10 (lowest) of how important it is for the service provider to correct the Service Problem.
problemEscalation	A string. Indicates if this service problem has been escalated or not. Possible values are 0 to 10. A value of zero means no escalation. The meanings of values 1-10 are to be determined by the user of the interface, but they show increasing levels of escalation.
reason	A string. Free text or optionally structured text. It can be Unknown.
relatedEntity	A list of related entities (RelatedEntity [*]). List of entities associated with this problem.
relatedEvent	A list of event references (EventRef [*]). List of events associated to this problem.
relatedParty	A list of related parties (RelatedParty [*]). List of parties or party roles playing a role within the service problem.
resolutionDate	A date time (DateTime). Time the problem was resolved.
responsibleParty	A related party (RelatedParty). Individual or organization responsible for handling this problem.
rootCauseResource	A list of resource references (ResourceRef [*]). Resource(s) that are associated to the underlying service problems that are the Root Cause of this one if any (used only if applicable).
rootCauseService	A list of service references (ServiceRef [*]). Service(s) that are associated to the underlying service problems that are the Root Cause of this one if any (used only if applicable).
slaViolation	A list of s l a violation references (SLAViolationRef [*]). A List of SLA violations associated with this problem.
status	A service problem state type (ServiceProblemStateType). Possible values for the state of the ServiceProblem.
statusChangeDate	A date time (DateTime). Time the problem was last status changed.

© TM Forum 2021. All Rights Reserved

statusChangeReason	A string. The reason of state change.
trackingRecord	A list of tracking records (TrackingRecord [*]). List of tracking records that allow the tracking of modifications on the problem.The tracking records should not be embedded in the problem to allow retrieving the problem without the tracking records.
troubleTicket	A list of trouble ticket references (TroubleTicketRef [*]). A list of trouble tickets associated with this problem.
underlyingAlarm	A list of resource alarm references (ResourceAlarmRef [*]). A list of alarms underlying this problem.
underlyingProblem	A list of service problem references (ServiceProblemRef [*]). A list of underlying problems. Relevant only if this problem is derived from other problems.

Characteristic sub-resource

Describes a given characteristic of an object or entity through a name/value pair.

@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
characteristicRelationship	A list of characteristic relationships (CharacteristicRelationship [*]). Another Characteristic that is related to the current Characteristic;.
id	A string. Unique identifier of the characteristic.
name	A string. Name of the characteristic.
value	An any (Any). The value of the characteristic.
valueType	A string. Data type of the value of the characteristic.

CharacteristicRelationship sub-resource

Another Characteristic that is related to the current Characteristic;.

href	An uri (Uri). Hyperlink reference.
id	A string. Unique identifier of the characteristic.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.

© TM Forum 2021. All Rights Reserved

@type	A string. When sub-classing, this defines the sub-class Extensible name.
relationshipType	A string. The type of relationship.

ExternalIdentifier sub-resource

An identification of an entity that is owned by or originates in a software system different from the current system, for example a ProductOrder handed off from a commerce platform into an order handling system. The structure identifies the system itself, the nature of the entity within the system (e.g. class name) and the unique ID of the entity within the system. It is anticipated that multiple external IDs can be held for a single entity, e.g. if the entity passed through multiple systems on the way to the current system. In this case the consumer is expected to sequence the IDs in the array in reverse order of provenance, i.e. most recent system first in the list.

href	An uri (Uri). Hyperlink reference.
id	A string. identification of the entity within the external system.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
externalIdentifierType	A string. Type of the identification, typically would be the type of the entity within the external system.
owner	A string. Name of the external system that owns the entity.

ImpactPattern sub-resource

Define the patterns of impact (optional), such as other service characteristics- Used when defining impact through another pattern than the pre-defined attributes.

href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
characteristic	A list of characteristics (Characteristic [*]). A generic list of any type of elements. Used for extensions or loose element encapsulation from other namespaces.
description	A string. Basic description of the impact pattern.

Note sub-resource

© TM Forum 2021. All Rights Reserved

Extra information about a given entity.

@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
author	A string. Author of the note.
date	A date time (DateTime). Date of the note.
id	A string. Identifier of the note within its containing entity.
text	A string. Text of the note.

RelatedEntity sub-resource

A reference to an entity, where the type of the entity is not known in advance.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
role	A string. The role of an entity.

RelatedParty sub-resource

Related Party reference. A related party defines party or party role linked to a specific entity.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.

© TM Forum 2021. All Rights Reserved

@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
role	A string. Role played by the related party.

RelatedPlace sub-resource

A Place and an associated role as installation adress, delivery address, etc....

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. A user-friendly name for the place, such as [Paris Store], [London Store], [Main Home].
href	A string. Unique reference of the place.
id	A string. Unique identifier of the place.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
role	A string. Role of the place, such as: [home delivery], [shop retrieval]).

TrackingRecord sub-resource

Tracking records allow the tracking of modifications on the problem. The tracking records should not be embedded in the problem to allow retrieving the problem without the tracking records.

@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
characteristic	A list of characteristics (Characteristic [*]). A generic list of any type of elements. Used for vendor Extensions or loose element encapsulation from other namespaces.
description	A string. Describes the action being done, such as: ack, clear.
id	A string. Identifier of the TrackingRecord.
systemId	A string. Describes the system Id from which the action was done.
time	A date time (DateTime). Describes the time at which the action was done.

© TM Forum 2021. All Rights Reserved

user	A string. Describes the user doing the action.

EntityRef relationship

Entity reference schema to be use for all entityRef class.

href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.

EventRef relationship

Events linked with service problem.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
eventTime	A date time (DateTime). Time the event occurred.

ResourceAlarmRef relationship

A set of alarm ids identifying the alarms that are underlying this problem.

href	A string. Reference of the Alarm.
id	A string. Unique identifier of the Alarm.
@baseType	A string. When sub-classing, this defines the super-class.

© TM Forum 2021. All Rights Reserved

@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
@referredType	A string. The actual type of the target instance when needed for disambiguation.
changeRequest	An entity reference (EntityRef). Entity reference schema to be use for all entityRef class.

ResourceRef relationship

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.

SLAViolationRef relationship

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.

ServiceProblemRef relationship

© TM Forum 2021. All Rights Reserved

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.

ServiceRef relationship

Service reference, for when Service is used by other entities.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.

TroubleTicketRef relationship

TroubleTicket reference, for when a Trouble Ticket is used by other entities.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.

© TM Forum 2021. All Rights Reserved

Json representation sample

We provide below the json representation of an example of a 'ServiceProblem' resource object


```
{
  "id": "problemxxxx0000",
  "correlationId": "543251",
  "originatingSystem": "System_001",
  "category": "supplier.originated",
  "href": "https://mycsp.com:8080/tmf-api/serviceProblemManagement/v4/serviceProblem/problemxxxx0000",
  "impactImportanceFactor": "0",
  "priority": "1",
  "description": "connection failure between Tokyo and Osaka",
  "problemEscalation": "0",
  "timeRaised": "2017-10-25T12:14:16.361Z",
  "timeChanged": "2017-10-30T12:13:16.361Z",
  "statusChangeDate": "2017-10-29T12:00:00.361Z",
  "statusChangeReason": "problem analysis has been completed in NP1",
  "resolutionDate": "2017-10-29T12:00:00.361Z",
  "status": "resolved",
  "reason": "Failure of resource NP1_Resource_1 in NP1",
  "affectedNumberOfServices": "2",
  "firstAlert": {
    "alertType": "TroubleTicket",
    "id": "NP1_TT_0000000",
    "href": "https://mycsp.com:8080/tmf-api/troubleTicketManagement/v4/troubleTicket/NP1_TT_000000"
  },
  "responsibleParty": {
    "role": "Supplier",
    "id": "NP1",
    "href": "https://mycsp.com:8080/tmf-api/partyManagement/v4/party/NP1",
    "@referredType": "Organization"
  },
  "originatorParty": {
    "role": "Supplier",
    "id": "NP1",
    "href": "https://mycsp.com:8080/tmf-api/partyManagement/v4/party/NP1",
    "@referredType": "Organization"
  },
  "relatedParty": [
    {
      "role": "Supplier",
      "id": "NP1",
      "href": "https://mycsp.com:8080/tmf-api/partyManagement/v4/party/NP1",
      "@referredType": "Organization"
    },
    {
      "role": "Partner",
      "id": "SP1",
      "href": "https://mycsp.com:8080/tmf-api/partyManagement/v4/party/SP1",
      "@referredType": "Organization"
    },
    {
      "role": "Partner",
      "id": "SP3",
```


© TM Forum 2021. All Rights Reserved

```
        "href": "https://mycsp.com:8080/tmf-api/partyManagement/v4/party/SP3",
        "@referredType": "Organization"
      }
    ],
    "affectedService": [
      {
        "id": "NP1_Tokyo_Osaka",
        "href": "https://mycsp.com:8080/tmf-api/serviceInventoryManagement/v4/service/NW1_Tokyo_Osaka"
      },
      {
        "id": "NP1_Tokyo_xxxx",
        "href": "https://mycsp.com:8080/tmf-api/serviceInventoryManagement/v4/service/NW1_Tokyo_xxxx"
      }
    ],
    "affectedResource": [
      {
        "id": "NP1_RES_0001",
        "href": "https://mycsp.com:8080/tmf-api/resourceInventoryManagement/v4/resource/NW1_RES_0001"
      }
    ],
    "affectedLocation": [
      {
        "id": "Loc000000",
        "href": "https://mycsp.com:8080/tmf-api/geographicAddressManagement/v4/geographicAddress/Loc000000/",
        "name": "144 Main Street Tokyo 51663556",
        "role": "VPN Endpoint",
        "@type": "GeographicAddress"
      },
      {
        "id": "Loc000001",
        "href": "https://mycsp.com:8080/tmf-api/geographicAddressManagement/v4/geographicAddress/Loc000001/",
        "name": "351 Main Street Osaka 3546365",
        "role": "VPN Endpoint",
        "@type": "GeographicAddress"
      }
    ],
    "associatedTroubleTicket": [
      {
        "id": "NP1_TT_0000000",
        "href": "https://mycsp.com:8080/tmf-api/troubleTicketManagement/v4/TroubleTicketRef/NP1_TT_000000"
      }
    ],
    "underlyingAlarm": [
      {
        "id": "NP1_A_0000000",
        "href": "https://mycsp.com:8080/tmf-api/alarmManagement/v4/resourceAlarm/NP1_A_000000",
        "@referredType": "ResourceAlarm"
      }
    ],
    "associatedSLAViolation": [
      {
        "id": "NP1_SLA_0000000",
        "href": "https://mycsp.com:8080/tmf-api/SLAManagement/v4/SLAViolationRef/NP1_SLA_000000"
      }
    ],
    "relatedEvent": [
```


© TM Forum 2021. All Rights Reserved

```
    {
      "eventType": "prediction",
      "id": "prediction_0001",
      "href": "https://mycsp.com:8080/tmf-api/eventManagement/v4/event/prediction_0001",
      "eventTime": "2014-12-20T17:00:00Z"
    }
  ],
  "relatedObject": [
    {
      "id": "product0001",
      "href": "https://mycsp.com:8080/tmf-api/productInventoryManagement/v4/product/product0001",
      "@referredType": "Product"
    }
  ],
  "rootCauseService": [
    {
      "id": "NP1_Tokyo_Osaka",
      "href": "https://mycsp.com:8080/tmf-api/serviceInventoryManagement/v4/service/NW1_Tokyo_Osaka"
    }
  ],
  "rootCauseResource": [
    {
      "id": "NP1_Resource_1",
      "href": "https://mycsp.com:8080/tmf-api/resourceInventoryManagement/v4/resource/NP1_Resource_1"
    }
  ],
  "parentProblem": [
    {
      "id": "problemxxxx0001",
      "correlationId": "xxxxxxxx",
      "href": "https://mycsp.com:8080/tmf-api/serviceProblemManagement/v4/serviceProblem/problemxxxx0001"
    }
  ],
  "underlyingProblem": [
    {
      "id": "problemxxxx0001",
      "correlationId": "xxxxxxxx",
      "href": "https://mycsp.com:8080/tmf-api/serviceProblemManagement/v4/serviceProblem/problemxxxx0001"
    }
  ],
  "trackingRecord": [
    {
      "description": "yyy cleared the problem",
      "systemId": "xxxx",
      "time": "2014-12-20T17:00:00Z",
      "user": "NPUSER1",
      "id": "TR_xxxx",
      "extensionInfo": [
        {
          "name": "vendorComment",
          "valueType": "string",
          "value": "Watch out for the dog"
        }
      ]
    }
  ],
```


© TM Forum 2021. All Rights Reserved

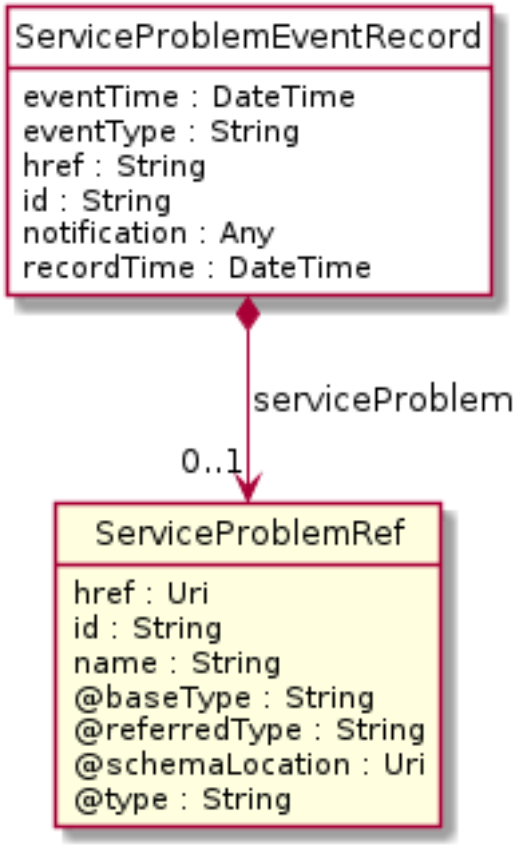
```
  "comment": [
    {
      "author": {
        "id": "SPM_handler_01",
        "href": "https://mycsp.com:8080/tmf-api/partyManagement/v4/party/SPM_handler_01",
        "@referredType": "Individual"
      },
      "date": "2014-12-20T17:00:00Z",
      "systemId": "System_002",
      "text": "receive trouble ticket from NP1, and create this Service Problem"
    },
    {
      "author": {
        "id": "NP1_handler_11",
        "href": "https://mycsp.com:8080/tmf-api/partyManagement/v4/party/NP1_handler_11",
        "@referredType": "Individual"
      },
      "date": "2014-12-20T17:00:00Z",
      "systemId": "System_002",
      "text": "status changed to Progress-Held"
    }
  ],
  "impactPatterns": {
    "description": "Many services are at risk in this problem",
    "extensionInfo": [
      {
        "name": "ImpactProbability",
        "valueType": "boolean",
        "value": true
      }
    ]
  },
  "extensionInfo": [
    {
      "name": "EstimatedCost",
      "valueType": "integer",
      "value": 20
    }
  ]
}
```


Service Problem Event Record resource

A record of an event (related to a service problem) received from another system.

Resource model

© TM Forum 2021. All Rights Reserved



Field descriptions

ServiceProblemEventRecord fields

eventTime	A date time (DateTime). Time at which the event occurred.
eventType	A string. Type of the recorded event.
href	A string. reference to this resource.
id	A string. Identifier of the service problem event record.
notification	An any (Any). A notification from the possible notifications for Service Problem (such as creation, status change, information required, change).
recordTime	A date time (DateTime). Time at which the record was created.
serviceProblem	A service problem reference (ServiceProblemRef). The service problem to which this record applies.

ServiceProblemRef relationship

@referredType	A string. The actual type of the target instance when needed for disambiguation.

© TM Forum 2021. All Rights Reserved

name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.

Json representation sample

We provide below the json representation of an example of a 'ServiceProblemEventRecord' resource object

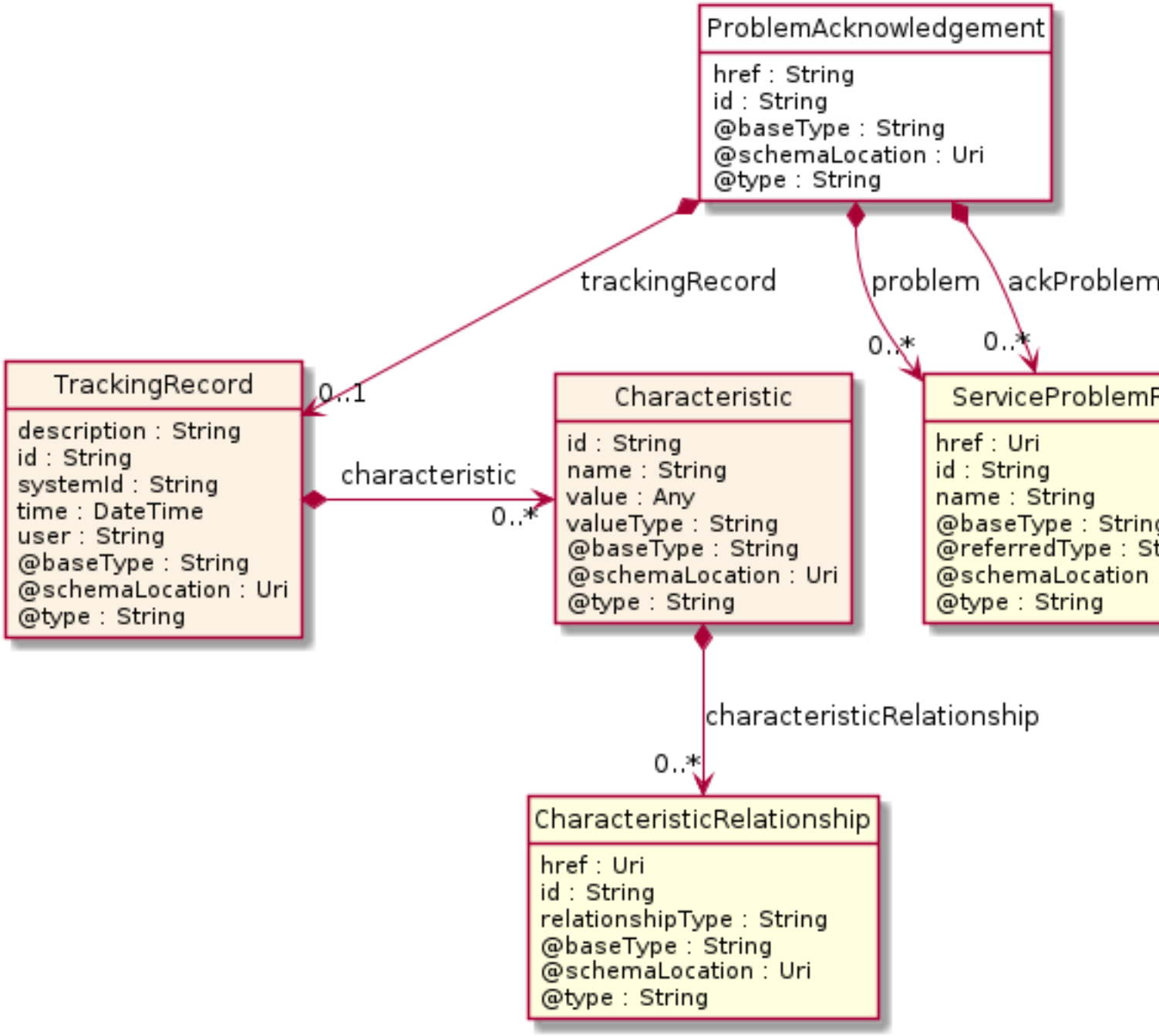

```
{
  "id": "42",
  "href": "https://mycsp.com:8080/tmf-api/serviceProblemManagement/v4/serviceProblemEventRecord/42",
  "recordTime": "2016-08-08T10:45:30.0Z",
  "eventType": "ServiceProblemCreationNotification",
  "eventTime": "2016-08-08T10:45:25.0Z",
  "serviceProblemId": "SP001",
  "notification": {
    "eventType": "ServiceProblemCreationNotification",
    "eventTime": "2016-08-08T10:45:25.0Z",
    "eventId": "92775",
    "event": {}
  },
  "serviceProblem": {
    "id": "problemxxxx0000",
    "href": "https://mycsp.com:8080/tmf-api/serviceProblemManagement/v4/serviceProblem/problemxxxx0000"
  }
}
```


Problem Acknowledgement resource

Task resource that requests acknowledgement of problems by the problem handler.

Resource model

© TM Forum 2021. All Rights Reserved



Field descriptions

ProblemAcknowledgement fields

href	A string. Reference to this task resource.
id	A string. Unique identifier of this task resource.
@baseType	A string. When sub-classing, this defines the super-class.

© TM Forum 2021. All Rights Reserved

@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
ackProblem	A list of service problem references (ServiceProblemRef [*]). The problems that were acknowledged, populated in the output to this task.
href	A string. Reference to this task resource.
id	A string. Unique identifier of this task resource.
problem	A list of service problem references (ServiceProblemRef [*]). The problems to be acknowledged, relevant in the input to this task.
trackingRecord	A tracking record (TrackingRecord). A record of the action taken in this acknowledgement.

Characteristic sub-resource

Describes a given characteristic of an object or entity through a name/value pair.

@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
characteristicRelationship	A list of characteristic relationships (CharacteristicRelationship [*]). Another Characteristic that is related to the current Characteristic;.
id	A string. Unique identifier of the characteristic.
name	A string. Name of the characteristic.
value	An any (Any). The value of the characteristic.
valueType	A string. Data type of the value of the characteristic.

CharacteristicRelationship sub-resource

Another Characteristic that is related to the current Characteristic;.

href	An uri (Uri). Hyperlink reference.
id	A string. Unique identifier of the characteristic.
@baseType	A string. When sub-classing, this defines the super-class.

© TM Forum 2021. All Rights Reserved

@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
relationshipType	A string. The type of relationship.

TrackingRecord sub-resource

Tracking records allow the tracking of modifications on the problem. The tracking records should not be embedded in the problem to allow retrieving the problem without the tracking records.

@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
characteristic	A list of characteristics (Characteristic [*]). A generic list of any type of elements. Used for vendor Extensions or loose element encapsulation from other namespaces.
description	A string. Describes the action being done, such as: ack, clear.
id	A string. Identifier of the TrackingRecord.
systemId	A string. Describes the system Id from which the action was done.
time	A date time (DateTime). Describes the time at which the action was done.
user	A string. Describes the user doing the action.

ServiceProblemRef relationship

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.

Json representation sample

© TM Forum 2021. All Rights Reserved

We provide below the json representation of an example of a 'ProblemAcknowledgement' resource object

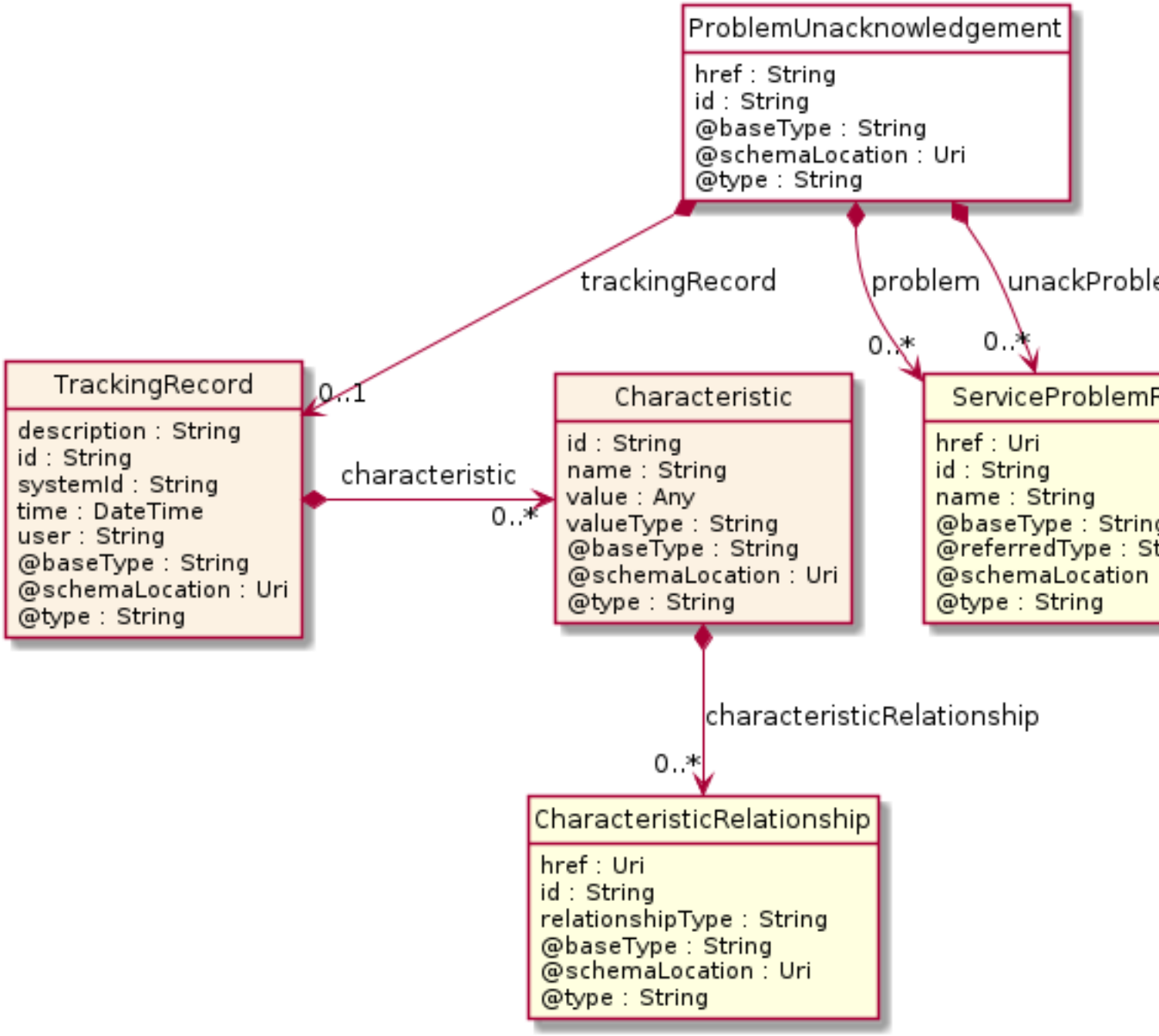

```
{
   "problem": [
      {
         "id": "41",
         "href": "https://mycsp.com:8080/tmf-api/serviceProblemManagement/v4/serviceProblem/41"
      },
      {
         "id": "42",
         "href": "https://mycsp.com:8080/tmf-api/serviceProblemManagement/v4/serviceProblem/42"
      },
      {
         "id": "43",
         "href": "https://mycsp.com:8080/tmf-api/serviceProblemManagement/v4/serviceProblem/43"
      }
   ],
   "trackingRecord": {
      "description": "yyy ack the problem",
      "systemId": "xxxx",
      "time": "2014-12-20T17:00:00Z",
      "user": "NP1"
   }
}
```


Problem Unacknowledgement resource

Task resource that requests unacknowledgement of problems, rolling back the status of the problems from Acknowledged to Submitted.

Resource model

© TM Forum 2021. All Rights Reserved



Field descriptions

ProblemUnacknowledgement fields

href	A string. Reference to this task resource.
id	A string. Unique identifier of this task resource.
@baseType	A string. When sub-classing, this defines the super-class.

© TM Forum 2021. All Rights Reserved

@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
href	A string. Reference to this task resource.
id	A string. Unique identifier of this task resource.
problem	A list of service problem references (ServiceProblemRef [*]). The problems to be unacknowledged, relevant in the input to this task.
trackingRecord	A tracking record (TrackingRecord). A record of the action taken in this acknowledgement.
unackProblem	A list of service problem references (ServiceProblemRef [*]). The problems that were unacknowledged, populated in the output to this task.

Characteristic sub-resource

Describes a given characteristic of an object or entity through a name/value pair.

@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
characteristicRelationship	A list of characteristic relationships (CharacteristicRelationship [*]). Another Characteristic that is related to the current Characteristic;.
id	A string. Unique identifier of the characteristic.
name	A string. Name of the characteristic.
value	An any (Any). The value of the characteristic.
valueType	A string. Data type of the value of the characteristic.

CharacteristicRelationship sub-resource

Another Characteristic that is related to the current Characteristic;.

href	An uri (Uri). Hyperlink reference.
id	A string. Unique identifier of the characteristic.
@baseType	A string. When sub-classing, this defines the super-class.

© TM Forum 2021. All Rights Reserved

@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
relationshipType	A string. The type of relationship.

TrackingRecord sub-resource

Tracking records allow the tracking of modifications on the problem. The tracking records should not be embedded in the problem to allow retrieving the problem without the tracking records.

@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
characteristic	A list of characteristics (Characteristic [*]). A generic list of any type of elements. Used for vendor Extensions or loose element encapsulation from other namespaces.
description	A string. Describes the action being done, such as: ack, clear.
id	A string. Identifier of the TrackingRecord.
systemId	A string. Describes the system Id from which the action was done.
time	A date time (DateTime). Describes the time at which the action was done.
user	A string. Describes the user doing the action.

ServiceProblemRef relationship

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.

Json representation sample

© TM Forum 2021. All Rights Reserved

We provide below the json representation of an example of a 'ProblemUnacknowledgement' resource object

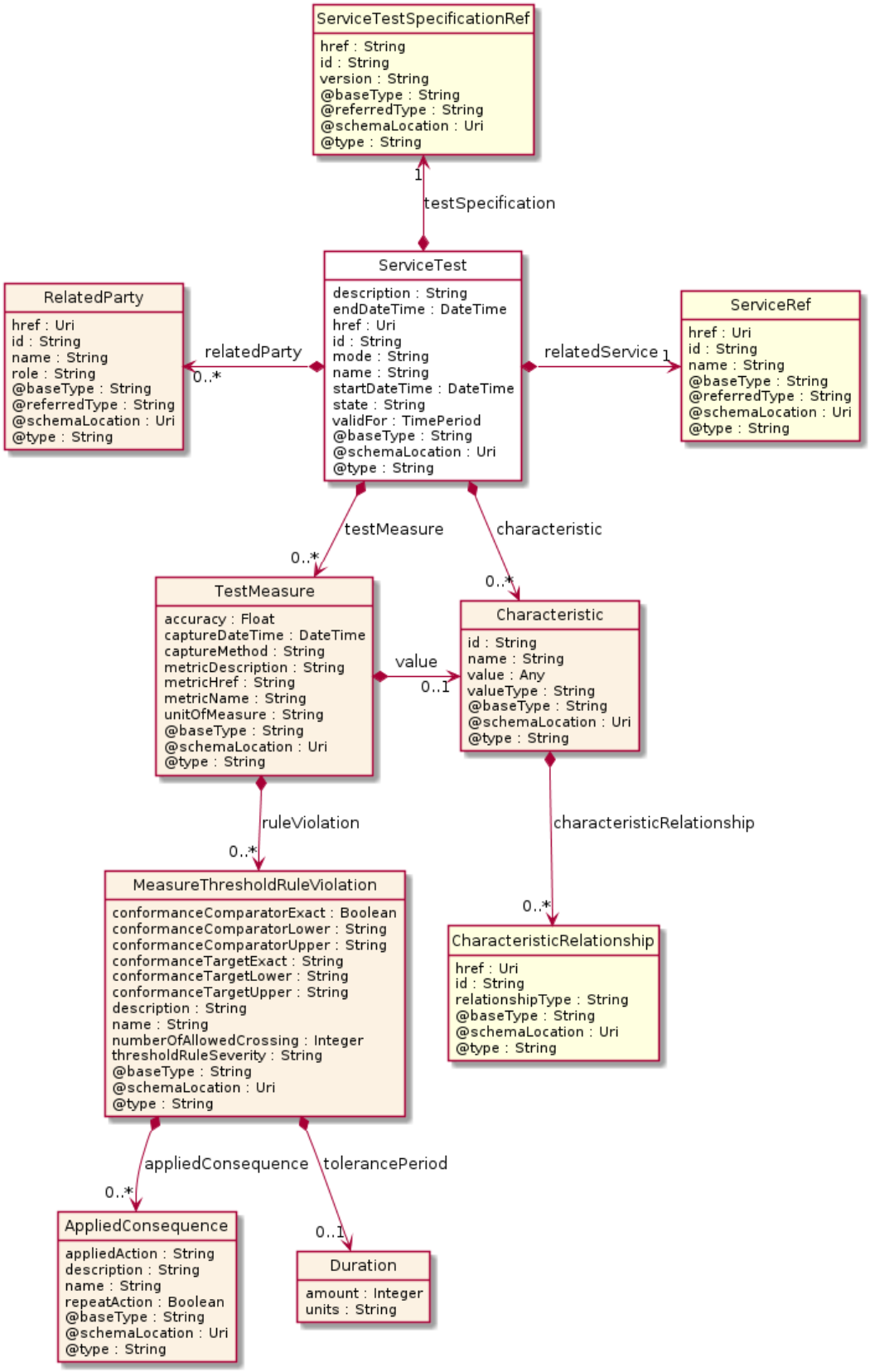

```
{
   "problem": [
      {
         "id": "41",
         "href": "https://mycsp.com:8080/tmf-api/serviceProblemManagement/v4/serviceProblem/41"
      },
      {
         "id": "42",
         "href": "https://mycsp.com:8080/tmf-api/serviceProblemManagement/v4/serviceProblem/42"
      },
      {
         "id": "43",
         "href": "https://mycsp.com:8080/tmf-api/serviceProblemManagement/v4/serviceProblem/43"
      }
   ],
   "trackingRecord": {
      "description": "yyy unack the problem",
      "systemId": "xxxx",
      "time": "2014-12-20T17:00:00Z",
      "user": "NP1"
   }
}
```


Service Test resource

A service test is an entity that exists for a controlled test invocation on a service. The service test is executed according to a schedule and contains service test configuration parameters that are to be applied at execution time, and service test measures that result.

Resource model

© TM Forum 2021. All Rights Reserved



Field descriptions

ServiceTest fields

© TM Forum 2021. All Rights Reserved

@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
characteristic	A list of characteristics (Characteristic [*]). List of characteristics with values that define the test run.
description	A string. Description of the service test.
endDateTime	A date time (DateTime). The end date and time of the service test.
mode	A string. An indication of whether the service test is running in "PROACTIVE" or "ONDEMAND" mode.
name	A string. The name of the service test.
relatedParty	A list of related parties (RelatedParty [*]). Party related to the test.
relatedService	A service reference (ServiceRef). The actual service being tested.
startDateTime	A date time (DateTime). The start date and time of the service test.
state	A string. The actual state the service test is in. Recommended states are found in ExecutionStateType schema possible values include acknowledged, rejected, pending, inProgress, cancelled, completed and failed.
testMeasure	A list of test measures (TestMeasure [*]). The results of the test in terms of the measured metrics.
testSpecification	A service test specification reference (ServiceTestSpecificationRef). The specification for this test.
validFor	A time period. The validity time for the test results.

AppliedConsequence sub-resource

An Applied Consequence defines the action (prescribed action or notification) to take when a MeasureThresholdRuleViolation occurs.

@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.

© TM Forum 2021. All Rights Reserved

@type	A string. When sub-classing, this defines the sub-class Extensible name.
appliedAction	A string. The action for a violated threshold. This could be a hyperlink to the action.
description	A string. A narrative that explains in detail what the consequence is.
name	A string. A word, term, or phrase by which Consequence is known and distinguished from other MetricDefMeasureConsequences.
repeatAction	A boolean. An indicator used to specify that a consequence should cease being applied if a value is in the same range as the previous value or continue being applied if a value is in the same range as the previous value. If the repeatAction is True, if the consequence is always applied as soon as the MetricMeasure value is in the range of values and if the repeatAction is False, the consequence is applied only if the previous MetricMeasure value was not in the same range.

Characteristic sub-resource

Describes a given characteristic of an object or entity through a name/value pair.

@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
characteristicRelationship	A list of characteristic relationships (CharacteristicRelationship [*]). Another Characteristic that is related to the current Characteristic;.
id	A string. Unique identifier of the characteristic.
name	A string. Name of the characteristic.
value	An any (Any). The value of the characteristic.
valueType	A string. Data type of the value of the characteristic.

CharacteristicRelationship sub-resource

Another Characteristic that is related to the current Characteristic;.

href	An uri (Uri). Hyperlink reference.
id	A string. Unique identifier of the characteristic.
@baseType	A string. When sub-classing, this defines the super-class.

© TM Forum 2021. All Rights Reserved

@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
relationshipType	A string. The type of relationship.

Duration sub-resource

A time interval in a given unit of time.

amount	An integer. Time interval (number of seconds, minutes, hours, etc.).
units	A string. Unit of time (seconds, minutes, hours, etc.).

MeasureThresholdRuleViolation sub-resource

A measureThresholdRuleViolation is a violation of a rule that defines the in the MericDefMeasureThresholdRule.

@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
appliedConsequence	A list of applied consequences (AppliedConsequence [*]). An Applied Consequence defines the action (prescribed action or notification) to take when a MeasureThresholdRuleViolation occurs.
conformanceComparatorExact	A boolean. An operator that when applied on a value specifies whether it is the same target value or not. This operator is used to compare with the conformanceTargetExact if used.
conformanceComparatorLower	A string. An operator that when applied on a value specifies whether a threshold is crossed or ceased to be crossed. This operator is used to compare with the conformanceTargetLower if used.
conformanceComparatorUpper	A string. An operator that when applied on a value specifies whether a threshold is crossed or ceased to be crossed. This operator is used to compare with the conformanceTargetUpper if used.
conformanceTargetExact	A string. To cater for values that are not numerical test metrics (e.g. a DSL line can be Synchronised or Unsynchronised. If the latter, the test should result in a rule violation).The allowed value can contain a REGEX expression.
conformanceTargetLower	A string. A value used to determine if the threshold is crossed or ceases to be crossed. It represents the lower limit. The value should be less than the

© TM Forum 2021. All Rights Reserved

	conformanceTargetUpper if used. The conformance comparators should also be logically defined so as to not lead to a logically impossible condition.
conformanceTargetUpper	A string. A value used to determine if the threshold is crossed or ceases to be crossed. It represents the Upper limit. The value should be greater than the conformanceTargetLower if used. The conformance comparators should also be logically defined so as to not lead to a logically impossible condition.
description	A string. Description for the MetricDefMeasureThresholdRule .
name	A string. Name for the MetricDefMeasureThresholdRule .
numberOfAllowedCrossing	An integer. The number of allowed crossing occurrences in reference to the tolerancePeriod without a consequence being initiated.
thresholdRuleSeverity	A string. A threshold can be generated in different severity levels. A crossing for each level may require a different condition and possibly trigger a different consequence.
tolerancePeriod	A duration (Duration). An interval of time of allowed crossing occurrences before a consequence being initiated.

RelatedParty sub-resource

Related Party reference. A related party defines party or party role linked to a specific entity.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
role	A string. Role played by the related party.

TestMeasure sub-resource

A TestMeasure specifies a measure of a specific aspect of a product, service, or resource test, such as lost packets or connectivity status.

@baseType	A string. When sub-classing, this defines the super-class.

© TM Forum 2021. All Rights Reserved

@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
accuracy	A float. The number of digits of accuracy captured for associated Metrics.
captureDateTime	A date time (DateTime). The date and time that the metric was captured.
captureMethod	A string. The method used to capture the Metrics (This may be replaced by a set of entities similar to the Performance Monitoring Ref).
metricDescription	A string. Brief description of the metric.
metricHref	A string. Hyperlink to access a metric for detail information.
metricName	A string. The name of the metric.
ruleViolation	A list of measure threshold rule violations (MeasureThresholdRuleViolation [*]). A list of rules that were violated in this test measure.
unitOfMeasure	A string. The unit of measure for the metric values, such as meters, cubic yards, kilograms [ISO 1000].
value	A characteristic (Characteristic). The value of Metric in the test. This also could be a document to record all values captured during the service test.

ServiceRef relationship

Service reference, for when Service is used by other entities.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.

ServiceTestSpecificationRef relationship

The service test specification used by the service test.

@baseType	A string. When sub-classing, this defines the super-class.

© TM Forum 2021. All Rights Reserved

@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
@referredType	A string. The actual type of the target instance when needed for disambiguation.
href	A string. Hyperlink to access a service test specification.
id	A string. Identifier of a service test specification.
version	A string. Version of a service test specification.

Json representation sample

We provide below the json representation of an example of a 'ServiceTest' resource object


```
{
  "href": "https://mycsp.com:8080/tmf-api/serviceTestManagement/v4/serviceTest/3112",
  "id": "3112",
  "description:": "This service test is a simple one",
  "endDateTime": "2016-03-02T24:00:00",
  "mode": "PROACTIVE",
  "startDateTime": "2016-03-02T00:00:00",
  "state": "Completed",
  "relatedService": {
    "href": "https://mycsp.com:8080/tmf-api/serviceManagement/v4/service/21",
    "id": "21",
    "name": "gameService"
  },
  "testSpecification": {
    "href": "https://mycsp.com:8080/tmf-api/serviceTestManagement/v4/serviceTestSpecification/22",
    "id": "22",
    "@referredType": "PerformanceTestSpecification"
  },
  "characteristic": [
    {
      "name": "number",
      "valueType": "integer",
      "value": 3
    },
    {
      "name": "restart",
      "valueType": "boolean",
      "value": true
    }
  ],
  "testMeasure": [
    {
      "accuracy": 0.0001,
      "captureDateTime": "2016-03-02T11:12:00",
      "captureMethod": "automatic capture by instrument",
      "metricDescription": "This is the flow speed in a service",
      "metricHref": "https://mycsp.com:8080/tmf-api/metricManagement/v4/metric/21",
      "metricName": "Flow speed",
```


© TM Forum 2021. All Rights Reserved

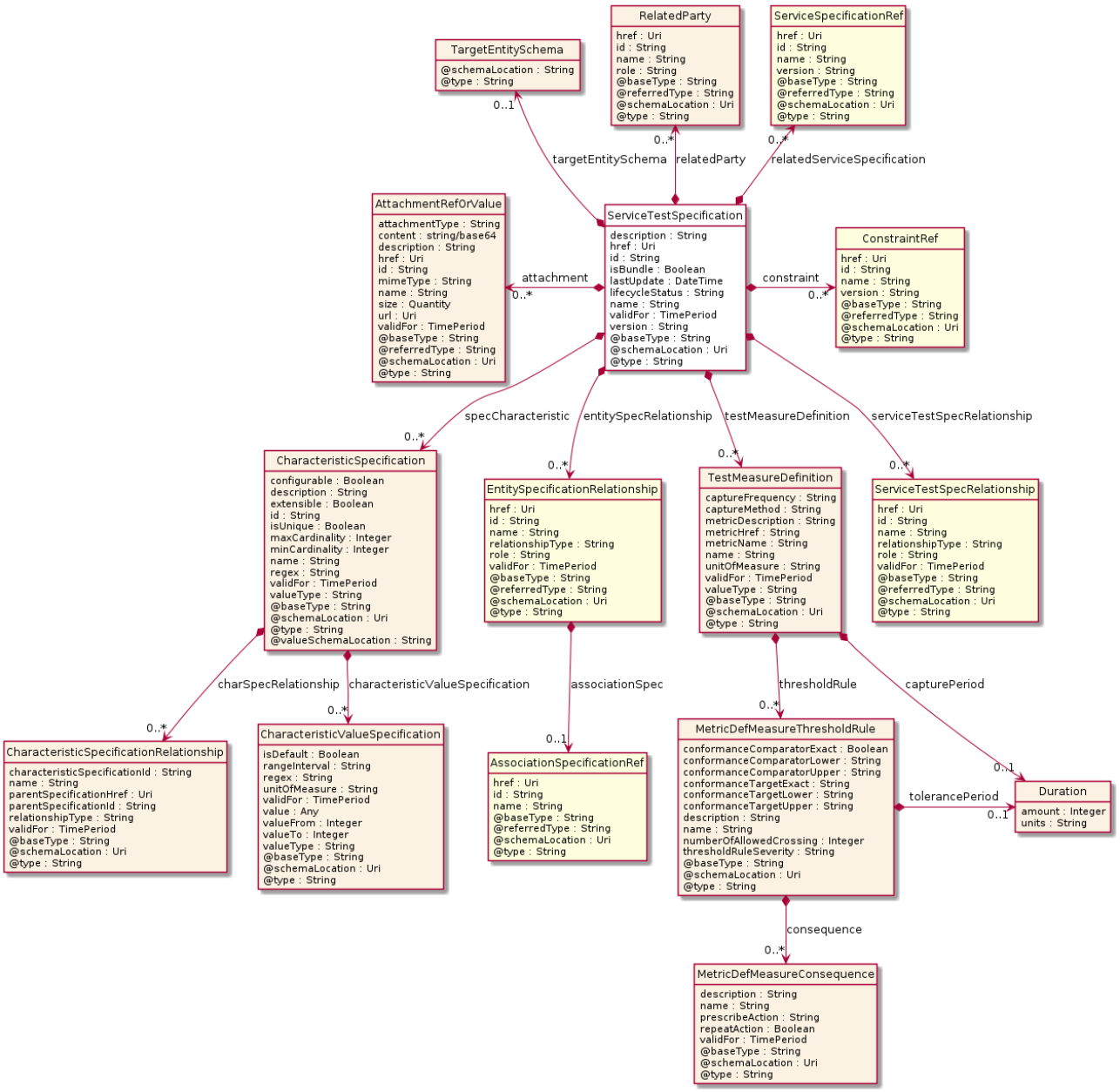
```
        "unitOfMeasure": "A second",
        "value": {
          "valueType": "bits",
          "value": {
            "@type": "integer",
            "value": 3000
          }
        },
        "ruleViolation": [
          {
            "conformanceTargetUpper": "2500",
            "conformanceComparatorUpper": "greater than ",
            "description": "This speed is greater than the target upper",
            "name": "tooMuchTraffic",
            "numberOfAllowedCrossing": 0,
            "thresholdRuleSeverity": "1",
            "appliedConsequence": [
              {
                "appliedAction": "sendWarning",
                "description": "send a warning to the system",
                "name": "sendWarning",
                "repeatAction": false
              },
              {
                "appliedAction": "generateReport",
                "description": "generate a waring report ",
                "name": "generateReport",
                "repeatAction": false
              }
            ]
          }
        ]
      }
    ],
    "@type": "PerformanceTest",
    "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Service/PerformanceTest.schema.json",
    "@baseType": "ServiceTest"
}
```


Service Test Specification resource

The service test specification describes the service test in terms of parameters to be configured and measures to be taken.

Resource model

© TM Forum 2021. All Rights Reserved



Field descriptions

ServiceTestSpecification fields

description	A string. Description of a service test specification.
isBundle	A boolean. isBundle determines whether specification represents a single specification (false), or a bundle of specifications (true).
lastUpdate	A date time (DateTime). Date and time of the last update of the specification.
lifecycleStatus	A string. Used to indicate the current lifecycle status of this catalog item.
name	A string. Name given to the specification.

© TM Forum 2021. All Rights Reserved

version	A string. specification version.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
description	A string. Description of a service test specification.
relatedServiceSpecification	A list of service specification references (ServiceSpecificationRef [*]). The related service specification may relate to more than one service specification.
serviceTestSpecRelationship	A list of service test spec relationships (ServiceTestSpecRelationship [*]). A list of service test specifications related to this specification e.g. dependency, substitution.
testMeasureDefinition	A list of test measure definitions (TestMeasureDefinition [*]). A list of definitions for the measurements for the test defined by this specification.
attachment	A list of attachment ref or values (AttachmentRefOrValue [*]). Attachments that may be of relevance to this specification, such as picture, document, media.
constraint	A list of constraint references (ConstraintRef [*]). This is a list of constraint references applied to this specification.
entitySpecRelationship	A list of entity specification relationships (EntitySpecificationRelationship [*]). Relationship to another specification.
relatedParty	A list of related parties (RelatedParty [*]). Parties who manage or otherwise have an interest in this specification.
specCharacteristic	A list of characteristic specifications (CharacteristicSpecification [*]). List of characteristics that the entity can take.
targetEntitySchema	A target entity schema (TargetEntitySchema). Pointer to a schema that defines the target entity.
validFor	A time period. The period for which this REST resource is valid.

AttachmentRefOrValue sub-resource

An attachment by value or by reference. An attachment complements the description of an element, for example through a document, a video, a picture.

© TM Forum 2021. All Rights Reserved

@referredType	A string. The actual type of the target instance when needed for disambiguation.
description	A string. A narrative text describing the content of the attachment.
href	An uri (Uri). URI for this Attachment.
id	A string. Unique identifier for this particular attachment.
url	An uri (Uri). Uniform Resource Locator, is a web page address (a subset of URI).
name	A string. The name of the attachment.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
attachmentType	A string. Attachment type such as video, picture.
content	A base 6 4 (Base64). The actual contents of the attachment object, if embedded, encoded as base64.
mimeType	A string. Attachment mime type such as extension file for video, picture and document.
size	A quantity (Quantity). The size of the attachment.
validFor	A time period. The period of time for which the attachment is valid.

CharacteristicSpecification sub-resource

This class defines a characteristic specification.

@valueSchemaLocation	A string. This (optional) field provides a link to the schema describing the value type.
configurable	A boolean. If true, the Boolean indicates that the target Characteristic is configurable.
description	A string. A narrative that explains the CharacteristicSpecification.
extensible	A boolean. An indicator that specifies that the values for the characteristic can be extended by adding new values when instantiating a characteristic for a resource.
id	A string. Unique ID for the characteristic.

© TM Forum 2021. All Rights Reserved

isUnique	A boolean. An indicator that specifies if a value is unique for the specification. Possible values are; "unique while value is in effect" and "unique whether value is in effect or not".
maxCardinality	An integer. The maximum number of instances a CharacteristicValue can take on. For example, zero to five phone numbers in a group calling plan, where five is the value for the maxCardinality.
minCardinality	An integer. The minimum number of instances a CharacteristicValue can take on. For example, zero to five phone numbers in a group calling plan, where zero is the value for the minCardinality.
name	A string. A word, term, or phrase by which this characteristic specification is known and distinguished from other characteristic specifications.
regex	A string. A rule or principle represented in regular expression used to derive the value of a characteristic value.
valueType	A string. A kind of value that the characteristic can take on, such as numeric, text and so forth.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
charSpecRelationship	A list of characteristic specification relationships (CharacteristicSpecificationRelationship [*]). An aggregation, migration, substitution, dependency or exclusivity relationship between/among Specification Characteristics.
characteristicValueSpecification	A list of characteristic value specifications (CharacteristicValueSpecification [*]). A CharacteristicValueSpecification object is used to define a set of attributes, each of which can be assigned to a corresponding set of attributes in a CharacteristicSpecification object. The values of the attributes in the CharacteristicValueSpecification object describe the values of the attributes that a corresponding Characteristic object can take on.
validFor	A time period. The period of time for which a characteristic is applicable.

CharacteristicSpecificationRelationship sub-resource

An aggregation, migration, substitution, dependency or exclusivity relationship between/among Characteristic specifications. The specification characteristic is embedded within the specification whose ID and href are in this entity, and identified by its ID.

© TM Forum 2021. All Rights Reserved

@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
characteristicSpecificationId	A string. Unique identifier of the characteristic within the specification.
name	A string. Name of the target characteristic within the specification.
parentSpecificationHref	An uri (Uri). Hyperlink reference to the parent specification containing the target characteristic.
parentSpecificationId	A string. Unique identifier of the parent specification containing the target characteristic.
relationshipType	A string. Type of relationship such as aggregation, migration, substitution, dependency, exclusivity.
validFor	A time period. The period for which the object is valid.

CharacteristicValueSpecification sub-resource

specification of a value (number or text or an object) that can be assigned to a Characteristic.

@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
isDefault	A boolean. If true, the Boolean Indicates if the value is the default value for a characteristic.
rangeInterval	A string. An indicator that specifies the inclusion or exclusion of the valueFrom and valueTo attributes. If applicable, possible values are "open", "closed", "closedBottom" and "closedTop".
regex	A string. A regular expression constraint for given value.
unitOfMeasure	A string. A length, surface, volume, dry measure, liquid measure, money, weight, time, and the like. In general, a determinate quantity or magnitude of the kind designated, taken as a standard of comparison for others of the same kind, in assigning to them numerical values, as 1 foot, 1 yard, 1 mile, 1 square foot.
validFor	A time period. The period of time for which a value is applicable.

© TM Forum 2021. All Rights Reserved

value	An any (Any). A discrete value that the characteristic can take on, or the actual value of the characteristic.
valueFrom	An integer. The low range value that a characteristic can take on.
valueTo	An integer. The upper range value that a characteristic can take on.
valueType	A string. A kind of value that the characteristic value can take on, such as numeric, text and so forth.

Duration sub-resource

A time interval in a given unit of time.

amount	An integer. Time interval (number of seconds, minutes, hours, etc.).
units	A string. Unit of time (seconds, minutes, hours, etc.).

EntitySpecificationRelationship sub-resource

A migration, substitution, dependency or exclusivity relationship between/among entity specifications.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
associationSpec	An association specification reference (AssociationSpecificationRef). A specification for an association used by this relationship.
relationshipType	A string. Type of relationship such as migration, substitution, dependency, exclusivity.
role	A string. The association role for this entity specification.
validFor	A time period. The period for which the entitySpecRelationship is valid.

MetricDefMeasureConsequence sub-resource

A MetricDefMeasureConsequence defines the action (prescribed action or notification) to take when a MetricDefMeasureThresholdRule is crossed.

© TM Forum 2021. All Rights Reserved

@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
description	A string. A narrative that explains in detail what the consequence is.
name	A string. A word, term, or phrase by which a MetricDefMeasureConsequence is known and distinguished from other MetricDefMeasureConsequences.
prescribeAction	A string. Recommended remedy for a violated threshold. This could be the hyperlink to the action.
repeatAction	A boolean. An indicator used to specify that a consequence should cease being applied if a value is in the same range as the previous value or continue being applied if a value is in the same range as the previous value. If the repeatAction is True, if the consequence is always applied as soon as the MetricMeasure value is in the range of values and if the repeatAction is False, the consequence is applied only if the previous MetricMeasure value was not in the same range.
validFor	A time period. A valid duration of a thing.

MetricDefMeasureThresholdRule sub-resource

A MetricDefMeasureThresholdRule is a rule that defines the condition (raise or clear) to achieve to apply consequences when a threshold is crossed or ceased to be crossed. It also defines the severity of the raise or clear of the threshold.

@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
conformanceComparatorExact	A boolean. An operator that when applied on a value specifies whether the value is the same or not. This operator is used to compare with the conformanceTargetExact if used.
conformanceComparatorLower	A string. An operator that when applied on a value specifies whether a threshold is crossed or ceased to be crossed. This operator is used to Service Test Management API REST Specification compare with the conformanceTargetLower if used.

© TM Forum 2021. All Rights Reserved

conformanceComparatorUpper	A string. An operator that when applied on a value specifies whether a threshold is crossed or ceased to be crossed. This operator is used to compare with the conformanceTargetUpper if used.
conformanceTargetExact	A string. To cater for values that are not numerical test metrics (e.g. a DSL line can be Synchronised or Unsynchronised. If the latter, the test should result in a rule violation). The allowed value can contain a REGEX expression.
conformanceTargetLower	A string. A value used to determine if the threshold is crossed or ceases to be crossed. It represents the lower limit. The value should be less than the conformanceTargetUpper if used. The conformance comparators should also be logically defined so as to not lead to a logically impossible condition.
conformanceTargetUpper	A string. A value used to determine if the threshold is crossed or ceases to be crossed. It represents the Upper limit. The value should be greater than the conformanceTargetLower if used. The conformance comparators should also be logically defined so as to not lead to a logically impossible condition.
consequence	A list of metric def measure consequences (MetricDefMeasureConsequence [*]). A list of consequences (actions, notifications) that will arise if the threshold is crossed.
description	A string. Description for the MetricDefMeasureThresholdRule .
name	A string. Name for the MetricDefMeasureThresholdRule .
numberOfAllowedCrossing	An integer. The number of allowed crossing occurrences in reference to the tolerancePeriod without a consequence being initiated.
thresholdRuleSeverity	A string. A threshold can be generated in different severity levels. A crossing for each level may require a different condition and possibly trigger a different consequence.
tolerancePeriod	A duration (Duration). An interval of time of allowed crossing occurrences before a consequence being initiated.

Quantity sub-resource

An amount in a given unit.

amount	A float. Numeric value in a given unit.
units	A string. Unit.

RelatedParty sub-resource

Related Party reference. A related party defines party or party role linked to a specific entity.

@referredType	A string. The actual type of the target instance when needed for disambiguation.

© TM Forum 2021. All Rights Reserved

name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
role	A string. Role played by the related party.

ServiceTestSpecRelationship sub-resource

A substitution, dependency or exclusivity relationship between/among service specifications.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
relationshipType	A string. Type of relationship such as substitution, dependency, exclusivity.
role	A string. The association role for this service test specification.
validFor	A time period. The period for which the serviceTestSpecRelationship is valid.

TargetEntitySchema sub-resource

The reference object to the schema and type of target entity which is described by a specification.

@schemaLocation	A string. This field provides a link to the schema describing the target entity.
@type	A string. Class type of the target entity.

TestMeasureDefinition sub-resource

A TestMeasureDefinition specifies a measure of a specific aspect of a product, service, or resource test, such as lost packets or connectivity status.

© TM Forum 2021. All Rights Reserved

@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
captureFrequency	A string. The frequency of capture for the metric. Note: This may be replaced by a set of entities similar to the Performance Monitoring.
captureMethod	A string. The method used to capture the Metric. Note: This may be replaced by a set of entities similar to the Performance Monitoring.
capturePeriod	A duration (Duration). A period of the capture.
metricDescription	A string. Brief description of the metric.
metricHref	A string. Hyperlink to access a metric for detail information.
metricName	A string. The name of a metric that in the test measure.
name	A string. The name of the TestMeasureDefinition.
thresholdRule	A list of metric def measure threshold rules (MetricDefMeasureThresholdRule [*]). The rule(s) associated with the measure threshold.
unitOfMeasure	A string. Name of a service test specification.
validFor	A time period. The time period for which this definition is valid.
valueType	A string. A kind of value that the Metric value can take on, such as numeric, text, and so forth.

AssociationSpecificationRef relationship

reference to an AssociationSpecification object.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.

© TM Forum 2021. All Rights Reserved

ConstraintRef relationship

Constraint reference. The Constraint resource represents a policy/rule applied to an entity or entity spec.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
version	A string. constraint version.

ServiceSpecificationRef relationship

Service specification reference: ServiceSpecification(s) required to realize a ProductSpecification.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
version	A string. Service specification version.

Json representation sample

We provide below the json representation of an example of a 'ServiceTestSpecification' resource object


```
{
  "description": "a service description",
  "href": "https://mycsp.com:8080/tmf-api/serviceTestManagement/v4/serviceTestSpecification/12",
  "id": "12",
  "version": "1.0",
  "name": "monkey test",
  "validFor": {
```


© TM Forum 2021. All Rights Reserved

```
    "endDateTime": "2017-03-01T00:00:00",
    "startDateTime": "2016-03-01T00:00:00"
  },
  "testMeasureDefinition": [
    {
      "captureFrequency": "once",
      "captureMethod": "automatic capture by instrument",
      "metricDescription": "This is the flow speed in a service",
      "metricHref": "https://mycsp.com:8080/tmf-api/metricManagement/v4/metric/21",
      "metricName": "Flow speed",
      "name": "Definition of Flow speed",
      "unitOfMeasure": "A second",
      "valueType": "bits",
      "capturePeriod": {
        "amount": "5",
        "units": "seconds"
      },
      "thresholdRule": [
        {
          "conformanceTargetUpper": "2500",
          "conformanceComparatorUpper": "greater than ",
          "description": "This speed is greater than the target upper",
          "name": "tooMuchTraffic",
          "numberOfAllowedCrossing": 0,
          "thresholdRuleSeverity": "1",
          "tolerancePeriod": {
            "amount": "15",
            "units": "minutes"
          },
          "consequence": [
            {
              "prescribeAction": "sendWarning",
              "description": "send a warning to the system",
              "name": "sendWarning",
              "repeatAction": false,
              "validFor": {
                "endDateTime": "2017-03-01T00:00:00",
                "startDateTime": "2016-03-01T00:00:00"
              }
            },
            {
              "prescribeAction": "generateReport",
              "description": "generate a waring report ",
              "name": "generateReport",
              "repeatAction": false,
              "validFor": {
                "endDateTime": "2017-03-01T00:00:00",
                "startDateTime": "2016-03-01T00:00:00"
              }
            }
          ]
        }
      ],
      "validFor": {
        "endDateTime": "2017-03-01T00:00:00",
        "startDateTime": "2016-03-01T00:00:00"
```


© TM Forum 2021. All Rights Reserved

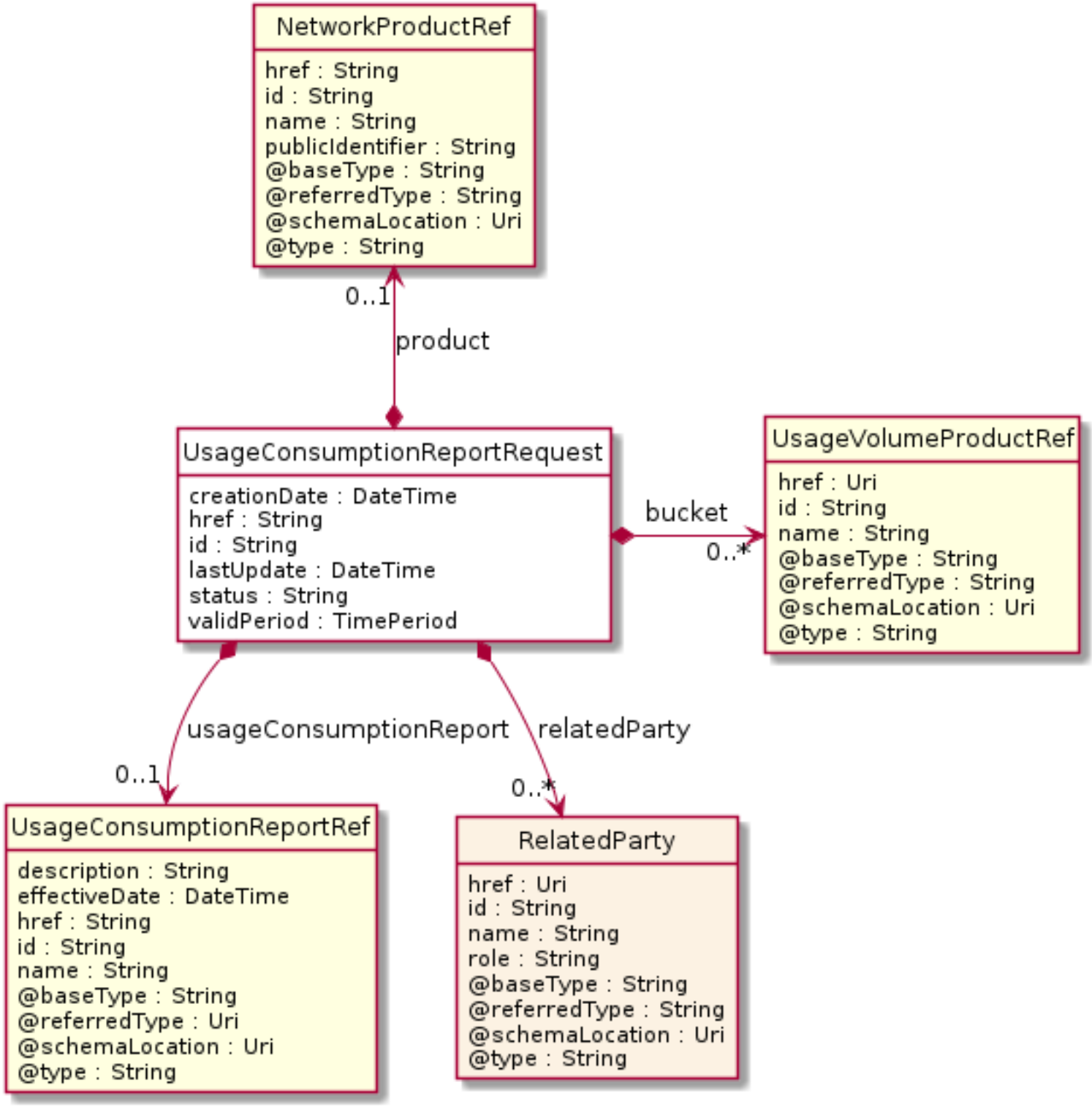
```
        }
      }
    ],
    "relatedServiceSpecification": {
      "href": "https://mycsp.com:8080/tmf-api/serviceCatalogManagement/v4/serviceSpecification/31",
      "id": "31",
      "@referredType": "vCPEServiceSpecification"
    },
    "@baseType": "ServiceTestSpecification",
    "@type": "PerformanceTestSpecification",
    "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/service/PerformanceTestSpecification.schema.json"
}
```


Usage Consumption Report Request resource

An UsageConsumptionReportRequest allows to manage the calculation request of an usage consumption report.

Resource model

© TM Forum 2021. All Rights Reserved



Field descriptions

UsageConsumptionReportRequest fields

bucket	A list of usage volume product references (UsageVolumeProductRef [*]). Reference of the buckets for which the usage consumption report is requested.

© TM Forum 2021. All Rights Reserved

creationDate	A date time (DateTime). Date and time of the request creation.
href	A string. Reference of the usage consumption report request.
id	A string. Unique identifier of the usage consumption report request given by the server.
lastUpdate	A date time (DateTime). Date when the status was last changed.
product	A network product reference (NetworkProductRef). Network product reference for which the usage consumption report is requested.
relatedParty	A list of related parties (RelatedParty [*]). Reference and role of the related parties for which the usage consumption report is requested.
status	A string. Status of the usage consumption report request (InProgress or done).
usageConsumptionReport	An usage consumption report reference (UsageConsumptionReportRef). References of the usage consumption report (given when it has been calculated and the status of the request is done).
validPeriod	A time period. Validity period.

RelatedParty sub-resource

Related Party reference. A related party defines party or party role linked to a specific entity.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
role	A string. Role played by the related party.

NetworkProductRef relationship

Reference of a product.

href	A string. Reference to the network product.
id	A string. Unique identifier of the network product.

© TM Forum 2021. All Rights Reserved

@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
@referredType	A string. Generic attribute indicating the name of the class type of the referred resource entity.
name	A string. Network product name.
publicIdentifier	A string. Public number associated to the product (msisdn number for mobile line for example).

UsageConsumptionReportRef relationship

Reference of an usage consumption report.

href	A string. Hyperlink to access the usage consumption report.
id	A string. Unique identifier of the usage consumption report given by the server.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
description	A string. Free short text describing the usage consumption report content.
effectiveDate	A date time (DateTime). Date and time when the usage consumption report was calculated and generated.
name	A string. Usage consumption report name.

UsageVolumeProductRef relationship

Reference of a bucket.

href	An uri (Uri). Hyperlink reference.
id	A string. Unique identifier of the bucket.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.

© TM Forum 2021. All Rights Reserved

@referredType	A string. Generic attribute indicating the name of the class type of the referred resource entity.
name	A string. Bucket name.

Json representation sample

We provide below the json representation of an example of a 'UsageConsumptionReportRequest' resource object

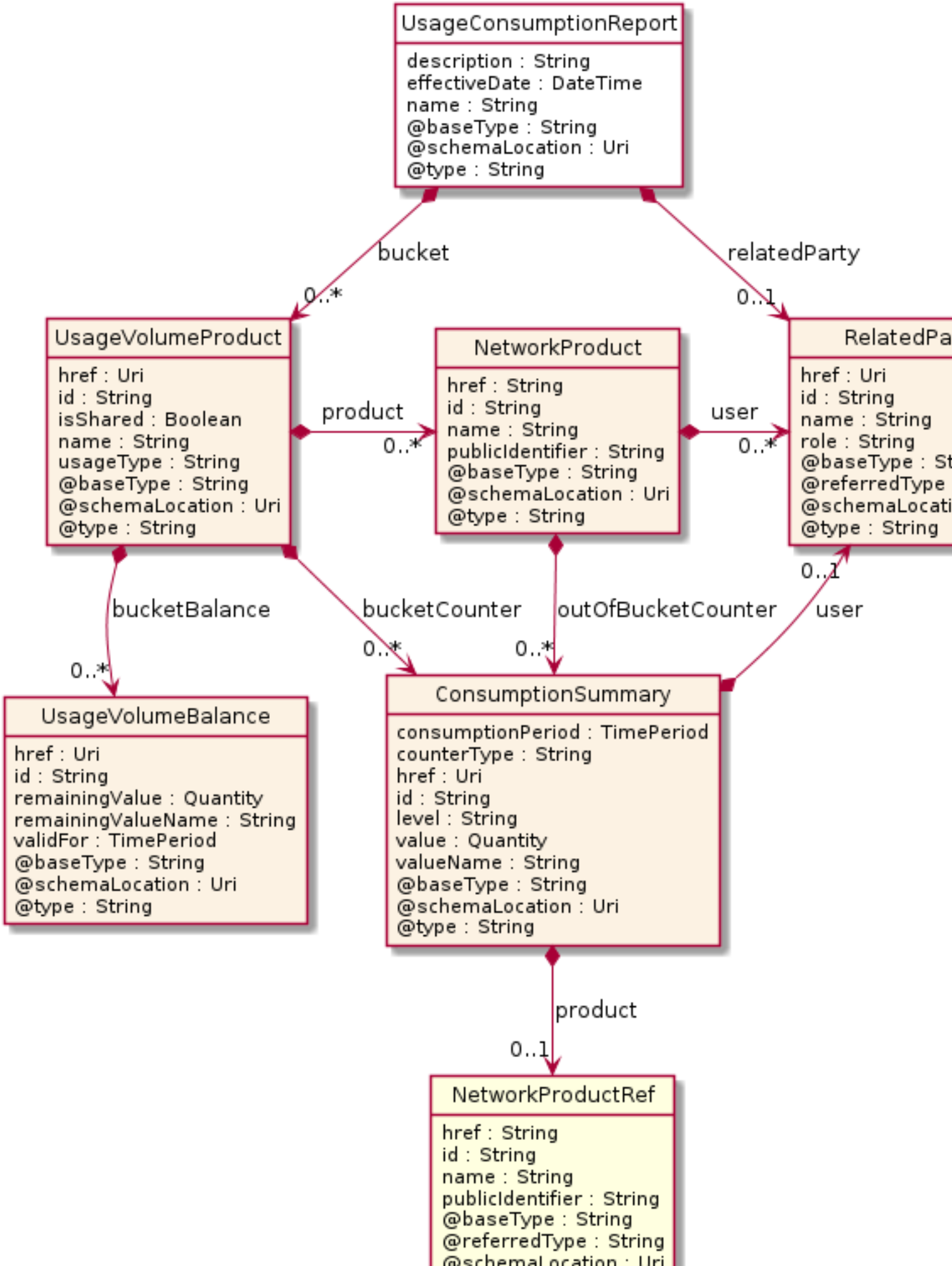

```
{
   "id": "ucrreq001",
   "href": "https://serverRoot/tmf-api/usageConsumption/usageConsumptionReportRequest/ucrreq001",
   "creationDate": "2018-03-15T:15:44:28",
   "status": "inProgress",
   "lastUpdate": "2018-03-15T:15:44:28",
   "product": {
      "publicIdentifier ": "33601010101"
   }
}
```


Usage Consumption Report resource

An usage consumption report enables to know at a given point the balances and the consumption counters related to various buckets (SMS, Voice, Data for example). It could be calculated for a device identified by a public key (msisdn number for a mobile device for example or PSTN or VOIP number for a fix device), for a subscribed offer or option or for an user.

Resource model

© TM Forum 2021. All Rights Reserved


UsageConsumptionReport
description : String
effectiveDate : DateTime
name : String
@baseType : String
@schemaLocation : Uri
@type : String
bucket
0..*
relatedParty
0..1
UsageVolumeProduct
href : Uri
id : String
isShared : Boolean
name : String
usageType : String
@baseType : String
@schemaLocation : Uri
@type : String
product
0..*
NetworkProduct
href : String
id : String
name : String
publicIdentifier : String
@baseType : String
@schemaLocation : Uri
@type : String
user
0..*
RelatedPa
href : Uri
id : String
name : String
role : String
@baseType : St
@referredType
@schemaLocati
@type : String
0..1
bucketBalance
0..*
bucketCounter
0..*
outOfBucketCounter
0..*
user
UsageVolumeBalance
href : Uri
id : String
remainingValue : Quantity
remainingValueName : String
validFor : TimePeriod
@baseType : String
@schemaLocation : Uri
@type : String
ConsumptionSummary
consumptionPeriod : TimePeriod
counterType : String
href : Uri
id : String
level : String
value : Quantity
valueName : String
@baseType : String
@schemaLocation : Uri
@type : String
product
0..1
NetworkProductRef
href : String
id : String
name : String
publicIdentifier : String
@baseType : String
@referredType : String

Field descriptions

UsageConsumptionReport fields

@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
bucket	A list of usage volume products (UsageVolumeProduct [*]). Bucket(s) included in the offer or option subscribed.
description	A string. Free short text describing the usage consumption report content.
effectiveDate	A date time (DateTime). Date and time when the usage consumption report was calculated and generated.
name	A string. Usage consumption report name.
relatedParty	A related party (RelatedParty). Related Party reference. A related party defines party or party role linked to a specific entity.

ConsumptionSummary sub-resource

The consumption counters (called ConsumptionSummary in the SID model) detail for example the different kind of consumption done on the bucket.

href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
consumptionPeriod	A time period. Consumption counter period between a start date time and an end date time. For prepaid bucket, the period of counters is between the start date of the bucket and the effective date of the usage consumption report generation. For postpaid bucket, the period of counters is between the last bill date and the effective date of the usage consumption report generation.
counterType	A string. Type of the consumption counter. We can give for example a counter of the used value for a bucket (counterType=used for example) or the value of the consumption done out of the bucket(s) (counterType=outOfBucket for example).

© TM Forum 2021. All Rights Reserved

level	A string. Counter level. The counter can be given globally for the bucket or detailed by user or by network product for example in case of shared bucket.
product	A network product reference (NetworkProductRef). Public identifier associated to the product for which the consumption counter is detailed. If the level of the counter is global (i.e. concerning all the network products), the list of them is not specified. It can be found through the network products listed in the bucket sub-resource.
user	A related party (RelatedParty). Reference of the user for which the consumption counter is detailed. If the level of the counter is global (i.e. concerning all the users), the list of them is not specified. It can be found through the users listed in the network products of the bucket sub-resource.
value	A quantity (Quantity). Numeric value of the bucket counter in a given unit.
valueName	A string. Value of the counter in a formatted string used for display needs for example.

NetworkProduct sub-resource

An instantiated network product (specialization of a product) subscribed by a customer.

href	A string. Reference to the network product.
id	A string. Unique identifier of the network product.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
name	A string. Network product name.
outOfBucketCounter	A list of consumption summaries (ConsumptionSummary [*]). Counters detailing usage consumption out of the buckets.
publicIdentifier	A string. Public number associated to the network product (msisdn number for mobile line for example).
user	A list of related parties (RelatedParty [*]). References of the users of the network product.

Quantity sub-resource

An amount in a given unit.

amount	A float. Numeric value in a given unit.

© TM Forum 2021. All Rights Reserved

units	A string. Unit.

RelatedParty sub-resource

Related Party reference. A related party defines party or party role linked to a specific entity.

@referredType	A string. The actual type of the target instance when needed for disambiguation.
name	A string. Name of the related entity.
href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
role	A string. Role played by the related party.

UsageVolumeBalance sub-resource

The balance (called UsageVolumeBalance in the SID model) defines the remaining allowed product usage quantity in terms of volume, time, currency or events. It corresponds to the initial allowed usage quantity minus the usage consumed on the bucket.

href	An uri (Uri). Hyperlink reference.
id	A string. unique identifier.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
remainingValue	A quantity (Quantity). Numeric remaining value for the bucket given in the unit (for example 1.9). This numeric value could be used for calculation for example.
remainingValueName	A string. Remaining value in a formatted string for the bucket given in the balance unit (for example 1.9 Gb). This formatted string could be used for display needs for example.
validFor	A time period. Balance period between a start date time and an end date time. For prepaid bucket, the period of the balance is between the effective date of the usage consumption report generation and the end date of the bucket. For postpaid bucket,

© TM Forum 2021. All Rights Reserved

	the period of the balance is between the effective date of the usage consumption report generation and the next bill date.

UsageVolumeProduct sub-resource

A bucket (called UsageVolumeProduct in the SID model) represents a quantity of usage, as 2 hours national calls or 50 sms for example. It could be either a quantity or an amount in a currency (i.e. It could represent a fixed number of SMS, MMS, minutes of calls, quantity of data, number of events as well as a specific amount in a given currency). It requires one or more network products from which usages will debit the bucket.

href	An uri (Uri). Hyperlink reference.
id	A string. Unique identifier of the bucket.
@baseType	A string. When sub-classing, this defines the super-class.
@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
bucketBalance	A list of usage volume balances (UsageVolumeBalance [*]). The balance (called UsageVolumeBalance in the SID model) defines the remaining allowed product usage quantity in terms of volume, time, currency or events. It corresponds to the initial allowed usage quantity minus the usage consumed on the bucket.
bucketCounter	A list of consumption summaries (ConsumptionSummary [*]). The consumption counters (called ConsumptionSummary in the SID model) detail for example the different kind of consumption done on the bucket.
isShared	A boolean. True if the bucket is shared between several devices or users.
name	A string. Bucket name.
product	A list of network products (NetworkProduct [*]). An instantiated network product (specialization of a product) subscribed by a customer.
usageType	A string. Type of usage concerned by the bucket, such as voice, sms, data.

NetworkProductRef relationship

Reference of a product.

href	A string. Reference to the network product.
id	A string. Unique identifier of the network product.
@baseType	A string. When sub-classing, this defines the super-class.

© TM Forum 2021. All Rights Reserved

@schemaLocation	An uri (Uri). A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A string. When sub-classing, this defines the sub-class Extensible name.
@referredType	A string. Generic attribute indicating the name of the class type of the referred resource entity.
name	A string. Network product name.
publicIdentifier	A string. Public number associated to the product (msisdn number for mobile line for example).

Json representation sample

We provide below the json representation of an example of a 'UsageConsumptionReport' resource object


```
{
  "id": "ucr0001",
  "href": "https://serverRoot/tmf-api/usageConsumption/usageConsumptionReport/ucr0001",
  "name": "Usage consumption report ucr0001",
  "description": "Usage consumption Report for Kate smartphone",
  "effectiveDate": "2018-03-15T:15:44:28",
  "bucket": [
    {
      "id": "bkt001",
      "name": "Main offer - data",
      "usageType": "data",
      "isShared": false,
      "product": [
        {
          "id": "product1",
          "name": "Kate smartphone",
          "publicIdentifier": "33601010101",
          "user": [
            {
              "id": "usr1",
              "name": "Kate",
              "role": "user",
              "@referredType": "individual"
            }
          ],
          "outOfBucketCounter": [
            {
              "counterType": "outOfBucket",
              "level": "global",
              "value": {
                "amount": 20.0,
                "units": "USD"
              },
              "valueName": "20 USD",
              "consumptionPeriod": {
                "startDateTime": "2018-03-01T:00:00:00",
                "endDateTime": "2018-03-15T:00:00:00"
```


© TM Forum 2021. All Rights Reserved

```
            }
          }
        ]
      }
    ],
    "bucketBalance": [
      {
        "remainingValue": {
          "amount": 1.8,
          "units": "Go"
        },
        "remainingValueName": "it remains 1.8 Go",
        "validFor": {
          "startDateTime": "2018-03-15T:00:00:00",
          "endDateTime": "2018-03-30T:00:00:00"
        }
      }
    ],
    "bucketCounter": [
      {
        "counterType": "used",
        "level": "global",
        "value": {
          "amount": 1.2,
          "units": "Go"
        },
        "valueName": "1.2 Go used",
        "consumptionPeriod": {
          "startDateTime": "2018-03-01T:00:00:00",
          "endDateTime": "2018-03-15T:00:00:00"
        }
      }
    ]
  },
  {
    "id": "bkt002",
    "name": "Main offer - national voice",
    "usageType": "national voice",
    "isShared": false,
    "product": [
      {
        "id": "product1",
        "name": "Kate smartphone",
        "publicIdentifier": "33601010101",
        "user": [
          {
            "id": "usr1",
            "name": "Kate",
            "role": "user",
            "@referredType": "individual"
          }
        ]
      }
    ],
    "bucketBalance": [
      {
```


© TM Forum 2021. All Rights Reserved

```
            "remainingValue": {
               "amount": 80.0,
               "units": "mins"
            },
            "remainingValueName": "it remains 80 minutes",
            "validFor": {
               "startDateTime": "2018-03-15T:00:00:00",
               "endDateTime": "2018-03-30T:00:00:00"
            }
         }
      ],
      "bucketCounter": [
         {
            "counterType": "used",
            "level": "global",
            "value": {
               "amount": 40.0,
               "units": "mins"
            },
            "valueName": "40 mins used",
            "consumptionPeriod": {
               "startDateTime": "2018-03-01T:00:00:00",
               "endDateTime": "2018-03-15T:00:00:00"
            }
         }
      ]
   },
   {
      "id": "bkt003",
      "name": "Main offer - sms",
      "usageType": "sms",
      "isShared": false,
      "product": [
         {
            "id": "product1",
            "name": "Kate smartphone",
            "publicIdentifier": "33601010101",
            "user": [
               {
                  "id": "usr1",
                  "name": "Kate",
                  "role": "user",
                  "@referredType": "individual"
               }
            ]
         }
      ],
      "bucketBalance": [
         {
            "remainingValue": {
               "amount": 95.0,
               "units": "sms"
            },
            "remainingValueName": "it remains 95 sms",
            "validFor": {
               "startDateTime": "2018-03-15T:00:00:00",
```


© TM Forum 2021. All Rights Reserved

```
            "endDateTime": "2018-03-30T:00:00:00"
          }
        }
      ],
      "bucketCounter": [
        {
          "counterType": "used",
          "level": "global",
          "value": {
            "amount": 25.0,
            "units": "sms"
          },
          "valueName": "25 sms used",
          "consumptionPeriod": {
            "startDateTime": "2018-03-01T:00:00:00",
            "endDateTime": "2018-03-15T:00:00:00"
          }
        }
      ]
    },
    {
      "id": "bkt004",
      "name": "Option Canada/USA - voice",
      "usageType": "voice",
      "isShared": false,
      "product": [
        {
          "id": "product1",
          "name": "Kate smartphone",
          "publicIdentifier": "33601010101",
          "user": [
            {
              "id": "usr1",
              "name": "Kate",
              "role": "user",
              "@referredType": "individual"
            }
          ]
        }
      ],
      "bucketBalance": [
        {
          "remainingValue": {
            "amount": 10.0,
            "units": "mins"
          },
          "remainingValueName": "it remains 10 minutes",
          "validFor": {
            "startDateTime": "2018-03-15T:00:00:00",
            "endDateTime": "2018-03-30T:00:00:00"
          }
        }
      ],
      "bucketCounter": [
        {
          "counterType": "used",
```


© TM Forum 2021. All Rights Reserved

```
            "level": "global",
            "value": {
               "amount": 20.0,
               "units": "mins"
            },
            "valueName": "20 mins used",
            "consumptionPeriod": {
               "startDateTime": "2018-03-01T:00:00:00",
               "endDateTime": "2018-03-15T:00:00:00"
            }
         }
      ]
   },
   {
      "id": "bkt005",
      "name": "Option Canada/USA - sms",
      "usageType": "sms",
      "isShared": false,
      "product": [
         {
            "id": "product1",
            "name": "Kate smartphone",
            "publicIdentifier": "33601010101",
            "user": [
               {
                  "id": "usr1",
                  "name": "Kate",
                  "role": "user",
                  "@referredType": "individual"
               }
            ]
         }
      ],
      "bucketBalance": [
         {
            "remainingValue": {
               "amount": 0.0,
               "units": "sms"
            },
            "remainingValueName": "Sms bucket exhausted",
            "validFor": {
               "startDateTime": "2018-03-15T:00:00:00",
               "endDateTime": "2018-03-30T:00:00:00"
            }
         }
      ],
      "bucketCounter": [
         {
            "counterType": "used",
            "level": "global",
            "value": {
               "amount": 10.0,
               "units": "sms"
            },
            "valueName": "10 sms used",
            "consumptionPeriod": {
```


© TM Forum 2021. All Rights Reserved

```
                "startDateTime": "2018-03-01T:00:00:00",
                "endDateTime": "2018-03-15T:00:00:00"
            }
        }
    ]
  }
 ]
}
```


Notification Resource Models

51 notifications are defined for this API

Notifications related to Service:
- ServiceCreateEvent
- ServiceAttributeValueChangeEvent
- ServiceStateChangeEvent
- ServiceDeleteEvent

Notifications related to Monitor:
- MonitorCreateEvent
- MonitorAttributeValueChangeEvent
- MonitorStateChangeEvent
- MonitorDeleteEvent

Notifications related to ServiceCatalog:
- ServiceCatalogCreateEvent
- ServiceCatalogChangeEvent
- ServiceCatalogBatchEvent
- ServiceCatalogDeleteEvent

Notifications related to ServiceCategory:
- ServiceCategoryCreateEvent
- ServiceCategoryChangeEvent
- ServiceCategoryDeleteEvent

Notifications related to ServiceCandidate:
- ServiceCandidateCreateEvent
- ServiceCandidateChangeEvent
- ServiceCandidateDeleteEvent

Notifications related to ServiceSpecification:
- ServiceSpecificationCreateEvent
- ServiceSpecificationChangeEvent
- ServiceSpecificationDeleteEvent

© TM Forum 2021. All Rights Reserved

Notifications related to ServiceOrder:
- ServiceOrderCreateEvent
- ServiceOrderAttributeValueChangeEvent
- ServiceOrderStateChangeEvent
- ServiceOrderDeleteEvent
- ServiceOrderInformationRequiredEvent
- ServiceOrderMilestoneEvent
- ServiceOrderJeopardyEvent

Notifications related to CancelServiceOrder:
- CancelServiceOrderCreateEvent
- CancelServiceOrderStateChangeEvent
- CancelServiceOrderInformationRequiredEvent

Notifications related to CheckServiceQualification:
- CheckServiceQualificationCreateEvent
- CheckServiceQualificationAttributeValueChangeEvent
- CheckServiceQualificationStateChangeEvent
- CheckServiceQualificationDeleteEvent
- CheckServiceQualificationInformationRequiredEvent

Notifications related to QueryServiceQualification:
- QueryServiceQualificationCreateEvent
- QueryServiceQualificationStateChangeEvent
- QueryServiceQualificationDeleteEvent

Notifications related to ServiceProblem:
- ServiceProblemCreateEvent
- ServiceProblemStateChangeEvent
- ServiceProblemAttributeValueChangeEvent
- ServiceProblemInformationRequiredEvent

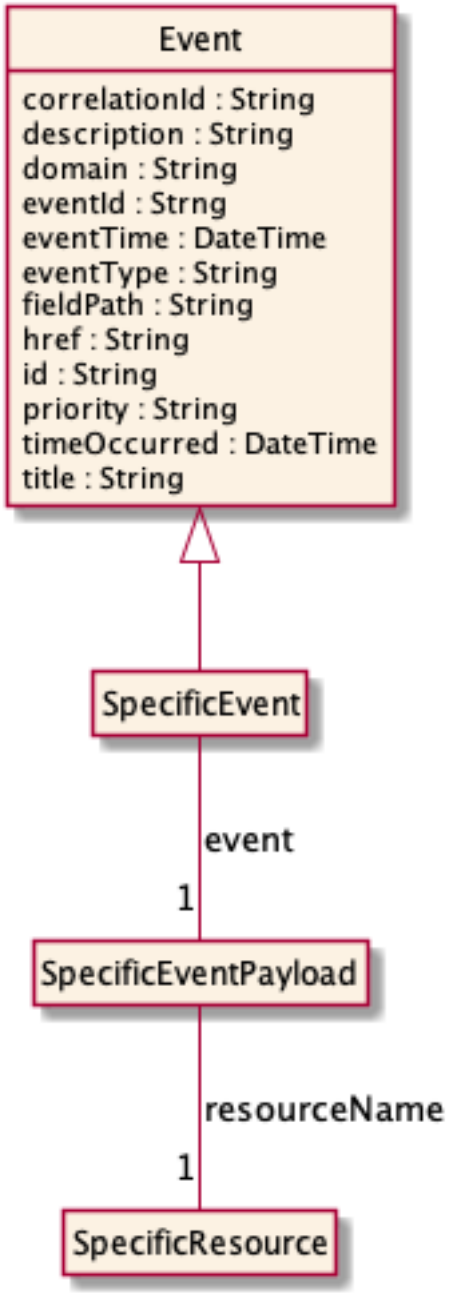
Notifications related to ServiceTest:
- ServiceTestCreateEvent
- ServiceTestAttributeValueChangeEvent
- ServiceTestStateChangeEvent
- ServiceTestDeleteEvent

Notifications related to ServiceTestSpecification:
- ServiceTestSpecificationCreateEvent
- ServiceTestSpecificationAttributeValueChangeEvent
- ServiceTestSpecificationDeleteEvent

Notifications related to UsageConsumptionReportRequest:
- UsageConsumptionReportRequestStateChangeEvent

© TM Forum 2021. All Rights Reserved

The notification structure for all notifications in this API follow the pattern depicted by the figure below.
A notification event resource (depicted by "SpecificEvent" placeholder) is a sub class of a generic Event structure containing at least an id of the event occurrence (eventId), an event timestamp (eventTime), and the name of the resource (eventType).
This notification structure owns an event payload structure ("SpecificEventPayload" placeholder) linked to the resource concerned by the notification using the resource name as access field ("resourceName" placeholder).



Service Create Event

Notification ServiceCreateEvent case for resource Service

Json representation sample

© TM Forum 2021. All Rights Reserved

We provide below the json representation of an example of a 'ServiceCreateEvent' notification event object


```
{
   "eventId":"00001",
   "eventTime":"2015-11-16T16:42:25-04:00",
   "eventType":"ServiceCreateEvent",
    "event": {
      "service" :
         {-- SEE Service RESOURCE SAMPLE --}
   }
}
```


Service Attribute Value Change Event

Notification ServiceAttributeValueChangeEvent case for resource Service

Json representation sample

We provide below the json representation of an example of a 'ServiceAttributeValueChangeEvent' notification event object


```
{
   "eventId":"00001",
   "eventTime":"2015-11-16T16:42:25-04:00",
   "eventType":"ServiceAttributeValueChangeEvent",
    "event": {
      "service" :
         {-- SEE Service RESOURCE SAMPLE --}
   }
}
```


Service State Change Event

Notification ServiceStateChangeEvent case for resource Service

Json representation sample

We provide below the json representation of an example of a 'ServiceStateChangeEvent' notification event object


```
{
   "eventId":"00001",
   "eventTime":"2015-11-16T16:42:25-04:00",
   "eventType":"ServiceStateChangeEvent",
    "event": {
      "service" :
         {-- SEE Service RESOURCE SAMPLE --}
   }
}
```


Service Delete Event

© TM Forum 2021. All Rights Reserved

Notification ServiceDeleteEvent case for resource Service

Json representation sample

We provide below the json representation of an example of a 'ServiceDeleteEvent' notification event object


```
{
   "eventId":"00001",
   "eventTime":"2015-11-16T16:42:25-04:00",
   "eventType":"ServiceDeleteEvent",
    "event": {
      "service" :
         {-- SEE Service RESOURCE SAMPLE --}
   }
}
```


Monitor Create Event

Notification MonitorCreateEvent case for resource Monitor

Json representation sample

We provide below the json representation of an example of a 'MonitorCreateEvent' notification event object


```
{
   "eventId":"00001",
   "eventTime":"2015-11-16T16:42:25-04:00",
   "eventType":"MonitorCreateEvent",
    "event": {
      "monitor" :
         {-- SEE Monitor RESOURCE SAMPLE --}
   }
}
```


Monitor Attribute Value Change Event

Notification MonitorAttributeValueChangeEvent case for resource Monitor

Json representation sample

We provide below the json representation of an example of a 'MonitorAttributeValueChangeEvent' notification event object


```
{
   "eventId":"00001",
   "eventTime":"2015-11-16T16:42:25-04:00",
   "eventType":"MonitorAttributeValueChangeEvent",
    "event": {
      "monitor" :
         {-- SEE Monitor RESOURCE SAMPLE --}
   }
```


© TM Forum 2021. All Rights Reserved

```
}
```

Monitor State Change Event

Notification MonitorStateChangeEvent case for resource Monitor

Json representation sample

We provide below the json representation of an example of a 'MonitorStateChangeEvent' notification event object

```
{
   "eventId":"00001",
   "eventTime":"2015-11-16T16:42:25-04:00",
   "eventType":"MonitorStateChangeEvent",
    "event": {
      "monitor" :
         {-- SEE Monitor RESOURCE SAMPLE --}
   }
}
```

Monitor Delete Event

Notification MonitorDeleteEvent case for resource Monitor

Json representation sample

We provide below the json representation of an example of a 'MonitorDeleteEvent' notification event object

```
{
   "eventId":"00001",
   "eventTime":"2015-11-16T16:42:25-04:00",
   "eventType":"MonitorDeleteEvent",
    "event": {
      "monitor" :
         {-- SEE Monitor RESOURCE SAMPLE --}
   }
}
```

Service Catalog Create Event

Notification ServiceCatalogCreateEvent case for resource ServiceCatalog

Json representation sample

We provide below the json representation of an example of a 'ServiceCatalogCreateEvent' notification event object

```
{
   "eventId":"00001",
   "eventTime":"2015-11-16T16:42:25-04:00",
   "eventType":"ServiceCatalogCreateEvent",
    "event": {
```

© TM Forum 2021. All Rights Reserved

```
    "serviceCatalog" :
      {-- SEE ServiceCatalog RESOURCE SAMPLE --}
  }
}
```

Service Catalog Change Event

Notification ServiceCatalogChangeEvent case for resource ServiceCatalog

Json representation sample

We provide below the json representation of an example of a 'ServiceCatalogChangeEvent' notification event object

```
{
  "eventId":"00001",
  "eventTime":"2015-11-16T16:42:25-04:00",
  "eventType":"ServiceCatalogChangeEvent",
   "event": {
     "serviceCatalog" :
       {-- SEE ServiceCatalog RESOURCE SAMPLE --}
  }
}
```

Service Catalog Batch Event

Notification ServiceCatalogBatchEvent case for resource ServiceCatalog

Json representation sample

We provide below the json representation of an example of a 'ServiceCatalogBatchEvent' notification event object

```
{
  "eventId":"00001",
  "eventTime":"2015-11-16T16:42:25-04:00",
  "eventType":"ServiceCatalogBatchEvent",
   "event": {
     "serviceCatalog" :
       {-- SEE ServiceCatalog RESOURCE SAMPLE --}
  }
}
```

Service Catalog Delete Event

Notification ServiceCatalogDeleteEvent case for resource ServiceCatalog

Json representation sample

We provide below the json representation of an example of a 'ServiceCatalogDeleteEvent' notification event object

© TM Forum 2021. All Rights Reserved

```
{
  "eventId":"00001",
  "eventTime":"2015-11-16T16:42:25-04:00",
  "eventType":"ServiceCatalogDeleteEvent",
   "event": {
     "serviceCatalog" :
        {-- SEE ServiceCatalog RESOURCE SAMPLE --}
  }
}
```

Service Category Create Event

Notification ServiceCategoryCreateEvent case for resource ServiceCategory

Json representation sample

We provide below the json representation of an example of a 'ServiceCategoryCreateEvent' notification event object

```
{
  "eventId":"00001",
  "eventTime":"2015-11-16T16:42:25-04:00",
  "eventType":"ServiceCategoryCreateEvent",
   "event": {
     "serviceCategory" :
        {-- SEE ServiceCategory RESOURCE SAMPLE --}
  }
}
```

Service Category Change Event

Notification ServiceCategoryChangeEvent case for resource ServiceCategory

Json representation sample

We provide below the json representation of an example of a 'ServiceCategoryChangeEvent' notification event object

```
{
  "eventId":"00001",
  "eventTime":"2015-11-16T16:42:25-04:00",
  "eventType":"ServiceCategoryChangeEvent",
   "event": {
     "serviceCategory" :
        {-- SEE ServiceCategory RESOURCE SAMPLE --}
  }
}
```

Service Category Delete Event

© TM Forum 2021. All Rights Reserved

Notification ServiceCategoryDeleteEvent case for resource ServiceCategory

Json representation sample

We provide below the json representation of an example of a 'ServiceCategoryDeleteEvent' notification event object


```
{
  "eventId":"00001",
  "eventTime":"2015-11-16T16:42:25-04:00",
  "eventType":"ServiceCategoryDeleteEvent",
   "event": {
     "serviceCategory" :
       {-- SEE ServiceCategory RESOURCE SAMPLE --}
  }
}
```


Service Candidate Create Event

Notification ServiceCandidateCreateEvent case for resource ServiceCandidate

Json representation sample

We provide below the json representation of an example of a 'ServiceCandidateCreateEvent' notification event object


```
{
  "eventId":"00001",
  "eventTime":"2015-11-16T16:42:25-04:00",
  "eventType":"ServiceCandidateCreateEvent",
   "event": {
     "serviceCandidate" :
       {-- SEE ServiceCandidate RESOURCE SAMPLE --}
  }
}
```


Service Candidate Change Event

Notification ServiceCandidateChangeEvent case for resource ServiceCandidate

Json representation sample

We provide below the json representation of an example of a 'ServiceCandidateChangeEvent' notification event object


```
{
  "eventId":"00001",
  "eventTime":"2015-11-16T16:42:25-04:00",
  "eventType":"ServiceCandidateChangeEvent",
   "event": {
     "serviceCandidate" :
```


© TM Forum 2021. All Rights Reserved

```
        {-- SEE ServiceCandidate RESOURCE SAMPLE --}
   }
}
```

Service Candidate Delete Event

Notification ServiceCandidateDeleteEvent case for resource ServiceCandidate

Json representation sample

We provide below the json representation of an example of a 'ServiceCandidateDeleteEvent' notification event object

```
{
   "eventId":"00001",
   "eventTime":"2015-11-16T16:42:25-04:00",
   "eventType":"ServiceCandidateDeleteEvent",
    "event": {
      "serviceCandidate" :
        {-- SEE ServiceCandidate RESOURCE SAMPLE --}
   }
}
```

Service Specification Create Event

Notification ServiceSpecificationCreateEvent case for resource ServiceSpecification

Json representation sample

We provide below the json representation of an example of a 'ServiceSpecificationCreateEvent' notification event object

```
{
   "eventId":"00001",
   "eventTime":"2015-11-16T16:42:25-04:00",
   "eventType":"ServiceSpecificationCreateEvent",
    "event": {
      "serviceSpecification" :
        {-- SEE ServiceSpecification RESOURCE SAMPLE --}
   }
}
```

Service Specification Change Event

Notification ServiceSpecificationChangeEvent case for resource ServiceSpecification

Json representation sample

We provide below the json representation of an example of a 'ServiceSpecificationChangeEvent' notification event object

© TM Forum 2021. All Rights Reserved

```
{
   "eventId":"00001",
   "eventTime":"2015-11-16T16:42:25-04:00",
   "eventType":"ServiceSpecificationChangeEvent",
    "event": {
      "serviceSpecification" :
         {-- SEE ServiceSpecification RESOURCE SAMPLE --}
   }
}
```

Service Specification Delete Event

Notification ServiceSpecificationDeleteEvent case for resource ServiceSpecification

Json representation sample

We provide below the json representation of an example of a 'ServiceSpecificationDeleteEvent' notification event object

```
{
   "eventId":"00001",
   "eventTime":"2015-11-16T16:42:25-04:00",
   "eventType":"ServiceSpecificationDeleteEvent",
    "event": {
      "serviceSpecification" :
         {-- SEE ServiceSpecification RESOURCE SAMPLE --}
   }
}
```

Service Order Create Event

Notification ServiceOrderCreateEvent case for resource ServiceOrder

Json representation sample

We provide below the json representation of an example of a 'ServiceOrderCreateEvent' notification event object

```
{
   "eventId":"00001",
   "eventTime":"2015-11-16T16:42:25-04:00",
   "eventType":"ServiceOrderCreateEvent",
    "event": {
      "serviceOrder" :
         {-- SEE ServiceOrder RESOURCE SAMPLE --}
   }
}
```

Service Order Attribute Value Change Event

Notification ServiceOrderAttributeValueChangeEvent case for resource ServiceOrder

© TM Forum 2021. All Rights Reserved

Json representation sample

We provide below the json representation of an example of a 'ServiceOrderAttributeValueChangeEvent' notification event object


```
{
   "eventId":"00001",
   "eventTime":"2015-11-16T16:42:25-04:00",
   "eventType":"ServiceOrderAttributeValueChangeEvent",
    "event": {
      "serviceOrder" :
         {-- SEE ServiceOrder RESOURCE SAMPLE --}
   }
}
```


Service Order State Change Event

Notification ServiceOrderStateChangeEvent case for resource ServiceOrder

Json representation sample

We provide below the json representation of an example of a 'ServiceOrderStateChangeEvent' notification event object


```
{
   "eventId":"00001",
   "eventTime":"2015-11-16T16:42:25-04:00",
   "eventType":"ServiceOrderStateChangeEvent",
    "event": {
      "serviceOrder" :
         {-- SEE ServiceOrder RESOURCE SAMPLE --}
   }
}
```


Service Order Delete Event

Notification ServiceOrderDeleteEvent case for resource ServiceOrder

Json representation sample

We provide below the json representation of an example of a 'ServiceOrderDeleteEvent' notification event object


```
{
   "eventId":"00001",
   "eventTime":"2015-11-16T16:42:25-04:00",
   "eventType":"ServiceOrderDeleteEvent",
    "event": {
      "serviceOrder" :
         {-- SEE ServiceOrder RESOURCE SAMPLE --}
   }
}
```


© TM Forum 2021. All Rights Reserved

Service Order Information Required Event

Notification ServiceOrderInformationRequiredEvent case for resource ServiceOrder

Json representation sample

We provide below the json representation of an example of a 'ServiceOrderInformationRequiredEvent' notification event object

```
{
   "eventId":"00001",
   "eventTime":"2015-11-16T16:42:25-04:00",
   "eventType":"ServiceOrderInformationRequiredEvent",
    "event": {
      "serviceOrder" :
         {-- SEE ServiceOrder RESOURCE SAMPLE --}
   }
}
```

Service Order Milestone Event

Notification ServiceOrderMilestoneEvent case for resource ServiceOrder

Json representation sample

We provide below the json representation of an example of a 'ServiceOrderMilestoneEvent' notification event object

```
{
   "eventId":"00001",
   "eventTime":"2015-11-16T16:42:25-04:00",
   "eventType":"ServiceOrderMilestoneEvent",
    "event": {
      "serviceOrder" :
         {-- SEE ServiceOrder RESOURCE SAMPLE --}
   }
}
```

Service Order Jeopardy Event

Notification ServiceOrderJeopardyEvent case for resource ServiceOrder

Json representation sample

We provide below the json representation of an example of a 'ServiceOrderJeopardyEvent' notification event object

```
{
   "eventId":"00001",
   "eventTime":"2015-11-16T16:42:25-04:00",
   "eventType":"ServiceOrderJeopardyEvent",
```

© TM Forum 2021. All Rights Reserved

```
    "event": {
      "serviceOrder" :
         {-- SEE ServiceOrder RESOURCE SAMPLE --}
    }
}
```

Cancel Service Order Create Event

Notification CancelServiceOrderCreateEvent case for resource CancelServiceOrder

Json representation sample

We provide below the json representation of an example of a 'CancelServiceOrderCreateEvent' notification event object

```
{
   "eventId":"00001",
   "eventTime":"2015-11-16T16:42:25-04:00",
   "eventType":"CancelServiceOrderCreateEvent",
    "event": {
      "cancelServiceOrder" :
         {-- SEE CancelServiceOrder RESOURCE SAMPLE --}
    }
}
```

Cancel Service Order State Change Event

Notification CancelServiceOrderStateChangeEvent case for resource CancelServiceOrder

Json representation sample

We provide below the json representation of an example of a 'CancelServiceOrderStateChangeEvent' notification event object

```
{
   "eventId":"00001",
   "eventTime":"2015-11-16T16:42:25-04:00",
   "eventType":"CancelServiceOrderStateChangeEvent",
    "event": {
      "cancelServiceOrder" :
         {-- SEE CancelServiceOrder RESOURCE SAMPLE --}
    }
}
```

Cancel Service Order Information Required Event

Notification CancelServiceOrderInformationRequiredEvent case for resource CancelServiceOrder

Json representation sample

© TM Forum 2021. All Rights Reserved

We provide below the json representation of an example of a 'CancelServiceOrderInformationRequiredEvent' notification event object


```
{
   "eventId":"00001",
   "eventTime":"2015-11-16T16:42:25-04:00",
   "eventType":"CancelServiceOrderInformationRequiredEvent",
    "event": {
      "cancelServiceOrder" :
         {-- SEE CancelServiceOrder RESOURCE SAMPLE --}
   }
}
```


Check Service Qualification Create Event

Notification CheckServiceQualificationCreateEvent case for resource CheckServiceQualification

Json representation sample

We provide below the json representation of an example of a 'CheckServiceQualificationCreateEvent' notification event object


```
{
   "eventId":"00001",
   "eventTime":"2015-11-16T16:42:25-04:00",
   "eventType":"CheckServiceQualificationCreateEvent",
    "event": {
      "checkServiceQualification" :
         {-- SEE CheckServiceQualification RESOURCE SAMPLE --}
   }
}
```


Check Service Qualification Attribute Value Change Event

Notification CheckServiceQualificationAttributeValueChangeEvent case for resource CheckServiceQualification

Json representation sample

We provide below the json representation of an example of a 'CheckServiceQualificationAttributeValueChangeEvent' notification event object


```
{
   "eventId":"00001",
   "eventTime":"2015-11-16T16:42:25-04:00",
   "eventType":"CheckServiceQualificationAttributeValueChangeEvent",
    "event": {
      "checkServiceQualification" :
         {-- SEE CheckServiceQualification RESOURCE SAMPLE --}
   }
}
```


© TM Forum 2021. All Rights Reserved

Check Service Qualification State Change Event

Notification CheckServiceQualificationStateChangeEvent case for resource CheckServiceQualification

Json representation sample

We provide below the json representation of an example of a 'CheckServiceQualificationStateChangeEvent' notification event object

```
{
   "eventId":"00001",
   "eventTime":"2015-11-16T16:42:25-04:00",
   "eventType":"CheckServiceQualificationStateChangeEvent",
    "event": {
      "checkServiceQualification" :
         {-- SEE CheckServiceQualification RESOURCE SAMPLE --}
   }
}
```

Check Service Qualification Delete Event

Notification CheckServiceQualificationDeleteEvent case for resource CheckServiceQualification

Json representation sample

We provide below the json representation of an example of a 'CheckServiceQualificationDeleteEvent' notification event object

```
{
   "eventId":"00001",
   "eventTime":"2015-11-16T16:42:25-04:00",
   "eventType":"CheckServiceQualificationDeleteEvent",
    "event": {
      "checkServiceQualification" :
         {-- SEE CheckServiceQualification RESOURCE SAMPLE --}
   }
}
```

Check Service Qualification Information Required Event

Notification CheckServiceQualificationInformationRequiredEvent case for resource CheckServiceQualification

Json representation sample

We provide below the json representation of an example of a 'CheckServiceQualificationInformationRequiredEvent' notification event object

```
{
   "eventId":"00001",
   "eventTime":"2015-11-16T16:42:25-04:00",
   "eventType":"CheckServiceQualificationInformationRequiredEvent",
```

© TM Forum 2021. All Rights Reserved

```
    "event": {
      "checkServiceQualification" :
         {-- SEE CheckServiceQualification RESOURCE SAMPLE --}
    }
}
```

Query Service Qualification Create Event

Notification QueryServiceQualificationCreateEvent case for resource QueryServiceQualification

Json representation sample

We provide below the json representation of an example of a 'QueryServiceQualificationCreateEvent' notification event object

```
{
   "eventId":"00001",
   "eventTime":"2015-11-16T16:42:25-04:00",
   "eventType":"QueryServiceQualificationCreateEvent",
    "event": {
      "queryServiceQualification" :
         {-- SEE QueryServiceQualification RESOURCE SAMPLE --}
    }
}
```

Query Service Qualification State Change Event

Notification QueryServiceQualificationStateChangeEvent case for resource QueryServiceQualification

Json representation sample

We provide below the json representation of an example of a 'QueryServiceQualificationStateChangeEvent' notification event object

```
{
   "eventId":"00001",
   "eventTime":"2015-11-16T16:42:25-04:00",
   "eventType":"QueryServiceQualificationStateChangeEvent",
    "event": {
      "queryServiceQualification" :
         {-- SEE QueryServiceQualification RESOURCE SAMPLE --}
    }
}
```

Query Service Qualification Delete Event

Notification QueryServiceQualificationDeleteEvent case for resource QueryServiceQualification

Json representation sample

© TM Forum 2021. All Rights Reserved

We provide below the json representation of an example of a 'QueryServiceQualificationDeleteEvent' notification event object


```
{
   "eventId":"00001",
   "eventTime":"2015-11-16T16:42:25-04:00",
   "eventType":"QueryServiceQualificationDeleteEvent",
    "event": {
      "queryServiceQualification" :
         {-- SEE QueryServiceQualification RESOURCE SAMPLE --}
   }
}
```


Service Problem Create Event

Notification ServiceProblemCreateEvent case for resource ServiceProblem

Json representation sample

We provide below the json representation of an example of a 'ServiceProblemCreateEvent' notification event object


```
{
   "eventId":"00001",
   "eventTime":"2015-11-16T16:42:25-04:00",
   "eventType":"ServiceProblemCreateEvent",
    "event": {
      "serviceProblem" :
         {-- SEE ServiceProblem RESOURCE SAMPLE --}
   }
}
```


Service Problem State Change Event

Notification ServiceProblemStateChangeEvent case for resource ServiceProblem

Json representation sample

We provide below the json representation of an example of a 'ServiceProblemStateChangeEvent' notification event object


```
{
   "eventId":"00001",
   "eventTime":"2015-11-16T16:42:25-04:00",
   "eventType":"ServiceProblemStateChangeEvent",
    "event": {
      "serviceProblem" :
         {-- SEE ServiceProblem RESOURCE SAMPLE --}
   }
}
```


© TM Forum 2021. All Rights Reserved

Service Problem Attribute Value Change Event

Notification ServiceProblemAttributeValueChangeEvent case for resource ServiceProblem

Json representation sample

We provide below the json representation of an example of a 'ServiceProblemAttributeValueChangeEvent' notification event object

```
{
   "eventId":"00001",
   "eventTime":"2015-11-16T16:42:25-04:00",
   "eventType":"ServiceProblemAttributeValueChangeEvent",
    "event": {
      "serviceProblem" :
         {-- SEE ServiceProblem RESOURCE SAMPLE --}
   }
}
```

Service Problem Information Required Event

Notification ServiceProblemInformationRequiredEvent case for resource ServiceProblem

Json representation sample

We provide below the json representation of an example of a 'ServiceProblemInformationRequiredEvent' notification event object

```
{
   "eventId":"00001",
   "eventTime":"2015-11-16T16:42:25-04:00",
   "eventType":"ServiceProblemInformationRequiredEvent",
    "event": {
      "serviceProblem" :
         {-- SEE ServiceProblem RESOURCE SAMPLE --}
   }
}
```

Service Test Create Event

Notification ServiceTestCreateEvent case for resource ServiceTest

Json representation sample

We provide below the json representation of an example of a 'ServiceTestCreateEvent' notification event object

```
{
   "eventId":"00001",
   "eventTime":"2015-11-16T16:42:25-04:00",
   "eventType":"ServiceTestCreateEvent",
    "event": {
      "serviceTest" :
```

© TM Forum 2021. All Rights Reserved

```
        {-- SEE ServiceTest RESOURCE SAMPLE --}
   }
}
```

Service Test Attribute Value Change Event

Notification ServiceTestAttributeValueChangeEvent case for resource ServiceTest

Json representation sample

We provide below the json representation of an example of a 'ServiceTestAttributeValueChangeEvent' notification event object

```
{
   "eventId":"00001",
   "eventTime":"2015-11-16T16:42:25-04:00",
   "eventType":"ServiceTestAttributeValueChangeEvent",
    "event": {
      "serviceTest" :
        {-- SEE ServiceTest RESOURCE SAMPLE --}
   }
}
```

Service Test State Change Event

Notification ServiceTestStateChangeEvent case for resource ServiceTest

Json representation sample

We provide below the json representation of an example of a 'ServiceTestStateChangeEvent' notification event object

```
{
   "eventId":"00001",
   "eventTime":"2015-11-16T16:42:25-04:00",
   "eventType":"ServiceTestStateChangeEvent",
    "event": {
      "serviceTest" :
        {-- SEE ServiceTest RESOURCE SAMPLE --}
   }
}
```

Service Test Delete Event

Notification ServiceTestDeleteEvent case for resource ServiceTest

Json representation sample

We provide below the json representation of an example of a 'ServiceTestDeleteEvent' notification event object

© TM Forum 2021. All Rights Reserved

```
{
  "eventId":"00001",
  "eventTime":"2015-11-16T16:42:25-04:00",
  "eventType":"ServiceTestDeleteEvent",
   "event": {
     "serviceTest" :
       {-- SEE ServiceTest RESOURCE SAMPLE --}
  }
}
```

Service Test Specification Create Event

Notification ServiceTestSpecificationCreateEvent case for resource ServiceTestSpecification

Json representation sample

We provide below the json representation of an example of a 'ServiceTestSpecificationCreateEvent' notification event object

```
{
  "eventId":"00001",
  "eventTime":"2015-11-16T16:42:25-04:00",
  "eventType":"ServiceTestSpecificationCreateEvent",
   "event": {
     "serviceTestSpecification" :
       {-- SEE ServiceTestSpecification RESOURCE SAMPLE --}
  }
}
```

Service Test Specification Attribute Value Change Event

Notification ServiceTestSpecificationAttributeValueChangeEvent case for resource ServiceTestSpecification

Json representation sample

We provide below the json representation of an example of a 'ServiceTestSpecificationAttributeValueChangeEvent' notification event object

```
{
  "eventId":"00001",
  "eventTime":"2015-11-16T16:42:25-04:00",
  "eventType":"ServiceTestSpecificationAttributeValueChangeEvent",
   "event": {
     "serviceTestSpecification" :
       {-- SEE ServiceTestSpecification RESOURCE SAMPLE --}
  }
}
```

Service Test Specification Delete Event

© TM Forum 2021. All Rights Reserved

Notification ServiceTestSpecificationDeleteEvent case for resource ServiceTestSpecification

Json representation sample

We provide below the json representation of an example of a 'ServiceTestSpecificationDeleteEvent' notification event object

```
{
   "eventId":"00001",
   "eventTime":"2015-11-16T16:42:25-04:00",
   "eventType":"ServiceTestSpecificationDeleteEvent",
    "event": {
      "serviceTestSpecification" :
         {-- SEE ServiceTestSpecification RESOURCE SAMPLE --}
   }
}
```

Usage Consumption Report Request State Change Event

Notification UsageConsumptionReportRequestStateChangeEvent case for resource UsageConsumptionReportRequest

Json representation sample

We provide below the json representation of an example of a 'UsageConsumptionReportRequestStateChangeEvent' notification event object

```
{
   "eventId":"00001",
   "eventTime":"2015-11-16T16:42:25-04:00",
   "eventType":"UsageConsumptionReportRequestStateChangeEvent",
    "event": {
      "usageConsumptionReportRequest" :
         {-- SEE UsageConsumptionReportRequest RESOURCE SAMPLE --}
   }
}
```

© TM Forum 2021. All Rights Reserved

API OPERATIONS

Remember the following Uniform Contract:

Operation on Entities	Uniform API Operation	Description
Query Entities	GET Resource	GET must be used to retrieve a representation of a resource.
Create Entity	POST Resource	POST must be used to create a new resource
Partial Update of an Entity	PATCH Resource	PATCH must be used to partially update a resource
Complete Update of an Entity	PUT Resource	PUT must be used to completely update a resource identified by its resource URI
Remove an Entity	DELETE Resource	DELETE must be used to remove a resource
Execute an Action on an Entity	POST on TASK Resource	POST must be used to execute Task Resources
Other Request Methods	POST on TASK Resource	GET and POST must not be used to tunnel other request methods.

Filtering and attribute selection rules are described in the TMF630 REST Design Guidelines Part 1 document.

Notifications are also described in a subsequent section.

© TM Forum 2021. All Rights Reserved

Operations on Service

List services

GET /service?fields=...&{filtering}

Description

This operation list service entities.
Attribute selection is enabled for all first level attributes.
Filtering may be available depending on the compliance level supported by an implementation.

Usage Samples

Here's an example of a request for retrieving multiple Service.

Request

```
GET /tmf-api/naas/v4/service
Accept: application/json
```

Response

```
200

[
  {
    "id": "5351",
    "href": "https://mycsp.com:8080/tmf-api/ServiceActivationAndConfiguration/v4/service/5351",
    "serviceType": "Cloud",
    "state": "Active",
    "serviceDate": "2018-01-15T12:26:11.747Z",
    "startDate": "2018-01-15T12:26:11.747Z",
    "serviceCharacteristic": [
      {
        "name": "vCPE_IP",
        "valueType": "object",
        "value": {
          "@type": "IPAddress",
          "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Service/IPAddress.schema.json",
          "address": "193.218.236.21"
        }
      }
    ],
    "@type": "vCPE",
    "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Service/vCPE.schema.json",
    "@baseType": "Service"
  }
```

© TM Forum 2021. All Rights Reserved

```
]
```

Retrieve service

GET /service/{id}?fields=...&{filtering}

Description

This operation retrieves a service entity.
Attribute selection is enabled for all first level attributes.
Filtering on sub-resources may be available depending on the compliance level supported by an implementation.

Usage Samples

Here's an example of a request for retrieving a specific Service.

Request

```
GET /tmf-api/naas/v4/service/5351
Accept: application/json
```

Response

```
200

{
  "id": "5351",
  "href": "https://mycsp.com:8080/tmf-api/ServiceActivationAndConfiguration/v4/service/5351",
  "serviceType": "Cloud",
  "state": "Active",
  "serviceDate": "2018-01-15T12:26:11.747Z",
  "startDate": "2018-01-15T12:26:11.747Z",
  "serviceCharacteristic": [
    {
      "name": "vCPE_IP",
      "valueType": "object",
      "value": {
        "@type": "IPAddress",
        "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Service/IPAddress.schema.json",
        "address": "193.218.236.21"
      }
    }
  ],
  "@type": "vCPE",
  "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Service/vCPE.schema.json",
  "@baseType": "Service"
```

© TM Forum 2021. All Rights Reserved

```
}
```

Create service

POST /service

Description

This operation creates a service entity.

Mandatory and Non Mandatory Attributes

The following tables provide the list of mandatory and non mandatory attributes when creating a Service, including any possible rule conditions and applicable default values. Notice that it is up to an implementer to add additional mandatory attributes.

Mandatory Attributes	Rule
state	
serviceSpecification.id	

Non Mandatory Attributes	Rule
@baseType	
@schemaLocation	
@type	
category	
description	
endDate	
feature	
hasStarted	
isBundle	
isServiceEnabled	
isStateful	
name	
note	
place	
relatedEntity	
relatedParty	
serviceCharacteristic	
serviceDate	
serviceOrderItem	
serviceRelationship	
serviceSpecification	
serviceType	
startDate	
startMode	
supportingResource	
supportingService	

© TM Forum 2021. All Rights Reserved

Usage Samples

Here's an example of a request for creating a specific Service.

Request

```
POST /tmf-api/naas/v4/service
Content-Type: application/json

{
  "serviceType": "Cloud",
  "state": "Active",
  "serviceDate": "2018-01-15T12:26:11.747Z",
  "startDate": "2018-01-15T12:26:11.747Z",
  "serviceCharacteristic": [
    {
      "name": "vCPE_IP",
      "valueType": "object",
      "value": {
        "@type": "IPAddress",
        "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Service/IPAddress.schema.json",
        "address": "193.218.236.21"
      }
    }
  ],
  "@type": "vCPE",
  "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Service/vCPE.schema.json",
  "@baseType": "Service"
}
```

Response

```
201

{
  "id": "5351",
  "href": "https://mycsp.com:8080/tmf-api/ServiceActivationAndConfiguration/v4/service/5351",
  "serviceType": "Cloud",
  "state": "Active",
  "serviceDate": "2018-01-15T12:26:11.747Z",
  "startDate": "2018-01-15T12:26:11.747Z",
  "serviceCharacteristic": [
    {
      "name": "vCPE_IP",
      "valueType": "object",
      "value": {
        "@type": "IPAddress",
        "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Service/IPAddress.schema.json",
        "address": "193.218.236.21"
```

© TM Forum 2021. All Rights Reserved

```
        }
      }
    ],
    "@type": "vCPE",
    "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Service/vCPE.schema.json",
    "@baseType": "Service"
}
```


Patch service

PATCH /service/{id}

Description

This operation allows partial updates of a service entity. Support of json/merge (https://tools.ietf.org/html/rfc7386) is mandatory, support of json/patch (http://tools.ietf.org/html/rfc5789) is optional.

Note: If the update operation yields to the creation of sub-resources or relationships, the same rules concerning mandatory sub-resource attributes and default value settings in the POST operation applies to the PATCH operation. Hence these tables are not repeated here.

Patchable and Non Patchable Attributes

The tables below provide the list of patchable and non patchable attributes, including constraint rules on their usage.

Patchable Attributes	Rule
@baseType	
@schemaLocation	
@type	
category	
description	
endDate	
feature	
hasStarted	
isBundle	
isServiceEnabled	
isStateful	
name	
note	
place	
relatedEntity	
relatedParty	
serviceCharacteristic	
serviceOrderItem	
serviceRelationship	
serviceSpecification	

© TM Forum 2021. All Rights Reserved

serviceType	
startDate	
startMode	
state	
supportingResource	
supportingService	

Non Patchable Attributes	Rule
id	
href	
serviceDate	

Usage Samples

Here's an example of a request for patching a Service.

Request

```
PATCH /tmf-api/naas/v4/service/5351
Content-Type: application/merge-patch+json

{
   "serviceType": "Cloud",
   "state": "Active",
   "serviceDate": "2018-01-15T12:26:11.747Z",
   "startDate": "2018-01-15T12:26:11.747Z",
   "serviceCharacteristic": [
      {
         "name": "vCPE_IP",
         "valueType": "object",
         "value": {
            "@type": "IPAddress",
            "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Service/IPAddress.schema.json",
            "address": "193.218.236.21"
         }
      }
   ],
   "@type": "vCPE",
   "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Service/vCPE.schema.json",
   "@baseType": "Service"
}
```

Response

```
200

{
   "id": "5351",
```

© TM Forum 2021. All Rights Reserved

```
  "href": "https://mycsp.com:8080/tmf-api/ServiceActivationAndConfiguration/v4/service/5351",
  "serviceType": "Cloud",
  "state": "Active",
  "serviceDate": "2018-01-15T12:26:11.747Z",
  "startDate": "2018-01-15T12:26:11.747Z",
  "serviceCharacteristic": [
    {
      "name": "vCPE_IP",
      "valueType": "object",
      "value": {
        "@type": "IPAddress",
        "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Service/IPAddress.schema.json",
        "address": "193.218.236.21"
      }
    }
  ],
  "@type": "vCPE",
  "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Service/vCPE.schema.json",
  "@baseType": "Service"
}
```


Delete service

`DELETE /service/{id}`

Description

This operation deletes a service entity.

Usage Samples

Here's an example of a request for patching a Service.

Request
DELETE /tmf-api/naas/v4/service/5351
Response
204

Operations on Monitor

List monitors

© TM Forum 2021. All Rights Reserved

GET /monitor?fields=...&{filtering}

Description

This operation list monitor entities.
Attribute selection is enabled for all first level attributes.
Filtering may be available depending on the compliance level supported by an implementation.

Usage Samples

Here's an example of a request for retrieving multiple Monitor.

Request

```
GET /tmf-api/naas/v4/monitor
Accept: application/json
```

Response

```
200

[
  {
    "id": "44555",
    "href": "https://mycsp.com:8080/tmf-api/ServiceActivationAndConfiguration/v4/monitor/44555",
    "state": "In Progress",
    "request": {
      "method": "http",
      "to": "https://mycsp.com:5050/netAPIs/activate",
      "body": "ACTIVATE BBAND 144445",
      "header": [
        {
          "name": "locale",
          "value": "en-us"
        }
      ]
    },
    "response": {
      "statusCode": "344",
      "body": "ACTIVATE BBAND 144445 REQUEST RECEIVED PENDING",
      "header": [
        {
          "name": "locale",
          "value": "en-us"
        }
      ]
    },
    "sourceHref": "https://mycsp.com:8080/tmf-api/ServiceActivationAndConfiguration/v4/service/5351",
```

© TM Forum 2021. All Rights Reserved

```
    "@type": "Monitor"
  }
]
```

Retrieve monitor

GET /monitor/{id}?fields=...&{filtering}

Description

This operation retrieves a monitor entity.
Attribute selection is enabled for all first level attributes.
Filtering on sub-resources may be available depending on the compliance level supported by an implementation.

Usage Samples

Here's an example of a request for retrieving a specific Monitor.

Request

```
GET /tmf-api/naas/v4/monitor/44555
Accept: application/json
```

Response

```
200

{
  "id": "44555",
  "href": "https://mycsp.com:8080/tmf-api/ServiceActivationAndConfiguration/v4/monitor/44555",
  "state": "In Progress",
  "request": {
    "method": "http",
    "to": "https://mycsp.com:5050/netAPIs/activate",
    "body": "ACTIVATE BBAND 144445",
    "header": [
      {
        "name": "locale",
        "value": "en-us"
      }
    ]
  },
  "response": {
    "statusCode": "344",
    "body": "ACTIVATE BBAND 144445 REQUEST RECEIVED PENDING",
    "header": [
      {
```

© TM Forum 2021. All Rights Reserved

```
            "name": "locale",
            "value": "en-us"
          }
      ]
    },
    "sourceHref": "https://mycsp.com:8080/tmf-api/ServiceActivationAndConfiguration/v4/service/5351",
    "@type": "Monitor"
}
```

Operations on Service Catalog

List service catalogs

`GET /serviceCatalog?fields=...&{filtering}`

Description

This operation list service catalog entities.
Attribute selection is enabled for all first level attributes.
Filtering may be available depending on the compliance level supported by an implementation.

Usage Samples

Here's an example of a request for retrieving multiple ServiceCatalog.

Request

```
GET /tmf-api/naas/v4/serviceCatalog
Accept: application/json
```

Response

```
200

[
  {
    "id": "3830",
    "href": "https://mycsp.com:8080/tmf-api/serviceCatalogManagement/v4/serviceCatalog/3830",
    "name": "Catalog Wholesale Business",
    "description": "This service catalog describes services that address the wholesale business segment.",
    "version": "1.0",
    "validFor": {
      "startDateTime": "2017-08-29T00:00:00Z",
      "endDateTime": "2018-03-25T00:00:00Z"
    },
    "lastUpdate": "2017-08-27T00:00:00Z",
    "lifecycleStatus": "Active",
```

© TM Forum 2021. All Rights Reserved

```
    "relatedParty": [
      {
        "href": "https://mycsp.com:8080/tmf-api/partyManagement/v4/organization/3426",
        "id": "3426",
        "name": "Broadly Broad Ltd",
        "role": "vendor"
      }
    ],
    "category": [
      {
        "href": "https://mycsp.com:8080/tmf-api/serviceCatalogManagement/v4/category/7752",
        "id": "7752",
        "name": "IoT"
      }
    ],
    "@type": "ServiceCatalog",
    "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Service/ServiceCatalog.schema.json ",
    "@baseType": "Catalog"
  }
]
```


Retrieve service catalog

GET /serviceCatalog/{id}?fields=...&{filtering}

Description

This operation retrieves a service catalog entity.
Attribute selection is enabled for all first level attributes.
Filtering on sub-resources may be available depending on the compliance level supported by an implementation.

Usage Samples

Here's an example of a request for retrieving a specific ServiceCatalog.

Request

```
GET /tmf-api/naas/v4/serviceCatalog/3830
Accept: application/json
```

Response


```
200

{
  "id": "3830",
  "href": "https://mycsp.com:8080/tmf-api/serviceCatalogManagement/v4/serviceCatalog/3830",
```


© TM Forum 2021. All Rights Reserved

```
  "name": "Catalog Wholesale Business",
  "description": "This service catalog describes services that address the wholesale business segment.",
  "version": "1.0",
  "validFor": {
     "startDateTime": "2017-08-29T00:00:00Z",
     "endDateTime": "2018-03-25T00:00:00Z"
  },
  "lastUpdate": "2017-08-27T00:00:00Z",
  "lifecycleStatus": "Active",
  "relatedParty": [
     {
        "href": "https://mycsp.com:8080/tmf-api/partyManagement/v4/organization/3426",
        "id": "3426",
        "name": "Broadly Broad Ltd",
        "role": "vendor"
     }
  ],
  "category": [
     {
        "href": "https://mycsp.com:8080/tmf-api/serviceCatalogManagement/v4/category/7752",
        "id": "7752",
        "name": "IoT"
     }
  ],
  "@type": "ServiceCatalog",
  "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Service/ServiceCatalog.schema.json ",
  "@baseType": "Catalog"
}
```


Create service catalog

POST /serviceCatalog

Description

This operation creates a service catalog entity.

Mandatory and Non Mandatory Attributes

The following tables provide the list of mandatory and non mandatory attributes when creating a ServiceCatalog, including any possible rule conditions and applicable default values. Notice that it is up to an implementer to add additional mandatory attributes.

Mandatory Attributes	Rule
name	

Non Mandatory Attributes	Rule
@baseType	
@schemaLocation	
@type	
category	

© TM Forum 2021. All Rights Reserved

description	
lastUpdate	
lifecycleStatus	
relatedParty	
validFor	
version	

Usage Samples

Here's an example of a request for creating a specific ServiceCatalog.

Request

```
POST /tmf-api/naas/v4/serviceCatalog
Content-Type: application/json

{
  "name": "Catalog Wholesale Business",
  "description": "This service catalog describes services that address the wholesale business segment.",
  "version": "1.0",
  "validFor": {
    "startDateTime": "2017-08-29T00:00:00Z",
    "endDateTime": "2018-03-25T00:00:00Z"
  },
  "lastUpdate": "2017-08-27T00:00:00Z",
  "lifecycleStatus": "Active",
  "relatedParty": [
    {
      "href": "https://mycsp.com:8080/tmf-api/partyManagement/v4/organization/3426",
      "id": "3426",
      "name": "Broadly Broad Ltd",
      "role": "vendor"
    }
  ],
  "category": [
    {
      "href": "https://mycsp.com:8080/tmf-api/serviceCatalogManagement/v4/category/7752",
      "id": "7752",
      "name": "IoT"
    }
  ],
  "@type": "ServiceCatalog",
  "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Service/ServiceCatalog.schema.json ",
  "@baseType": "Catalog"
}
```

Response

201

© TM Forum 2021. All Rights Reserved

```
{
  "id": "3830",
  "href": "https://mycsp.com:8080/tmf-api/serviceCatalogManagement/v4/serviceCatalog/3830",
  "name": "Catalog Wholesale Business",
  "description": "This service catalog describes services that address the wholesale business segment.",
  "version": "1.0",
  "validFor": {
    "startDateTime": "2017-08-29T00:00:00Z",
    "endDateTime": "2018-03-25T00:00:00Z"
  },
  "lastUpdate": "2017-08-27T00:00:00Z",
  "lifecycleStatus": "Active",
  "relatedParty": [
    {
      "href": "https://mycsp.com:8080/tmf-api/partyManagement/v4/organization/3426",
      "id": "3426",
      "name": "Broadly Broad Ltd",
      "role": "vendor"
    }
  ],
  "category": [
    {
      "href": "https://mycsp.com:8080/tmf-api/serviceCatalogManagement/v4/category/7752",
      "id": "7752",
      "name": "IoT"
    }
  ],
  "@type": "ServiceCatalog",
  "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Service/ServiceCatalog.schema.json ",
  "@baseType": "Catalog"
}
```


Patch service catalog

`PATCH /serviceCatalog/{id}`

Description

This operation allows partial updates of a service catalog entity. Support of json/merge (https://tools.ietf.org/html/rfc7386) is mandatory, support of json/patch (http://tools.ietf.org/html/rfc5789) is optional.

Note: If the update operation yields to the creation of sub-resources or relationships, the same rules concerning mandatory sub-resource attributes and default value settings in the POST operation applies to the PATCH operation. Hence these tables are not repeated here.

Patchable and Non Patchable Attributes

The tables below provide the list of patchable and non patchable attributes, including constraint rules on their usage.

© TM Forum 2021. All Rights Reserved

Patchable Attributes	Rule
@baseType	
@schemaLocation	
@type	
category	
description	
lifecycleStatus	
name	
relatedParty	
validFor	
version	

Non Patchable Attributes	Rule
id	
href	
lastUpdate	

Usage Samples

Here's an example of a request for patching a ServiceCatalog.

Request

```
PATCH /tmf-api/naas/v4/serviceCatalog/3830
Content-Type: application/merge-patch+json

{
  "name": "Catalog Wholesale Business",
  "description": "This service catalog describes services that address the wholesale business segment.",
  "version": "1.0",
  "validFor": {
    "startDateTime": "2017-08-29T00:00:00Z",
    "endDateTime": "2018-03-25T00:00:00Z"
  },
  "lastUpdate": "2017-08-27T00:00:00Z",
  "lifecycleStatus": "Active",
  "relatedParty": [
    {
      "href": "https://mycsp.com:8080/tmf-api/partyManagement/v4/organization/3426",
      "id": "3426",
      "name": "Broadly Broad Ltd",
      "role": "vendor"
    }
  ],
  "category": [
    {
      "href": "https://mycsp.com:8080/tmf-api/serviceCatalogManagement/v4/category/7752",
      "id": "7752",
      "name": "IoT"
    }
```

© TM Forum 2021. All Rights Reserved

```
  ],
  "@type": "ServiceCatalog",
  "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Service/ServiceCatalog.schema.json ",
  "@baseType": "Catalog"
}
```

Response

200

```
{
  "id": "3830",
  "href": "https://mycsp.com:8080/tmf-api/serviceCatalogManagement/v4/serviceCatalog/3830",
  "name": "Catalog Wholesale Business",
  "description": "This service catalog describes services that address the wholesale business segment.",
  "version": "1.0",
  "validFor": {
    "startDateTime": "2017-08-29T00:00:00Z",
    "endDateTime": "2018-03-25T00:00:00Z"
  },
  "lastUpdate": "2017-08-27T00:00:00Z",
  "lifecycleStatus": "Active",
  "relatedParty": [
    {
      "href": "https://mycsp.com:8080/tmf-api/partyManagement/v4/organization/3426",
      "id": "3426",
      "name": "Broadly Broad Ltd",
      "role": "vendor"
    }
  ],
  "category": [
    {
      "href": "https://mycsp.com:8080/tmf-api/serviceCatalogManagement/v4/category/7752",
      "id": "7752",
      "name": "IoT"
    }
  ],
  "@type": "ServiceCatalog",
  "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Service/ServiceCatalog.schema.json ",
  "@baseType": "Catalog"
}
```

Delete service catalog

`DELETE /serviceCatalog/{id}`

Description

This operation deletes a service catalog entity.

© TM Forum 2021. All Rights Reserved

Usage Samples

Here's an example of a request for patching a ServiceCatalog.

Request
DELETE /tmf-api/naas/v4/serviceCatalog/3830
Response
204

Operations on Service Category

List service categories

`GET /serviceCategory?fields=...&{filtering}`

Description

This operation list service category entities.
Attribute selection is enabled for all first level attributes.
Filtering may be available depending on the compliance level supported by an implementation.

Usage Samples

Here's an example of a request for retrieving multiple ServiceCategory.

Request
GET /tmf-api/naas/v4/serviceCategory Accept: application/json
Response
200 [{ "id": "1708", "href": "https://mycsp.com:8080/tmf-api/serviceCatalogManagement/v4/serviceCategory/1708", "name": "Cloud Services",

© TM Forum 2021. All Rights Reserved

```
    "description": "A category to hold all available cloud service offers",
    "version": "1.0",
    "validFor": {
      "startDateTime": "2017-08-24T00:00:00Z",
      "endDateTime": "2018-03-25T00:00:00Z"
    },
    "lifecycleStatus": "Active",
    "lastUpdate": "2017-08-27T00:00:00Z",
    "isRoot": true,
    "serviceCandidate": [
      {
        "href": "https://mycsp.com:8080/tmf-api/serviceCatalogManagement/v4/serviceCandidate/5850",
        "id": "5850",
        "name": "Speed Max",
        "version": "1.1",
        "@referredType": "ServiceCandidate"
      }
    ],
    "category": [
      {
        "href": "https://mycsp.com:8080/tmf-api/serviceCatalogManagement/v4/category/6086",
        "id": "6086",
        "name": "Cloud",
        "@referredType": "ServiceCategory"
      }
    ],
    "@type": "ServiceCategory",
    "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Service/ServiceCategory.schema.json",
    "@baseType": "Category"
  }
]
```


Retrieve service category

`GET /serviceCategory/{id}?fields=...&{filtering}`

Description

This operation retrieves a service category entity.
Attribute selection is enabled for all first level attributes.
Filtering on sub-resources may be available depending on the compliance level supported by an implementation.

Usage Samples

Here's an example of a request for retrieving a specific ServiceCategory.

Request

© TM Forum 2021. All Rights Reserved

```
GET /tmf-api/naas/v4/serviceCategory/1708
Accept: application/json
```

Response

200


```
{
  "id": "1708",
  "href": "https://mycsp.com:8080/tmf-api/serviceCatalogManagement/v4/serviceCategory/1708",
  "name": "Cloud Services",
  "description": "A category to hold all available cloud service offers",
  "version": "1.0",
  "validFor": {
    "startDateTime": "2017-08-24T00:00:00Z",
    "endDateTime": "2018-03-25T00:00:00Z"
  },
  "lifecycleStatus": "Active",
  "lastUpdate": "2017-08-27T00:00:00Z",
  "isRoot": true,
  "serviceCandidate": [
    {
      "href": "https://mycsp.com:8080/tmf-api/serviceCatalogManagement/v4/serviceCandidate/5850",
      "id": "5850",
      "name": "Speed Max",
      "version": "1.1",
      "@referredType": "ServiceCandidate"
    }
  ],
  "category": [
    {
      "href": "https://mycsp.com:8080/tmf-api/serviceCatalogManagement/v4/category/6086",
      "id": "6086",
      "name": "Cloud",
      "@referredType": "ServiceCategory"
    }
  ],
  "@type": "ServiceCategory",
  "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Service/ServiceCategory.schema.json",
  "@baseType": "Category"
}
```


Create service category

`POST /serviceCategory`

Description

This operation creates a service category entity.

Mandatory and Non Mandatory Attributes

© TM Forum 2021. All Rights Reserved

The following tables provide the list of mandatory and non mandatory attributes when creating a ServiceCategory, including any possible rule conditions and applicable default values. Notice that it is up to an implementer to add additional mandatory attributes.

Mandatory Attributes	Rule
name	

Non Mandatory Attributes	Rule
@baseType	
@schemaLocation	
@type	
category	
description	
isRoot	
lastUpdate	
lifecycleStatus	
parentId	
serviceCandidate	
validFor	
version	

Usage Samples

Here's an example of a request for creating a specific ServiceCategory.

Request

```
POST /tmf-api/naas/v4/serviceCategory
Content-Type: application/json

{
  "name": "Cloud Services",
  "description": "A category to hold all available cloud service offers",
  "version": "1.0",
  "validFor": {
    "startDateTime": "2017-08-24T00:00:00Z",
    "endDateTime": "2018-03-25T00:00:00Z"
  },
  "lifecycleStatus": "Active",
  "lastUpdate": "2017-08-27T00:00:00Z",
  "isRoot": true,
  "serviceCandidate": [
    {
      "href": "https://mycsp.com:8080/tmf-api/serviceCatalogManagement/v4/serviceCandidate/5850",
      "id": "5850",
      "name": "Speed Max",
      "version": "1.1",
      "@referredType": "ServiceCandidate"
    }
```

© TM Forum 2021. All Rights Reserved

```
  ],
  "category": [
    {
      "href": "https://mycsp.com:8080/tmf-api/serviceCatalogManagement/v4/category/6086",
      "id": "6086",
      "name": "Cloud",
      "@referredType": "ServiceCategory"
    }
  ],
  "@type": "ServiceCategory",
  "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Service/ServiceCategory.schema.json",
  "@baseType": "Category"
}
```


Response

201


```
{
  "id": "1708",
  "href": "https://mycsp.com:8080/tmf-api/serviceCatalogManagement/v4/serviceCategory/1708",
  "name": "Cloud Services",
  "description": "A category to hold all available cloud service offers",
  "version": "1.0",
  "validFor": {
    "startDateTime": "2017-08-24T00:00:00Z",
    "endDateTime": "2018-03-25T00:00:00Z"
  },
  "lifecycleStatus": "Active",
  "lastUpdate": "2017-08-27T00:00:00Z",
  "isRoot": true,
  "serviceCandidate": [
    {
      "href": "https://mycsp.com:8080/tmf-api/serviceCatalogManagement/v4/serviceCandidate/5850",
      "id": "5850",
      "name": "Speed Max",
      "version": "1.1",
      "@referredType": "ServiceCandidate"
    }
  ],
  "category": [
    {
      "href": "https://mycsp.com:8080/tmf-api/serviceCatalogManagement/v4/category/6086",
      "id": "6086",
      "name": "Cloud",
      "@referredType": "ServiceCategory"
    }
  ],
  "@type": "ServiceCategory",
  "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Service/ServiceCategory.schema.json",
  "@baseType": "Category"
}
```


© TM Forum 2021. All Rights Reserved

Patch service category

PATCH /serviceCategory/{id}

Description

This operation allows partial updates of a service category entity. Support of json/merge (https://tools.ietf.org/html/rfc7386) is mandatory, support of json/patch (http://tools.ietf.org/html/rfc5789) is optional.

Note: If the update operation yields to the creation of sub-resources or relationships, the same rules concerning mandatory sub-resource attributes and default value settings in the POST operation applies to the PATCH operation. Hence these tables are not repeated here.

Patchable and Non Patchable Attributes

The tables below provide the list of patchable and non patchable attributes, including constraint rules on their usage.

Patchable Attributes	Rule
@baseType	
@schemaLocation	
@type	
category	
description	
isRoot	
lifecycleStatus	
name	
parentId	
serviceCandidate	
validFor	
version	

Non Patchable Attributes	Rule
id	
href	
lastUpdate	

Usage Samples

Here's an example of a request for patching a ServiceCategory.

Request
PATCH /tmf-api/naas/v4/serviceCategory/1708 Content-Type: application/merge-patch+json

© TM Forum 2021. All Rights Reserved

```
{
   "name": "Cloud Services",
   "description": "A category to hold all available cloud service offers",
   "version": "1.0",
   "validFor": {
      "startDateTime": "2017-08-24T00:00:00Z",
      "endDateTime": "2018-03-25T00:00:00Z"
   },
   "lifecycleStatus": "Active",
   "lastUpdate": "2017-08-27T00:00:00Z",
   "isRoot": true,
   "serviceCandidate": [
      {
         "href": "https://mycsp.com:8080/tmf-api/serviceCatalogManagement/v4/serviceCandidate/5850",
         "id": "5850",
         "name": "Speed Max",
         "version": "1.1",
         "@referredType": "ServiceCandidate"
      }
   ],
   "category": [
      {
         "href": "https://mycsp.com:8080/tmf-api/serviceCatalogManagement/v4/category/6086",
         "id": "6086",
         "name": "Cloud",
         "@referredType": "ServiceCategory"
      }
   ],
   "@type": "ServiceCategory",
   "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Service/ServiceCategory.schema.json",
   "@baseType": "Category"
}
```


Response

200


```
{
   "id": "1708",
   "href": "https://mycsp.com:8080/tmf-api/serviceCatalogManagement/v4/serviceCategory/1708",
   "name": "Cloud Services",
   "description": "A category to hold all available cloud service offers",
   "version": "1.0",
   "validFor": {
      "startDateTime": "2017-08-24T00:00:00Z",
      "endDateTime": "2018-03-25T00:00:00Z"
   },
   "lifecycleStatus": "Active",
   "lastUpdate": "2017-08-27T00:00:00Z",
   "isRoot": true,
   "serviceCandidate": [
      {
         "href": "https://mycsp.com:8080/tmf-api/serviceCatalogManagement/v4/serviceCandidate/5850",
```


© TM Forum 2021. All Rights Reserved

```
        "id": "5850",
        "name": "Speed Max",
        "version": "1.1",
        "@referredType": "ServiceCandidate"
      }
    ],
    "category": [
      {
        "href": "https://mycsp.com:8080/tmf-api/serviceCatalogManagement/v4/category/6086",
        "id": "6086",
        "name": "Cloud",
        "@referredType": "ServiceCategory"
      }
    ],
    "@type": "ServiceCategory",
    "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Service/ServiceCategory.schema.json",
    "@baseType": "Category"
}
```


Delete service category

`DELETE /serviceCategory/{id}`

Description

This operation deletes a service category entity.

Usage Samples

Here's an example of a request for patching a ServiceCategory.

Request
DELETE /tmf-api/naas/v4/serviceCategory/1708
Response
204

Operations on Service Candidate

List service candidates

`GET /serviceCandidate?fields=...&{filtering}`

© TM Forum 2021. All Rights Reserved

Description

This operation list service candidate entities.
Attribute selection is enabled for all first level attributes.
Filtering may be available depending on the compliance level supported by an implementation.

Usage Samples

Here's an example of a request for retrieving multiple ServiceCandidate.

Request

```
GET /tmf-api/naas/v4/serviceCandidate
Accept: application/json
```

Response

```
200

[
  {
    "id": "4994",
    "href": "https://mycsp.com:8080/tmf-api/serviceCatalogManagement/v4/serviceCandidate/4994",
    "name": "TVServiceCandidate",
    "description": "This service candidate allows provision of TV service",
    "version": "2.1",
    "validFor": {
      "startDateTime": "2017-08-23T00:00:00Z",
      "endDateTime": "2018-03-25T00:00:00Z"
    },
    "lastUpdate": "2017-08-27T00:00:00Z",
    "lifecycleStatus": "Active",
    "category": [
      {
        "id": "5980",
        "href": "https://mycsp.com:8080/tmf-api/serviceCatalogManagement/v4/serviceCategory/5980",
        "version": "3.2",
        "name": "TV"
      }
    ],
    "serviceSpecification": {
      "id": "9600",
      "href": "https://mycsp.com:8080/tmf-api/serviceCatalogManagement/v4/serviceSpecification/9600",
      "version": "2.1",
      "name": "CFSS_TV",
      "@type": "CustomerFacingServiceSpecification"
    },
    "@type": "ServiceCandidate",
    "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Service/ServiceCandidate.schema.json ",
```

© TM Forum 2021. All Rights Reserved

```
    "@baseType": ""
  }
]
```

Retrieve service candidate

GET /serviceCandidate/{id}?fields=...&{filtering}

Description

This operation retrieves a service candidate entity.
Attribute selection is enabled for all first level attributes.
Filtering on sub-resources may be available depending on the compliance level supported by an implementation.

Usage Samples

Here's an example of a request for retrieving a specific ServiceCandidate.

Request

```
GET /tmf-api/naas/v4/serviceCandidate/4994
Accept: application/json
```

Response

```
200

{
  "id": "4994",
  "href": "https://mycsp.com:8080/tmf-api/serviceCatalogManagement/v4/serviceCandidate/4994",
  "name": "TVServiceCandidate",
  "description": "This service candidate allows provision of TV service",
  "version": "2.1",
  "validFor": {
    "startDateTime": "2017-08-23T00:00:00Z",
    "endDateTime": "2018-03-25T00:00:00Z"
  },
  "lastUpdate": "2017-08-27T00:00:00Z",
  "lifecycleStatus": "Active",
  "category": [
    {
      "id": "5980",
      "href": "https://mycsp.com:8080/tmf-api/serviceCatalogManagement/v4/serviceCategory/5980",
      "version": "3.2",
      "name": "TV"
    }
  ],
```

© TM Forum 2021. All Rights Reserved

```
  "serviceSpecification": {
    "id": "9600",
    "href": "https://mycsp.com:8080/tmf-api/serviceCatalogManagement/v4/serviceSpecification/9600",
    "version": "2.1",
    "name": "CFSS_TV",
    "@type": "CustomerFacingServiceSpecification"
  },
  "@type": "ServiceCandidate",
  "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Service/ServiceCandidate.schema.json ",
  "@baseType": ""
}
```


Create service candidate

POST /serviceCandidate

Description

This operation creates a service candidate entity.

Mandatory and Non Mandatory Attributes

The following tables provide the list of mandatory and non mandatory attributes when creating a ServiceCandidate, including any possible rule conditions and applicable default values. Notice that it is up to an implementer to add additional mandatory attributes.

Mandatory Attributes	Rule
name	
serviceSpecification	

Non Mandatory Attributes	Rule
@baseType	
@schemaLocation	
@type	
category	
description	
lastUpdate	
lifecycleStatus	
validFor	
version	

Usage Samples

Here's an example of a request for creating a specific ServiceCandidate.

Request

© TM Forum 2021. All Rights Reserved

POST /tmf-api/naas/v4/serviceCandidate
Content-Type: application/json


```
{
   "name": "TVServiceCandidate",
   "description": "This service candidate allows provision of TV service",
   "version": "2.1",
   "validFor": {
      "startDateTime": "2017-08-23T00:00:00Z",
      "endDateTime": "2018-03-25T00:00:00Z"
   },
   "lastUpdate": "2017-08-27T00:00:00Z",
   "lifecycleStatus": "Active",
   "category": [
      {
         "id": "5980",
         "href": "https://mycsp.com:8080/tmf-api/serviceCatalogManagement/v4/serviceCategory/5980",
         "version": "3.2",
         "name": "TV"
      }
   ],
   "serviceSpecification": {
      "id": "9600",
      "href": "https://mycsp.com:8080/tmf-api/serviceCatalogManagement/v4/serviceSpecification/9600",
      "version": "2.1",
      "name": "CFSS_TV",
      "@type": "CustomerFacingServiceSpecification"
   },
   "@type": "ServiceCandidate",
   "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Service/ServiceCandidate.schema.json ",
   "@baseType": ""
}
```


Response

201


```
{
   "id": "4994",
   "href": "https://mycsp.com:8080/tmf-api/serviceCatalogManagement/v4/serviceCandidate/4994",
   "name": "TVServiceCandidate",
   "description": "This service candidate allows provision of TV service",
   "version": "2.1",
   "validFor": {
      "startDateTime": "2017-08-23T00:00:00Z",
      "endDateTime": "2018-03-25T00:00:00Z"
   },
   "lastUpdate": "2017-08-27T00:00:00Z",
   "lifecycleStatus": "Active",
   "category": [
      {
         "id": "5980",
         "href": "https://mycsp.com:8080/tmf-api/serviceCatalogManagement/v4/serviceCategory/5980",
```


© TM Forum 2021. All Rights Reserved

```
        "version": "3.2",
        "name": "TV"
      }
   ],
   "serviceSpecification": {
      "id": "9600",
      "href": "https://mycsp.com:8080/tmf-api/serviceCatalogManagement/v4/serviceSpecification/9600",
      "version": "2.1",
      "name": "CFSS_TV",
      "@type": "CustomerFacingServiceSpecification"
   },
   "@type": "ServiceCandidate",
   "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Service/ServiceCandidate.schema.json ",
   "@baseType": ""
}
```


Patch service candidate

PATCH /serviceCandidate/{id}

Description

This operation allows partial updates of a service candidate entity. Support of json/merge (https://tools.ietf.org/html/rfc7386) is mandatory, support of json/patch (http://tools.ietf.org/html/rfc5789) is optional.

Note: If the update operation yields to the creation of sub-resources or relationships, the same rules concerning mandatory sub-resource attributes and default value settings in the POST operation applies to the PATCH operation. Hence these tables are not repeated here.

Patchable and Non Patchable Attributes

The tables below provide the list of patchable and non patchable attributes, including constraint rules on their usage.

Patchable Attributes	Rule
@baseType	
@schemaLocation	
@type	
category	
description	
lifecycleStatus	
name	
serviceSpecification	
validFor	
version	

Non Patchable Attributes	Rule
id	

© TM Forum 2021. All Rights Reserved

href	
lastUpdate	

Usage Samples

Here's an example of a request for patching a ServiceCandidate.

Request

```
PATCH /tmf-api/naas/v4/serviceCandidate/4994
Content-Type: application/merge-patch+json

{
   "name": "TVServiceCandidate",
   "description": "This service candidate allows provision of TV service",
   "version": "2.1",
   "validFor": {
      "startDateTime": "2017-08-23T00:00:00Z",
      "endDateTime": "2018-03-25T00:00:00Z"
   },
   "lastUpdate": "2017-08-27T00:00:00Z",
   "lifecycleStatus": "Active",
   "category": [
      {
         "id": "5980",
         "href": "https://mycsp.com:8080/tmf-api/serviceCatalogManagement/v4/serviceCategory/5980",
         "version": "3.2",
         "name": "TV"
      }
   ],
   "serviceSpecification": {
      "id": "9600",
      "href": "https://mycsp.com:8080/tmf-api/serviceCatalogManagement/v4/serviceSpecification/9600",
      "version": "2.1",
      "name": "CFSS_TV",
      "@type": "CustomerFacingServiceSpecification"
   },
   "@type": "ServiceCandidate",
   "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Service/ServiceCandidate.schema.json ",
   "@baseType": ""
}
```

Response

```
200

{
   "id": "4994",
   "href": "https://mycsp.com:8080/tmf-api/serviceCatalogManagement/v4/serviceCandidate/4994",
   "name": "TVServiceCandidate",
```

© TM Forum 2021. All Rights Reserved

```
  "description": "This service candidate allows provision of TV service",
  "version": "2.1",
  "validFor": {
    "startDateTime": "2017-08-23T00:00:00Z",
    "endDateTime": "2018-03-25T00:00:00Z"
  },
  "lastUpdate": "2017-08-27T00:00:00Z",
  "lifecycleStatus": "Active",
  "category": [
    {
      "id": "5980",
      "href": "https://mycsp.com:8080/tmf-api/serviceCatalogManagement/v4/serviceCategory/5980",
      "version": "3.2",
      "name": "TV"
    }
  ],
  "serviceSpecification": {
    "id": "9600",
    "href": "https://mycsp.com:8080/tmf-api/serviceCatalogManagement/v4/serviceSpecification/9600",
    "version": "2.1",
    "name": "CFSS_TV",
    "@type": "CustomerFacingServiceSpecification"
  },
  "@type": "ServiceCandidate",
  "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Service/ServiceCandidate.schema.json ",
  "@baseType": ""
}
```


Delete service candidate

`DELETE /serviceCandidate/{id}`

Description

This operation deletes a service candidate entity.

Usage Samples

Here's an example of a request for patching a ServiceCandidate.

Request
DELETE /tmf-api/naas/v4/serviceCandidate/4994
Response

© TM Forum 2021. All Rights Reserved

204

Operations on Service Specification

List service specifications

GET /serviceSpecification?fields=...&{filtering}

Description

This operation list service specification entities.
Attribute selection is enabled for all first level attributes.
Filtering may be available depending on the compliance level supported by an implementation.

Usage Samples

Here's an example of a request for retrieving multiple ServiceSpecification.

Request

```
GET /tmf-api/naas/v4/serviceSpecification
Accept: application/json
```

Response

200


```
[
  {
    "id": "7655",
    "href": "https://mycsp.com:8080/tmf-api/serviceCatalogManagement/v4/serviceSpecification/7655",
    "name": "Firewall Service",
    "description": "This service specification describes a firewall service that can be deployed in customer premise.",
    "version": "2.1",
    "validFor": {
      "startDateTime": "2017-08-23T00:00:00Z",
      "endDateTime": "2018-03-25T00:00:00Z"
    },
    "lastUpdate": "2017-08-27T00:00:00Z",
    "lifecycleStatus": "Active",
    "isBundle": false,
    "resourceSpecification": [
      {
        "href": "https://mycsp.com:8080/tmf-api/resourceCatalogManagement/v4/resourceSpecification/42",
        "id": "42",
        "name": "Firewall",
```


© TM Forum 2021. All Rights Reserved

```
        "version": "1.0",
        "@referredType": "ResourceFunctionSpec"
      }
    ],
    "attachment": [
      {
        "description": "This attachment gives a block diagram of the firewall.",
        "href": "https://mycsp.com:8080/tmf-api/documentManagement/v4/attachment/22",
        "id": "22",
        "url": "https://mycsp.com:7070/docloader?docnum=3534536"
      }
    ],
    "serviceSpecCharacteristic": [
      {
        "name": "OperatingSystem",
        "description": "This characteristic describes the operating system run by the service",
        "valueType": "string",
        "configurable": true,
        "validFor": {
          "startDateTime": "2017-08-12T00:00:00Z",
          "endDateTime": "2018-03-07T00:00:00Z"
        },
        "@valueSchemaLocation": "",
        "minCardinality": 0,
        "maxCardinality": 1,
        "isUnique": true,
        "regex": "",
        "extensible": false,
        "serviceSpecCharacteristicValue": [
          {
            "valueType": "string",
            "isDefault": true,
            "value": "Android KitKat",
            "validFor": {
              "startDateTime": "2017-08-06T00:00:00Z",
              "endDateTime": "2018-03-07T00:00:00Z"
            }
          }
        ],
        "serviceSpecCharRelationship": [
          {
            "relationshipType": "dependency",
            "role": "",
            "id": "4690",
            "href": "https://mycsp.com:8080/tmf-
api/serviceCatalogManagement/v4/serviceSpecificationCharacteristic/4690",
            "name": "OperatingSystem",
            "validFor": {
              "startDateTime": "2017-08-11T00:00:00Z",
              "endDateTime": "2018-03-07T00:00:00Z"
            }
          }
        ],
        "@type": "ServiceSpecCharacteristic",
        "@schemaLocation": "https://mycsp.com:8080/tmf-
```


© TM Forum 2021. All Rights Reserved

```
api/schema/Service/ServiceSpecCharacteristic.schema.json"
        },
        {
          "name": "Scalability",
          "description": "Scalability parameters for this resource facing service spec ",
          "valueType": "CapabilityScalable",
          "configurable": true,
          "validFor": {
            "startDateTime": "2017-08-17T00:00:00Z",
            "endDateTime": "2018-03-12T00:00:00Z"
          },
          "@valueSchemaLocation": "https://mycsp.com:8080/tmf-api/schema/Service/CapabilityScalable.schema.json",
          "minCardinality": 0,
          "maxCardinality": 1,
          "isUnique": true,
          "regex": "",
          "extensible": true,
          "serviceSpecCharacteristicValue": [
            {
              "valueType": "object",
              "isDefault": true,
              "value": {
                "minInstances": 1,
                "maxInstances": 1000
              },
              "validFor": {
                "startDateTime": "2017-08-17T00:00:00Z",
                "endDateTime": "2018-03-12T00:00:00Z"
              },
              "@type": "CapabilityScalable",
              "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Service/CapabilityScalable.schema.json"
            }
          ],
          "serviceSpecCharRelationship": [],
          "@type": "ServiceSpecCharacteristic",
          "@schemaLocation": "https://mycsp.com:8080/tmf-
api/schema/Service/ServiceSpecCharacteristic.schema.json"
        }
      ],
      "relatedParty": [
        {
          "href": "https://mycsp.com:8080/tmf-api/partyManagement/v4/organization/3643",
          "id": "3643",
          "name": "Firewall Express",
          "role": "Supplier"
        }
      ],
      "serviceSpecRelationship": [
        {
          "relationshipType": "dependency",
          "role": "dependent",
          "id": "5563",
          "href": "https://mycsp.com:8080/tmf-api/serviceCatalogManagement/v4/serviceSpecification/5563",
          "name": "Points to the Deep Packet Inspection service on which this Firewall service depends",
          "validFor": {
```


© TM Forum 2021. All Rights Reserved

```
            "startDateTime": "2017-08-25T00:00:00Z",
            "endDateTime": "2018-03-25T00:00:00Z"
          }
        }
      ],
      "serviceLevelSpecification": [
        {
          "href": "https://mycsp.com:8080/tmf-api/serviceQualityManagement/v4/serviceLevelSpecification/8899",
          "id": "8899",
          "name": "Platinum"
        }
      ],
      "targetServiceSchema": {
        "@type": "RFS",
        "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Service/RFS.schema.json"
      },
      "@type": "ResourceFacingServiceSpecification",
      "@schemaLocation": "https://mycsp.com:8080/tmf-
api/schema/Service/ResourceFacingServiceSpecification.schema.json",
      "@baseType": "ServiceSpecification"
    }
]
```


Retrieve service specification

`GET /serviceSpecification/{id}?fields=...&{filtering}`

Description

This operation retrieves a service specification entity.
Attribute selection is enabled for all first level attributes.
Filtering on sub-resources may be available depending on the compliance level supported by an implementation.

Usage Samples

Here's an example of a request for retrieving a specific ServiceSpecification.

Request
GET /tmf-api/naas/v4/serviceSpecification/7655 Accept: application/json
Response
200 {

© TM Forum 2021. All Rights Reserved

```
"id": "7655",
"href": "https://mycsp.com:8080/tmf-api/serviceCatalogManagement/v4/serviceSpecification/7655",
"name": "Firewall Service",
"description": "This service specification describes a firewall service that can be deployed in customer premise.",
"version": "2.1",
"validFor": {
   "startDateTime": "2017-08-23T00:00:00Z",
   "endDateTime": "2018-03-25T00:00:00Z"
},
"lastUpdate": "2017-08-27T00:00:00Z",
"lifecycleStatus": "Active",
"isBundle": false,
"resourceSpecification": [
   {
      "href": "https://mycsp.com:8080/tmf-api/resourceCatalogManagement/v4/resourceSpecification/42",
      "id": "42",
      "name": "Firewall",
      "version": "1.0",
      "@referredType": "ResourceFunctionSpec"
   }
],
"attachment": [
   {
      "description": "This attachment gives a block diagram of the firewall.",
      "href": "https://mycsp.com:8080/tmf-api/documentManagement/v4/attachment/22",
      "id": "22",
      "url": "https://mycsp.com:7070/docloader?docnum=3534536"
   }
],
"serviceSpecCharacteristic": [
   {
      "name": "OperatingSystem",
      "description": "This characteristic describes the operating system run by the service",
      "valueType": "string",
      "configurable": true,
      "validFor": {
         "startDateTime": "2017-08-12T00:00:00Z",
         "endDateTime": "2018-03-07T00:00:00Z"
      },
      "@valueSchemaLocation": "",
      "minCardinality": 0,
      "maxCardinality": 1,
      "isUnique": true,
      "regex": "",
      "extensible": false,
      "serviceSpecCharacteristicValue": [
         {
            "valueType": "string",
            "isDefault": true,
            "value": "Android KitKat",
            "validFor": {
               "startDateTime": "2017-08-06T00:00:00Z",
               "endDateTime": "2018-03-07T00:00:00Z"
            }
         }
```


© TM Forum 2021. All Rights Reserved

```
        ],
        "serviceSpecCharRelationship": [
          {
            "relationshipType": "dependency",
            "role": "",
            "id": "4690",
            "href": "https://mycsp.com:8080/tmf-
api/serviceCatalogManagement/v4/serviceSpecificationCharacteristic/4690",
            "name": "OperatingSystem",
            "validFor": {
              "startDateTime": "2017-08-11T00:00:00Z",
              "endDateTime": "2018-03-07T00:00:00Z"
            }
          }
        ],
        "@type": "ServiceSpecCharacteristic",
        "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Service/ServiceSpecCharacteristic.schema.json"
      },
      {
        "name": "Scalability",
        "description": "Scalability parameters for this resource facing service spec ",
        "valueType": "CapabilityScalable",
        "configurable": true,
        "validFor": {
          "startDateTime": "2017-08-17T00:00:00Z",
          "endDateTime": "2018-03-12T00:00:00Z"
        },
        "@valueSchemaLocation": "https://mycsp.com:8080/tmf-api/schema/Service/CapabilityScalable.schema.json",
        "minCardinality": 0,
        "maxCardinality": 1,
        "isUnique": true,
        "regex": "",
        "extensible": true,
        "serviceSpecCharacteristicValue": [
          {
            "valueType": "object",
            "isDefault": true,
            "value": {
              "minInstances": 1,
              "maxInstances": 1000
            },
            "validFor": {
              "startDateTime": "2017-08-17T00:00:00Z",
              "endDateTime": "2018-03-12T00:00:00Z"
            },
            "@type": "CapabilityScalable",
            "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Service/CapabilityScalable.schema.json"
          }
        ],
        "serviceSpecCharRelationship": [],
        "@type": "ServiceSpecCharacteristic",
        "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Service/ServiceSpecCharacteristic.schema.json"
      }
    ],
    "relatedParty": [
```


© TM Forum 2021. All Rights Reserved

```
    {
      "href": "https://mycsp.com:8080/tmf-api/partyManagement/v4/organization/3643",
      "id": "3643",
      "name": "Firewall Express",
      "role": "Supplier"
    }
  ],
  "serviceSpecRelationship": [
    {
      "relationshipType": "dependency",
      "role": "dependent",
      "id": "5563",
      "href": "https://mycsp.com:8080/tmf-api/serviceCatalogManagement/v4/serviceSpecification/5563",
      "name": "Points to the Deep Packet Inspection service on which this Firewall service depends",
      "validFor": {
        "startDateTime": "2017-08-25T00:00:00Z",
        "endDateTime": "2018-03-25T00:00:00Z"
      }
    }
  ],
  "serviceLevelSpecification": [
    {
      "href": "https://mycsp.com:8080/tmf-api/serviceQualityManagement/v4/serviceLevelSpecification/8899",
      "id": "8899",
      "name": "Platinum"
    }
  ],
  "targetServiceSchema": {
    "@type": "RFS",
    "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Service/RFS.schema.json"
  },
  "@type": "ResourceFacingServiceSpecification",
  "@schemaLocation": "https://mycsp.com:8080/tmf-
api/schema/Service/ResourceFacingServiceSpecification.schema.json",
  "@baseType": "ServiceSpecification"
}
```


Create service specification

POST /serviceSpecification

Description

This operation creates a service specification entity.

Mandatory and Non Mandatory Attributes

The following tables provide the list of mandatory and non mandatory attributes when creating a ServiceSpecification, including any possible rule conditions and applicable default values. Notice that it is up to an implementer to add additional mandatory attributes.

Mandatory Attributes	Rule
name	

© TM Forum 2021. All Rights Reserved

Non Mandatory Attributes	Rule
description	
isBundle	
lastUpdate	
lifecycleStatus	
version	
@baseType	
@schemaLocation	
@type	
featureSpecification	
resourceSpecification	
serviceLevelSpecification	
serviceSpecRelationship	
attachment	
constraint	
entitySpecRelationship	
relatedParty	
specCharacteristic	
targetEntitySchema	
validFor	

Usage Samples

Here's an example of a request for creating a specific ServiceSpecification.

Request

```
POST /tmf-api/naas/v4/serviceSpecification
Content-Type: application/json

{
  "name": "Firewall Service",
  "description": "This service specification describes a firewall service that can be deployed in customer premise.",
  "version": "2.1",
  "validFor": {
    "startDateTime": "2017-08-23T00:00:00Z",
    "endDateTime": "2018-03-25T00:00:00Z"
  },
  "lastUpdate": "2017-08-27T00:00:00Z",
  "lifecycleStatus": "Active",
  "isBundle": false,
  "resourceSpecification": [
    {
      "href": "https://mycsp.com:8080/tmf-api/resourceCatalogManagement/v4/resourceSpecification/42",
      "id": "42",
      "name": "Firewall",
      "version": "1.0",
      "@referredType": "ResourceFunctionSpec"
```

© TM Forum 2021. All Rights Reserved

```
    }
  ],
  "attachment": [
    {
      "description": "This attachment gives a block diagram of the firewall.",
      "href": "https://mycsp.com:8080/tmf-api/documentManagement/v4/attachment/22",
      "id": "22",
      "url": "https://mycsp.com:7070/docloader?docnum=3534536"
    }
  ],
  "serviceSpecCharacteristic": [
    {
      "name": "OperatingSystem",
      "description": "This characteristic describes the operating system run by the service",
      "valueType": "string",
      "configurable": true,
      "validFor": {
        "startDateTime": "2017-08-12T00:00:00Z",
        "endDateTime": "2018-03-07T00:00:00Z"
      },
      "@valueSchemaLocation": "",
      "minCardinality": 0,
      "maxCardinality": 1,
      "isUnique": true,
      "regex": "",
      "extensible": false,
      "serviceSpecCharacteristicValue": [
        {
          "valueType": "string",
          "isDefault": true,
          "value": "Android KitKat",
          "validFor": {
            "startDateTime": "2017-08-06T00:00:00Z",
            "endDateTime": "2018-03-07T00:00:00Z"
          }
        }
      ],
      "serviceSpecCharRelationship": [
        {
          "relationshipType": "dependency",
          "role": "",
          "id": "4690",
          "href": "https://mycsp.com:8080/tmf-
api/serviceCatalogManagement/v4/serviceSpecificationCharacteristic/4690",
          "name": "OperatingSystem",
          "validFor": {
            "startDateTime": "2017-08-11T00:00:00Z",
            "endDateTime": "2018-03-07T00:00:00Z"
          }
        }
      ],
      "@type": "ServiceSpecCharacteristic",
      "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Service/ServiceSpecCharacteristic.schema.json"
    },
    {
```


© TM Forum 2021. All Rights Reserved

```
        "name": "Scalability",
        "description": "Scalability parameters for this resource facing service spec ",
        "valueType": "CapabilityScalable",
        "configurable": true,
        "validFor": {
          "startDateTime": "2017-08-17T00:00:00Z",
          "endDateTime": "2018-03-12T00:00:00Z"
        },
        "@valueSchemaLocation": "https://mycsp.com:8080/tmf-api/schema/Service/CapabilityScalable.schema.json",
        "minCardinality": 0,
        "maxCardinality": 1,
        "isUnique": true,
        "regex": "",
        "extensible": true,
        "serviceSpecCharacteristicValue": [
          {
            "valueType": "object",
            "isDefault": true,
            "value": {
              "minInstances": 1,
              "maxInstances": 1000
            },
            "validFor": {
              "startDateTime": "2017-08-17T00:00:00Z",
              "endDateTime": "2018-03-12T00:00:00Z"
            },
            "@type": "CapabilityScalable",
            "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Service/CapabilityScalable.schema.json"
          }
        ],
        "serviceSpecCharRelationship": [],
        "@type": "ServiceSpecCharacteristic",
        "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Service/ServiceSpecCharacteristic.schema.json"
      }
    ],
    "relatedParty": [
      {
        "href": "https://mycsp.com:8080/tmf-api/partyManagement/v4/organization/3643",
        "id": "3643",
        "name": "Firewall Express",
        "role": "Supplier"
      }
    ],
    "serviceSpecRelationship": [
      {
        "relationshipType": "dependency",
        "role": "dependent",
        "id": "5563",
        "href": "https://mycsp.com:8080/tmf-api/serviceCatalogManagement/v4/serviceSpecification/5563",
        "name": "Points to the Deep Packet Inspection service on which this Firewall service depends",
        "validFor": {
          "startDateTime": "2017-08-25T00:00:00Z",
          "endDateTime": "2018-03-25T00:00:00Z"
        }
      }
```


© TM Forum 2021. All Rights Reserved

```
  ],
  "serviceLevelSpecification": [
    {
      "href": "https://mycsp.com:8080/tmf-api/serviceQualityManagement/v4/serviceLevelSpecification/8899",
      "id": "8899",
      "name": "Platinum"
    }
  ],
  "targetServiceSchema": {
    "@type": "RFS",
    "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Service/RFS.schema.json"
  },
  "@type": "ResourceFacingServiceSpecification",
  "@schemaLocation": "https://mycsp.com:8080/tmf-
api/schema/Service/ResourceFacingServiceSpecification.schema.json",
  "@baseType": "ServiceSpecification"
}
```


Response

201


```
{
  "id": "7655",
  "href": "https://mycsp.com:8080/tmf-api/serviceCatalogManagement/v4/serviceSpecification/7655",
  "name": "Firewall Service",
  "description": "This service specification describes a firewall service that can be deployed in customer premise.",
  "version": "2.1",
  "validFor": {
    "startDateTime": "2017-08-23T00:00:00Z",
    "endDateTime": "2018-03-25T00:00:00Z"
  },
  "lastUpdate": "2017-08-27T00:00:00Z",
  "lifecycleStatus": "Active",
  "isBundle": false,
  "resourceSpecification": [
    {
      "href": "https://mycsp.com:8080/tmf-api/resourceCatalogManagement/v4/resourceSpecification/42",
      "id": "42",
      "name": "Firewall",
      "version": "1.0",
      "@referredType": "ResourceFunctionSpec"
    }
  ],
  "attachment": [
    {
      "description": "This attachment gives a block diagram of the firewall.",
      "href": "https://mycsp.com:8080/tmf-api/documentManagement/v4/attachment/22",
      "id": "22",
      "url": "https://mycsp.com:7070/docloader?docnum=3534536"
    }
  ],
  "serviceSpecCharacteristic": [
```


© TM Forum 2021. All Rights Reserved

```
    {
      "name": "OperatingSystem",
      "description": "This characteristic describes the operating system run by the service",
      "valueType": "string",
      "configurable": true,
      "validFor": {
        "startDateTime": "2017-08-12T00:00:00Z",
        "endDateTime": "2018-03-07T00:00:00Z"
      },
      "@valueSchemaLocation": "",
      "minCardinality": 0,
      "maxCardinality": 1,
      "isUnique": true,
      "regex": "",
      "extensible": false,
      "serviceSpecCharacteristicValue": [
        {
          "valueType": "string",
          "isDefault": true,
          "value": "Android KitKat",
          "validFor": {
            "startDateTime": "2017-08-06T00:00:00Z",
            "endDateTime": "2018-03-07T00:00:00Z"
          }
        }
      ],
      "serviceSpecCharRelationship": [
        {
          "relationshipType": "dependency",
          "role": "",
          "id": "4690",
          "href": "https://mycsp.com:8080/tmf-
api/serviceCatalogManagement/v4/serviceSpecificationCharacteristic/4690",
          "name": "OperatingSystem",
          "validFor": {
            "startDateTime": "2017-08-11T00:00:00Z",
            "endDateTime": "2018-03-07T00:00:00Z"
          }
        }
      ],
      "@type": "ServiceSpecCharacteristic",
      "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Service/ServiceSpecCharacteristic.schema.json"
    },
    {
      "name": "Scalability",
      "description": "Scalability parameters for this resource facing service spec ",
      "valueType": "CapabilityScalable",
      "configurable": true,
      "validFor": {
        "startDateTime": "2017-08-17T00:00:00Z",
        "endDateTime": "2018-03-12T00:00:00Z"
      },
      "@valueSchemaLocation": "https://mycsp.com:8080/tmf-api/schema/Service/CapabilityScalable.schema.json",
      "minCardinality": 0,
      "maxCardinality": 1,
```


© TM Forum 2021. All Rights Reserved

```
        "isUnique": true,
        "regex": "",
        "extensible": true,
        "serviceSpecCharacteristicValue": [
          {
            "valueType": "object",
            "isDefault": true,
            "value": {
              "minInstances": 1,
              "maxInstances": 1000
            },
            "validFor": {
              "startDateTime": "2017-08-17T00:00:00Z",
              "endDateTime": "2018-03-12T00:00:00Z"
            },
            "@type": "CapabilityScalable",
            "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Service/CapabilityScalable.schema.json"
          }
        ],
        "serviceSpecCharRelationship": [],
        "@type": "ServiceSpecCharacteristic",
        "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Service/ServiceSpecCharacteristic.schema.json"
      }
    ],
    "relatedParty": [
      {
        "href": "https://mycsp.com:8080/tmf-api/partyManagement/v4/organization/3643",
        "id": "3643",
        "name": "Firewall Express",
        "role": "Supplier"
      }
    ],
    "serviceSpecRelationship": [
      {
        "relationshipType": "dependency",
        "role": "dependent",
        "id": "5563",
        "href": "https://mycsp.com:8080/tmf-api/serviceCatalogManagement/v4/serviceSpecification/5563",
        "name": "Points to the Deep Packet Inspection service on which this Firewall service depends",
        "validFor": {
          "startDateTime": "2017-08-25T00:00:00Z",
          "endDateTime": "2018-03-25T00:00:00Z"
        }
      }
    ],
    "serviceLevelSpecification": [
      {
        "href": "https://mycsp.com:8080/tmf-api/serviceQualityManagement/v4/serviceLevelSpecification/8899",
        "id": "8899",
        "name": "Platinum"
      }
    ],
    "targetServiceSchema": {
      "@type": "RFS",
      "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Service/RFS.schema.json"
```


© TM Forum 2021. All Rights Reserved

```
  },
  "@type": "ResourceFacingServiceSpecification",
  "@schemaLocation": "https://mycsp.com:8080/tmf-
api/schema/Service/ResourceFacingServiceSpecification.schema.json",
  "@baseType": "ServiceSpecification"
}
```

Patch service specification

PATCH /serviceSpecification/{id}

Description

This operation allows partial updates of a service specification entity. Support of json/merge (https://tools.ietf.org/html/rfc7386) is mandatory, support of json/patch (http://tools.ietf.org/html/rfc5789) is optional.

Note: If the update operation yields to the creation of sub-resources or relationships, the same rules concerning mandatory sub-resource attributes and default value settings in the POST operation applies to the PATCH operation. Hence these tables are not repeated here.

Patchable and Non Patchable Attributes

The tables below provide the list of patchable and non patchable attributes, including constraint rules on their usage.

Patchable Attributes	Rule
description	
isBundle	
lifecycleStatus	
name	
version	
@baseType	
@schemaLocation	
@type	
featureSpecification	
resourceSpecification	
serviceLevelSpecification	
serviceSpecRelationship	
attachment	
constraint	
entitySpecRelationship	
relatedParty	
specCharacteristic	
targetEntitySchema	
validFor	

© TM Forum 2021. All Rights Reserved

Non Patchable Attributes	Rule
id	
href	
lastUpdate	

Usage Samples

Here's an example of a request for patching a ServiceSpecification.

Request

```
PATCH /tmf-api/naas/v4/serviceSpecification/7655
Content-Type: application/merge-patch+json

{
  "name": "Firewall Service",
  "description": "This service specification describes a firewall service that can be deployed in customer premise.",
  "version": "2.1",
  "validFor": {
    "startDateTime": "2017-08-23T00:00:00Z",
    "endDateTime": "2018-03-25T00:00:00Z"
  },
  "lastUpdate": "2017-08-27T00:00:00Z",
  "lifecycleStatus": "Active",
  "isBundle": false,
  "resourceSpecification": [
    {
      "href": "https://mycsp.com:8080/tmf-api/resourceCatalogManagement/v4/resourceSpecification/42",
      "id": "42",
      "name": "Firewall",
      "version": "1.0",
      "@referredType": "ResourceFunctionSpec"
    }
  ],
  "attachment": [
    {
      "description": "This attachment gives a block diagram of the firewall.",
      "href": "https://mycsp.com:8080/tmf-api/documentManagement/v4/attachment/22",
      "id": "22",
      "url": "https://mycsp.com:7070/docloader?docnum=3534536"
    }
  ],
  "serviceSpecCharacteristic": [
    {
      "name": "OperatingSystem",
      "description": "This characteristic describes the operating system run by the service",
      "valueType": "string",
      "configurable": true,
      "validFor": {
        "startDateTime": "2017-08-12T00:00:00Z",
        "endDateTime": "2018-03-07T00:00:00Z"
      },
```

© TM Forum 2021. All Rights Reserved

```
        "@valueSchemaLocation": "",
        "minCardinality": 0,
        "maxCardinality": 1,
        "isUnique": true,
        "regex": "",
        "extensible": false,
        "serviceSpecCharacteristicValue": [
          {
            "valueType": "string",
            "isDefault": true,
            "value": "Android KitKat",
            "validFor": {
              "startDateTime": "2017-08-06T00:00:00Z",
              "endDateTime": "2018-03-07T00:00:00Z"
            }
          }
        ],
        "serviceSpecCharRelationship": [
          {
            "relationshipType": "dependency",
            "role": "",
            "id": "4690",
            "href": "https://mycsp.com:8080/tmf-
api/serviceCatalogManagement/v4/serviceSpecificationCharacteristic/4690",
            "name": "OperatingSystem",
            "validFor": {
              "startDateTime": "2017-08-11T00:00:00Z",
              "endDateTime": "2018-03-07T00:00:00Z"
            }
          }
        ],
        "@type": "ServiceSpecCharacteristic",
        "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Service/ServiceSpecCharacteristic.schema.json"
      },
      {
        "name": "Scalability",
        "description": "Scalability parameters for this resource facing service spec ",
        "valueType": "CapabilityScalable",
        "configurable": true,
        "validFor": {
          "startDateTime": "2017-08-17T00:00:00Z",
          "endDateTime": "2018-03-12T00:00:00Z"
        },
        "@valueSchemaLocation": "https://mycsp.com:8080/tmf-api/schema/Service/CapabilityScalable.schema.json",
        "minCardinality": 0,
        "maxCardinality": 1,
        "isUnique": true,
        "regex": "",
        "extensible": true,
        "serviceSpecCharacteristicValue": [
          {
            "valueType": "object",
            "isDefault": true,
            "value": {
              "minInstances": 1,
```


© TM Forum 2021. All Rights Reserved

```
                "maxInstances": 1000
              },
              "validFor": {
                "startDateTime": "2017-08-17T00:00:00Z",
                "endDateTime": "2018-03-12T00:00:00Z"
              },
              "@type": "CapabilityScalable",
              "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Service/CapabilityScalable.schema.json"
            }
          ],
          "serviceSpecCharRelationship": [],
          "@type": "ServiceSpecCharacteristic",
          "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Service/ServiceSpecCharacteristic.schema.json"
        }
      ],
      "relatedParty": [
        {
          "href": "https://mycsp.com:8080/tmf-api/partyManagement/v4/organization/3643",
          "id": "3643",
          "name": "Firewall Express",
          "role": "Supplier"
        }
      ],
      "serviceSpecRelationship": [
        {
          "relationshipType": "dependency",
          "role": "dependent",
          "id": "5563",
          "href": "https://mycsp.com:8080/tmf-api/serviceCatalogManagement/v4/serviceSpecification/5563",
          "name": "Points to the Deep Packet Inspection service on which this Firewall service depends",
          "validFor": {
            "startDateTime": "2017-08-25T00:00:00Z",
            "endDateTime": "2018-03-25T00:00:00Z"
          }
        }
      ],
      "serviceLevelSpecification": [
        {
          "href": "https://mycsp.com:8080/tmf-api/serviceQualityManagement/v4/serviceLevelSpecification/8899",
          "id": "8899",
          "name": "Platinum"
        }
      ],
      "targetServiceSchema": {
        "@type": "RFS",
        "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Service/RFS.schema.json"
      },
      "@type": "ResourceFacingServiceSpecification",
      "@schemaLocation": "https://mycsp.com:8080/tmf-
api/schema/Service/ResourceFacingServiceSpecification.schema.json",
      "@baseType": "ServiceSpecification"
}
```


© TM Forum 2021. All Rights Reserved

Response

200


```
{
  "id": "7655",
  "href": "https://mycsp.com:8080/tmf-api/serviceCatalogManagement/v4/serviceSpecification/7655",
  "name": "Firewall Service",
  "description": "This service specification describes a firewall service that can be deployed in customer premise.",
  "version": "2.1",
  "validFor": {
    "startDateTime": "2017-08-23T00:00:00Z",
    "endDateTime": "2018-03-25T00:00:00Z"
  },
  "lastUpdate": "2017-08-27T00:00:00Z",
  "lifecycleStatus": "Active",
  "isBundle": false,
  "resourceSpecification": [
    {
      "href": "https://mycsp.com:8080/tmf-api/resourceCatalogManagement/v4/resourceSpecification/42",
      "id": "42",
      "name": "Firewall",
      "version": "1.0",
      "@referredType": "ResourceFunctionSpec"
    }
  ],
  "attachment": [
    {
      "description": "This attachment gives a block diagram of the firewall.",
      "href": "https://mycsp.com:8080/tmf-api/documentManagement/v4/attachment/22",
      "id": "22",
      "url": "https://mycsp.com:7070/docloader?docnum=3534536"
    }
  ],
  "serviceSpecCharacteristic": [
    {
      "name": "OperatingSystem",
      "description": "This characteristic describes the operating system run by the service",
      "valueType": "string",
      "configurable": true,
      "validFor": {
        "startDateTime": "2017-08-12T00:00:00Z",
        "endDateTime": "2018-03-07T00:00:00Z"
      },
      "@valueSchemaLocation": "",
      "minCardinality": 0,
      "maxCardinality": 1,
      "isUnique": true,
      "regex": "",
      "extensible": false,
      "serviceSpecCharacteristicValue": [
        {
          "valueType": "string",
          "isDefault": true,
```


© TM Forum 2021. All Rights Reserved

```
            "value": "Android KitKat",
            "validFor": {
              "startDateTime": "2017-08-06T00:00:00Z",
              "endDateTime": "2018-03-07T00:00:00Z"
            }
          }
        ],
        "serviceSpecCharRelationship": [
          {
            "relationshipType": "dependency",
            "role": "",
            "id": "4690",
            "href": "https://mycsp.com:8080/tmf-
api/serviceCatalogManagement/v4/serviceSpecificationCharacteristic/4690",
            "name": "OperatingSystem",
            "validFor": {
              "startDateTime": "2017-08-11T00:00:00Z",
              "endDateTime": "2018-03-07T00:00:00Z"
            }
          }
        ],
        "@type": "ServiceSpecCharacteristic",
        "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Service/ServiceSpecCharacteristic.schema.json"
      },
      {
        "name": "Scalability",
        "description": "Scalability parameters for this resource facing service spec ",
        "valueType": "CapabilityScalable",
        "configurable": true,
        "validFor": {
          "startDateTime": "2017-08-17T00:00:00Z",
          "endDateTime": "2018-03-12T00:00:00Z"
        },
        "@valueSchemaLocation": "https://mycsp.com:8080/tmf-api/schema/Service/CapabilityScalable.schema.json",
        "minCardinality": 0,
        "maxCardinality": 1,
        "isUnique": true,
        "regex": "",
        "extensible": true,
        "serviceSpecCharacteristicValue": [
          {
            "valueType": "object",
            "isDefault": true,
            "value": {
              "minInstances": 1,
              "maxInstances": 1000
            },
            "validFor": {
              "startDateTime": "2017-08-17T00:00:00Z",
              "endDateTime": "2018-03-12T00:00:00Z"
            },
            "@type": "CapabilityScalable",
            "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Service/CapabilityScalable.schema.json"
          }
        ],
```


© TM Forum 2021. All Rights Reserved

```
        "serviceSpecCharRelationship": [],
        "@type": "ServiceSpecCharacteristic",
        "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Service/ServiceSpecCharacteristic.schema.json"
      }
    ],
    "relatedParty": [
      {
        "href": "https://mycsp.com:8080/tmf-api/partyManagement/v4/organization/3643",
        "id": "3643",
        "name": "Firewall Express",
        "role": "Supplier"
      }
    ],
    "serviceSpecRelationship": [
      {
        "relationshipType": "dependency",
        "role": "dependent",
        "id": "5563",
        "href": "https://mycsp.com:8080/tmf-api/serviceCatalogManagement/v4/serviceSpecification/5563",
        "name": "Points to the Deep Packet Inspection service on which this Firewall service depends",
        "validFor": {
          "startDateTime": "2017-08-25T00:00:00Z",
          "endDateTime": "2018-03-25T00:00:00Z"
        }
      }
    ],
    "serviceLevelSpecification": [
      {
        "href": "https://mycsp.com:8080/tmf-api/serviceQualityManagement/v4/serviceLevelSpecification/8899",
        "id": "8899",
        "name": "Platinum"
      }
    ],
    "targetServiceSchema": {
      "@type": "RFS",
      "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Service/RFS.schema.json"
    },
    "@type": "ResourceFacingServiceSpecification",
    "@schemaLocation": "https://mycsp.com:8080/tmf-
api/schema/Service/ResourceFacingServiceSpecification.schema.json",
    "@baseType": "ServiceSpecification"
}
```


Delete service specification

`DELETE /serviceSpecification/{id}`

Description

This operation deletes a service specification entity.

© TM Forum 2021. All Rights Reserved

Usage Samples

Here's an example of a request for patching a ServiceSpecification.

Request
DELETE /tmf-api/naas/v4/serviceSpecification/7655
Response
204

Operations on Service Order

List service orders

`GET /serviceOrder?fields=...&{filtering}`

Description

This operation list service order entities.
Attribute selection is enabled for all first level attributes.
Filtering may be available depending on the compliance level supported by an implementation.

Usage Samples

Here's an example of a request for retrieving multiple ServiceOrder.

Request
GET /tmf-api/naas/v4/serviceOrder Accept: application/json
Response
200 [{ "id": "42", "href": "http://serverlocation:port/serviceOrderingManagement/v4/serviceOrder/42", "externalId": "OrangeBSS747",

© TM Forum 2021. All Rights Reserved

```
    "priority": "1",
    "description": "Service order description",
    "category": "TMF resource illustration",
    "state": "acknowledged",
    "orderDate": "2018-01-12T09:37:40.508Z",
    "completionDate": "",
    "requestedStartDate": "2018-01-15T09:37:40.508Z",
    "requestedCompletionDate": "2018-01-15T09:37:40.508Z",
    "expectedCompletionDate": "2018-01-15T09:37:40.508Z",
    "startDate": "2018-01-12T09:37:40.508Z",
    "@type": "ServiceOrder",
    "note": [
      {
        "date": "2018-01-15T09:37:40.508Z",
        "author": "Jean Pontus",
        "text": "Some text"
      }
    ],
    "relatedParty": [
      {
        "id": "456",
        "href": "http://serverlocation:port/partyManagement/v4/party/456",
        "role": "requester",
        "name": "Jean Pontus",
        "@referredType": "Individual"
      }
    ],
    "orderItem": [
      {
        "id": "1",
        "action": "add",
        "state": "acknowledged",
        "service": {
          "@type": "vCPE",
          "@schemaLocation": "http//my.schemas/vCpe.schema.json",
          "state": "active",
          "serviceType": "CFS",
          "serviceCharacteristic": [
            {
              "name": "vCPE_IP",
              "valueType": "object",
              "value": {
                "@type": "IPAddress",
                "@schemaLocation": "http://my.schemas/IPAddress.schema.json",
                "vCPE_IP": "193.218.236.21"
              }
            }
          ],
          "serviceSpecification": {
            "@type": "ONAPServiceSpec",
            "@schemaLocation": "http//my.schemas/OnapServiceSpec.schema.json",
            "id": "12",
            "href": "http://.../serviceSpecification/12",
            "name": "vCPE",
            "version": "1",
```


© TM Forum 2021. All Rights Reserved

```
            "invariantUUID": "456-852-357",
            "toscaModelURL": "http//...",
            "targetServiceSchema": {
              "@type": "vCPE",
              "@schemaLocation": "http//my.schemas/vCpe.schema.json"
            }
          }
        }
      },
      {
        "id": "2",
        "action": "modify",
        "state": "acknowledged",
        "service": {
          "id": "456",
          "href": "http://serverlocation:port/serviceInventoryManagement/v4/service/456",
          "state": "active",
          "serviceType": "CFS",
          "serviceCharacteristic": [
            {
              "name": "Characteristic1",
              "value": "newValue"
            }
          ],
          "supportingResource": [
            {
              "id": "3456_DFG5-H690",
              "href": "http://...",
              "@referredType": "CloudResource"
            }
          ]
        }
      },
      {
        "id": "3",
        "action": "add",
        "state": "acknowledged",
        "service": {
          "state": "active",
          "serviceType": "CFS",
          "serviceRelationship": [
            {
              "relationshipType": "reliesOn",
              "service": {
                "href": "https://.../serviceManagement/v4/service/45",
                "id": "45"
              }
            }
          ],
          "serviceSpecification": {
            "id": "48",
            "href": "http//..../48",
            "name": "genericService48",
            "version": "2"
          }
```


© TM Forum 2021. All Rights Reserved

```
        }
      },
      {
        "id": "4",
        "action": "modify",
        "state": "acknowledged",
        "service": {
          "id": "12",
          "href": "http://serverlocation:port/serviceInventoryManagement/v4/service/12",
          "state": "inactive"
        }
      }
    ]
  }
]
```


Retrieve service order

`GET /serviceOrder/{id}?fields=...&{filtering}`

Description

This operation retrieves a service order entity.
Attribute selection is enabled for all first level attributes.
Filtering on sub-resources may be available depending on the compliance level supported by an implementation.

Usage Samples

Here's an example of a request for retrieving a specific ServiceOrder.

Request

```
GET /tmf-api/naas/v4/serviceOrder/42
Accept: application/json
```

Response


```
200

{
  "id": "42",
  "href": "http://serverlocation:port/serviceOrderingManagement/v4/serviceOrder/42",
  "externalId": "OrangeBSS747",
  "priority": "1",
  "description": "Service order description",
  "category": "TMF resource illustration",
  "state": "acknowledged",
```


© TM Forum 2021. All Rights Reserved

```
"orderDate": "2018-01-12T09:37:40.508Z",
"completionDate": "",
"requestedStartDate": "2018-01-15T09:37:40.508Z",
"requestedCompletionDate": "2018-01-15T09:37:40.508Z",
"expectedCompletionDate": "2018-01-15T09:37:40.508Z",
"startDate": "2018-01-12T09:37:40.508Z",
"@type": "ServiceOrder",
"note": [
  {
    "date": "2018-01-15T09:37:40.508Z",
    "author": "Jean Pontus",
    "text": "Some text"
  }
],
"relatedParty": [
  {
    "id": "456",
    "href": "http://serverlocation:port/partyManagement/v4/party/456",
    "role": "requester",
    "name": "Jean Pontus",
    "@referredType": "Individual"
  }
],
"orderItem": [
  {
    "id": "1",
    "action": "add",
    "state": "acknowledged",
    "service": {
      "@type": "vCPE",
      "@schemaLocation": "http//my.schemas/vCpe.schema.json",
      "state": "active",
      "serviceType": "CFS",
      "serviceCharacteristic": [
        {
          "name": "vCPE_IP",
          "valueType": "object",
          "value": {
            "@type": "IPAddress",
            "@schemaLocation": "http://my.schemas/IPAddress.schema.json",
            "vCPE_IP": "193.218.236.21"
          }
        }
      ],
      "serviceSpecification": {
        "@type": "ONAPServiceSpec",
        "@schemaLocation": "http//my.schemas/OnapServiceSpec.schema.json",
        "id": "12",
        "href": "http://.../serviceSpecification/12",
        "name": "vCPE",
        "version": "1",
        "invariantUUID": "456-852-357",
        "toscaModelURL": "http//...",
        "targetServiceSchema": {
          "@type": "vCPE",
```


© TM Forum 2021. All Rights Reserved

```
            "@schemaLocation": "http//my.schemas/vCpe.schema.json"
          }
        }
      }
    },
    {
      "id": "2",
      "action": "modify",
      "state": "acknowledged",
      "service": {
        "id": "456",
        "href": "http://serverlocation:port/serviceInventoryManagement/v4/service/456",
        "state": "active",
        "serviceType": "CFS",
        "serviceCharacteristic": [
          {
            "name": "Characteristic1",
            "value": "newValue"
          }
        ],
        "supportingResource": [
          {
            "id": "3456_DFG5-H690",
            "href": "http://...",
            "@referredType": "CloudResource"
          }
        ]
      }
    },
    {
      "id": "3",
      "action": "add",
      "state": "acknowledged",
      "service": {
        "state": "active",
        "serviceType": "CFS",
        "serviceRelationship": [
          {
            "relationshipType": "reliesOn",
            "service": {
              "href": "https://.../serviceManagement/v4/service/45",
              "id": "45"
            }
          }
        ],
        "serviceSpecification": {
          "id": "48",
          "href": "http//..../48",
          "name": "genericService48",
          "version": "2"
        }
      }
    },
    {
      "id": "4",
```


© TM Forum 2021. All Rights Reserved

```
        "action": "modify",
        "state": "acknowledged",
        "service": {
          "id": "12",
          "href": "http://serverlocation:port/serviceInventoryManagement/v4/service/12",
          "state": "inactive"
        }
      }
   ]
}
```


Create service order

`POST /serviceOrder`

Description

This operation creates a service order entity.

Mandatory and Non Mandatory Attributes

The following tables provide the list of mandatory and non mandatory attributes when creating a ServiceOrder, including any possible rule conditions and applicable default values. Notice that it is up to an implementer to add additional mandatory attributes.

Mandatory Attributes	Rule
serviceOrderItem	

Non Mandatory Attributes	Rule
@baseType	
@schemaLocation	
@type	
cancellationDate	
cancellationReason	
category	
completionDate	
description	
errorMessage	
expectedCompletionDate	
externalId	
externalReference	
jeopardyAlert	
milestone	
note	
notificationContact	
orderDate	
orderRelationship	
priority	
relatedParty	

© TM Forum 2021. All Rights Reserved

requestedCompletionDate	
requestedStartDate	
startDate	
state	

Additional Rules

The following table provides additional rules indicating mandatory fields in sub-resources or relationships when creating a ServiceOrder resource.

Context	Mandatory Sub-Attributes
serviceOrderItem	id, action, service
relatedParty	id, @type
note	text
orderRelationship	id, relationshipType
serviceOrderItem.service.supportingResource	id
serviceOrderItem.service.serviceCharacteristic	name, value
serviceOrderItem.serviceOrderItemRelationship	relationshipType
serviceOrderItem.serviceOrderItemRelationship.orderItem	itemId, serviceOrderId
serviceOrderItem.appointment	id
externalReference	name
serviceOrderItem.service.place	role, @type
serviceOrderItem.service.relatedEntity	role, @type

The following pre-conditions apply for this operation.

Pre-conditions
When creating a service order (post) an order must not have the state, cancellationDate, cancellationReason, completionDate, orderDate fields valued
When creating a service order (post) an order item must not have the state field valued
In relatedParty, the role is mandatory if the id identified a party - optional if it is a party role.

Usage Samples

Here's an example of a request for creating a specific ServiceOrder.

Request

```
POST /tmf-api/naas/v4/serviceOrder
Content-Type: application/json

{
   "externalId": "OrangeBSS747",
   "priority": "1",
   "description": "Service order description",
   "category": "TMF resource illustration",
   "state": "acknowledged",
```

© TM Forum 2021. All Rights Reserved

```
"orderDate": "2018-01-12T09:37:40.508Z",
"completionDate": "",
"requestedStartDate": "2018-01-15T09:37:40.508Z",
"requestedCompletionDate": "2018-01-15T09:37:40.508Z",
"expectedCompletionDate": "2018-01-15T09:37:40.508Z",
"startDate": "2018-01-12T09:37:40.508Z",
"@type": "ServiceOrder",
"note": [
  {
    "date": "2018-01-15T09:37:40.508Z",
    "author": "Jean Pontus",
    "text": "Some text"
  }
],
"relatedParty": [
  {
    "id": "456",
    "href": "http://serverlocation:port/partyManagement/v4/party/456",
    "role": "requester",
    "name": "Jean Pontus",
    "@referredType": "Individual"
  }
],
"orderItem": [
  {
    "id": "1",
    "action": "add",
    "state": "acknowledged",
    "service": {
      "@type": "vCPE",
      "@schemaLocation": "http//my.schemas/vCpe.schema.json",
      "state": "active",
      "serviceType": "CFS",
      "serviceCharacteristic": [
        {
          "name": "vCPE_IP",
          "valueType": "object",
          "value": {
            "@type": "IPAddress",
            "@schemaLocation": "http://my.schemas/IPAddress.schema.json",
            "vCPE_IP": "193.218.236.21"
          }
        }
      ],
      "serviceSpecification": {
        "@type": "ONAPServiceSpec",
        "@schemaLocation": "http//my.schemas/OnapServiceSpec.schema.json",
        "id": "12",
        "href": "http://.../serviceSpecification/12",
        "name": "vCPE",
        "version": "1",
        "invariantUUID": "456-852-357",
        "toscaModelURL": "http//...",
        "targetServiceSchema": {
          "@type": "vCPE",
```


© TM Forum 2021. All Rights Reserved

```
            "@schemaLocation": "http//my.schemas/vCpe.schema.json"
          }
        }
      }
    },
    {
      "id": "2",
      "action": "modify",
      "state": "acknowledged",
      "service": {
        "id": "456",
        "href": "http://serverlocation:port/serviceInventoryManagement/v4/service/456",
        "state": "active",
        "serviceType": "CFS",
        "serviceCharacteristic": [
          {
            "name": "Characteristic1",
            "value": "newValue"
          }
        ],
        "supportingResource": [
          {
            "id": "3456_DFG5-H690",
            "href": "http://...",
            "@referredType": "CloudResource"
          }
        ]
      }
    },
    {
      "id": "3",
      "action": "add",
      "state": "acknowledged",
      "service": {
        "state": "active",
        "serviceType": "CFS",
        "serviceRelationship": [
          {
            "relationshipType": "reliesOn",
            "service": {
              "href": "https://.../serviceManagement/v4/service/45",
              "id": "45"
            }
          }
        ],
        "serviceSpecification": {
          "id": "48",
          "href": "http//..../48",
          "name": "genericService48",
          "version": "2"
        }
      }
    },
    {
      "id": "4",
```


© TM Forum 2021. All Rights Reserved

```
        "action": "modify",
        "state": "acknowledged",
        "service": {
          "id": "12",
          "href": "http://serverlocation:port/serviceInventoryManagement/v4/service/12",
          "state": "inactive"
        }
      }
  ]
}
```


Response

201


```
{
  "id": "42",
  "href": "http://serverlocation:port/serviceOrderingManagement/v4/serviceOrder/42",
  "externalId": "OrangeBSS747",
  "priority": "1",
  "description": "Service order description",
  "category": "TMF resource illustration",
  "state": "acknowledged",
  "orderDate": "2018-01-12T09:37:40.508Z",
  "completionDate": "",
  "requestedStartDate": "2018-01-15T09:37:40.508Z",
  "requestedCompletionDate": "2018-01-15T09:37:40.508Z",
  "expectedCompletionDate": "2018-01-15T09:37:40.508Z",
  "startDate": "2018-01-12T09:37:40.508Z",
  "@type": "ServiceOrder",
  "note": [
    {
      "date": "2018-01-15T09:37:40.508Z",
      "author": "Jean Pontus",
      "text": "Some text"
    }
  ],
  "relatedParty": [
    {
      "id": "456",
      "href": "http://serverlocation:port/partyManagement/v4/party/456",
      "role": "requester",
      "name": "Jean Pontus",
      "@referredType": "Individual"
    }
  ],
  "orderItem": [
    {
      "id": "1",
      "action": "add",
      "state": "acknowledged",
      "service": {
        "@type": "vCPE",
```


© TM Forum 2021. All Rights Reserved

```
        "@schemaLocation": "http//my.schemas/vCpe.schema.json",
        "state": "active",
        "serviceType": "CFS",
        "serviceCharacteristic": [
          {
            "name": "vCPE_IP",
            "valueType": "object",
            "value": {
              "@type": "IPAddress",
              "@schemaLocation": "http://my.schemas/IPAddress.schema.json",
              "vCPE_IP": "193.218.236.21"
            }
          }
        ],
        "serviceSpecification": {
          "@type": "ONAPServiceSpec",
          "@schemaLocation": "http//my.schemas/OnapServiceSpec.schema.json",
          "id": "12",
          "href": "http://.../serviceSpecification/12",
          "name": "vCPE",
          "version": "1",
          "invariantUUID": "456-852-357",
          "toscaModelURL": "http//...",
          "targetServiceSchema": {
            "@type": "vCPE",
            "@schemaLocation": "http//my.schemas/vCpe.schema.json"
          }
        }
      }
    },
    {
      "id": "2",
      "action": "modify",
      "state": "acknowledged",
      "service": {
        "id": "456",
        "href": "http://serverlocation:port/serviceInventoryManagement/v4/service/456",
        "state": "active",
        "serviceType": "CFS",
        "serviceCharacteristic": [
          {
            "name": "Characteristic1",
            "value": "newValue"
          }
        ],
        "supportingResource": [
          {
            "id": "3456_DFG5-H690",
            "href": "http://...",
            "@referredType": "CloudResource"
          }
        ]
      }
    },
    {
```


© TM Forum 2021. All Rights Reserved

```
        "id": "3",
        "action": "add",
        "state": "acknowledged",
        "service": {
          "state": "active",
          "serviceType": "CFS",
          "serviceRelationship": [
            {
              "relationshipType": "reliesOn",
              "service": {
                "href": "https://.../serviceManagement/v4/service/45",
                "id": "45"
              }
            }
          ],
          "serviceSpecification": {
            "id": "48",
            "href": "http//..../48",
            "name": "genericService48",
            "version": "2"
          }
        }
      },
      {
        "id": "4",
        "action": "modify",
        "state": "acknowledged",
        "service": {
          "id": "12",
          "href": "http://serverlocation:port/serviceInventoryManagement/v4/service/12",
          "state": "inactive"
        }
      }
    ]
}
```


Patch service order

`PATCH /serviceOrder/{id}`

Description

This operation allows partial updates of a service order entity. Support of json/merge (https://tools.ietf.org/html/rfc7386) is mandatory, support of json/patch (http://tools.ietf.org/html/rfc5789) is optional.

Note: If the update operation yields to the creation of sub-resources or relationships, the same rules concerning mandatory sub-resource attributes and default value settings in the POST operation applies to the PATCH operation. Hence these tables are not repeated here.

Patchable and Non Patchable Attributes

© TM Forum 2021. All Rights Reserved

The tables below provide the list of patchable and non patchable attributes, including constraint rules on their usage.

Patchable Attributes	Rule
description	
expectedCompletionDate	
externalId	
externalReference	
note	
notificationContact	
orderRelationship	
priority	
relatedParty	
requestedCompletionDate	
requestedStartDate	
serviceOrderItem	
state	

Non Patchable Attributes	Rule
id	
href	
orderDate	
jeopardyAlert	
errorMessage	
milestone	
@baseType	
@schemaLocation	
@type	
cancellationDate	
cancellationReason	
category	
completionDate	
startDate	

Additional Rules

The following pre-conditions apply for this operation.

Pre-conditions
When creating a service order (post) an order must not have the state, cancellationDate, cancellationReason, completionDate, orderDate fields valued
When creating a service order (post) an order item must not have the state field valued
In relatedParty, the role is mandatory if the id identified a party - optional if it is a party role.

Usage Samples

Here's an example of a request for patching a ServiceOrder.

© TM Forum 2021. All Rights Reserved

Request

```
PATCH /tmf-api/naas/v4/serviceOrder/42
Content-Type: application/merge-patch+json

{
  "externalId": "OrangeBSS747",
  "priority": "1",
  "description": "Service order description",
  "category": "TMF resource illustration",
  "state": "acknowledged",
  "orderDate": "2018-01-12T09:37:40.508Z",
  "completionDate": "",
  "requestedStartDate": "2018-01-15T09:37:40.508Z",
  "requestedCompletionDate": "2018-01-15T09:37:40.508Z",
  "expectedCompletionDate": "2018-01-15T09:37:40.508Z",
  "startDate": "2018-01-12T09:37:40.508Z",
  "@type": "ServiceOrder",
  "note": [
    {
      "date": "2018-01-15T09:37:40.508Z",
      "author": "Jean Pontus",
      "text": "Some text"
    }
  ],
  "relatedParty": [
    {
      "id": "456",
      "href": "http://serverlocation:port/partyManagement/v4/party/456",
      "role": "requester",
      "name": "Jean Pontus",
      "@referredType": "Individual"
    }
  ],
  "orderItem": [
    {
      "id": "1",
      "action": "add",
      "state": "acknowledged",
      "service": {
        "@type": "vCPE",
        "@schemaLocation": "http//my.schemas/vCpe.schema.json",
        "state": "active",
        "serviceType": "CFS",
        "serviceCharacteristic": [
          {
            "name": "vCPE_IP",
            "valueType": "object",
            "value": {
              "@type": "IPAddress",
              "@schemaLocation": "http://my.schemas/IPAddress.schema.json",
              "vCPE_IP": "193.218.236.21"
            }
          }
```

© TM Forum 2021. All Rights Reserved

```
        ],
        "serviceSpecification": {
          "@type": "ONAPServiceSpec",
          "@schemaLocation": "http//my.schemas/OnapServiceSpec.schema.json",
          "id": "12",
          "href": "http://.../serviceSpecification/12",
          "name": "vCPE",
          "version": "1",
          "invariantUUID": "456-852-357",
          "toscaModelURL": "http//...",
          "targetServiceSchema": {
            "@type": "vCPE",
            "@schemaLocation": "http//my.schemas/vCpe.schema.json"
          }
        }
      }
    },
    {
      "id": "2",
      "action": "modify",
      "state": "acknowledged",
      "service": {
        "id": "456",
        "href": "http://serverlocation:port/serviceInventoryManagement/v4/service/456",
        "state": "active",
        "serviceType": "CFS",
        "serviceCharacteristic": [
          {
            "name": "Characteristic1",
            "value": "newValue"
          }
        ],
        "supportingResource": [
          {
            "id": "3456_DFG5-H690",
            "href": "http://...",
            "@referredType": "CloudResource"
          }
        ]
      }
    },
    {
      "id": "3",
      "action": "add",
      "state": "acknowledged",
      "service": {
        "state": "active",
        "serviceType": "CFS",
        "serviceRelationship": [
          {
            "relationshipType": "reliesOn",
            "service": {
              "href": "https://.../serviceManagement/v4/service/45",
              "id": "45"
            }
```


© TM Forum 2021. All Rights Reserved

```
            }
          ],
          "serviceSpecification": {
            "id": "48",
            "href": "http//..../48",
            "name": "genericService48",
            "version": "2"
          }
        }
      },
      {
        "id": "4",
        "action": "modify",
        "state": "acknowledged",
        "service": {
          "id": "12",
          "href": "http://serverlocation:port/serviceInventoryManagement/v4/service/12",
          "state": "inactive"
        }
      }
    ]
}
```


Response

200


```
{
  "id": "42",
  "href": "http://serverlocation:port/serviceOrderingManagement/v4/serviceOrder/42",
  "externalId": "OrangeBSS747",
  "priority": "1",
  "description": "Service order description",
  "category": "TMF resource illustration",
  "state": "acknowledged",
  "orderDate": "2018-01-12T09:37:40.508Z",
  "completionDate": "",
  "requestedStartDate": "2018-01-15T09:37:40.508Z",
  "requestedCompletionDate": "2018-01-15T09:37:40.508Z",
  "expectedCompletionDate": "2018-01-15T09:37:40.508Z",
  "startDate": "2018-01-12T09:37:40.508Z",
  "@type": "ServiceOrder",
  "note": [
    {
      "date": "2018-01-15T09:37:40.508Z",
      "author": "Jean Pontus",
      "text": "Some text"
    }
  ],
  "relatedParty": [
    {
      "id": "456",
      "href": "http://serverlocation:port/partyManagement/v4/party/456",
```


© TM Forum 2021. All Rights Reserved

```
      "role": "requester",
      "name": "Jean Pontus",
      "@referredType": "Individual"
    }
  ],
  "orderItem": [
    {
      "id": "1",
      "action": "add",
      "state": "acknowledged",
      "service": {
        "@type": "vCPE",
        "@schemaLocation": "http//my.schemas/vCpe.schema.json",
        "state": "active",
        "serviceType": "CFS",
        "serviceCharacteristic": [
          {
            "name": "vCPE_IP",
            "valueType": "object",
            "value": {
              "@type": "IPAddress",
              "@schemaLocation": "http://my.schemas/IPAddress.schema.json",
              "vCPE_IP": "193.218.236.21"
            }
          }
        ],
        "serviceSpecification": {
          "@type": "ONAPServiceSpec",
          "@schemaLocation": "http//my.schemas/OnapServiceSpec.schema.json",
          "id": "12",
          "href": "http://.../serviceSpecification/12",
          "name": "vCPE",
          "version": "1",
          "invariantUUID": "456-852-357",
          "toscaModelURL": "http//...",
          "targetServiceSchema": {
            "@type": "vCPE",
            "@schemaLocation": "http//my.schemas/vCpe.schema.json"
          }
        }
      }
    },
    {
      "id": "2",
      "action": "modify",
      "state": "acknowledged",
      "service": {
        "id": "456",
        "href": "http://serverlocation:port/serviceInventoryManagement/v4/service/456",
        "state": "active",
        "serviceType": "CFS",
        "serviceCharacteristic": [
          {
            "name": "Characteristic1",
            "value": "newValue"
```


© TM Forum 2021. All Rights Reserved

```
            }
          ],
          "supportingResource": [
            {
              "id": "3456_DFG5-H690",
              "href": "http://...",
              "@referredType": "CloudResource"
            }
          ]
        }
      },
      {
        "id": "3",
        "action": "add",
        "state": "acknowledged",
        "service": {
          "state": "active",
          "serviceType": "CFS",
          "serviceRelationship": [
            {
              "relationshipType": "reliesOn",
              "service": {
                "href": "https://.../serviceManagement/v4/service/45",
                "id": "45"
              }
            }
          ],
          "serviceSpecification": {
            "id": "48",
            "href": "http//..../48",
            "name": "genericService48",
            "version": "2"
          }
        }
      },
      {
        "id": "4",
        "action": "modify",
        "state": "acknowledged",
        "service": {
          "id": "12",
          "href": "http://serverlocation:port/serviceInventoryManagement/v4/service/12",
          "state": "inactive"
        }
      }
    ]
}
```


Delete service order

`DELETE /serviceOrder/{id}`

Description

© TM Forum 2021. All Rights Reserved

This operation deletes a service order entity.

Usage Samples

Here's an example of a request for patching a ServiceOrder.

Request
DELETE /tmf-api/naas/v4/serviceOrder/42
Response
204

Operations on Cancel Service Order

List cancel service orders

`GET /cancelServiceOrder?fields=...&{filtering}`

Description

This operation list cancel service order entities.
Attribute selection is enabled for all first level attributes.
Filtering may be available depending on the compliance level supported by an implementation.

Usage Samples

Here's an example of a request for retrieving multiple CancelServiceOrder.

Request
GET /tmf-api/naas/v4/cancelServiceOrder Accept: application/json
Response
200

© TM Forum 2021. All Rights Reserved

```
[
  {
    "id": "789-fsds5-kjp",
    "href": "https://host:port/serviceOrdering/v4/cancelServiceOrder/789-fsds5-kjp",
    "cancellationReason": "Duplicate service order",
    "effectiveCancellationDate": "2020-08-27T12:58:35.489Z",
    "requestedCancellationDate": "2020-08-27T12:56:21.931Z",
    "serviceOrder": {
      "id": "45f-98f-ss45",
      "href": "https://host:port/serviceOrdering/v4/serviceOrder/45f-98f-ss45",
      "@referredType": "ServiceOrder"
    },
    "state": "done",
    "@type": "CancelServiceOrder"
  }
]
```


Retrieve cancel service order

`GET /cancelServiceOrder/{id}?fields=...&{filtering}`

Description

This operation retrieves a cancel service order entity.
Attribute selection is enabled for all first level attributes.
Filtering on sub-resources may be available depending on the compliance level supported by an implementation.

Usage Samples

Here's an example of a request for retrieving a specific CancelServiceOrder.

Request

```
GET /tmf-api/naas/v4/cancelServiceOrder/789-fsds5-kjp
Accept: application/json
```

Response

200


```
{
  "id": "789-fsds5-kjp",
  "href": "https://host:port/serviceOrdering/v4/cancelServiceOrder/789-fsds5-kjp",
  "cancellationReason": "Duplicate service order",
  "effectiveCancellationDate": "2020-08-27T12:58:35.489Z",
  "requestedCancellationDate": "2020-08-27T12:56:21.931Z",
  "serviceOrder": {
```


© TM Forum 2021. All Rights Reserved

```
      "id": "45f-98f-ss45",
      "href": "https://host:port/serviceOrdering/v4/serviceOrder/45f-98f-ss45",
      "@referredType": "ServiceOrder"
   },
   "state": "done",
   "@type": "CancelServiceOrder"
}
```


Create cancel service order

POST /cancelServiceOrder

Description

This operation creates a cancel service order entity.

Mandatory and Non Mandatory Attributes

The following tables provide the list of mandatory and non mandatory attributes when creating a CancelServiceOrder, including any possible rule conditions and applicable default values. Notice that it is up to an implementer to add additional mandatory attributes.

Mandatory Attributes	Rule
ServiceOrder	

Non Mandatory Attributes	Rule
cancellationReason	
effectiveCancellationDate	
requestedCancellationDate	
@baseType	
@schemaLocation	
@type	
completionMessage	
errorMessage	
serviceOrder	
state	

Additional Rules

The following table provides additional rules indicating mandatory fields in sub-resources or relationships when creating a CancelServiceOrder resource.

Context	Mandatory Sub-Attributes
ServiceOrder	id

The following pre-conditions apply for this operation.

© TM Forum 2021. All Rights Reserved

Pre-conditions
service order id must exist
When creating a cancel service order (post) the id, href, effectiveCancellationDate and state fields must not be valued in the request

Usage Samples

Here's an example of a request for creating a specific CancelServiceOrder.

Request

```
POST /tmf-api/naas/v4/cancelServiceOrder
Content-Type: application/json

{
   "cancellationReason": "Duplicate service order",
   "effectiveCancellationDate": "2020-08-27T12:58:35.489Z",
   "requestedCancellationDate": "2020-08-27T12:56:21.931Z",
   "serviceOrder": {
      "id": "45f-98f-ss45",
      "href": "https://host:port/serviceOrdering/v4/serviceOrder/45f-98f-ss45",
      "@referredType": "ServiceOrder"
   },
   "state": "done",
   "@type": "CancelServiceOrder"
}
```

Response

```
201

{
   "id": "789-fsds5-kjp",
   "href": "https://host:port/serviceOrdering/v4/cancelServiceOrder/789-fsds5-kjp",
   "cancellationReason": "Duplicate service order",
   "effectiveCancellationDate": "2020-08-27T12:58:35.489Z",
   "requestedCancellationDate": "2020-08-27T12:56:21.931Z",
   "serviceOrder": {
      "id": "45f-98f-ss45",
      "href": "https://host:port/serviceOrdering/v4/serviceOrder/45f-98f-ss45",
      "@referredType": "ServiceOrder"
   },
   "state": "done",
   "@type": "CancelServiceOrder"
}
```

Operations on Check Service Qualification

© TM Forum 2021. All Rights Reserved

List check service qualifications

GET /checkServiceQualification?fields=...&{filtering}

Description

This operation list check service qualification entities.
Attribute selection is enabled for all first level attributes.
Filtering may be available depending on the compliance level supported by an implementation.

Usage Samples

Here's an example of a request for retrieving multiple CheckServiceQualification.

Request

```
GET /tmf-api/naas/v4/checkServiceQualification
Accept: application/json
```

Response

```
200

[
  {
    "id": "44",
    "href": "http://serverlocation:port/serviceQualificationManagement/v4/checkServiceQualification/456",
    "checkServiceQualificationDate": "2020-04-10T19:20:39.239Z",
    "description": "check service qualification illustration",
    "effectiveQualificationDate": "2020-04-11T15:20:39.239Z",
    "expectedQualificationDate": "2020-04-11T15:20:39.239Z",
    "expirationDate": "2020-04-24T19:20:39.239Z",
    "externalId": "CSQ1",
    "provideAlternative": true,
    "provideUnavailabilityReason": true,
    "instantSyncQualification": false,
    "qualificationResult": "alternate",
    "relatedParty": [
      {
        "id": "456",
        "href": "http://serverlocation:port/partyManagement/v4/party/456",
        "name": "Jean Pontus",
        "role": "Buyer contact",
        "@type": "RelatedParty",
        "@referredType": "Individual"
      }
    ],
    "serviceQualificationItem": [
```

© TM Forum 2021. All Rights Reserved

```
{
  "id": "1",
  "expectedActivationDate": "2020-04-16T19:20:39.239Z",
  "expectedServiceAvailabilityDate": "2020-04-11T19:20:39.239Z",
  "expirationDate": "2020-04-24T19:20:39.239Z",
  "qualificationResult": "alternate",
  "state": "done",
  "alternateServiceProposal": [
    {
      "id": "1",
      "alternateServiceAvailabilityDate": "2020-04-15T19:20:39.239Z",
      "alternateService": {
        "serviceCharacteristic": [
          {
            "id": "1",
            "name": "BroadcastQuality",
            "valueType": "string",
            "value": "HD",
            "@type": "ServiceCharacteristic"
          }
        ],
        "serviceSpecification": {
          "id": "23",
          "href": "string",
          "name": "TVAccess",
          "@type": "ServiceSpecification"
        },
        "@type": "Service"
      },
      "@type": "AlternateServiceProposal"
    }
  ],
  "eligibilityUnavailabilityReason": [
    {
      "code": "26-001",
      "label": "not enough bandwith at this place",
      "@baseType": "string",
      "@schemaLocation": "string",
      "@type": "string"
    }
  ],
  "service": {
    "place": [
      {
        "id": "11",
        "href": "http://serverlocation:port/geographicSiteManagement/v4/geographicSite/11",
        "name": "Jean Pontus home",
        "role": "Installation place",
        "@referredType": "GeographicSite"
      }
    ],
    "serviceCharacteristic": [
      {
        "id": "1",
        "name": "BroadcastQuality",
```


© TM Forum 2021. All Rights Reserved

```
                    "valueType": "string",
                    "value": "4K"
                  }
               ],
               "serviceSpecification": {
                  "id": "23",
                  "href": "string",
                  "name": "TVAccess",
                  "@type": "ServiceSpecification"
               },
               "@type": "Service"
            },
            "@type": "ServiceQualificationItem"
         }
      ],
      "state": "done",
      "@type": "CheckServiceQualification"
   }
]
```


Retrieve check service qualification

`GET /checkServiceQualification/{id}?fields=...&{filtering}`

Description

This operation retrieves a check service qualification entity.
Attribute selection is enabled for all first level attributes.
Filtering on sub-resources may be available depending on the compliance level supported by an implementation.

Usage Samples

Here's an example of a request for retrieving a specific CheckServiceQualification.

Request

```
GET /tmf-api/naas/v4/checkServiceQualification/44
Accept: application/json
```

Response

200


```
{
   "id": "44",
   "href": "http://serverlocation:port/serviceQualificationManagement/v4/checkServiceQualification/456",
   "checkServiceQualificationDate": "2020-04-10T19:20:39.239Z",
```


© TM Forum 2021. All Rights Reserved

```
  "description": "check service qualification illustration",
  "effectiveQualificationDate": "2020-04-11T15:20:39.239Z",
  "expectedQualificationDate": "2020-04-11T15:20:39.239Z",
  "expirationDate": "2020-04-24T19:20:39.239Z",
  "externalId": "CSQ1",
  "provideAlternative": true,
  "provideUnavailabilityReason": true,
  "instantSyncQualification": false,
  "qualificationResult": "alternate",
  "relatedParty": [
    {
      "id": "456",
      "href": "http://serverlocation:port/partyManagement/v4/party/456",
      "name": "Jean Pontus",
      "role": "Buyer contact",
      "@type": "RelatedParty",
      "@referredType": "Individual"
    }
  ],
  "serviceQualificationItem": [
    {
      "id": "1",
      "expectedActivationDate": "2020-04-16T19:20:39.239Z",
      "expectedServiceAvailabilityDate": "2020-04-11T19:20:39.239Z",
      "expirationDate": "2020-04-24T19:20:39.239Z",
      "qualificationResult": "alternate",
      "state": "done",
      "alternateServiceProposal": [
        {
          "id": "1",
          "alternateServiceAvailabilityDate": "2020-04-15T19:20:39.239Z",
          "alternateService": {
            "serviceCharacteristic": [
              {
                "id": "1",
                "name": "BroadcastQuality",
                "valueType": "string",
                "value": "HD",
                "@type": "ServiceCharacteristic"
              }
            ],
            "serviceSpecification": {
              "id": "23",
              "href": "string",
              "name": "TVAccess",
              "@type": "ServiceSpecification"
            },
            "@type": "Service"
          },
          "@type": "AlternateServiceProposal"
        }
      ],
      "eligibilityUnavailabilityReason": [
        {
          "code": "26-001",
```


© TM Forum 2021. All Rights Reserved

```
            "label": "not enough bandwith at this place",
            "@baseType": "string",
            "@schemaLocation": "string",
            "@type": "string"
          }
        ],
        "service": {
          "place": [
            {
              "id": "11",
              "href": "http://serverlocation:port/geographicSiteManagement/v4/geographicSite/11",
              "name": "Jean Pontus home",
              "role": "Installation place",
              "@referredType": "GeographicSite"
            }
          ],
          "serviceCharacteristic": [
            {
              "id": "1",
              "name": "BroadcastQuality",
              "valueType": "string",
              "value": "4K"
            }
          ],
          "serviceSpecification": {
            "id": "23",
            "href": "string",
            "name": "TVAccess",
            "@type": "ServiceSpecification"
          },
          "@type": "Service"
        },
        "@type": "ServiceQualificationItem"
      }
    ],
    "state": "done",
    "@type": "CheckServiceQualification"
}
```


Create check service qualification

POST /checkServiceQualification

Description

This operation creates a check service qualification entity.

Mandatory and Non Mandatory Attributes

The following tables provide the list of mandatory and non mandatory attributes when creating a CheckServiceQualification, including any possible rule conditions and applicable default values. Notice that it is up to an implementer to add additional mandatory attributes.

© TM Forum 2021. All Rights Reserved

Mandatory Attributes	Rule
serviceQualificationItem	

Non Mandatory Attributes	Rule
description	
effectiveQualificationDate	
estimatedResponseDate	
expectedQualificationDate	
expirationDate	
externalId	
instantSyncQualification	
@baseType	
@schemaLocation	
@type	
checkServiceQualificationDate	
provideAlternative	
provideUnavailabilityReason	
qualificationResult	
relatedParty	
state	

Usage Samples

Here's an example of a request for creating a specific CheckServiceQualification.

Request

```
POST /tmf-api/naas/v4/checkServiceQualification
Content-Type: application/json

{
  "checkServiceQualificationDate": "2020-04-10T19:20:39.239Z",
  "description": "check service qualification illustration",
  "effectiveQualificationDate": "2020-04-11T15:20:39.239Z",
  "expectedQualificationDate": "2020-04-11T15:20:39.239Z",
  "expirationDate": "2020-04-24T19:20:39.239Z",
  "externalId": "CSQ1",
  "provideAlternative": true,
  "provideUnavailabilityReason": true,
  "instantSyncQualification": false,
  "qualificationResult": "alternate",
  "relatedParty": [
    {
      "id": "456",
      "href": "http://serverlocation:port/partyManagement/v4/party/456",
      "name": "Jean Pontus",
      "role": "Buyer contact",
      "@type": "RelatedParty",
      "@referredType": "Individual"
```

© TM Forum 2021. All Rights Reserved

```
    }
  ],
  "serviceQualificationItem": [
    {
      "id": "1",
      "expectedActivationDate": "2020-04-16T19:20:39.239Z",
      "expectedServiceAvailabilityDate": "2020-04-11T19:20:39.239Z",
      "expirationDate": "2020-04-24T19:20:39.239Z",
      "qualificationResult": "alternate",
      "state": "done",
      "alternateServiceProposal": [
        {
          "id": "1",
          "alternateServiceAvailabilityDate": "2020-04-15T19:20:39.239Z",
          "alternateService": {
            "serviceCharacteristic": [
              {
                "id": "1",
                "name": "BroadcastQuality",
                "valueType": "string",
                "value": "HD",
                "@type": "ServiceCharacteristic"
              }
            ],
            "serviceSpecification": {
              "id": "23",
              "href": "string",
              "name": "TVAccess",
              "@type": "ServiceSpecification"
            },
            "@type": "Service"
          },
          "@type": "AlternateServiceProposal"
        }
      ],
      "eligibilityUnavailabilityReason": [
        {
          "code": "26-001",
          "label": "not enough bandwith at this place",
          "@baseType": "string",
          "@schemaLocation": "string",
          "@type": "string"
        }
      ],
      "service": {
        "place": [
          {
            "id": "11",
            "href": "http://serverlocation:port/geographicSiteManagement/v4/geographicSite/11",
            "name": "Jean Pontus home",
            "role": "Installation place",
            "@referredType": "GeographicSite"
          }
        ],
        "serviceCharacteristic": [
```


© TM Forum 2021. All Rights Reserved

```
            {
               "id": "1",
               "name": "BroadcastQuality",
               "valueType": "string",
               "value": "4K"
            }
         ],
         "serviceSpecification": {
            "id": "23",
            "href": "string",
            "name": "TVAccess",
            "@type": "ServiceSpecification"
         },
         "@type": "Service"
      },
      "@type": "ServiceQualificationItem"
   }
  ],
  "state": "done",
  "@type": "CheckServiceQualification"
}
```


Response

201


```
{
  "id": "44",
  "href": "http://serverlocation:port/serviceQualificationManagement/v4/checkServiceQualification/456",
  "checkServiceQualificationDate": "2020-04-10T19:20:39.239Z",
  "description": "check service qualification illustration",
  "effectiveQualificationDate": "2020-04-11T15:20:39.239Z",
  "expectedQualificationDate": "2020-04-11T15:20:39.239Z",
  "expirationDate": "2020-04-24T19:20:39.239Z",
  "externalId": "CSQ1",
  "provideAlternative": true,
  "provideUnavailabilityReason": true,
  "instantSyncQualification": false,
  "qualificationResult": "alternate",
  "relatedParty": [
    {
      "id": "456",
      "href": "http://serverlocation:port/partyManagement/v4/party/456",
      "name": "Jean Pontus",
      "role": "Buyer contact",
      "@type": "RelatedParty",
      "@referredType": "Individual"
    }
  ],
  "serviceQualificationItem": [
    {
      "id": "1",
      "expectedActivationDate": "2020-04-16T19:20:39.239Z",
```


© TM Forum 2021. All Rights Reserved

```
"expectedServiceAvailabilityDate": "2020-04-11T19:20:39.239Z",
"expirationDate": "2020-04-24T19:20:39.239Z",
"qualificationResult": "alternate",
"state": "done",
"alternateServiceProposal": [
  {
    "id": "1",
    "alternateServiceAvailabilityDate": "2020-04-15T19:20:39.239Z",
    "alternateService": {
      "serviceCharacteristic": [
        {
          "id": "1",
          "name": "BroadcastQuality",
          "valueType": "string",
          "value": "HD",
          "@type": "ServiceCharacteristic"
        }
      ],
      "serviceSpecification": {
        "id": "23",
        "href": "string",
        "name": "TVAccess",
        "@type": "ServiceSpecification"
      },
      "@type": "Service"
    },
    "@type": "AlternateServiceProposal"
  }
],
"eligibilityUnavailabilityReason": [
  {
    "code": "26-001",
    "label": "not enough bandwith at this place",
    "@baseType": "string",
    "@schemaLocation": "string",
    "@type": "string"
  }
],
"service": {
  "place": [
    {
      "id": "11",
      "href": "http://serverlocation:port/geographicSiteManagement/v4/geographicSite/11",
      "name": "Jean Pontus home",
      "role": "Installation place",
      "@referredType": "GeographicSite"
    }
  ],
  "serviceCharacteristic": [
    {
      "id": "1",
      "name": "BroadcastQuality",
      "valueType": "string",
      "value": "4K"
    }
```


© TM Forum 2021. All Rights Reserved

```
          ],
          "serviceSpecification": {
            "id": "23",
            "href": "string",
            "name": "TVAccess",
            "@type": "ServiceSpecification"
          },
          "@type": "Service"
        },
        "@type": "ServiceQualificationItem"
      }
    ],
    "state": "done",
    "@type": "CheckServiceQualification"
}
```


Patch check service qualification

PATCH /checkServiceQualification/{id}

Description

This operation allows partial updates of a check service qualification entity. Support of json/merge (https://tools.ietf.org/html/rfc7386) is mandatory, support of json/patch (http://tools.ietf.org/html/rfc5789) is optional.

Note: If the update operation yields to the creation of sub-resources or relationships, the same rules concerning mandatory sub-resource attributes and default value settings in the POST operation applies to the PATCH operation. Hence these tables are not repeated here.

Patchable and Non Patchable Attributes

The tables below provide the list of patchable and non patchable attributes, including constraint rules on their usage.

Patchable Attributes	Rule
description	
effectiveQualificationDate	
estimatedResponseDate	
expectedQualificationDate	
expirationDate	
externalId	
instantSyncQualification	
provideAlternative	
provideUnavailabilityReason	
qualificationResult	
serviceQualificationItem	
relatedParty	
state	

© TM Forum 2021. All Rights Reserved

Non Patchable Attributes	Rule
id	
href	
checkServiceQualificationDate	
@type	
@schemaLocation	
@baseType	

Usage Samples

Here's an example of a request for patching a CheckServiceQualification.

Request

```
PATCH /tmf-api/naas/v4/checkServiceQualification/44
Content-Type: application/merge-patch+json

{
  "checkServiceQualificationDate": "2020-04-10T19:20:39.239Z",
  "description": "check service qualification illustration",
  "effectiveQualificationDate": "2020-04-11T15:20:39.239Z",
  "expectedQualificationDate": "2020-04-11T15:20:39.239Z",
  "expirationDate": "2020-04-24T19:20:39.239Z",
  "externalId": "CSQ1",
  "provideAlternative": true,
  "provideUnavailabilityReason": true,
  "instantSyncQualification": false,
  "qualificationResult": "alternate",
  "relatedParty": [
    {
      "id": "456",
      "href": "http://serverlocation:port/partyManagement/v4/party/456",
      "name": "Jean Pontus",
      "role": "Buyer contact",
      "@type": "RelatedParty",
      "@referredType": "Individual"
    }
  ],
  "serviceQualificationItem": [
    {
      "id": "1",
      "expectedActivationDate": "2020-04-16T19:20:39.239Z",
      "expectedServiceAvailabilityDate": "2020-04-11T19:20:39.239Z",
      "expirationDate": "2020-04-24T19:20:39.239Z",
      "qualificationResult": "alternate",
      "state": "done",
      "alternateServiceProposal": [
        {
          "id": "1",
          "alternateServiceAvailabilityDate": "2020-04-15T19:20:39.239Z",
          "alternateService": {
            "serviceCharacteristic": [
```

© TM Forum 2021. All Rights Reserved

```
                {
                  "id": "1",
                  "name": "BroadcastQuality",
                  "valueType": "string",
                  "value": "HD",
                  "@type": "ServiceCharacteristic"
                }
              ],
              "serviceSpecification": {
                "id": "23",
                "href": "string",
                "name": "TVAccess",
                "@type": "ServiceSpecification"
              },
              "@type": "Service"
            },
            "@type": "AlternateServiceProposal"
          }
        ],
        "eligibilityUnavailabilityReason": [
          {
            "code": "26-001",
            "label": "not enough bandwith at this place",
            "@baseType": "string",
            "@schemaLocation": "string",
            "@type": "string"
          }
        ],
        "service": {
          "place": [
            {
              "id": "11",
              "href": "http://serverlocation:port/geographicSiteManagement/v4/geographicSite/11",
              "name": "Jean Pontus home",
              "role": "Installation place",
              "@referredType": "GeographicSite"
            }
          ],
          "serviceCharacteristic": [
            {
              "id": "1",
              "name": "BroadcastQuality",
              "valueType": "string",
              "value": "4K"
            }
          ],
          "serviceSpecification": {
            "id": "23",
            "href": "string",
            "name": "TVAccess",
            "@type": "ServiceSpecification"
          },
          "@type": "Service"
        },
        "@type": "ServiceQualificationItem"
```


© TM Forum 2021. All Rights Reserved

```
      }
   ],
   "state": "done",
   "@type": "CheckServiceQualification"
}
```


Response


```
200

{
   "id": "44",
   "href": "http://serverlocation:port/serviceQualificationManagement/v4/checkServiceQualification/456",
   "checkServiceQualificationDate": "2020-04-10T19:20:39.239Z",
   "description": "check service qualification illustration",
   "effectiveQualificationDate": "2020-04-11T15:20:39.239Z",
   "expectedQualificationDate": "2020-04-11T15:20:39.239Z",
   "expirationDate": "2020-04-24T19:20:39.239Z",
   "externalId": "CSQ1",
   "provideAlternative": true,
   "provideUnavailabilityReason": true,
   "instantSyncQualification": false,
   "qualificationResult": "alternate",
   "relatedParty": [
      {
         "id": "456",
         "href": "http://serverlocation:port/partyManagement/v4/party/456",
         "name": "Jean Pontus",
         "role": "Buyer contact",
         "@type": "RelatedParty",
         "@referredType": "Individual"
      }
   ],
   "serviceQualificationItem": [
      {
         "id": "1",
         "expectedActivationDate": "2020-04-16T19:20:39.239Z",
         "expectedServiceAvailabilityDate": "2020-04-11T19:20:39.239Z",
         "expirationDate": "2020-04-24T19:20:39.239Z",
         "qualificationResult": "alternate",
         "state": "done",
         "alternateServiceProposal": [
            {
               "id": "1",
               "alternateServiceAvailabilityDate": "2020-04-15T19:20:39.239Z",
               "alternateService": {
                  "serviceCharacteristic": [
                     {
                        "id": "1",
                        "name": "BroadcastQuality",
                        "valueType": "string",
                        "value": "HD",
                        "@type": "ServiceCharacteristic"
```


© TM Forum 2021. All Rights Reserved

```
                    }
                ],
                "serviceSpecification": {
                    "id": "23",
                    "href": "string",
                    "name": "TVAccess",
                    "@type": "ServiceSpecification"
                },
                "@type": "Service"
            },
            "@type": "AlternateServiceProposal"
        }
    ],
    "eligibilityUnavailabilityReason": [
        {
            "code": "26-001",
            "label": "not enough bandwith at this place",
            "@baseType": "string",
            "@schemaLocation": "string",
            "@type": "string"
        }
    ],
    "service": {
        "place": [
            {
                "id": "11",
                "href": "http://serverlocation:port/geographicSiteManagement/v4/geographicSite/11",
                "name": "Jean Pontus home",
                "role": "Installation place",
                "@referredType": "GeographicSite"
            }
        ],
        "serviceCharacteristic": [
            {
                "id": "1",
                "name": "BroadcastQuality",
                "valueType": "string",
                "value": "4K"
            }
        ],
        "serviceSpecification": {
            "id": "23",
            "href": "string",
            "name": "TVAccess",
            "@type": "ServiceSpecification"
        },
        "@type": "Service"
    },
    "@type": "ServiceQualificationItem"
  }
 ],
 "state": "done",
 "@type": "CheckServiceQualification"
}
```


© TM Forum 2021. All Rights Reserved

Delete check service qualification

DELETE /checkServiceQualification/{id}

Description

This operation deletes a check service qualification entity.

Usage Samples

Here's an example of a request for patching a CheckServiceQualification.

Request
DELETE /tmf-api/naas/v4/checkServiceQualification/44
Response
204

Operations on Query Service Qualification

List query service qualifications

GET /queryServiceQualification?fields=...&{filtering}

Description

This operation list query service qualification entities.
Attribute selection is enabled for all first level attributes.
Filtering may be available depending on the compliance level supported by an implementation.

Usage Samples

Here's an example of a request for retrieving multiple QueryServiceQualification.

Request
GET /tmf-api/naas/v4/queryServiceQualification Accept: application/json

© TM Forum 2021. All Rights Reserved

Response

200


```
[
  {
    "id": "54",
    "href": "http://serverlocation:port/serviceQualificationManagement/v4/queryServiceQualification/54",
    "description": "Query Service Qualification Illustration - 2 serviceSpec available at this place for this category",
    "effectiveQualificationDate": "2020-04-13T13:51:41.218Z",
    "estimatedResponseDate": "2020-04-13T14:00:00.218Z",
    "expectedQualificationDate": "2020-04-13T13:51:41.218Z",
    "expirationDate": "2020-04-15T13:51:41.218Z",
    "externalId": "QSQ-456",
    "instantSyncQualification": true,
    "queryServiceQualificationDate": "2020-04-13T13:51:41.218Z",
    "relatedParty": [
      {
        "id": "456",
        "href": "http://serverlocation:port/partyManagement/v4/party/456",
        "name": "Jean Pontus",
        "role": "Buyer contact",
        "@type": "RelatedParty",
        "@referredType": "Individual"
      }
    ],
    "searchCriteria": {
      "id": "1",
      "category": {
        "id": "1",
        "href": "http://serverlocation:port/serviceCatalogManagement/v4/category/1",
        "name": "Fiber Access",
        "@type": "Category"
      },
      "service": {
        "place": [
          {
            "role": "Installation Place",
            "@type": "GeographicAddress",
            "city": "San Francisco",
            "postcode": "94121",
            "streetName": "25th",
            "streetType": "Avenue",
            "streetNr": "437",
            "geographicAddress": [
              {
                "levelType": "floor",
                "levelNumber": "3"
              }
            ]
          }
        ]
```


© TM Forum 2021. All Rights Reserved

```
        },
        "@type": "SearchCriteria"
      },
      "serviceQualificationItem": [
        {
          "id": "1",
          "service": {
            "serviceSpecification": {
              "id": "78",
              "href": "http://serverlocation:port/serviceCatalogManagement/v4/serviceSpecification/78",
              "name": "X-Fiber Service",
              "@type": "ServiceSpecification"
            },
            "@type": "Service"
          },
          "@type": "ServiceQualificationItem"
        },
        {
          "id": "2",
          "service": {
            "serviceSpecification": {
              "id": "87",
              "href": "http://serverlocation:port/serviceCatalogManagement/v4/serviceSpecification/87",
              "name": "Standard Fiber Service",
              "@type": "ServiceSpecification"
            },
            "@type": "Service"
          },
          "@type": "ServiceQualificationItem"
        }
      ],
      "state": "done",
      "@type": "QueryServiceQualification"
    }
]
```


Retrieve query service qualification

`GET /queryServiceQualification/{id}?fields=...&{filtering}`

Description

This operation retrieves a query service qualification entity.
Attribute selection is enabled for all first level attributes.
Filtering on sub-resources may be available depending on the compliance level supported by an implementation.

Usage Samples

Here's an example of a request for retrieving a specific QueryServiceQualification.

© TM Forum 2021. All Rights Reserved

Request

GET /tmf-api/naas/v4/queryServiceQualification/54
Accept: application/json

Response

200


```
{
  "id": "54",
  "href": "http://serverlocation:port/serviceQualificationManagement/v4/queryServiceQualification/54",
  "description": "Query Service Qualification Illustration - 2 serviceSpec available at this place for this category",
  "effectiveQualificationDate": "2020-04-13T13:51:41.218Z",
  "estimatedResponseDate": "2020-04-13T14:00:00.218Z",
  "expectedQualificationDate": "2020-04-13T13:51:41.218Z",
  "expirationDate": "2020-04-15T13:51:41.218Z",
  "externalId": "QSQ-456",
  "instantSyncQualification": true,
  "queryServiceQualificationDate": "2020-04-13T13:51:41.218Z",
  "relatedParty": [
    {
      "id": "456",
      "href": "http://serverlocation:port/partyManagement/v4/party/456",
      "name": "Jean Pontus",
      "role": "Buyer contact",
      "@type": "RelatedParty",
      "@referredType": "Individual"
    }
  ],
  "searchCriteria": {
    "id": "1",
    "category": {
      "id": "1",
      "href": "http://serverlocation:port/serviceCatalogManagement/v4/category/1",
      "name": "Fiber Access",
      "@type": "Category"
    },
    "service": {
      "place": [
        {
          "role": "Installation Place",
          "@type": "GeographicAddress",
          "city": "San Francisco",
          "postcode": "94121",
          "streetName": "25th",
          "streetType": "Avenue",
          "streetNr": "437",
          "geographicAddress": [
            {
              "levelType": "floor",
              "levelNumber": "3"
```


© TM Forum 2021. All Rights Reserved

```
              }
            ]
          }
        ]
      },
      "@type": "SearchCriteria"
    },
    "serviceQualificationItem": [
      {
        "id": "1",
        "service": {
          "serviceSpecification": {
            "id": "78",
            "href": "http://serverlocation:port/serviceCatalogManagement/v4/serviceSpecification/78",
            "name": "X-Fiber Service",
            "@type": "ServiceSpecification"
          },
          "@type": "Service"
        },
        "@type": "ServiceQualificationItem"
      },
      {
        "id": "2",
        "service": {
          "serviceSpecification": {
            "id": "87",
            "href": "http://serverlocation:port/serviceCatalogManagement/v4/serviceSpecification/87",
            "name": "Standard Fiber Service",
            "@type": "ServiceSpecification"
          },
          "@type": "Service"
        },
        "@type": "ServiceQualificationItem"
      }
    ],
    "state": "done",
    "@type": "QueryServiceQualification"
}
```


Create query service qualification

`POST /queryServiceQualification`

Description

This operation creates a query service qualification entity.

Mandatory and Non Mandatory Attributes

The following tables provide the list of mandatory and non mandatory attributes when creating a QueryServiceQualification, including any possible rule conditions and applicable default values. Notice that it is up to an implementer to add additional mandatory attributes.

© TM Forum 2021. All Rights Reserved

Mandatory Attributes	Rule
searchCriteria	

Non Mandatory Attributes	Rule
description	
effectiveQualificationDate	
estimatedResponseDate	
expectedQualificationDate	
expirationDate	
externalId	
instantSyncQualification	
@baseType	
@schemaLocation	
@type	
queryServiceQualificationDate	
serviceQualificationItem	
relatedParty	
state	

Usage Samples

Here's an example of a request for creating a specific QueryServiceQualification.

Request

```
POST /tmf-api/naas/v4/queryServiceQualification
Content-Type: application/json

{
  "description": "Query Service Qualification Illustration - 2 serviceSpec available at this place for this category",
  "effectiveQualificationDate": "2020-04-13T13:51:41.218Z",
  "estimatedResponseDate": "2020-04-13T14:00:00.218Z",
  "expectedQualificationDate": "2020-04-13T13:51:41.218Z",
  "expirationDate": "2020-04-15T13:51:41.218Z",
  "externalId": "QSQ-456",
  "instantSyncQualification": true,
  "queryServiceQualificationDate": "2020-04-13T13:51:41.218Z",
  "relatedParty": [
    {
      "id": "456",
      "href": "http://serverlocation:port/partyManagement/v4/party/456",
      "name": "Jean Pontus",
      "role": "Buyer contact",
      "@type": "RelatedParty",
      "@referredType": "Individual"
    }
  ],
  "searchCriteria": {
    "id": "1",
```

© TM Forum 2021. All Rights Reserved

```
    "category": {
      "id": "1",
      "href": "http://serverlocation:port/serviceCatalogManagement/v4/category/1",
      "name": "Fiber Access",
      "@type": "Category"
    },
    "service": {
      "place": [
        {
          "role": "Installation Place",
          "@type": "GeographicAddress",
          "city": "San Francisco",
          "postcode": "94121",
          "streetName": "25th",
          "streetType": "Avenue",
          "streetNr": "437",
          "geographicAddress": [
            {
              "levelType": "floor",
              "levelNumber": "3"
            }
          ]
        }
      ]
    },
    "@type": "SearchCriteria"
  },
  "serviceQualificationItem": [
    {
      "id": "1",
      "service": {
        "serviceSpecification": {
          "id": "78",
          "href": "http://serverlocation:port/serviceCatalogManagement/v4/serviceSpecification/78",
          "name": "X-Fiber Service",
          "@type": "ServiceSpecification"
        },
        "@type": "Service"
      },
      "@type": "ServiceQualificationItem"
    },
    {
      "id": "2",
      "service": {
        "serviceSpecification": {
          "id": "87",
          "href": "http://serverlocation:port/serviceCatalogManagement/v4/serviceSpecification/87",
          "name": "Standard Fiber Service",
          "@type": "ServiceSpecification"
        },
        "@type": "Service"
      },
      "@type": "ServiceQualificationItem"
    }
  ],
```


© TM Forum 2021. All Rights Reserved

```
  "state": "done",
  "@type": "QueryServiceQualification"
}
```


Response

201


```
{
  "id": "54",
  "href": "http://serverlocation:port/serviceQualificationManagement/v4/queryServiceQualification/54",
  "description": "Query Service Qualification Illustration - 2 serviceSpec available at this place for this category",
  "effectiveQualificationDate": "2020-04-13T13:51:41.218Z",
  "estimatedResponseDate": "2020-04-13T14:00:00.218Z",
  "expectedQualificationDate": "2020-04-13T13:51:41.218Z",
  "expirationDate": "2020-04-15T13:51:41.218Z",
  "externalId": "QSQ-456",
  "instantSyncQualification": true,
  "queryServiceQualificationDate": "2020-04-13T13:51:41.218Z",
  "relatedParty": [
    {
      "id": "456",
      "href": "http://serverlocation:port/partyManagement/v4/party/456",
      "name": "Jean Pontus",
      "role": "Buyer contact",
      "@type": "RelatedParty",
      "@referredType": "Individual"
    }
  ],
  "searchCriteria": {
    "id": "1",
    "category": {
      "id": "1",
      "href": "http://serverlocation:port/serviceCatalogManagement/v4/category/1",
      "name": "Fiber Access",
      "@type": "Category"
    },
    "service": {
      "place": [
        {
          "role": "Installation Place",
          "@type": "GeographicAddress",
          "city": "San Francisco",
          "postcode": "94121",
          "streetName": "25th",
          "streetType": "Avenue",
          "streetNr": "437",
          "geographicAddress": [
            {
              "levelType": "floor",
              "levelNumber": "3"
            }
          ]
```


© TM Forum 2021. All Rights Reserved

```
          }
        ]
      },
      "@type": "SearchCriteria"
    },
    "serviceQualificationItem": [
      {
        "id": "1",
        "service": {
          "serviceSpecification": {
            "id": "78",
            "href": "http://serverlocation:port/serviceCatalogManagement/v4/serviceSpecification/78",
            "name": "X-Fiber Service",
            "@type": "ServiceSpecification"
          },
          "@type": "Service"
        },
        "@type": "ServiceQualificationItem"
      },
      {
        "id": "2",
        "service": {
          "serviceSpecification": {
            "id": "87",
            "href": "http://serverlocation:port/serviceCatalogManagement/v4/serviceSpecification/87",
            "name": "Standard Fiber Service",
            "@type": "ServiceSpecification"
          },
          "@type": "Service"
        },
        "@type": "ServiceQualificationItem"
      }
    ],
    "state": "done",
    "@type": "QueryServiceQualification"
}
```


Patch query service qualification

`PATCH /queryServiceQualification/{id}`

Description

This operation allows partial updates of a query service qualification entity. Support of json/merge (https://tools.ietf.org/html/rfc7386) is mandatory, support of json/patch (http://tools.ietf.org/html/rfc5789) is optional.

Note: If the update operation yields to the creation of sub-resources or relationships, the same rules concerning mandatory sub-resource attributes and default value settings in the POST operation applies to the PATCH operation. Hence these tables are not repeated here.

Patchable and Non Patchable Attributes

© TM Forum 2021. All Rights Reserved

The tables below provide the list of patchable and non patchable attributes, including constraint rules on their usage.

Patchable Attributes	Rule
description	
effectiveQualificationDate	
estimatedResponseDate	
expectedQualificationDate	
expirationDate	
externalId	
instantSyncQualification	
searchCriteria	
serviceQualificationItem	
relatedParty	
state	

Non Patchable Attributes	Rule
id	
href	
queryServiceQualificationDate	
@type	
@schemaLocation	
@baseType	

Usage Samples

Here's an example of a request for patching a QueryServiceQualification.

Request

```
PATCH /tmf-api/naas/v4/queryServiceQualification/54
Content-Type: application/merge-patch+json

{
   "description": "Query Service Qualification Illustration - 2 serviceSpec available at this place for this category",
   "effectiveQualificationDate": "2020-04-13T13:51:41.218Z",
   "estimatedResponseDate": "2020-04-13T14:00:00.218Z",
   "expectedQualificationDate": "2020-04-13T13:51:41.218Z",
   "expirationDate": "2020-04-15T13:51:41.218Z",
   "externalId": "QSQ-456",
   "instantSyncQualification": true,
   "queryServiceQualificationDate": "2020-04-13T13:51:41.218Z",
   "relatedParty": [
      {
         "id": "456",
         "href": "http://serverlocation:port/partyManagement/v4/party/456",
         "name": "Jean Pontus",
         "role": "Buyer contact",
         "@type": "RelatedParty",
```

© TM Forum 2021. All Rights Reserved

```
        "@referredType": "Individual"
      }
    ],
    "searchCriteria": {
      "id": "1",
      "category": {
        "id": "1",
        "href": "http://serverlocation:port/serviceCatalogManagement/v4/category/1",
        "name": "Fiber Access",
        "@type": "Category"
      },
      "service": {
        "place": [
          {
            "role": "Installation Place",
            "@type": "GeographicAddress",
            "city": "San Francisco",
            "postcode": "94121",
            "streetName": "25th",
            "streetType": "Avenue",
            "streetNr": "437",
            "geographicAddress": [
              {
                "levelType": "floor",
                "levelNumber": "3"
              }
            ]
          }
        ]
      },
      "@type": "SearchCriteria"
    },
    "serviceQualificationItem": [
      {
        "id": "1",
        "service": {
          "serviceSpecification": {
            "id": "78",
            "href": "http://serverlocation:port/serviceCatalogManagement/v4/serviceSpecification/78",
            "name": "X-Fiber Service",
            "@type": "ServiceSpecification"
          },
          "@type": "Service"
        },
        "@type": "ServiceQualificationItem"
      },
      {
        "id": "2",
        "service": {
          "serviceSpecification": {
            "id": "87",
            "href": "http://serverlocation:port/serviceCatalogManagement/v4/serviceSpecification/87",
            "name": "Standard Fiber Service",
            "@type": "ServiceSpecification"
          },
```


© TM Forum 2021. All Rights Reserved

```
            "@type": "Service"
          },
          "@type": "ServiceQualificationItem"
        }
    ],
    "state": "done",
    "@type": "QueryServiceQualification"
}
```


Response

200


```
{
    "id": "54",
    "href": "http://serverlocation:port/serviceQualificationManagement/v4/queryServiceQualification/54",
    "description": "Query Service Qualification Illustration - 2 serviceSpec available at this place for this category",
    "effectiveQualificationDate": "2020-04-13T13:51:41.218Z",
    "estimatedResponseDate": "2020-04-13T14:00:00.218Z",
    "expectedQualificationDate": "2020-04-13T13:51:41.218Z",
    "expirationDate": "2020-04-15T13:51:41.218Z",
    "externalId": "QSQ-456",
    "instantSyncQualification": true,
    "queryServiceQualificationDate": "2020-04-13T13:51:41.218Z",
    "relatedParty": [
      {
        "id": "456",
        "href": "http://serverlocation:port/partyManagement/v4/party/456",
        "name": "Jean Pontus",
        "role": "Buyer contact",
        "@type": "RelatedParty",
        "@referredType": "Individual"
      }
    ],
    "searchCriteria": {
      "id": "1",
      "category": {
        "id": "1",
        "href": "http://serverlocation:port/serviceCatalogManagement/v4/category/1",
        "name": "Fiber Access",
        "@type": "Category"
      },
      "service": {
        "place": [
          {
            "role": "Installation Place",
            "@type": "GeographicAddress",
            "city": "San Francisco",
            "postcode": "94121",
            "streetName": "25th",
            "streetType": "Avenue",
            "streetNr": "437",
            "geographicAddress": [
```


© TM Forum 2021. All Rights Reserved

```
                {
                  "levelType": "floor",
                  "levelNumber": "3"
                }
              ]
            }
          ]
        },
        "@type": "SearchCriteria"
      },
      "serviceQualificationItem": [
        {
          "id": "1",
          "service": {
            "serviceSpecification": {
              "id": "78",
              "href": "http://serverlocation:port/serviceCatalogManagement/v4/serviceSpecification/78",
              "name": "X-Fiber Service",
              "@type": "ServiceSpecification"
            },
            "@type": "Service"
          },
          "@type": "ServiceQualificationItem"
        },
        {
          "id": "2",
          "service": {
            "serviceSpecification": {
              "id": "87",
              "href": "http://serverlocation:port/serviceCatalogManagement/v4/serviceSpecification/87",
              "name": "Standard Fiber Service",
              "@type": "ServiceSpecification"
            },
            "@type": "Service"
          },
          "@type": "ServiceQualificationItem"
        }
      ],
      "state": "done",
      "@type": "QueryServiceQualification"
    }
```


Delete query service qualification

`DELETE /queryServiceQualification/{id}`

Description

This operation deletes a query service qualification entity.

Usage Samples

© TM Forum 2021. All Rights Reserved

Here's an example of a request for patching a QueryServiceQualification.

Request
DELETE /tmf-api/naas/v4/queryServiceQualification/54
Response
204

Operations on Service Problem

List service problems

`GET /serviceProblem?fields=...&{filtering}`

Description

This operation list service problem entities.
Attribute selection is enabled for all first level attributes.
Filtering may be available depending on the compliance level supported by an implementation.

Usage Samples

Here's an example of a request for retrieving multiple ServiceProblem.

Request
GET /tmf-api/naas/v4/serviceProblem Accept: application/json
Response
200 [{ "id": "problemxxxx0000", "correlationId": "543251", "originatingSystem": "System_001", "category": "supplier.originated", "href": "https://mycsp.com:8080/tmf-api/serviceProblemManagement/v4/serviceProblem/problemxxxx0000",

© TM Forum 2021. All Rights Reserved

```
    "impactImportanceFactor": "0",
    "priority": "1",
    "description": "connection failure between Tokyo and Osaka",
    "problemEscalation": "0",
    "timeRaised": "2017-10-25T12:14:16.361Z",
    "timeChanged": "2017-10-30T12:13:16.361Z",
    "statusChangeDate": "2017-10-29T12:00:00.361Z",
    "statusChangeReason": "problem analysis has been completed in NP1",
    "resolutionDate": "2017-10-29T12:00:00.361Z",
    "status": "resolved",
    "reason": "Failure of resource NP1_Resource_1 in NP1",
    "affectedNumberOfServices": "2",
    "firstAlert": {
      "alertType": "TroubleTicket",
      "id": "NP1_TT_0000000",
      "href": "https://mycsp.com:8080/tmf-api/troubleTicketManagement/v4/troubleTicket/NP1_TT_000000"
    },
    "responsibleParty": {
      "role": "Supplier",
      "id": "NP1",
      "href": "https://mycsp.com:8080/tmf-api/partyManagement/v4/party/NP1",
      "@referredType": "Organization"
    },
    "originatorParty": {
      "role": "Supplier",
      "id": "NP1",
      "href": "https://mycsp.com:8080/tmf-api/partyManagement/v4/party/NP1",
      "@referredType": "Organization"
    },
    "relatedParty": [
      {
        "role": "Supplier",
        "id": "NP1",
        "href": "https://mycsp.com:8080/tmf-api/partyManagement/v4/party/NP1",
        "@referredType": "Organization"
      },
      {
        "role": "Partner",
        "id": "SP1",
        "href": "https://mycsp.com:8080/tmf-api/partyManagement/v4/party/SP1",
        "@referredType": "Organization"
      },
      {
        "role": "Partner",
        "id": "SP3",
        "href": "https://mycsp.com:8080/tmf-api/partyManagement/v4/party/SP3",
        "@referredType": "Organization"
      }
    ],
    "affectedService": [
      {
        "id": "NP1_Tokyo_Osaka",
        "href": "https://mycsp.com:8080/tmf-api/serviceInventoryManagement/v4/service/NW1_Tokyo_Osaka"
      },
      {
```


© TM Forum 2021. All Rights Reserved

```
            "id": "NP1_Tokyo_xxxx",
            "href": "https://mycsp.com:8080/tmf-api/serviceInventoryManagement/v4/service/NW1_Tokyo_xxxx"
          }
        ],
        "affectedResource": [
          {
            "id": "NP1_RES_0001",
            "href": "https://mycsp.com:8080/tmf-api/resourceInventoryManagement/v4/resource/NW1_RES_0001"
          }
        ],
        "affectedLocation": [
          {
            "id": "Loc000000",
            "href": "https://mycsp.com:8080/tmf-
api/geographicAddressManagement/v4/geographicAddress/Loc000000/",
            "name": "144 Main Street Tokyo 51663556",
            "role": "VPN Endpoint",
            "@type": "GeographicAddress"
          },
          {
            "id": "Loc000001",
            "href": "https://mycsp.com:8080/tmf-
api/geographicAddressManagement/v4/geographicAddress/Loc000001/",
            "name": "351 Main Street Osaka 3546365",
            "role": "VPN Endpoint",
            "@type": "GeographicAddress"
          }
        ],
        "associatedTroubleTicket": [
          {
            "id": "NP1_TT_0000000",
            "href": "https://mycsp.com:8080/tmf-api/troubleTicketManagement/v4/TroubleTicketRef/NP1_TT_000000"
          }
        ],
        "underlyingAlarm": [
          {
            "id": "NP1_A_0000000",
            "href": "https://mycsp.com:8080/tmf-api/alarmManagement/v4/resourceAlarm/NP1_A_000000",
            "@referredType": "ResourceAlarm"
          }
        ],
        "associatedSLAViolation": [
          {
            "id": "NP1_SLA_0000000",
            "href": "https://mycsp.com:8080/tmf-api/SLAManagement/v4/SLAViolationRef/NP1_SLA_000000"
          }
        ],
        "relatedEvent": [
          {
            "eventType": "prediction",
            "id": "prediction_0001",
            "href": "https://mycsp.com:8080/tmf-api/eventManagement/v4/event/prediction_0001",
            "eventTime": "2014-12-20T17:00:00Z"
          }
        ],
```


© TM Forum 2021. All Rights Reserved

```
"relatedObject": [
  {
    "id": "product0001",
    "href": "https://mycsp.com:8080/tmf-api/productInventoryManagement/v4/product/product0001",
    "@referredType": "Product"
  }
],
"rootCauseService": [
  {
    "id": "NP1_Tokyo_Osaka",
    "href": "https://mycsp.com:8080/tmf-api/serviceInventoryManagement/v4/service/NW1_Tokyo_Osaka"
  }
],
"rootCauseResource": [
  {
    "id": "NP1_Resource_1",
    "href": "https://mycsp.com:8080/tmf-api/resourceInventoryManagement/v4/resource/NP1_Resource_1"
  }
],
"parentProblem": [
  {
    "id": "problemxxxx0001",
    "correlationId": "xxxxxxxx",
    "href": "https://mycsp.com:8080/tmf-api/serviceProblemManagement/v4/serviceProblem/problemxxxx0001"
  }
],
"underlyingProblem": [
  {
    "id": "problemxxxx0001",
    "correlationId": "xxxxxxxx",
    "href": "https://mycsp.com:8080/tmf-api/serviceProblemManagement/v4/serviceProblem/problemxxxx0001"
  }
],
"trackingRecord": [
  {
    "description": "yyy cleared the problem",
    "systemId": "xxxx",
    "time": "2014-12-20T17:00:00Z",
    "user": "NPUSER1",
    "id": "TR_xxxx",
    "extensionInfo": [
      {
        "name": "vendorComment",
        "valueType": "string",
        "value": "Watch out for the dog"
      }
    ]
  }
],
"comment": [
  {
    "author": {
      "id": "SPM_handler_01",
      "href": "https://mycsp.com:8080/tmf-api/partyManagement/v4/party/SPM_handler_01",
      "@referredType": "Individual"
```


© TM Forum 2021. All Rights Reserved

```
          },
          "date": "2014-12-20T17:00:00Z",
          "systemId": "System_002",
          "text": "receive trouble ticket from NP1, and create this Service Problem"
        },
        {
          "author": {
            "id": "NP1_handler_11",
            "href": "https://mycsp.com:8080/tmf-api/partyManagement/v4/party/NP1_handler_11",
            "@referredType": "Individual"
          },
          "date": "2014-12-20T17:00:00Z",
          "systemId": "System_002",
          "text": "status changed to Progress-Held"
        }
      ],
      "impactPatterns": {
        "description": "Many services are at risk in this problem",
        "extensionInfo": [
          {
            "name": "ImpactProbability",
            "valueType": "boolean",
            "value": true
          }
        ]
      },
      "extensionInfo": [
        {
          "name": "EstimatedCost",
          "valueType": "integer",
          "value": 20
        }
      ]
    }
]
```


Retrieve service problem

`GET /serviceProblem/{id}?fields=...&{filtering}`

Description

This operation retrieves a service problem entity.
Attribute selection is enabled for all first level attributes.
Filtering on sub-resources may be available depending on the compliance level supported by an implementation.

Usage Samples

Here's an example of a request for retrieving a specific ServiceProblem.

© TM Forum 2021. All Rights Reserved

Request

GET /tmf-api/naas/v4/serviceProblem/problemxxxx0000
Accept: application/json

Response

200


```
{
  "id": "problemxxxx0000",
  "correlationId": "543251",
  "originatingSystem": "System_001",
  "category": "supplier.originated",
  "href": "https://mycsp.com:8080/tmf-api/serviceProblemManagement/v4/serviceProblem/problemxxxx0000",
  "impactImportanceFactor": "0",
  "priority": "1",
  "description": "connection failure between Tokyo and Osaka",
  "problemEscalation": "0",
  "timeRaised": "2017-10-25T12:14:16.361Z",
  "timeChanged": "2017-10-30T12:13:16.361Z",
  "statusChangeDate": "2017-10-29T12:00:00.361Z",
  "statusChangeReason": "problem analysis has been completed in NP1",
  "resolutionDate": "2017-10-29T12:00:00.361Z",
  "status": "resolved",
  "reason": "Failure of resource NP1_Resource_1 in NP1",
  "affectedNumberOfServices": "2",
  "firstAlert": {
    "alertType": "TroubleTicket",
    "id": "NP1_TT_0000000",
    "href": "https://mycsp.com:8080/tmf-api/troubleTicketManagement/v4/troubleTicket/NP1_TT_000000"
  },
  "responsibleParty": {
    "role": "Supplier",
    "id": "NP1",
    "href": "https://mycsp.com:8080/tmf-api/partyManagement/v4/party/NP1",
    "@referredType": "Organization"
  },
  "originatorParty": {
    "role": "Supplier",
    "id": "NP1",
    "href": "https://mycsp.com:8080/tmf-api/partyManagement/v4/party/NP1",
    "@referredType": "Organization"
  },
  "relatedParty": [
    {
      "role": "Supplier",
      "id": "NP1",
      "href": "https://mycsp.com:8080/tmf-api/partyManagement/v4/party/NP1",
      "@referredType": "Organization"
    },
    {
```


© TM Forum 2021. All Rights Reserved

```
        "role": "Partner",
        "id": "SP1",
        "href": "https://mycsp.com:8080/tmf-api/partyManagement/v4/party/SP1",
        "@referredType": "Organization"
      },
      {
        "role": "Partner",
        "id": "SP3",
        "href": "https://mycsp.com:8080/tmf-api/partyManagement/v4/party/SP3",
        "@referredType": "Organization"
      }
    ],
    "affectedService": [
      {
        "id": "NP1_Tokyo_Osaka",
        "href": "https://mycsp.com:8080/tmf-api/serviceInventoryManagement/v4/service/NW1_Tokyo_Osaka"
      },
      {
        "id": "NP1_Tokyo_xxxx",
        "href": "https://mycsp.com:8080/tmf-api/serviceInventoryManagement/v4/service/NW1_Tokyo_xxxx"
      }
    ],
    "affectedResource": [
      {
        "id": "NP1_RES_0001",
        "href": "https://mycsp.com:8080/tmf-api/resourceInventoryManagement/v4/resource/NW1_RES_0001"
      }
    ],
    "affectedLocation": [
      {
        "id": "Loc000000",
        "href": "https://mycsp.com:8080/tmf-api/geographicAddressManagement/v4/geographicAddress/Loc000000/",
        "name": "144 Main Street Tokyo 51663556",
        "role": "VPN Endpoint",
        "@type": "GeographicAddress"
      },
      {
        "id": "Loc000001",
        "href": "https://mycsp.com:8080/tmf-api/geographicAddressManagement/v4/geographicAddress/Loc000001/",
        "name": "351 Main Street Osaka 3546365",
        "role": "VPN Endpoint",
        "@type": "GeographicAddress"
      }
    ],
    "associatedTroubleTicket": [
      {
        "id": "NP1_TT_0000000",
        "href": "https://mycsp.com:8080/tmf-api/troubleTicketManagement/v4/TroubleTicketRef/NP1_TT_000000"
      }
    ],
    "underlyingAlarm": [
      {
        "id": "NP1_A_0000000",
        "href": "https://mycsp.com:8080/tmf-api/alarmManagement/v4/resourceAlarm/NP1_A_000000",
        "@referredType": "ResourceAlarm"
```


© TM Forum 2021. All Rights Reserved

```
    }
  ],
  "associatedSLAViolation": [
    {
      "id": "NP1_SLA_0000000",
      "href": "https://mycsp.com:8080/tmf-api/SLAManagement/v4/SLAViolationRef/NP1_SLA_000000"
    }
  ],
  "relatedEvent": [
    {
      "eventType": "prediction",
      "id": "prediction_0001",
      "href": "https://mycsp.com:8080/tmf-api/eventManagement/v4/event/prediction_0001",
      "eventTime": "2014-12-20T17:00:00Z"
    }
  ],
  "relatedObject": [
    {
      "id": "product0001",
      "href": "https://mycsp.com:8080/tmf-api/productInventoryManagement/v4/product/product0001",
      "@referredType": "Product"
    }
  ],
  "rootCauseService": [
    {
      "id": "NP1_Tokyo_Osaka",
      "href": "https://mycsp.com:8080/tmf-api/serviceInventoryManagement/v4/service/NW1_Tokyo_Osaka"
    }
  ],
  "rootCauseResource": [
    {
      "id": "NP1_Resource_1",
      "href": "https://mycsp.com:8080/tmf-api/resourceInventoryManagement/v4/resource/NP1_Resource_1"
    }
  ],
  "parentProblem": [
    {
      "id": "problemxxxx0001",
      "correlationId": "xxxxxxxx",
      "href": "https://mycsp.com:8080/tmf-api/serviceProblemManagement/v4/serviceProblem/problemxxxx0001"
    }
  ],
  "underlyingProblem": [
    {
      "id": "problemxxxx0001",
      "correlationId": "xxxxxxxx",
      "href": "https://mycsp.com:8080/tmf-api/serviceProblemManagement/v4/serviceProblem/problemxxxx0001"
    }
  ],
  "trackingRecord": [
    {
      "description": "yyy cleared the problem",
      "systemId": "xxxx",
      "time": "2014-12-20T17:00:00Z",
      "user": "NPUSER1",
```


© TM Forum 2021. All Rights Reserved

```
        "id": "TR_xxxx",
        "extensionInfo": [
          {
            "name": "vendorComment",
            "valueType": "string",
            "value": "Watch out for the dog"
          }
        ]
      }
    ],
    "comment": [
      {
        "author": {
          "id": "SPM_handler_01",
          "href": "https://mycsp.com:8080/tmf-api/partyManagement/v4/party/SPM_handler_01",
          "@referredType": "Individual"
        },
        "date": "2014-12-20T17:00:00Z",
        "systemId": "System_002",
        "text": "receive trouble ticket from NP1, and create this Service Problem"
      },
      {
        "author": {
          "id": "NP1_handler_11",
          "href": "https://mycsp.com:8080/tmf-api/partyManagement/v4/party/NP1_handler_11",
          "@referredType": "Individual"
        },
        "date": "2014-12-20T17:00:00Z",
        "systemId": "System_002",
        "text": "status changed to Progress-Held"
      }
    ],
    "impactPatterns": {
      "description": "Many services are at risk in this problem",
      "extensionInfo": [
        {
          "name": "ImpactProbability",
          "valueType": "boolean",
          "value": true
        }
      ]
    },
    "extensionInfo": [
      {
        "name": "EstimatedCost",
        "valueType": "integer",
        "value": 20
      }
    ]
}
```


Create service problem

POST /serviceProblem

© TM Forum 2021. All Rights Reserved

Description

This operation creates a service problem entity.

Mandatory and Non Mandatory Attributes

The following tables provide the list of mandatory and non mandatory attributes when creating a ServiceProblem, including any possible rule conditions and applicable default values. Notice that it is up to an implementer to add additional mandatory attributes.

Mandatory Attributes	Rule
category	
priority	
description	
reason	
originatorParty	

Non Mandatory Attributes	Rule
@baseType	
@schemaLocation	
@type	
affectedNumberOfServices	
affectedResource	
affectedService	
characteristic	
creationDate	
externalIdentifier	
firstAlert	
impactImportanceFactor	
impactPattern	
lastUpdate	
name	
note	
originatingSystem	
parentProblem	
place	
problemEscalation	
relatedEntity	
relatedEvent	
relatedParty	
resolutionDate	
responsibleParty	
rootCauseResource	
rootCauseService	
slaViolation	
status	
statusChangeDate	
statusChangeReason	

© TM Forum 2021. All Rights Reserved

trackingRecord	
troubleTicket	
underlyingAlarm	
underlyingProblem	

Usage Samples

Here's an example of a request for creating a specific ServiceProblem.

Request

```
POST /tmf-api/naas/v4/serviceProblem
Content-Type: application/json

{
  "correlationId": "543251",
  "originatingSystem": "System_001",
  "category": "supplier.originated",
  "impactImportanceFactor": "0",
  "priority": "1",
  "description": "connection failure between Tokyo and Osaka",
  "problemEscalation": "0",
  "timeRaised": "2017-10-25T12:14:16.361Z",
  "timeChanged": "2017-10-30T12:13:16.361Z",
  "statusChangeDate": "2017-10-29T12:00:00.361Z",
  "statusChangeReason": "problem analysis has been completed in NP1",
  "resolutionDate": "2017-10-29T12:00:00.361Z",
  "status": "resolved",
  "reason": "Failure of resource NP1_Resource_1 in NP1",
  "affectedNumberOfServices": "2",
  "firstAlert": {
    "alertType": "TroubleTicket",
    "id": "NP1_TT_0000000",
    "href": "https://mycsp.com:8080/tmf-api/troubleTicketManagement/v4/troubleTicket/NP1_TT_000000"
  },
  "responsibleParty": {
    "role": "Supplier",
    "id": "NP1",
    "href": "https://mycsp.com:8080/tmf-api/partyManagement/v4/party/NP1",
    "@referredType": "Organization"
  },
  "originatorParty": {
    "role": "Supplier",
    "id": "NP1",
    "href": "https://mycsp.com:8080/tmf-api/partyManagement/v4/party/NP1",
    "@referredType": "Organization"
  },
  "relatedParty": [
    {
      "role": "Supplier",
      "id": "NP1",
      "href": "https://mycsp.com:8080/tmf-api/partyManagement/v4/party/NP1",
```

© TM Forum 2021. All Rights Reserved

```
      "@referredType": "Organization"
    },
    {
      "role": "Partner",
      "id": "SP1",
      "href": "https://mycsp.com:8080/tmf-api/partyManagement/v4/party/SP1",
      "@referredType": "Organization"
    },
    {
      "role": "Partner",
      "id": "SP3",
      "href": "https://mycsp.com:8080/tmf-api/partyManagement/v4/party/SP3",
      "@referredType": "Organization"
    }
  ],
  "affectedService": [
    {
      "id": "NP1_Tokyo_Osaka",
      "href": "https://mycsp.com:8080/tmf-api/serviceInventoryManagement/v4/service/NW1_Tokyo_Osaka"
    },
    {
      "id": "NP1_Tokyo_xxxx",
      "href": "https://mycsp.com:8080/tmf-api/serviceInventoryManagement/v4/service/NW1_Tokyo_xxxx"
    }
  ],
  "affectedResource": [
    {
      "id": "NP1_RES_0001",
      "href": "https://mycsp.com:8080/tmf-api/resourceInventoryManagement/v4/resource/NW1_RES_0001"
    }
  ],
  "affectedLocation": [
    {
      "id": "Loc000000",
      "href": "https://mycsp.com:8080/tmf-api/geographicAddressManagement/v4/geographicAddress/Loc000000/",
      "name": "144 Main Street Tokyo 51663556",
      "role": "VPN Endpoint",
      "@type": "GeographicAddress"
    },
    {
      "id": "Loc000001",
      "href": "https://mycsp.com:8080/tmf-api/geographicAddressManagement/v4/geographicAddress/Loc000001/",
      "name": "351 Main Street Osaka 3546365",
      "role": "VPN Endpoint",
      "@type": "GeographicAddress"
    }
  ],
  "associatedTroubleTicket": [
    {
      "id": "NP1_TT_0000000",
      "href": "https://mycsp.com:8080/tmf-api/troubleTicketManagement/v4/TroubleTicketRef/NP1_TT_000000"
    }
  ],
  "underlyingAlarm": [
    {
```


© TM Forum 2021. All Rights Reserved

```
        "id": "NP1_A_0000000",
        "href": "https://mycsp.com:8080/tmf-api/alarmManagement/v4/resourceAlarm/NP1_A_000000",
        "@referredType": "ResourceAlarm"
      }
    ],
    "associatedSLAViolation": [
      {
        "id": "NP1_SLA_0000000",
        "href": "https://mycsp.com:8080/tmf-api/SLAManagement/v4/SLAViolationRef/NP1_SLA_000000"
      }
    ],
    "relatedEvent": [
      {
        "eventType": "prediction",
        "id": "prediction_0001",
        "href": "https://mycsp.com:8080/tmf-api/eventManagement/v4/event/prediction_0001",
        "eventTime": "2014-12-20T17:00:00Z"
      }
    ],
    "relatedObject": [
      {
        "id": "product0001",
        "href": "https://mycsp.com:8080/tmf-api/productInventoryManagement/v4/product/product0001",
        "@referredType": "Product"
      }
    ],
    "rootCauseService": [
      {
        "id": "NP1_Tokyo_Osaka",
        "href": "https://mycsp.com:8080/tmf-api/serviceInventoryManagement/v4/service/NW1_Tokyo_Osaka"
      }
    ],
    "rootCauseResource": [
      {
        "id": "NP1_Resource_1",
        "href": "https://mycsp.com:8080/tmf-api/resourceInventoryManagement/v4/resource/NP1_Resource_1"
      }
    ],
    "parentProblem": [
      {
        "id": "problemxxxx0001",
        "correlationId": "xxxxxxxx",
        "href": "https://mycsp.com:8080/tmf-api/serviceProblemManagement/v4/serviceProblem/problemxxxx0001"
      }
    ],
    "underlyingProblem": [
      {
        "id": "problemxxxx0001",
        "correlationId": "xxxxxxxx",
        "href": "https://mycsp.com:8080/tmf-api/serviceProblemManagement/v4/serviceProblem/problemxxxx0001"
      }
    ],
    "trackingRecord": [
      {
        "description": "yyy cleared the problem",
```


© TM Forum 2021. All Rights Reserved

```
        "systemId": "xxxx",
        "time": "2014-12-20T17:00:00Z",
        "user": "NPUSER1",
        "id": "TR_xxxx",
        "extensionInfo": [
          {
            "name": "vendorComment",
            "valueType": "string",
            "value": "Watch out for the dog"
          }
        ]
      }
  ],
  "comment": [
    {
      "author": {
        "id": "SPM_handler_01",
        "href": "https://mycsp.com:8080/tmf-api/partyManagement/v4/party/SPM_handler_01",
        "@referredType": "Individual"
      },
      "date": "2014-12-20T17:00:00Z",
      "systemId": "System_002",
      "text": "receive trouble ticket from NP1, and create this Service Problem"
    },
    {
      "author": {
        "id": "NP1_handler_11",
        "href": "https://mycsp.com:8080/tmf-api/partyManagement/v4/party/NP1_handler_11",
        "@referredType": "Individual"
      },
      "date": "2014-12-20T17:00:00Z",
      "systemId": "System_002",
      "text": "status changed to Progress-Held"
    }
  ],
  "impactPatterns": {
    "description": "Many services are at risk in this problem",
    "extensionInfo": [
      {
        "name": "ImpactProbability",
        "valueType": "boolean",
        "value": true
      }
    ]
  },
  "extensionInfo": [
    {
      "name": "EstimatedCost",
      "valueType": "integer",
      "value": 20
    }
  ]
}
```


© TM Forum 2021. All Rights Reserved

Response

201


```
{
  "id": "problemxxxx0000",
  "correlationId": "543251",
  "originatingSystem": "System_001",
  "category": "supplier.originated",
  "href": "https://mycsp.com:8080/tmf-api/serviceProblemManagement/v4/serviceProblem/problemxxxx0000",
  "impactImportanceFactor": "0",
  "priority": "1",
  "description": "connection failure between Tokyo and Osaka",
  "problemEscalation": "0",
  "timeRaised": "2017-10-25T12:14:16.361Z",
  "timeChanged": "2017-10-30T12:13:16.361Z",
  "statusChangeDate": "2017-10-29T12:00:00.361Z",
  "statusChangeReason": "problem analysis has been completed in NP1",
  "resolutionDate": "2017-10-29T12:00:00.361Z",
  "status": "resolved",
  "reason": "Failure of resource NP1_Resource_1 in NP1",
  "affectedNumberOfServices": "2",
  "firstAlert": {
    "alertType": "TroubleTicket",
    "id": "NP1_TT_0000000",
    "href": "https://mycsp.com:8080/tmf-api/troubleTicketManagement/v4/troubleTicket/NP1_TT_000000"
  },
  "responsibleParty": {
    "role": "Supplier",
    "id": "NP1",
    "href": "https://mycsp.com:8080/tmf-api/partyManagement/v4/party/NP1",
    "@referredType": "Organization"
  },
  "originatorParty": {
    "role": "Supplier",
    "id": "NP1",
    "href": "https://mycsp.com:8080/tmf-api/partyManagement/v4/party/NP1",
    "@referredType": "Organization"
  },
  "relatedParty": [
    {
      "role": "Supplier",
      "id": "NP1",
      "href": "https://mycsp.com:8080/tmf-api/partyManagement/v4/party/NP1",
      "@referredType": "Organization"
    },
    {
      "role": "Partner",
      "id": "SP1",
      "href": "https://mycsp.com:8080/tmf-api/partyManagement/v4/party/SP1",
      "@referredType": "Organization"
    },
    {
      "role": "Partner",
```


© TM Forum 2021. All Rights Reserved

```
        "id": "SP3",
        "href": "https://mycsp.com:8080/tmf-api/partyManagement/v4/party/SP3",
        "@referredType": "Organization"
      }
    ],
    "affectedService": [
      {
        "id": "NP1_Tokyo_Osaka",
        "href": "https://mycsp.com:8080/tmf-api/serviceInventoryManagement/v4/service/NW1_Tokyo_Osaka"
      },
      {
        "id": "NP1_Tokyo_xxxx",
        "href": "https://mycsp.com:8080/tmf-api/serviceInventoryManagement/v4/service/NW1_Tokyo_xxxx"
      }
    ],
    "affectedResource": [
      {
        "id": "NP1_RES_0001",
        "href": "https://mycsp.com:8080/tmf-api/resourceInventoryManagement/v4/resource/NW1_RES_0001"
      }
    ],
    "affectedLocation": [
      {
        "id": "Loc000000",
        "href": "https://mycsp.com:8080/tmf-api/geographicAddressManagement/v4/geographicAddress/Loc000000/",
        "name": "144 Main Street Tokyo 51663556",
        "role": "VPN Endpoint",
        "@type": "GeographicAddress"
      },
      {
        "id": "Loc000001",
        "href": "https://mycsp.com:8080/tmf-api/geographicAddressManagement/v4/geographicAddress/Loc000001/",
        "name": "351 Main Street Osaka 3546365",
        "role": "VPN Endpoint",
        "@type": "GeographicAddress"
      }
    ],
    "associatedTroubleTicket": [
      {
        "id": "NP1_TT_0000000",
        "href": "https://mycsp.com:8080/tmf-api/troubleTicketManagement/v4/TroubleTicketRef/NP1_TT_000000"
      }
    ],
    "underlyingAlarm": [
      {
        "id": "NP1_A_0000000",
        "href": "https://mycsp.com:8080/tmf-api/alarmManagement/v4/resourceAlarm/NP1_A_000000",
        "@referredType": "ResourceAlarm"
      }
    ],
    "associatedSLAViolation": [
      {
        "id": "NP1_SLA_0000000",
        "href": "https://mycsp.com:8080/tmf-api/SLAManagement/v4/SLAViolationRef/NP1_SLA_000000"
      }
```


© TM Forum 2021. All Rights Reserved

```
  ],
  "relatedEvent": [
    {
      "eventType": "prediction",
      "id": "prediction_0001",
      "href": "https://mycsp.com:8080/tmf-api/eventManagement/v4/event/prediction_0001",
      "eventTime": "2014-12-20T17:00:00Z"
    }
  ],
  "relatedObject": [
    {
      "id": "product0001",
      "href": "https://mycsp.com:8080/tmf-api/productInventoryManagement/v4/product/product0001",
      "@referredType": "Product"
    }
  ],
  "rootCauseService": [
    {
      "id": "NP1_Tokyo_Osaka",
      "href": "https://mycsp.com:8080/tmf-api/serviceInventoryManagement/v4/service/NW1_Tokyo_Osaka"
    }
  ],
  "rootCauseResource": [
    {
      "id": "NP1_Resource_1",
      "href": "https://mycsp.com:8080/tmf-api/resourceInventoryManagement/v4/resource/NP1_Resource_1"
    }
  ],
  "parentProblem": [
    {
      "id": "problemxxxx0001",
      "correlationId": "xxxxxxxx",
      "href": "https://mycsp.com:8080/tmf-api/serviceProblemManagement/v4/serviceProblem/problemxxxx0001"
    }
  ],
  "underlyingProblem": [
    {
      "id": "problemxxxx0001",
      "correlationId": "xxxxxxxx",
      "href": "https://mycsp.com:8080/tmf-api/serviceProblemManagement/v4/serviceProblem/problemxxxx0001"
    }
  ],
  "trackingRecord": [
    {
      "description": "yyy cleared the problem",
      "systemId": "xxxx",
      "time": "2014-12-20T17:00:00Z",
      "user": "NPUSER1",
      "id": "TR_xxxx",
      "extensionInfo": [
        {
          "name": "vendorComment",
          "valueType": "string",
          "value": "Watch out for the dog"
        }
```


© TM Forum 2021. All Rights Reserved

```
        ]
      }
    ],
    "comment": [
      {
        "author": {
          "id": "SPM_handler_01",
          "href": "https://mycsp.com:8080/tmf-api/partyManagement/v4/party/SPM_handler_01",
          "@referredType": "Individual"
        },
        "date": "2014-12-20T17:00:00Z",
        "systemId": "System_002",
        "text": "receive trouble ticket from NP1, and create this Service Problem"
      },
      {
        "author": {
          "id": "NP1_handler_11",
          "href": "https://mycsp.com:8080/tmf-api/partyManagement/v4/party/NP1_handler_11",
          "@referredType": "Individual"
        },
        "date": "2014-12-20T17:00:00Z",
        "systemId": "System_002",
        "text": "status changed to Progress-Held"
      }
    ],
    "impactPatterns": {
      "description": "Many services are at risk in this problem",
      "extensionInfo": [
        {
          "name": "ImpactProbability",
          "valueType": "boolean",
          "value": true
        }
      ]
    },
    "extensionInfo": [
      {
        "name": "EstimatedCost",
        "valueType": "integer",
        "value": 20
      }
    ]
}
```


Patch service problem

PATCH /serviceProblem/{id}

Description

This operation allows partial updates of a service problem entity. Support of json/merge (https://tools.ietf.org/html/rfc7386) is mandatory, support of json/patch (http://tools.ietf.org/html/rfc5789) is optional.

© TM Forum 2021. All Rights Reserved

Note: If the update operation yields to the creation of sub-resources or relationships, the same rules concerning mandatory sub-resource attributes and default value settings in the POST operation applies to the PATCH operation. Hence these tables are not repeated here.

Patchable and Non Patchable Attributes

The tables below provide the list of patchable and non patchable attributes, including constraint rules on their usage.

Patchable Attributes	Rule
@baseType	
@schemaLocation	
@type	
affectedNumberOfServices	
affectedResource	
affectedService	
category	
characteristic	
creationDate	
description	
externalIdentifier	
impactImportanceFactor	
impactPattern	
lastUpdate	
name	
note	
originatorParty	
parentProblem	
place	
priority	
problemEscalation	
reason	
relatedEntity	
relatedEvent	
relatedParty	
resolutionDate	
responsibleParty	
rootCauseResource	
rootCauseService	
slaViolation	
status	
statusChangeDate	
statusChangeReason	
troubleTicket	
underlyingAlarm	
underlyingProblem	

© TM Forum 2021. All Rights Reserved

Non Patchable Attributes	Rule
correlationId	
firstAlert	
href	
id	
originatingSystem	
timeRaised	
trackingRecord	

Usage Samples

Here's an example of a request for patching a ServiceProblem.

Request

```
PATCH /tmf-api/naas/v4/serviceProblem/problemxxxx0000
Content-Type: application/merge-patch+json

{
   "correlationId": "543251",
   "originatingSystem": "System_001",
   "category": "supplier.originated",
   "impactImportanceFactor": "0",
   "priority": "1",
   "description": "connection failure between Tokyo and Osaka",
   "problemEscalation": "0",
   "timeRaised": "2017-10-25T12:14:16.361Z",
   "timeChanged": "2017-10-30T12:13:16.361Z",
   "statusChangeDate": "2017-10-29T12:00:00.361Z",
   "statusChangeReason": "problem analysis has been completed in NP1",
   "resolutionDate": "2017-10-29T12:00:00.361Z",
   "status": "resolved",
   "reason": "Failure of resource NP1_Resource_1 in NP1",
   "affectedNumberOfServices": "2",
   "firstAlert": {
      "alertType": "TroubleTicket",
      "id": "NP1_TT_0000000",
      "href": "https://mycsp.com:8080/tmf-api/troubleTicketManagement/v4/troubleTicket/NP1_TT_000000"
   },
   "responsibleParty": {
      "role": "Supplier",
      "id": "NP1",
      "href": "https://mycsp.com:8080/tmf-api/partyManagement/v4/party/NP1",
      "@referredType": "Organization"
   },
   "originatorParty": {
      "role": "Supplier",
      "id": "NP1",
      "href": "https://mycsp.com:8080/tmf-api/partyManagement/v4/party/NP1",
      "@referredType": "Organization"
   },
   "relatedParty": [
```

© TM Forum 2021. All Rights Reserved

```
    {
      "role": "Supplier",
      "id": "NP1",
      "href": "https://mycsp.com:8080/tmf-api/partyManagement/v4/party/NP1",
      "@referredType": "Organization"
    },
    {
      "role": "Partner",
      "id": "SP1",
      "href": "https://mycsp.com:8080/tmf-api/partyManagement/v4/party/SP1",
      "@referredType": "Organization"
    },
    {
      "role": "Partner",
      "id": "SP3",
      "href": "https://mycsp.com:8080/tmf-api/partyManagement/v4/party/SP3",
      "@referredType": "Organization"
    }
  ],
  "affectedService": [
    {
      "id": "NP1_Tokyo_Osaka",
      "href": "https://mycsp.com:8080/tmf-api/serviceInventoryManagement/v4/service/NW1_Tokyo_Osaka"
    },
    {
      "id": "NP1_Tokyo_xxxx",
      "href": "https://mycsp.com:8080/tmf-api/serviceInventoryManagement/v4/service/NW1_Tokyo_xxxx"
    }
  ],
  "affectedResource": [
    {
      "id": "NP1_RES_0001",
      "href": "https://mycsp.com:8080/tmf-api/resourceInventoryManagement/v4/resource/NW1_RES_0001"
    }
  ],
  "affectedLocation": [
    {
      "id": "Loc000000",
      "href": "https://mycsp.com:8080/tmf-api/geographicAddressManagement/v4/geographicAddress/Loc000000/",
      "name": "144 Main Street Tokyo 51663556",
      "role": "VPN Endpoint",
      "@type": "GeographicAddress"
    },
    {
      "id": "Loc000001",
      "href": "https://mycsp.com:8080/tmf-api/geographicAddressManagement/v4/geographicAddress/Loc000001/",
      "name": "351 Main Street Osaka 3546365",
      "role": "VPN Endpoint",
      "@type": "GeographicAddress"
    }
  ],
  "associatedTroubleTicket": [
    {
      "id": "NP1_TT_0000000",
      "href": "https://mycsp.com:8080/tmf-api/troubleTicketManagement/v4/TroubleTicketRef/NP1_TT_000000"
```


© TM Forum 2021. All Rights Reserved

```
    }
  ],
  "underlyingAlarm": [
    {
      "id": "NP1_A_0000000",
      "href": "https://mycsp.com:8080/tmf-api/alarmManagement/v4/resourceAlarm/NP1_A_000000",
      "@referredType": "ResourceAlarm"
    }
  ],
  "associatedSLAViolation": [
    {
      "id": "NP1_SLA_0000000",
      "href": "https://mycsp.com:8080/tmf-api/SLAManagement/v4/SLAViolationRef/NP1_SLA_000000"
    }
  ],
  "relatedEvent": [
    {
      "eventType": "prediction",
      "id": "prediction_0001",
      "href": "https://mycsp.com:8080/tmf-api/eventManagement/v4/event/prediction_0001",
      "eventTime": "2014-12-20T17:00:00Z"
    }
  ],
  "relatedObject": [
    {
      "id": "product0001",
      "href": "https://mycsp.com:8080/tmf-api/productInventoryManagement/v4/product/product0001",
      "@referredType": "Product"
    }
  ],
  "rootCauseService": [
    {
      "id": "NP1_Tokyo_Osaka",
      "href": "https://mycsp.com:8080/tmf-api/serviceInventoryManagement/v4/service/NW1_Tokyo_Osaka"
    }
  ],
  "rootCauseResource": [
    {
      "id": "NP1_Resource_1",
      "href": "https://mycsp.com:8080/tmf-api/resourceInventoryManagement/v4/resource/NP1_Resource_1"
    }
  ],
  "parentProblem": [
    {
      "id": "problemxxxx0001",
      "correlationId": "xxxxxxxx",
      "href": "https://mycsp.com:8080/tmf-api/serviceProblemManagement/v4/serviceProblem/problemxxxx0001"
    }
  ],
  "underlyingProblem": [
    {
      "id": "problemxxxx0001",
      "correlationId": "xxxxxxxx",
      "href": "https://mycsp.com:8080/tmf-api/serviceProblemManagement/v4/serviceProblem/problemxxxx0001"
    }
```


© TM Forum 2021. All Rights Reserved

```
  ],
  "trackingRecord": [
    {
      "description": "yyy cleared the problem",
      "systemId": "xxxx",
      "time": "2014-12-20T17:00:00Z",
      "user": "NPUSER1",
      "id": "TR_xxxx",
      "extensionInfo": [
        {
          "name": "vendorComment",
          "valueType": "string",
          "value": "Watch out for the dog"
        }
      ]
    }
  ],
  "comment": [
    {
      "author": {
        "id": "SPM_handler_01",
        "href": "https://mycsp.com:8080/tmf-api/partyManagement/v4/party/SPM_handler_01",
        "@referredType": "Individual"
      },
      "date": "2014-12-20T17:00:00Z",
      "systemId": "System_002",
      "text": "receive trouble ticket from NP1, and create this Service Problem"
    },
    {
      "author": {
        "id": "NP1_handler_11",
        "href": "https://mycsp.com:8080/tmf-api/partyManagement/v4/party/NP1_handler_11",
        "@referredType": "Individual"
      },
      "date": "2014-12-20T17:00:00Z",
      "systemId": "System_002",
      "text": "status changed to Progress-Held"
    }
  ],
  "impactPatterns": {
    "description": "Many services are at risk in this problem",
    "extensionInfo": [
      {
        "name": "ImpactProbability",
        "valueType": "boolean",
        "value": true
      }
    ]
  },
  "extensionInfo": [
    {
      "name": "EstimatedCost",
      "valueType": "integer",
      "value": 20
    }
```


© TM Forum 2021. All Rights Reserved

```
  ]
}
```

Response

200


```
{
  "id": "problemxxxx0000",
  "correlationId": "543251",
  "originatingSystem": "System_001",
  "category": "supplier.originated",
  "href": "https://mycsp.com:8080/tmf-api/serviceProblemManagement/v4/serviceProblem/problemxxxx0000",
  "impactImportanceFactor": "0",
  "priority": "1",
  "description": "connection failure between Tokyo and Osaka",
  "problemEscalation": "0",
  "timeRaised": "2017-10-25T12:14:16.361Z",
  "timeChanged": "2017-10-30T12:13:16.361Z",
  "statusChangeDate": "2017-10-29T12:00:00.361Z",
  "statusChangeReason": "problem analysis has been completed in NP1",
  "resolutionDate": "2017-10-29T12:00:00.361Z",
  "status": "resolved",
  "reason": "Failure of resource NP1_Resource_1 in NP1",
  "affectedNumberOfServices": "2",
  "firstAlert": {
    "alertType": "TroubleTicket",
    "id": "NP1_TT_0000000",
    "href": "https://mycsp.com:8080/tmf-api/troubleTicketManagement/v4/troubleTicket/NP1_TT_000000"
  },
  "responsibleParty": {
    "role": "Supplier",
    "id": "NP1",
    "href": "https://mycsp.com:8080/tmf-api/partyManagement/v4/party/NP1",
    "@referredType": "Organization"
  },
  "originatorParty": {
    "role": "Supplier",
    "id": "NP1",
    "href": "https://mycsp.com:8080/tmf-api/partyManagement/v4/party/NP1",
    "@referredType": "Organization"
  },
  "relatedParty": [
    {
      "role": "Supplier",
      "id": "NP1",
      "href": "https://mycsp.com:8080/tmf-api/partyManagement/v4/party/NP1",
      "@referredType": "Organization"
    },
    {
      "role": "Partner",
      "id": "SP1",
      "href": "https://mycsp.com:8080/tmf-api/partyManagement/v4/party/SP1",
```


© TM Forum 2021. All Rights Reserved

```
        "@referredType": "Organization"
      },
      {
        "role": "Partner",
        "id": "SP3",
        "href": "https://mycsp.com:8080/tmf-api/partyManagement/v4/party/SP3",
        "@referredType": "Organization"
      }
    ],
    "affectedService": [
      {
        "id": "NP1_Tokyo_Osaka",
        "href": "https://mycsp.com:8080/tmf-api/serviceInventoryManagement/v4/service/NW1_Tokyo_Osaka"
      },
      {
        "id": "NP1_Tokyo_xxxx",
        "href": "https://mycsp.com:8080/tmf-api/serviceInventoryManagement/v4/service/NW1_Tokyo_xxxx"
      }
    ],
    "affectedResource": [
      {
        "id": "NP1_RES_0001",
        "href": "https://mycsp.com:8080/tmf-api/resourceInventoryManagement/v4/resource/NW1_RES_0001"
      }
    ],
    "affectedLocation": [
      {
        "id": "Loc000000",
        "href": "https://mycsp.com:8080/tmf-api/geographicAddressManagement/v4/geographicAddress/Loc000000/",
        "name": "144 Main Street Tokyo 51663556",
        "role": "VPN Endpoint",
        "@type": "GeographicAddress"
      },
      {
        "id": "Loc000001",
        "href": "https://mycsp.com:8080/tmf-api/geographicAddressManagement/v4/geographicAddress/Loc000001/",
        "name": "351 Main Street Osaka 3546365",
        "role": "VPN Endpoint",
        "@type": "GeographicAddress"
      }
    ],
    "associatedTroubleTicket": [
      {
        "id": "NP1_TT_0000000",
        "href": "https://mycsp.com:8080/tmf-api/troubleTicketManagement/v4/TroubleTicketRef/NP1_TT_000000"
      }
    ],
    "underlyingAlarm": [
      {
        "id": "NP1_A_0000000",
        "href": "https://mycsp.com:8080/tmf-api/alarmManagement/v4/resourceAlarm/NP1_A_000000",
        "@referredType": "ResourceAlarm"
      }
    ],
    "associatedSLAViolation": [
```


© TM Forum 2021. All Rights Reserved

```
    {
      "id": "NP1_SLA_0000000",
      "href": "https://mycsp.com:8080/tmf-api/SLAManagement/v4/SLAViolationRef/NP1_SLA_000000"
    }
  ],
  "relatedEvent": [
    {
      "eventType": "prediction",
      "id": "prediction_0001",
      "href": "https://mycsp.com:8080/tmf-api/eventManagement/v4/event/prediction_0001",
      "eventTime": "2014-12-20T17:00:00Z"
    }
  ],
  "relatedObject": [
    {
      "id": "product0001",
      "href": "https://mycsp.com:8080/tmf-api/productInventoryManagement/v4/product/product0001",
      "@referredType": "Product"
    }
  ],
  "rootCauseService": [
    {
      "id": "NP1_Tokyo_Osaka",
      "href": "https://mycsp.com:8080/tmf-api/serviceInventoryManagement/v4/service/NW1_Tokyo_Osaka"
    }
  ],
  "rootCauseResource": [
    {
      "id": "NP1_Resource_1",
      "href": "https://mycsp.com:8080/tmf-api/resourceInventoryManagement/v4/resource/NP1_Resource_1"
    }
  ],
  "parentProblem": [
    {
      "id": "problemxxxx0001",
      "correlationId": "xxxxxxxx",
      "href": "https://mycsp.com:8080/tmf-api/serviceProblemManagement/v4/serviceProblem/problemxxxx0001"
    }
  ],
  "underlyingProblem": [
    {
      "id": "problemxxxx0001",
      "correlationId": "xxxxxxxx",
      "href": "https://mycsp.com:8080/tmf-api/serviceProblemManagement/v4/serviceProblem/problemxxxx0001"
    }
  ],
  "trackingRecord": [
    {
      "description": "yyy cleared the problem",
      "systemId": "xxxx",
      "time": "2014-12-20T17:00:00Z",
      "user": "NPUSER1",
      "id": "TR_xxxx",
      "extensionInfo": [
        {
```


© TM Forum 2021. All Rights Reserved

```
            "name": "vendorComment",
            "valueType": "string",
            "value": "Watch out for the dog"
          }
        ]
      }
    ],
    "comment": [
      {
        "author": {
          "id": "SPM_handler_01",
          "href": "https://mycsp.com:8080/tmf-api/partyManagement/v4/party/SPM_handler_01",
          "@referredType": "Individual"
        },
        "date": "2014-12-20T17:00:00Z",
        "systemId": "System_002",
        "text": "receive trouble ticket from NP1, and create this Service Problem"
      },
      {
        "author": {
          "id": "NP1_handler_11",
          "href": "https://mycsp.com:8080/tmf-api/partyManagement/v4/party/NP1_handler_11",
          "@referredType": "Individual"
        },
        "date": "2014-12-20T17:00:00Z",
        "systemId": "System_002",
        "text": "status changed to Progress-Held"
      }
    ],
    "impactPatterns": {
      "description": "Many services are at risk in this problem",
      "extensionInfo": [
        {
          "name": "ImpactProbability",
          "valueType": "boolean",
          "value": true
        }
      ]
    },
    "extensionInfo": [
      {
        "name": "EstimatedCost",
        "valueType": "integer",
        "value": 20
      }
    ]
}
```


Delete service problem

`DELETE /serviceProblem/{id}`

Description

© TM Forum 2021. All Rights Reserved

This operation deletes a service problem entity.

Usage Samples

Here's an example of a request for patching a ServiceProblem.

Request
DELETE /tmf-api/naas/v4/serviceProblem/problemxxxx0000
Response
204

Operations on Service Problem Event Record

List service problem event records

`GET /serviceProblemEventRecord?fields=...&{filtering}`

Description

This operation list service problem event record entities.
Attribute selection is enabled for all first level attributes.
Filtering may be available depending on the compliance level supported by an implementation.

Usage Samples

Here's an example of a request for retrieving multiple ServiceProblemEventRecord.

Request
GET /tmf-api/naas/v4/serviceProblemEventRecord Accept: application/json
Response
200

© TM Forum 2021. All Rights Reserved

```
[
  {
    "id": "42",
    "href": "https://mycsp.com:8080/tmf-api/serviceProblemManagement/v4/serviceProblemEventRecord/42",
    "recordTime": "2016-08-08T10:45:30.0Z",
    "eventType": "ServiceProblemCreationNotification",
    "eventTime": "2016-08-08T10:45:25.0Z",
    "serviceProblemId": "SP001",
    "notification": {
      "eventType": "ServiceProblemCreationNotification",
      "eventTime": "2016-08-08T10:45:25.0Z",
      "eventId": "92775",
      "event": {}
    },
    "serviceProblem": {
      "id": "problemxxxx0000",
      "href": "https://mycsp.com:8080/tmf-api/serviceProblemManagement/v4/serviceProblem/problemxxxx0000"
    }
  }
]
```


Retrieve service problem event record

`GET /serviceProblemEventRecord/{id}?fields=...&{filtering}`

Description

This operation retrieves a service problem event record entity.
Attribute selection is enabled for all first level attributes.
Filtering on sub-resources may be available depending on the compliance level supported by an implementation.

Usage Samples

Here's an example of a request for retrieving a specific ServiceProblemEventRecord.

Request

```
GET /tmf-api/naas/v4/serviceProblemEventRecord/42
Accept: application/json
```

Response


```
200

{
  "id": "42",
  "href": "https://mycsp.com:8080/tmf-api/serviceProblemManagement/v4/serviceProblemEventRecord/42",
```


© TM Forum 2021. All Rights Reserved

```
  "recordTime": "2016-08-08T10:45:30.0Z",
  "eventType": "ServiceProblemCreationNotification",
  "eventTime": "2016-08-08T10:45:25.0Z",
  "serviceProblemId": "SP001",
  "notification": {
     "eventType": "ServiceProblemCreationNotification",
     "eventTime": "2016-08-08T10:45:25.0Z",
     "eventId": "92775",
     "event": {}
  },
  "serviceProblem": {
     "id": "problemxxxx0000",
     "href": "https://mycsp.com:8080/tmf-api/serviceProblemManagement/v4/serviceProblem/problemxxxx0000"
  }
}
```


Operations on Problem Acknowledgement

Create problem acknowledgement

`POST /problemAcknowledgement`

Description

This operation creates a problem acknowledgement entity.

Mandatory and Non Mandatory Attributes

The following tables provide the list of mandatory and non mandatory attributes when creating a ProblemAcknowledgement, including any possible rule conditions and applicable default values. Notice that it is up to an implementer to add additional mandatory attributes.

Mandatory Attributes	Rule
problem	

Non Mandatory Attributes	Rule
@baseType	
@schemaLocation	
@type	
ackProblem	
trackingRecord	

Usage Samples

Here's an example of a request for creating a specific ProblemAcknowledgement.

Request

© TM Forum 2021. All Rights Reserved

```
POST /tmf-api/naas/v4/problemAcknowledgement
Content-Type: application/json

{
  "problem": [
    {
      "id": "41",
      "href": "https://mycsp.com:8080/tmf-api/serviceProblemManagement/v4/serviceProblem/41"
    },
    {
      "id": "42",
      "href": "https://mycsp.com:8080/tmf-api/serviceProblemManagement/v4/serviceProblem/42"
    },
    {
      "id": "43",
      "href": "https://mycsp.com:8080/tmf-api/serviceProblemManagement/v4/serviceProblem/43"
    }
  ],
  "trackingRecord": {
    "description": "yyy ack the problem",
    "systemId": "xxxx",
    "time": "2014-12-20T17:00:00Z",
    "user": "NP1"
  }
}
```

Response

```
201

{
  "problem": [
    {
      "id": "41",
      "href": "https://mycsp.com:8080/tmf-api/serviceProblemManagement/v4/serviceProblem/41"
    },
    {
      "id": "42",
      "href": "https://mycsp.com:8080/tmf-api/serviceProblemManagement/v4/serviceProblem/42"
    },
    {
      "id": "43",
      "href": "https://mycsp.com:8080/tmf-api/serviceProblemManagement/v4/serviceProblem/43"
    }
  ],
  "trackingRecord": {
    "description": "yyy ack the problem",
    "systemId": "xxxx",
    "time": "2014-12-20T17:00:00Z",
    "user": "NP1"
  }
}
```

© TM Forum 2021. All Rights Reserved

Operations on Problem Unacknowledgement

Create problem unacknowledgement

`POST /problemUnacknowledgement`

Description

This operation creates a problem unacknowledgement entity.

Mandatory and Non Mandatory Attributes

The following tables provide the list of mandatory and non mandatory attributes when creating a ProblemUnacknowledgement, including any possible rule conditions and applicable default values. Notice that it is up to an implementer to add additional mandatory attributes.

Mandatory Attributes	Rule
problem	

Non Mandatory Attributes	Rule
@baseType	
@schemaLocation	
@type	
trackingRecord	
unackProblem	

Usage Samples

Here's an example of a request for creating a specific ProblemUnacknowledgement.

Request

```
POST /tmf-api/naas/v4/problemUnacknowledgement
Content-Type: application/json

{
  "problem": [
    {
      "id": "41",
      "href": "https://mycsp.com:8080/tmf-api/serviceProblemManagement/v4/serviceProblem/41"
    },
    {
      "id": "42",
      "href": "https://mycsp.com:8080/tmf-api/serviceProblemManagement/v4/serviceProblem/42"
    },
    {
      "id": "43",
      "href": "https://mycsp.com:8080/tmf-api/serviceProblemManagement/v4/serviceProblem/43"
    }
```

© TM Forum 2021. All Rights Reserved

```
  ],
  "trackingRecord": {
    "description": "yyy unack the problem",
    "systemId": "xxxx",
    "time": "2014-12-20T17:00:00Z",
    "user": "NP1"
  }
}
```


Response

201


```
{
  "problem": [
    {
      "id": "41",
      "href": "https://mycsp.com:8080/tmf-api/serviceProblemManagement/v4/serviceProblem/41"
    },
    {
      "id": "42",
      "href": "https://mycsp.com:8080/tmf-api/serviceProblemManagement/v4/serviceProblem/42"
    },
    {
      "id": "43",
      "href": "https://mycsp.com:8080/tmf-api/serviceProblemManagement/v4/serviceProblem/43"
    }
  ],
  "trackingRecord": {
    "description": "yyy unack the problem",
    "systemId": "xxxx",
    "time": "2014-12-20T17:00:00Z",
    "user": "NP1"
  }
}
```


Operations on Service Test

List service tests

`GET /serviceTest?fields=...&{filtering}`

Description

This operation list service test entities.
Attribute selection is enabled for all first level attributes.
Filtering may be available depending on the compliance level supported by an implementation.

© TM Forum 2021. All Rights Reserved

Usage Samples

Here's an example of a request for retrieving multiple ServiceTest.

Request

```
GET /tmf-api/naas/v4/serviceTest
Accept: application/json
```

Response

```
200

[
  {
    "href": "https://mycsp.com:8080/tmf-api/serviceTestManagement/v4/serviceTest/3112",
    "id": "3112",
    "description:": "This service test is a simple one",
    "endDateTime": "2016-03-02T24:00:00",
    "mode": "PROACTIVE",
    "startDateTime": "2016-03-02T00:00:00",
    "state": "Completed",
    "relatedService": {
      "href": "https://mycsp.com:8080/tmf-api/serviceManagement/v4/service/21",
      "id": "21",
      "name": "gameService"
    },
    "testSpecification": {
      "href": "https://mycsp.com:8080/tmf-api/serviceTestManagement/v4/serviceTestSpecification/22",
      "id": "22",
      "@referredType": "PerformanceTestSpecification"
    },
    "characteristic": [
      {
        "name": "number",
        "valueType": "integer",
        "value": 3
      },
      {
        "name": "restart",
        "valueType": "boolean",
        "value": true
      }
    ],
    "testMeasure": [
      {
        "accuracy": 0.0001,
        "captureDateTime": "2016-03-02T11:12:00",
        "captureMethod": "automatic capture by instrument",
        "metricDescription": "This is the flow speed in a service",
        "metricHref": "https://mycsp.com:8080/tmf-api/metricManagement/v4/metric/21",
```

© TM Forum 2021. All Rights Reserved

```
            "metricName": "Flow speed",
            "unitOfMeasure": "A second",
            "value": {
               "valueType": "bits",
               "value": {
                  "@type": "integer",
                  "value": 3000
               }
            },
            "ruleViolation": [
               {
                  "conformanceTargetUpper": "2500",
                  "conformanceComparatorUpper": "greater than ",
                  "description": "This speed is greater than the target upper",
                  "name": "tooMuchTraffic",
                  "numberOfAllowedCrossing": 0,
                  "thresholdRuleSeverity": "1",
                  "appliedConsequence": [
                     {
                        "appliedAction": "sendWarning",
                        "description": "send a warning to the system",
                        "name": "sendWarning",
                        "repeatAction": false
                     },
                     {
                        "appliedAction": "generateReport",
                        "description": "generate a waring report ",
                        "name": "generateReport",
                        "repeatAction": false
                     }
                  ]
               }
            ]
         }
      ],
      "@type": "PerformanceTest",
      "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Service/PerformanceTest.schema.json",
      "@baseType": "ServiceTest"
   }
]
```


Retrieve service test

`GET /serviceTest/{id}?fields=...&{filtering}`

Description

This operation retrieves a service test entity.
Attribute selection is enabled for all first level attributes.
Filtering on sub-resources may be available depending on the compliance level supported by an implementation.

© TM Forum 2021. All Rights Reserved

Usage Samples

Here's an example of a request for retrieving a specific ServiceTest.

Request

```
GET /tmf-api/naas/v4/serviceTest/3112
Accept: application/json
```

Response

```
200

{
  "href": "https://mycsp.com:8080/tmf-api/serviceTestManagement/v4/serviceTest/3112",
  "id": "3112",
  "description:": "This service test is a simple one",
  "endDateTime": "2016-03-02T24:00:00",
  "mode": "PROACTIVE",
  "startDateTime": "2016-03-02T00:00:00",
  "state": "Completed",
  "relatedService": {
    "href": "https://mycsp.com:8080/tmf-api/serviceManagement/v4/service/21",
    "id": "21",
    "name": "gameService"
  },
  "testSpecification": {
    "href": "https://mycsp.com:8080/tmf-api/serviceTestManagement/v4/serviceTestSpecification/22",
    "id": "22",
    "@referredType": "PerformanceTestSpecification"
  },
  "characteristic": [
    {
      "name": "number",
      "valueType": "integer",
      "value": 3
    },
    {
      "name": "restart",
      "valueType": "boolean",
      "value": true
    }
  ],
  "testMeasure": [
    {
      "accuracy": 0.0001,
      "captureDateTime": "2016-03-02T11:12:00",
      "captureMethod": "automatic capture by instrument",
      "metricDescription": "This is the flow speed in a service",
      "metricHref": "https://mycsp.com:8080/tmf-api/metricManagement/v4/metric/21",
      "metricName": "Flow speed",
```

© TM Forum 2021. All Rights Reserved

```
        "unitOfMeasure": "A second",
        "value": {
          "valueType": "bits",
          "value": {
            "@type": "integer",
            "value": 3000
          }
        },
        "ruleViolation": [
          {
            "conformanceTargetUpper": "2500",
            "conformanceComparatorUpper": "greater than ",
            "description": "This speed is greater than the target upper",
            "name": "tooMuchTraffic",
            "numberOfAllowedCrossing": 0,
            "thresholdRuleSeverity": "1",
            "appliedConsequence": [
              {
                "appliedAction": "sendWarning",
                "description": "send a warning to the system",
                "name": "sendWarning",
                "repeatAction": false
              },
              {
                "appliedAction": "generateReport",
                "description": "generate a waring report ",
                "name": "generateReport",
                "repeatAction": false
              }
            ]
          }
        ]
      }
    ],
    "@type": "PerformanceTest",
    "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Service/PerformanceTest.schema.json",
    "@baseType": "ServiceTest"
}
```


Create service test

POST /serviceTest

Description

This operation creates a service test entity.

Mandatory and Non Mandatory Attributes

The following tables provide the list of mandatory and non mandatory attributes when creating a ServiceTest, including any possible rule conditions and applicable default values. Notice that it is up to an implementer to add additional mandatory attributes.

© TM Forum 2021. All Rights Reserved

Mandatory Attributes	Rule
name	
relatedService	
testSpecification	

Non Mandatory Attributes	Rule
@baseType	
@schemaLocation	
@type	
characteristic	
description	
endDateTime	
mode	
relatedParty	
startDateTime	
state	
testMeasure	
validFor	

Usage Samples

Here's an example of a request for creating a specific ServiceTest.

Request

```
POST /tmf-api/naas/v4/serviceTest
Content-Type: application/json

{
   "description:": "This service test is a simple one",
   "endDateTime": "2016-03-02T24:00:00",
   "mode": "PROACTIVE",
   "startDateTime": "2016-03-02T00:00:00",
   "state": "Completed",
   "relatedService": {
      "href": "https://mycsp.com:8080/tmf-api/serviceManagement/v4/service/21",
      "id": "21",
      "name": "gameService"
   },
   "testSpecification": {
      "href": "https://mycsp.com:8080/tmf-api/serviceTestManagement/v4/serviceTestSpecification/22",
      "id": "22",
      "@referredType": "PerformanceTestSpecification"
   },
   "characteristic": [
      {
         "name": "number",
         "valueType": "integer",
         "value": 3
```

© TM Forum 2021. All Rights Reserved

```
      },
      {
        "name": "restart",
        "valueType": "boolean",
        "value": true
      }
    ],
    "testMeasure": [
      {
        "accuracy": 0.0001,
        "captureDateTime": "2016-03-02T11:12:00",
        "captureMethod": "automatic capture by instrument",
        "metricDescription": "This is the flow speed in a service",
        "metricHref": "https://mycsp.com:8080/tmf-api/metricManagement/v4/metric/21",
        "metricName": "Flow speed",
        "unitOfMeasure": "A second",
        "value": {
          "valueType": "bits",
          "value": {
            "@type": "integer",
            "value": 3000
          }
        },
        "ruleViolation": [
          {
            "conformanceTargetUpper": "2500",
            "conformanceComparatorUpper": "greater than ",
            "description": "This speed is greater than the target upper",
            "name": "tooMuchTraffic",
            "numberOfAllowedCrossing": 0,
            "thresholdRuleSeverity": "1",
            "appliedConsequence": [
              {
                "appliedAction": "sendWarning",
                "description": "send a warning to the system",
                "name": "sendWarning",
                "repeatAction": false
              },
              {
                "appliedAction": "generateReport",
                "description": "generate a waring report ",
                "name": "generateReport",
                "repeatAction": false
              }
            ]
          }
        ]
      }
    ],
    "@type": "PerformanceTest",
    "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Service/PerformanceTest.schema.json",
    "@baseType": "ServiceTest"
}
```


© TM Forum 2021. All Rights Reserved

Response

201


```
{
  "href": "https://mycsp.com:8080/tmf-api/serviceTestManagement/v4/serviceTest/3112",
  "id": "3112",
  "description:": "This service test is a simple one",
  "endDateTime": "2016-03-02T24:00:00",
  "mode": "PROACTIVE",
  "startDateTime": "2016-03-02T00:00:00",
  "state": "Completed",
  "relatedService": {
    "href": "https://mycsp.com:8080/tmf-api/serviceManagement/v4/service/21",
    "id": "21",
    "name": "gameService"
  },
  "testSpecification": {
    "href": "https://mycsp.com:8080/tmf-api/serviceTestManagement/v4/serviceTestSpecification/22",
    "id": "22",
    "@referredType": "PerformanceTestSpecification"
  },
  "characteristic": [
    {
      "name": "number",
      "valueType": "integer",
      "value": 3
    },
    {
      "name": "restart",
      "valueType": "boolean",
      "value": true
    }
  ],
  "testMeasure": [
    {
      "accuracy": 0.0001,
      "captureDateTime": "2016-03-02T11:12:00",
      "captureMethod": "automatic capture by instrument",
      "metricDescription": "This is the flow speed in a service",
      "metricHref": "https://mycsp.com:8080/tmf-api/metricManagement/v4/metric/21",
      "metricName": "Flow speed",
      "unitOfMeasure": "A second",
      "value": {
        "valueType": "bits",
        "value": {
          "@type": "integer",
          "value": 3000
        }
      },
      "ruleViolation": [
        {
          "conformanceTargetUpper": "2500",
          "conformanceComparatorUpper": "greater than ",
```


© TM Forum 2021. All Rights Reserved

```
            "description": "This speed is greater than the target upper",
            "name": "tooMuchTraffic",
            "numberOfAllowedCrossing": 0,
            "thresholdRuleSeverity": "1",
            "appliedConsequence": [
              {
                "appliedAction": "sendWarning",
                "description": "send a warning to the system",
                "name": "sendWarning",
                "repeatAction": false
              },
              {
                "appliedAction": "generateReport",
                "description": "generate a waring report ",
                "name": "generateReport",
                "repeatAction": false
              }
            ]
          }
        ]
      }
  ],
  "@type": "PerformanceTest",
  "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Service/PerformanceTest.schema.json",
  "@baseType": "ServiceTest"
}
```


Patch service test

PATCH /serviceTest/{id}

Description

This operation allows partial updates of a service test entity. Support of json/merge (https://tools.ietf.org/html/rfc7386) is mandatory, support of json/patch (http://tools.ietf.org/html/rfc5789) is optional.

Note: If the update operation yields to the creation of sub-resources or relationships, the same rules concerning mandatory sub-resource attributes and default value settings in the POST operation applies to the PATCH operation. Hence these tables are not repeated here.

Patchable and Non Patchable Attributes

The tables below provide the list of patchable and non patchable attributes, including constraint rules on their usage.

Patchable Attributes	Rule
characteristic	
description	
endDateTime	
mode	

© TM Forum 2021. All Rights Reserved

name	
relatedParty	
relatedService	
startDateTime	
state	
testMeasure	
testSpecification	
validFor	

Non Patchable Attributes	Rule
href	
id	
@type	
@schemaLocation	
@baseType	

Usage Samples

Here's an example of a request for patching a ServiceTest.

Request

```
PATCH /tmf-api/naas/v4/serviceTest/3112
Content-Type: application/merge-patch+json

{
   "description:": "This service test is a simple one",
   "endDateTime": "2016-03-02T24:00:00",
   "mode": "PROACTIVE",
   "startDateTime": "2016-03-02T00:00:00",
   "state": "Completed",
   "relatedService": {
      "href": "https://mycsp.com:8080/tmf-api/serviceManagement/v4/service/21",
      "id": "21",
      "name": "gameService"
   },
   "testSpecification": {
      "href": "https://mycsp.com:8080/tmf-api/serviceTestManagement/v4/serviceTestSpecification/22",
      "id": "22",
      "@referredType": "PerformanceTestSpecification"
   },
   "characteristic": [
      {
         "name": "number",
         "valueType": "integer",
         "value": 3
      },
      {
         "name": "restart",
         "valueType": "boolean",
```

© TM Forum 2021. All Rights Reserved

```
        "value": true
      }
    ],
    "testMeasure": [
      {
        "accuracy": 0.0001,
        "captureDateTime": "2016-03-02T11:12:00",
        "captureMethod": "automatic capture by instrument",
        "metricDescription": "This is the flow speed in a service",
        "metricHref": "https://mycsp.com:8080/tmf-api/metricManagement/v4/metric/21",
        "metricName": "Flow speed",
        "unitOfMeasure": "A second",
        "value": {
          "valueType": "bits",
          "value": {
            "@type": "integer",
            "value": 3000
          }
        },
        "ruleViolation": [
          {
            "conformanceTargetUpper": "2500",
            "conformanceComparatorUpper": "greater than ",
            "description": "This speed is greater than the target upper",
            "name": "tooMuchTraffic",
            "numberOfAllowedCrossing": 0,
            "thresholdRuleSeverity": "1",
            "appliedConsequence": [
              {
                "appliedAction": "sendWarning",
                "description": "send a warning to the system",
                "name": "sendWarning",
                "repeatAction": false
              },
              {
                "appliedAction": "generateReport",
                "description": "generate a waring report ",
                "name": "generateReport",
                "repeatAction": false
              }
            ]
          }
        ]
      }
    ],
    "@type": "PerformanceTest",
    "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Service/PerformanceTest.schema.json",
    "@baseType": "ServiceTest"
}
```


Response

© TM Forum 2021. All Rights Reserved

200


```
{
  "href": "https://mycsp.com:8080/tmf-api/serviceTestManagement/v4/serviceTest/3112",
  "id": "3112",
  "description:": "This service test is a simple one",
  "endDateTime": "2016-03-02T24:00:00",
  "mode": "PROACTIVE",
  "startDateTime": "2016-03-02T00:00:00",
  "state": "Completed",
  "relatedService": {
    "href": "https://mycsp.com:8080/tmf-api/serviceManagement/v4/service/21",
    "id": "21",
    "name": "gameService"
  },
  "testSpecification": {
    "href": "https://mycsp.com:8080/tmf-api/serviceTestManagement/v4/serviceTestSpecification/22",
    "id": "22",
    "@referredType": "PerformanceTestSpecification"
  },
  "characteristic": [
    {
      "name": "number",
      "valueType": "integer",
      "value": 3
    },
    {
      "name": "restart",
      "valueType": "boolean",
      "value": true
    }
  ],
  "testMeasure": [
    {
      "accuracy": 0.0001,
      "captureDateTime": "2016-03-02T11:12:00",
      "captureMethod": "automatic capture by instrument",
      "metricDescription": "This is the flow speed in a service",
      "metricHref": "https://mycsp.com:8080/tmf-api/metricManagement/v4/metric/21",
      "metricName": "Flow speed",
      "unitOfMeasure": "A second",
      "value": {
        "valueType": "bits",
        "value": {
          "@type": "integer",
          "value": 3000
        }
      },
      "ruleViolation": [
        {
          "conformanceTargetUpper": "2500",
          "conformanceComparatorUpper": "greater than ",
          "description": "This speed is greater than the target upper",
          "name": "tooMuchTraffic",
          "numberOfAllowedCrossing": 0,
```


© TM Forum 2021. All Rights Reserved

```
            "thresholdRuleSeverity": "1",
            "appliedConsequence": [
              {
                "appliedAction": "sendWarning",
                "description": "send a warning to the system",
                "name": "sendWarning",
                "repeatAction": false
              },
              {
                "appliedAction": "generateReport",
                "description": "generate a waring report ",
                "name": "generateReport",
                "repeatAction": false
              }
            ]
          }
        ]
      }
  ],
  "@type": "PerformanceTest",
  "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/Service/PerformanceTest.schema.json",
  "@baseType": "ServiceTest"
}
```


Delete service test

`DELETE /serviceTest/{id}`

Description

This operation deletes a service test entity.

Usage Samples

Here's an example of a request for patching a ServiceTest.

Request
DELETE /tmf-api/naas/v4/serviceTest/3112
Response
204

Operations on Service Test Specification

© TM Forum 2021. All Rights Reserved

List service test specifications

GET /serviceTestSpecification?fields=...&{filtering}

Description

This operation list service test specification entities.
Attribute selection is enabled for all first level attributes.
Filtering may be available depending on the compliance level supported by an implementation.

Usage Samples

Here's an example of a request for retrieving multiple ServiceTestSpecification.

Request

```
GET /tmf-api/naas/v4/serviceTestSpecification
Accept: application/json
```

Response

```
200

[
  {
    "description": "a service description",
    "href": "https://mycsp.com:8080/tmf-api/serviceTestManagement/v4/serviceTestSpecification/12",
    "id": "12",
    "version": "1.0",
    "name": "monkey test",
    "validFor": {
      "endDateTime": "2017-03-01T00:00:00",
      "startDateTime": "2016-03-01T00:00:00"
    },
    "testMeasureDefinition": [
      {
        "captureFrequency": "once",
        "captureMethod": "automatic capture by instrument",
        "metricDescription": "This is the flow speed in a service",
        "metricHref": "https://mycsp.com:8080/tmf-api/metricManagement/v4/metric/21",
        "metricName": "Flow speed",
        "name": "Definition of Flow speed",
        "unitOfMeasure": "A second",
        "valueType": "bits",
        "capturePeriod": {
          "amount": "5",
          "units": "seconds"
        },
```

© TM Forum 2021. All Rights Reserved

```
            "thresholdRule": [
              {
                "conformanceTargetUpper": "2500",
                "conformanceComparatorUpper": "greater than ",
                "description": "This speed is greater than the target upper",
                "name": "tooMuchTraffic",
                "numberOfAllowedCrossing": 0,
                "thresholdRuleSeverity": "1",
                "tolerancePeriod": {
                  "amount": "15",
                  "units": "minutes"
                },
                "consequence": [
                  {
                    "prescribeAction": "sendWarning",
                    "description": "send a warning to the system",
                    "name": "sendWarning",
                    "repeatAction": false,
                    "validFor": {
                      "endDateTime": "2017-03-01T00:00:00",
                      "startDateTime": "2016-03-01T00:00:00"
                    }
                  },
                  {
                    "prescribeAction": "generateReport",
                    "description": "generate a waring report ",
                    "name": "generateReport",
                    "repeatAction": false,
                    "validFor": {
                      "endDateTime": "2017-03-01T00:00:00",
                      "startDateTime": "2016-03-01T00:00:00"
                    }
                  }
                ]
              }
            ],
            "validFor": {
              "endDateTime": "2017-03-01T00:00:00",
              "startDateTime": "2016-03-01T00:00:00"
            }
          }
        ],
        "relatedServiceSpecification": {
          "href": "https://mycsp.com:8080/tmf-api/serviceCatalogManagement/v4/serviceSpecification/31",
          "id": "31",
          "@referredType": "vCPEServiceSpecification"
        },
        "@baseType": "ServiceTestSpecification",
        "@type": "PerformanceTestSpecification",
        "@schemaLocation": "https://mycsp.com:8080/tmf-
api/schema/service/PerformanceTestSpecification.schema.json"
      }
]
```


© TM Forum 2021. All Rights Reserved

Retrieve service test specification

GET /serviceTestSpecification/{id}?fields=...&{filtering}

Description

This operation retrieves a service test specification entity.
Attribute selection is enabled for all first level attributes.
Filtering on sub-resources may be available depending on the compliance level supported by an implementation.

Usage Samples

Here's an example of a request for retrieving a specific ServiceTestSpecification.

Request

```
GET /tmf-api/naas/v4/serviceTestSpecification/12
Accept: application/json
```

Response

```
200

{
  "description": "a service description",
  "href": "https://mycsp.com:8080/tmf-api/serviceTestManagement/v4/serviceTestSpecification/12",
  "id": "12",
  "version": "1.0",
  "name": "monkey test",
  "validFor": {
    "endDateTime": "2017-03-01T00:00:00",
    "startDateTime": "2016-03-01T00:00:00"
  },
  "testMeasureDefinition": [
    {
      "captureFrequency": "once",
      "captureMethod": "automatic capture by instrument",
      "metricDescription": "This is the flow speed in a service",
      "metricHref": "https://mycsp.com:8080/tmf-api/metricManagement/v4/metric/21",
      "metricName": "Flow speed",
      "name": "Definition of Flow speed",
      "unitOfMeasure": "A second",
      "valueType": "bits",
      "capturePeriod": {
        "amount": "5",
        "units": "seconds"
      },
      "thresholdRule": [
```

© TM Forum 2021. All Rights Reserved

```
          {
            "conformanceTargetUpper": "2500",
            "conformanceComparatorUpper": "greater than ",
            "description": "This speed is greater than the target upper",
            "name": "tooMuchTraffic",
            "numberOfAllowedCrossing": 0,
            "thresholdRuleSeverity": "1",
            "tolerancePeriod": {
              "amount": "15",
              "units": "minutes"
            },
            "consequence": [
              {
                "prescribeAction": "sendWarning",
                "description": "send a warning to the system",
                "name": "sendWarning",
                "repeatAction": false,
                "validFor": {
                  "endDateTime": "2017-03-01T00:00:00",
                  "startDateTime": "2016-03-01T00:00:00"
                }
              },
              {
                "prescribeAction": "generateReport",
                "description": "generate a waring report ",
                "name": "generateReport",
                "repeatAction": false,
                "validFor": {
                  "endDateTime": "2017-03-01T00:00:00",
                  "startDateTime": "2016-03-01T00:00:00"
                }
              }
            ]
          }
        ],
        "validFor": {
          "endDateTime": "2017-03-01T00:00:00",
          "startDateTime": "2016-03-01T00:00:00"
        }
      }
    ],
    "relatedServiceSpecification": {
      "href": "https://mycsp.com:8080/tmf-api/serviceCatalogManagement/v4/serviceSpecification/31",
      "id": "31",
      "@referredType": "vCPEServiceSpecification"
    },
    "@baseType": "ServiceTestSpecification",
    "@type": "PerformanceTestSpecification",
    "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/service/PerformanceTestSpecification.schema.json"
}
```


Create service test specification

POST /serviceTestSpecification

© TM Forum 2021. All Rights Reserved

Description

This operation creates a service test specification entity.

Mandatory and Non Mandatory Attributes

The following tables provide the list of mandatory and non mandatory attributes when creating a ServiceTestSpecification, including any possible rule conditions and applicable default values. Notice that it is up to an implementer to add additional mandatory attributes.

Mandatory Attributes	Rule
name	
relatedServiceSpecification	

Non Mandatory Attributes	Rule
description	
isBundle	
lastUpdate	
lifecycleStatus	
version	
@baseType	
@schemaLocation	
@type	
description	
serviceTestSpecRelationship	
testMeasureDefinition	
attachment	
constraint	
entitySpecRelationship	
relatedParty	
specCharacteristic	
targetEntitySchema	
validFor	

Usage Samples

Here's an example of a request for creating a specific ServiceTestSpecification.

Request

```
POST /tmf-api/naas/v4/serviceTestSpecification
Content-Type: application/json

{
   "description": "a service description",
   "version": "1.0",
   "name": "monkey test",
   "validFor": {
```

© TM Forum 2021. All Rights Reserved

```
    "endDateTime": "2017-03-01T00:00:00",
    "startDateTime": "2016-03-01T00:00:00"
  },
  "testMeasureDefinition": [
    {
      "captureFrequency": "once",
      "captureMethod": "automatic capture by instrument",
      "metricDescription": "This is the flow speed in a service",
      "metricHref": "https://mycsp.com:8080/tmf-api/metricManagement/v4/metric/21",
      "metricName": "Flow speed",
      "name": "Definition of Flow speed",
      "unitOfMeasure": "A second",
      "valueType": "bits",
      "capturePeriod": {
        "amount": "5",
        "units": "seconds"
      },
      "thresholdRule": [
        {
          "conformanceTargetUpper": "2500",
          "conformanceComparatorUpper": "greater than ",
          "description": "This speed is greater than the target upper",
          "name": "tooMuchTraffic",
          "numberOfAllowedCrossing": 0,
          "thresholdRuleSeverity": "1",
          "tolerancePeriod": {
            "amount": "15",
            "units": "minutes"
          },
          "consequence": [
            {
              "prescribeAction": "sendWarning",
              "description": "send a warning to the system",
              "name": "sendWarning",
              "repeatAction": false,
              "validFor": {
                "endDateTime": "2017-03-01T00:00:00",
                "startDateTime": "2016-03-01T00:00:00"
              }
            },
            {
              "prescribeAction": "generateReport",
              "description": "generate a waring report ",
              "name": "generateReport",
              "repeatAction": false,
              "validFor": {
                "endDateTime": "2017-03-01T00:00:00",
                "startDateTime": "2016-03-01T00:00:00"
              }
            }
          ]
        }
      ],
      "validFor": {
        "endDateTime": "2017-03-01T00:00:00",
```


© TM Forum 2021. All Rights Reserved

```
            "startDateTime": "2016-03-01T00:00:00"
          }
        }
    ],
    "relatedServiceSpecification": {
      "href": "https://mycsp.com:8080/tmf-api/serviceCatalogManagement/v4/serviceSpecification/31",
      "id": "31",
      "@referredType": "vCPEServiceSpecification"
    },
    "@baseType": "ServiceTestSpecification",
    "@type": "PerformanceTestSpecification",
    "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/service/PerformanceTestSpecification.schema.json"
}
```


Response


```
201

{
    "description": "a service description",
    "href": "https://mycsp.com:8080/tmf-api/serviceTestManagement/v4/serviceTestSpecification/12",
    "id": "12",
    "version": "1.0",
    "name": "monkey test",
    "validFor": {
      "endDateTime": "2017-03-01T00:00:00",
      "startDateTime": "2016-03-01T00:00:00"
    },
    "testMeasureDefinition": [
      {
        "captureFrequency": "once",
        "captureMethod": "automatic capture by instrument",
        "metricDescription": "This is the flow speed in a service",
        "metricHref": "https://mycsp.com:8080/tmf-api/metricManagement/v4/metric/21",
        "metricName": "Flow speed",
        "name": "Definition of Flow speed",
        "unitOfMeasure": "A second",
        "valueType": "bits",
        "capturePeriod": {
          "amount": "5",
          "units": "seconds"
        },
        "thresholdRule": [
          {
            "conformanceTargetUpper": "2500",
            "conformanceComparatorUpper": "greater than ",
            "description": "This speed is greater than the target upper",
            "name": "tooMuchTraffic",
            "numberOfAllowedCrossing": 0,
            "thresholdRuleSeverity": "1",
            "tolerancePeriod": {
              "amount": "15",
              "units": "minutes"
```


© TM Forum 2021. All Rights Reserved

```
            },
            "consequence": [
              {
                "prescribeAction": "sendWarning",
                "description": "send a warning to the system",
                "name": "sendWarning",
                "repeatAction": false,
                "validFor": {
                  "endDateTime": "2017-03-01T00:00:00",
                  "startDateTime": "2016-03-01T00:00:00"
                }
              },
              {
                "prescribeAction": "generateReport",
                "description": "generate a waring report ",
                "name": "generateReport",
                "repeatAction": false,
                "validFor": {
                  "endDateTime": "2017-03-01T00:00:00",
                  "startDateTime": "2016-03-01T00:00:00"
                }
              }
            ]
          }
        ],
        "validFor": {
          "endDateTime": "2017-03-01T00:00:00",
          "startDateTime": "2016-03-01T00:00:00"
        }
      }
  ],
  "relatedServiceSpecification": {
    "href": "https://mycsp.com:8080/tmf-api/serviceCatalogManagement/v4/serviceSpecification/31",
    "id": "31",
    "@referredType": "vCPEServiceSpecification"
  },
  "@baseType": "ServiceTestSpecification",
  "@type": "PerformanceTestSpecification",
  "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/service/PerformanceTestSpecification.schema.json"
}
```


Patch service test specification

`PATCH /serviceTestSpecification/{id}`

Description

This operation allows partial updates of a service test specification entity. Support of json/merge (https://tools.ietf.org/html/rfc7386) is mandatory, support of json/patch (http://tools.ietf.org/html/rfc5789) is optional.

Note: If the update operation yields to the creation of sub-resources or relationships, the same rules concerning

© TM Forum 2021. All Rights Reserved

mandatory sub-resource attributes and default value settings in the POST operation applies to the PATCH operation. Hence these tables are not repeated here.

Patchable and Non Patchable Attributes

The tables below provide the list of patchable and non patchable attributes, including constraint rules on their usage.

Patchable Attributes	Rule
description	
isBundle	
lastUpdate	
lifecycleStatus	
name	
version	
@baseType	
@schemaLocation	
@type	
description	
relatedServiceSpecification	
serviceTestSpecRelationship	
testMeasureDefinition	
attachment	
constraint	
entitySpecRelationship	
relatedParty	
specCharacteristic	
targetEntitySchema	

Non Patchable Attributes	Rule
href	
id	
validFor	

Usage Samples

Here's an example of a request for patching a ServiceTestSpecification.

Request

```
PATCH /tmf-api/naas/v4/serviceTestSpecification/12
Content-Type: application/merge-patch+json

{
   "description": "a service description",
   "version": "1.0",
   "name": "monkey test",
   "validFor": {
```

© TM Forum 2021. All Rights Reserved

```
    "endDateTime": "2017-03-01T00:00:00",
    "startDateTime": "2016-03-01T00:00:00"
  },
  "testMeasureDefinition": [
    {
      "captureFrequency": "once",
      "captureMethod": "automatic capture by instrument",
      "metricDescription": "This is the flow speed in a service",
      "metricHref": "https://mycsp.com:8080/tmf-api/metricManagement/v4/metric/21",
      "metricName": "Flow speed",
      "name": "Definition of Flow speed",
      "unitOfMeasure": "A second",
      "valueType": "bits",
      "capturePeriod": {
        "amount": "5",
        "units": "seconds"
      },
      "thresholdRule": [
        {
          "conformanceTargetUpper": "2500",
          "conformanceComparatorUpper": "greater than ",
          "description": "This speed is greater than the target upper",
          "name": "tooMuchTraffic",
          "numberOfAllowedCrossing": 0,
          "thresholdRuleSeverity": "1",
          "tolerancePeriod": {
            "amount": "15",
            "units": "minutes"
          },
          "consequence": [
            {
              "prescribeAction": "sendWarning",
              "description": "send a warning to the system",
              "name": "sendWarning",
              "repeatAction": false,
              "validFor": {
                "endDateTime": "2017-03-01T00:00:00",
                "startDateTime": "2016-03-01T00:00:00"
              }
            },
            {
              "prescribeAction": "generateReport",
              "description": "generate a waring report ",
              "name": "generateReport",
              "repeatAction": false,
              "validFor": {
                "endDateTime": "2017-03-01T00:00:00",
                "startDateTime": "2016-03-01T00:00:00"
              }
            }
          ]
        }
      ],
      "validFor": {
        "endDateTime": "2017-03-01T00:00:00",
```


© TM Forum 2021. All Rights Reserved

```
            "startDateTime": "2016-03-01T00:00:00"
          }
        }
    ],
    "relatedServiceSpecification": {
      "href": "https://mycsp.com:8080/tmf-api/serviceCatalogManagement/v4/serviceSpecification/31",
      "id": "31",
      "@referredType": "vCPEServiceSpecification"
    },
    "@baseType": "ServiceTestSpecification",
    "@type": "PerformanceTestSpecification",
    "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/service/PerformanceTestSpecification.schema.json"
}
```


Response

200


```
{
    "description": "a service description",
    "href": "https://mycsp.com:8080/tmf-api/serviceTestManagement/v4/serviceTestSpecification/12",
    "id": "12",
    "version": "1.0",
    "name": "monkey test",
    "validFor": {
      "endDateTime": "2017-03-01T00:00:00",
      "startDateTime": "2016-03-01T00:00:00"
    },
    "testMeasureDefinition": [
      {
        "captureFrequency": "once",
        "captureMethod": "automatic capture by instrument",
        "metricDescription": "This is the flow speed in a service",
        "metricHref": "https://mycsp.com:8080/tmf-api/metricManagement/v4/metric/21",
        "metricName": "Flow speed",
        "name": "Definition of Flow speed",
        "unitOfMeasure": "A second",
        "valueType": "bits",
        "capturePeriod": {
          "amount": "5",
          "units": "seconds"
        },
        "thresholdRule": [
          {
            "conformanceTargetUpper": "2500",
            "conformanceComparatorUpper": "greater than ",
            "description": "This speed is greater than the target upper",
            "name": "tooMuchTraffic",
            "numberOfAllowedCrossing": 0,
            "thresholdRuleSeverity": "1",
            "tolerancePeriod": {
              "amount": "15",
              "units": "minutes"
```


© TM Forum 2021. All Rights Reserved

```
            },
            "consequence": [
              {
                "prescribeAction": "sendWarning",
                "description": "send a warning to the system",
                "name": "sendWarning",
                "repeatAction": false,
                "validFor": {
                  "endDateTime": "2017-03-01T00:00:00",
                  "startDateTime": "2016-03-01T00:00:00"
                }
              },
              {
                "prescribeAction": "generateReport",
                "description": "generate a waring report ",
                "name": "generateReport",
                "repeatAction": false,
                "validFor": {
                  "endDateTime": "2017-03-01T00:00:00",
                  "startDateTime": "2016-03-01T00:00:00"
                }
              }
            ]
          }
        ],
        "validFor": {
          "endDateTime": "2017-03-01T00:00:00",
          "startDateTime": "2016-03-01T00:00:00"
        }
      }
    ],
    "relatedServiceSpecification": {
      "href": "https://mycsp.com:8080/tmf-api/serviceCatalogManagement/v4/serviceSpecification/31",
      "id": "31",
      "@referredType": "vCPEServiceSpecification"
    },
    "@baseType": "ServiceTestSpecification",
    "@type": "PerformanceTestSpecification",
    "@schemaLocation": "https://mycsp.com:8080/tmf-api/schema/service/PerformanceTestSpecification.schema.json"
}
```


Delete service test specification

`DELETE /serviceTestSpecification/{id}`

Description

This operation deletes a service test specification entity.

Usage Samples

© TM Forum 2021. All Rights Reserved

Here's an example of a request for patching a ServiceTestSpecification.

Request
DELETE /tmf-api/naas/v4/serviceTestSpecification/12
Response
204

Operations on Usage Consumption Report Request

List usage consumption report requests

`GET /usageConsumptionReportRequest?fields=...&{filtering}`

Description

This operation list usage consumption report request entities.
Attribute selection is enabled for all first level attributes.
Filtering may be available depending on the compliance level supported by an implementation.

Usage Samples

Here's an example of a request for retrieving multiple UsageConsumptionReportRequest.

Request
GET /tmf-api/naas/v4/usageConsumptionReportRequest Accept: application/json
Response
200 [{ "id": "ucrreq001", "href": "https://serverRoot/tmf-api/usageConsumption/usageConsumptionReportRequest/ucrreq001", "creationDate": "2018-03-15T:15:44:28", "status": "inProgress", "lastUpdate": "2018-03-15T:15:44:28",

© TM Forum 2021. All Rights Reserved

```
    "product": {
      "publicIdentifier ": "33601010101"
    }
  }
]
```

Retrieve usage consumption report request

GET /usageConsumptionReportRequest/{id}?fields=...&{filtering}

Description

This operation retrieves an usage consumption report request entity.
Attribute selection is enabled for all first level attributes.
Filtering on sub-resources may be available depending on the compliance level supported by an implementation.

Usage Samples

Here's an example of a request for retrieving a specific UsageConsumptionReportRequest.

Request

```
GET /tmf-api/naas/v4/usageConsumptionReportRequest/ucrreq001
Accept: application/json
```

Response

```
200

{
  "id": "ucrreq001",
  "href": "https://serverRoot/tmf-api/usageConsumption/usageConsumptionReportRequest/ucrreq001",
  "creationDate": "2018-03-15T:15:44:28",
  "status": "inProgress",
  "lastUpdate": "2018-03-15T:15:44:28",
  "product": {
    "publicIdentifier ": "33601010101"
  }
}
```

Create usage consumption report request

POST /usageConsumptionReportRequest

© TM Forum 2021. All Rights Reserved

Description

This operation creates an usage consumption report request entity.

Mandatory and Non Mandatory Attributes

The following tables provide the list of mandatory and non mandatory attributes when creating a UsageConsumptionReportRequest, including any possible rule conditions and applicable default values. Notice that it is up to an implementer to add additional mandatory attributes.

Mandatory Attributes	Rule

Non Mandatory Attributes	Rule
bucket	
creationDate	
lastUpdate	
product	
relatedParty	
status	
usageConsumptionReport	
validPeriod	

Usage Samples

Here's an example of a request for creating a specific UsageConsumptionReportRequest.

Request

```
POST /tmf-api/naas/v4/usageConsumptionReportRequest
Content-Type: application/json

{
   "creationDate": "2018-03-15T:15:44:28",
   "status": "inProgress",
   "lastUpdate": "2018-03-15T:15:44:28",
   "product": {
      "publicIdentifier ": "33601010101"
   }
}
```

Response

```
201

{
   "id": "ucrreq001",
   "href": "https://serverRoot/tmf-api/usageConsumption/usageConsumptionReportRequest/ucrreq001",
```

© TM Forum 2021. All Rights Reserved

```
  "creationDate": "2018-03-15T:15:44:28",
  "status": "inProgress",
  "lastUpdate": "2018-03-15T:15:44:28",
  "product": {
    "publicIdentifier ": "33601010101"
  }
}
```

Delete usage consumption report request

DELETE /usageConsumptionReportRequest/{id}

Description

This operation deletes an usage consumption report request entity.

Usage Samples

Here's an example of a request for patching a UsageConsumptionReportRequest.

Request
DELETE /tmf-api/naas/v4/usageConsumptionReportRequest/ucrreq001
Response
204

Operations on Usage Consumption Report

List usage consumption reports

GET /usageConsumptionReport?fields=...&{filtering}

Description

This operation list usage consumption report entities.
Attribute selection is enabled for all first level attributes.
Filtering may be available depending on the compliance level supported by an implementation.

Usage Samples

© TM Forum 2021. All Rights Reserved

Here's an example of a request for retrieving multiple UsageConsumptionReport.

Request

```
GET /tmf-api/naas/v4/usageConsumptionReport
Accept: application/json
```

Response

```
200

[
  {
    "id": "ucr0001",
    "href": "https://serverRoot/tmf-api/usageConsumption/usageConsumptionReport/ucr0001",
    "name": "Usage consumption report ucr0001",
    "description": "Usage consumption Report for Kate smartphone",
    "effectiveDate": "2018-03-15T:15:44:28",
    "bucket": [
      {
        "id": "bkt001",
        "name": "Main offer - data",
        "usageType": "data",
        "isShared": false,
        "product": [
          {
            "id": "product1",
            "name": "Kate smartphone",
            "publicIdentifier": "33601010101",
            "user": [
              {
                "id": "usr1",
                "name": "Kate",
                "role": "user",
                "@referredType": "individual"
              }
            ],
            "outOfBucketCounter": [
              {
                "counterType": "outOfBucket",
                "level": "global",
                "value": {
                  "amount": 20.0,
                  "units": "USD"
                },
                "valueName": "20 USD",
                "consumptionPeriod": {
                  "startDateTime": "2018-03-01T:00:00:00",
                  "endDateTime": "2018-03-15T:00:00:00"
                }
              }
            ]
```

© TM Forum 2021. All Rights Reserved

```
        }
      ],
      "bucketBalance": [
        {
          "remainingValue": {
            "amount": 1.8,
            "units": "Go"
          },
          "remainingValueName": "it remains 1.8 Go",
          "validFor": {
            "startDateTime": "2018-03-15T:00:00:00",
            "endDateTime": "2018-03-30T:00:00:00"
          }
        }
      ],
      "bucketCounter": [
        {
          "counterType": "used",
          "level": "global",
          "value": {
            "amount": 1.2,
            "units": "Go"
          },
          "valueName": "1.2 Go used",
          "consumptionPeriod": {
            "startDateTime": "2018-03-01T:00:00:00",
            "endDateTime": "2018-03-15T:00:00:00"
          }
        }
      ]
    },
    {
      "id": "bkt002",
      "name": "Main offer - national voice",
      "usageType": "national voice",
      "isShared": false,
      "product": [
        {
          "id": "product1",
          "name": "Kate smartphone",
          "publicIdentifier": "33601010101",
          "user": [
            {
              "id": "usr1",
              "name": "Kate",
              "role": "user",
              "@referredType": "individual"
            }
          ]
        }
      ],
      "bucketBalance": [
        {
          "remainingValue": {
            "amount": 80.0,
```


© TM Forum 2021. All Rights Reserved

```
            "units": "mins"
          },
          "remainingValueName": "it remains 80 minutes",
          "validFor": {
            "startDateTime": "2018-03-15T:00:00:00",
            "endDateTime": "2018-03-30T:00:00:00"
          }
        }
      ],
      "bucketCounter": [
        {
          "counterType": "used",
          "level": "global",
          "value": {
            "amount": 40.0,
            "units": "mins"
          },
          "valueName": "40 mins used",
          "consumptionPeriod": {
            "startDateTime": "2018-03-01T:00:00:00",
            "endDateTime": "2018-03-15T:00:00:00"
          }
        }
      ]
    },
    {
      "id": "bkt003",
      "name": "Main offer - sms",
      "usageType": "sms",
      "isShared": false,
      "product": [
        {
          "id": "product1",
          "name": "Kate smartphone",
          "publicIdentifier": "33601010101",
          "user": [
            {
              "id": "usr1",
              "name": "Kate",
              "role": "user",
              "@referredType": "individual"
            }
          ]
        }
      ],
      "bucketBalance": [
        {
          "remainingValue": {
            "amount": 95.0,
            "units": "sms"
          },
          "remainingValueName": "it remains 95 sms",
          "validFor": {
            "startDateTime": "2018-03-15T:00:00:00",
            "endDateTime": "2018-03-30T:00:00:00"
```


© TM Forum 2021. All Rights Reserved

```
            }
          }
        ],
        "bucketCounter": [
          {
            "counterType": "used",
            "level": "global",
            "value": {
              "amount": 25.0,
              "units": "sms"
            },
            "valueName": "25 sms used",
            "consumptionPeriod": {
              "startDateTime": "2018-03-01T:00:00:00",
              "endDateTime": "2018-03-15T:00:00:00"
            }
          }
        ]
      },
      {
        "id": "bkt004",
        "name": "Option Canada/USA - voice",
        "usageType": "voice",
        "isShared": false,
        "product": [
          {
            "id": "product1",
            "name": "Kate smartphone",
            "publicIdentifier": "33601010101",
            "user": [
              {
                "id": "usr1",
                "name": "Kate",
                "role": "user",
                "@referredType": "individual"
              }
            ]
          }
        ],
        "bucketBalance": [
          {
            "remainingValue": {
              "amount": 10.0,
              "units": "mins"
            },
            "remainingValueName": "it remains 10 minutes",
            "validFor": {
              "startDateTime": "2018-03-15T:00:00:00",
              "endDateTime": "2018-03-30T:00:00:00"
            }
          }
        ],
        "bucketCounter": [
          {
            "counterType": "used",
```


© TM Forum 2021. All Rights Reserved

```
            "level": "global",
            "value": {
               "amount": 20.0,
               "units": "mins"
            },
            "valueName": "20 mins used",
            "consumptionPeriod": {
               "startDateTime": "2018-03-01T:00:00:00",
               "endDateTime": "2018-03-15T:00:00:00"
            }
         }
      ]
   },
   {
      "id": "bkt005",
      "name": "Option Canada/USA - sms",
      "usageType": "sms",
      "isShared": false,
      "product": [
         {
            "id": "product1",
            "name": "Kate smartphone",
            "publicIdentifier": "33601010101",
            "user": [
               {
                  "id": "usr1",
                  "name": "Kate",
                  "role": "user",
                  "@referredType": "individual"
               }
            ]
         }
      ],
      "bucketBalance": [
         {
            "remainingValue": {
               "amount": 0.0,
               "units": "sms"
            },
            "remainingValueName": "Sms bucket exhausted",
            "validFor": {
               "startDateTime": "2018-03-15T:00:00:00",
               "endDateTime": "2018-03-30T:00:00:00"
            }
         }
      ],
      "bucketCounter": [
         {
            "counterType": "used",
            "level": "global",
            "value": {
               "amount": 10.0,
               "units": "sms"
            },
            "valueName": "10 sms used",
```


© TM Forum 2021. All Rights Reserved

```
            "consumptionPeriod": {
               "startDateTime": "2018-03-01T:00:00:00",
               "endDateTime": "2018-03-15T:00:00:00"
            }
         }
      ]
   }
 ]
}
]
```


Retrieve usage consumption report

`GET /usageConsumptionReport/{id}?fields=...&{filtering}`

Description

This operation retrieves an usage consumption report entity.
Attribute selection is enabled for all first level attributes.
Filtering on sub-resources may be available depending on the compliance level supported by an implementation.

Usage Samples

Here's an example of a request for retrieving a specific UsageConsumptionReport.

Request

```
GET /tmf-api/naas/v4/usageConsumptionReport/ucr0001
Accept: application/json
```

Response


```
200

{
  "id": "ucr0001",
  "href": "https://serverRoot/tmf-api/usageConsumption/usageConsumptionReport/ucr0001",
  "name": "Usage consumption report ucr0001",
  "description": "Usage consumption Report for Kate smartphone",
  "effectiveDate": "2018-03-15T:15:44:28",
  "bucket": [
    {
      "id": "bkt001",
      "name": "Main offer - data",
      "usageType": "data",
      "isShared": false,
      "product": [
```


© TM Forum 2021. All Rights Reserved

```
        {
          "id": "product1",
          "name": "Kate smartphone",
          "publicIdentifier": "33601010101",
          "user": [
            {
              "id": "usr1",
              "name": "Kate",
              "role": "user",
              "@referredType": "individual"
            }
          ],
          "outOfBucketCounter": [
            {
              "counterType": "outOfBucket",
              "level": "global",
              "value": {
                "amount": 20.0,
                "units": "USD"
              },
              "valueName": "20 USD",
              "consumptionPeriod": {
                "startDateTime": "2018-03-01T:00:00:00",
                "endDateTime": "2018-03-15T:00:00:00"
              }
            }
          ]
        }
      ],
      "bucketBalance": [
        {
          "remainingValue": {
            "amount": 1.8,
            "units": "Go"
          },
          "remainingValueName": "it remains 1.8 Go",
          "validFor": {
            "startDateTime": "2018-03-15T:00:00:00",
            "endDateTime": "2018-03-30T:00:00:00"
          }
        }
      ],
      "bucketCounter": [
        {
          "counterType": "used",
          "level": "global",
          "value": {
            "amount": 1.2,
            "units": "Go"
          },
          "valueName": "1.2 Go used",
          "consumptionPeriod": {
            "startDateTime": "2018-03-01T:00:00:00",
            "endDateTime": "2018-03-15T:00:00:00"
          }
```


© TM Forum 2021. All Rights Reserved

```
        }
      ]
    },
    {
      "id": "bkt002",
      "name": "Main offer - national voice",
      "usageType": "national voice",
      "isShared": false,
      "product": [
        {
          "id": "product1",
          "name": "Kate smartphone",
          "publicIdentifier": "33601010101",
          "user": [
            {
              "id": "usr1",
              "name": "Kate",
              "role": "user",
              "@referredType": "individual"
            }
          ]
        }
      ],
      "bucketBalance": [
        {
          "remainingValue": {
            "amount": 80.0,
            "units": "mins"
          },
          "remainingValueName": "it remains 80 minutes",
          "validFor": {
            "startDateTime": "2018-03-15T:00:00:00",
            "endDateTime": "2018-03-30T:00:00:00"
          }
        }
      ],
      "bucketCounter": [
        {
          "counterType": "used",
          "level": "global",
          "value": {
            "amount": 40.0,
            "units": "mins"
          },
          "valueName": "40 mins used",
          "consumptionPeriod": {
            "startDateTime": "2018-03-01T:00:00:00",
            "endDateTime": "2018-03-15T:00:00:00"
          }
        }
      ]
    },
    {
      "id": "bkt003",
      "name": "Main offer - sms",
```


© TM Forum 2021. All Rights Reserved

```
    "usageType": "sms",
    "isShared": false,
    "product": [
      {
        "id": "product1",
        "name": "Kate smartphone",
        "publicIdentifier": "33601010101",
        "user": [
          {
            "id": "usr1",
            "name": "Kate",
            "role": "user",
            "@referredType": "individual"
          }
        ]
      }
    ],
    "bucketBalance": [
      {
        "remainingValue": {
          "amount": 95.0,
          "units": "sms"
        },
        "remainingValueName": "it remains 95 sms",
        "validFor": {
          "startDateTime": "2018-03-15T:00:00:00",
          "endDateTime": "2018-03-30T:00:00:00"
        }
      }
    ],
    "bucketCounter": [
      {
        "counterType": "used",
        "level": "global",
        "value": {
          "amount": 25.0,
          "units": "sms"
        },
        "valueName": "25 sms used",
        "consumptionPeriod": {
          "startDateTime": "2018-03-01T:00:00:00",
          "endDateTime": "2018-03-15T:00:00:00"
        }
      }
    ]
  },
  {
    "id": "bkt004",
    "name": "Option Canada/USA - voice",
    "usageType": "voice",
    "isShared": false,
    "product": [
      {
        "id": "product1",
        "name": "Kate smartphone",
```


© TM Forum 2021. All Rights Reserved

```
            "publicIdentifier": "33601010101",
            "user": [
              {
                "id": "usr1",
                "name": "Kate",
                "role": "user",
                "@referredType": "individual"
              }
            ]
          }
        ],
        "bucketBalance": [
          {
            "remainingValue": {
              "amount": 10.0,
              "units": "mins"
            },
            "remainingValueName": "it remains 10 minutes",
            "validFor": {
              "startDateTime": "2018-03-15T:00:00:00",
              "endDateTime": "2018-03-30T:00:00:00"
            }
          }
        ],
        "bucketCounter": [
          {
            "counterType": "used",
            "level": "global",
            "value": {
              "amount": 20.0,
              "units": "mins"
            },
            "valueName": "20 mins used",
            "consumptionPeriod": {
              "startDateTime": "2018-03-01T:00:00:00",
              "endDateTime": "2018-03-15T:00:00:00"
            }
          }
        ]
      },
      {
        "id": "bkt005",
        "name": "Option Canada/USA - sms",
        "usageType": "sms",
        "isShared": false,
        "product": [
          {
            "id": "product1",
            "name": "Kate smartphone",
            "publicIdentifier": "33601010101",
            "user": [
              {
                "id": "usr1",
                "name": "Kate",
                "role": "user",
```


© TM Forum 2021. All Rights Reserved

```
                "@referredType": "individual"
              }
            ]
          }
        ],
        "bucketBalance": [
          {
            "remainingValue": {
              "amount": 0.0,
              "units": "sms"
            },
            "remainingValueName": "Sms bucket exhausted",
            "validFor": {
              "startDateTime": "2018-03-15T:00:00:00",
              "endDateTime": "2018-03-30T:00:00:00"
            }
          }
        ],
        "bucketCounter": [
          {
            "counterType": "used",
            "level": "global",
            "value": {
              "amount": 10.0,
              "units": "sms"
            },
            "valueName": "10 sms used",
            "consumptionPeriod": {
              "startDateTime": "2018-03-01T:00:00:00",
              "endDateTime": "2018-03-15T:00:00:00"
            }
          }
        ]
      }
    ]
}
```


Delete usage consumption report

`DELETE /usageConsumptionReport/{id}`

Description

This operation deletes an usage consumption report entity.

Usage Samples

Here's an example of a request for patching a UsageConsumptionReport.

© TM Forum 2021. All Rights Reserved

Request
DELETE /tmf-api/naas/v4/usageConsumptionReport/ucr0001
Response
204

© TM Forum 2021. All Rights Reserved

API NOTIFICATIONS

For every single of operation on the entities use the following templates and provide sample REST notification POST calls.

It is assumed that the Pub/Sub uses the Register and UnRegister mechanisms described in the REST Guidelines part 1. Refer to the guidelines for more details.

Register listener

`POST /hub`

Description

Sets the communication endpoint address the service instance must use to deliver information about its health state, execution state, failures and metrics. Subsequent POST calls will be rejected by the service if it does not support multiple listeners. In this case DELETE /api/hub/{id} must be called before an endpoint can be created again.

Behavior

Returns HTTP/1.1 status code 204 if the request was successful.

Returns HTTP/1.1 status code 409 if request is not successful.

Usage Samples

Here's an example of a request for registering a listener.

Request
POST /api/hub Accept: application/json {"callback": "http://in.listener.com"}
Response
201 Content-Type: application/json Location: /api/hub/42 {"id":"42","callback":"http://in.listener.com","query":null}

© TM Forum 2021. All Rights Reserved

© TM Forum 2021. All Rights Reserved

Unregister listener

DELETE /hub/{id}

Description

Clears the communication endpoint address that was set by creating the Hub.

Behavior

Returns HTTP/1.1 status code 204 if the request was successful.

Returns HTTP/1.1 status code 404 if the resource is not found.

Usage Samples

Here's an example of a request for un-registering a listener.

Request
DELETE /api/hub/42 Accept: application/json
Response
204

Publish Event to listener

POST /client/listener

Description

Clears the communication endpoint address that was set by creating the Hub.

Provides to a registered listener the description of the event that was raised. The /client/listener url is the callback url passed when registering the listener.

Behavior

Returns HTTP/1.1 status code 201 if the service is able to set the configuration.

Usage Samples

© TM Forum 2021. All Rights Reserved

Here's an example of a notification received by the listener. In this example “EVENT TYPE” should be replaced by one of the notification types supported by this API (see Notification resources Models section) and EVENT BODY refers to the data structure of the given notification type.

Request

```
POST /client/listener
Accept: application/json

{
   "event": {
            EVENT BODY
         },
   "eventType": "EVENT_TYPE"
}
```

Response

201

For detailed examples on the general TM Forum notification mechanism, see the TMF REST Design Guidelines Part 1 document.

© TM Forum 2021. All Rights Reserved

Acknowledgements

Version History

Release Number	Date	Release led by:	Description
Release 4.0	October 2021	Johanne Mayer	Updated user guide to be generated by the TMF tool

Release History

Release Number	Date	Release led by:	Description
21.5.0	October 2021	Johanne Mayer	New template for 4.0

Contributors to Document

Johanne Mayer	MayerConsult Inc.

© TM Forum 2021. All Rights Reserved